CreaseBench: question booklet

185 questions on flat-folded origami: reading crease patterns, folding, tree theory, flap design and defect finding. Each question is exactly the prompt a model receives. The model conventions are printed once below and apply to every question of that tier.

How to answer. Give the JSON answer each question asks for. Answers are graded by actually folding them, so partial sketches score nothing. For large inputs, use the files in `inputs/Qnnn/`; any tool you like is allowed (ruler, compass, paper, a computer) unless you are timing a closed-book attempt.

split	questions	n	contents
expert	Q001–Q034	34	Headline split, graded pass/fail. Expert-level design, image-only tree reading and inverse folding.
expert-analysis	Q035–Q045	11	Pass/fail, reported outside the headline: flap regions and repair of global conflicts.
hard	Q046–Q092	47	Design targets with at least 6 flaps; analysis of patterns with at least 100 creases.
core	Q093–Q136	44	The rest of the advanced tier.
warmup	Q137–Q185	49	The basic tier: small folds and simple crease patterns.

Conventions: advanced tier

You are taking CreaseBench, a test of spatial and analytical reasoning about flat-folded origami.

Conventions (all tasks):

- The paper is the unit square [0,1] x [0,1], seen from its top side. Lengths are in units of the paper side.
- Crease patterns (CPs) use the FOLD format: `vertices_coords`, `edges_vertices`, `edges_assignment` ("M" mountain, "V" valley, "B" paper boundary, "U" folded but direction unspecified) and `faces_vertices` (counter-clockwise vertex loops; face indices refer to this list).
- Valley fold: the paper on both sides of the crease rises toward the viewer, so the moving part ends up ON TOP. Mountain fold: the opposite (it ends up underneath).
- A crease pattern describes a flat-folded state in which EVERY crease is folded completely flat (by 180 degrees) and the paper is otherwise unbent. No paper may pass through other paper.
- Folded coordinates: report them as seen from above, without mirroring. Any position and rotation is fine (answers are aligned rigidly before grading), but the scale must be true.
- Layer order: a pair [f, g] in `face_above` means face f lies above face g (closer to the viewer) where they overlap, in the frame of your reported coordinates.
- Give your final answer as a single JSON object in a ```json fenced code block at the end of your reply. Numbers should have at least 6 significant digits.

Tree theory conventions (advanced tasks):

- A flat-folded base is described by its stick-figure TREE: project the folded model onto its axis (the line its flaps lie along) and contract every connected piece of paper at the same axis coordinate to a point. Leaves are flap tips; edge lengths are distances along the axis, in units of the paper side.
- A tree is given as JSON `{"edges": [[node, node, length], ...]}`; node names are arbitrary labels.
- A design realizes a target tree at SCALE m if its flap tips can be matched to the target's leaves so that for every pair of target leaves (i, j) the tree distance between the matched tips is at least m * d_target(i, j) (Lang's path condition: flaps may be shortened or bundled afterwards). The paper is ONE uncut unit square. Bigger m is better; it is compared with the best known circle/river packing.
- Design systems: box pleating on an N-grid = every crease at 0/45/90/135 degrees with vertices on the 1/N lattice; 22.5-degree design = every crease at a multiple of 22.5 degrees; a Pythagorean stretch = a box-pleated pattern that also uses lattice directions from a Pythagorean triple (a, b, c) (slopes b/a, a/(b+c), b/(a+c), ...) to place flaps that are diagonally offset; grafting = inserting strips of paper into an existing crease pattern.

Conventions: basic tier

You are taking CreaseBench, a test of spatial and analytical reasoning about flat-folded origami.

Conventions (all tasks):

- The paper is the unit square [0,1] x [0,1], seen from its top side. Lengths are in units of the paper side.
- Crease patterns (CPs) use the FOLD format: `vertices_coords`, `edges_vertices`, `edges_assignment` ("M" mountain, "V" valley, "B" paper boundary, "U" folded but direction unspecified) and `faces_vertices` (counter-clockwise vertex loops; face indices refer to this list).
- Valley fold: the paper on both sides of the crease rises toward the viewer, so the moving part ends up ON TOP. Mountain fold: the opposite (it ends up underneath).
- A crease pattern describes a flat-folded state in which EVERY crease is folded completely flat (by 180 degrees) and the paper is otherwise unbent. No paper may pass through other paper.
- Folded coordinates: report them as seen from above, without mirroring. Any position and rotation is fine (answers are aligned rigidly before grading), but the scale must be true.
- Layer order: a pair [f, g] in `face_above` means face f lies above face g (closer to the viewer) where they overlap, in the frame of your reported coordinates.
- Give your final answer as a single JSON object in a ```json fenced code block at the end of your reply. Numbers should have at least 6 significant digits.

Q001

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q001/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a crab: two claws, eight legs, eyes on stalks). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0957510 (one unit of tree length -> 0.0957510 of the paper side).

Input `tree`:

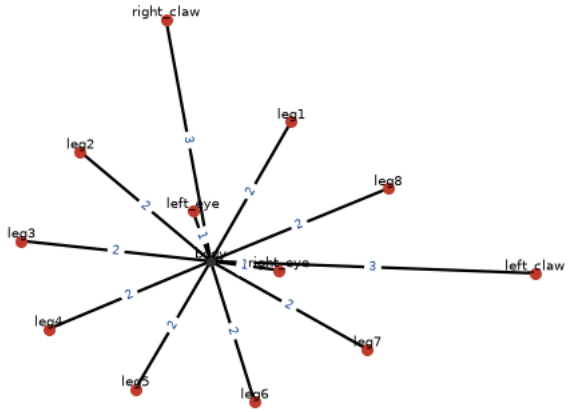

```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["body","leg6",2],["body","leg7",2],["body","leg8",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q002

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q002/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a dragon: head with two horns, two wings, four legs, long tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0687332 (one unit of tree length -> 0.0687332 of the paper side).

Input `tree`:

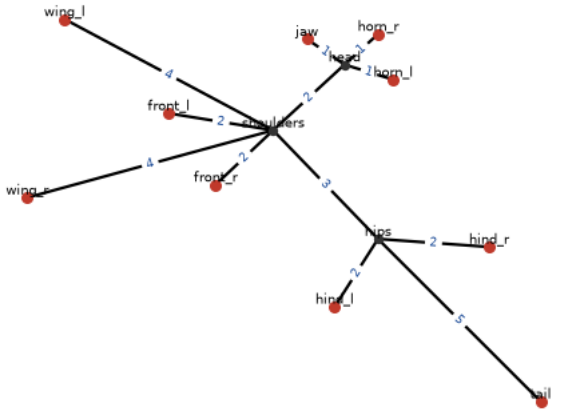

```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulders","wing_l",4],["shoulders","wing_r",4],["shoulder
s","front_l",2],["shoulders","front_r",2],["shoulders","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q003

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q003/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a lobster: two big claws, long antennae, antennules, eight walking legs, a three-lobed tail fan). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0453081 (one unit of tree length -> 0.0453081 of the paper side).

Input `tree`:

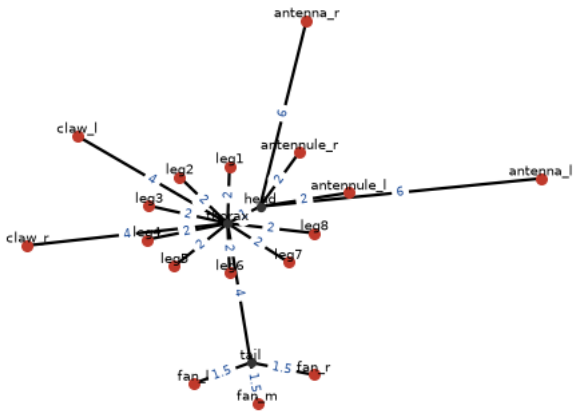

```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thorax","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thorax","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tail","fan_r",1.5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q004

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q004/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a praying mantis: antennae, raptorial forelegs, four walking legs, four wings, abdomen with two cerci). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0443844 (one unit of tree length -> 0.0443844 of the paper side).

Input `tree`:

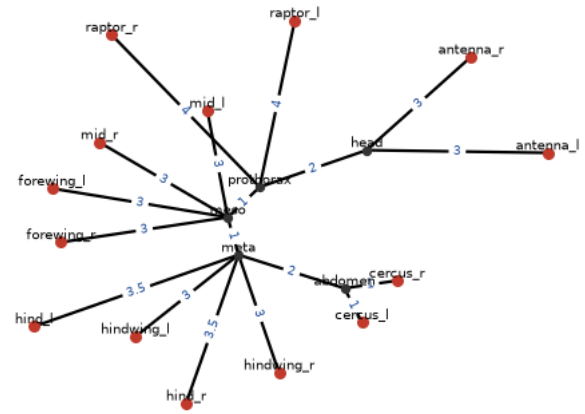

```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["meso","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],["abdomen","cercus_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q005

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q005/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a pegasus: head with two ears and a muzzle, wings with three primary feathers each, four legs, a tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0432756 (one unit of tree length -> 0.0432756 of the paper side).

Input `tree`:

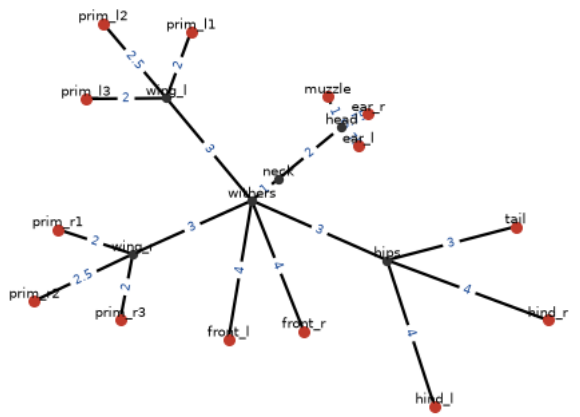

```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["neck","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","prim_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hips","hind_r",4],["hips","tail",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q006

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q006/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a wingless wyrm: a head with two horns, two forelegs, and twelve dorsal scales along a long body ending in a tail fin). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0609262 (one unit of tree length -> 0.0609262 of the paper side).

Input `tree`:

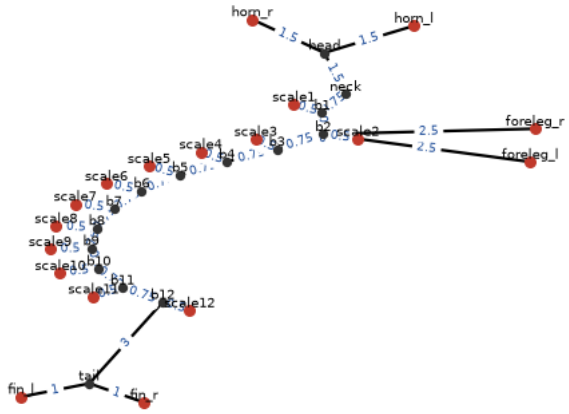

```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b
5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale1",0.5],
["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale
9",0.5],["b10","scale10",0.5],["b11","scale11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tail","fin_l",
1],["tail","fin_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q007

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q007/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a stag beetle: long mandibles, antennae, palps, six legs, two elytra (wings), abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0454431 (one unit of tree length -> 0.0454431 of the paper side).

Input `tree`:

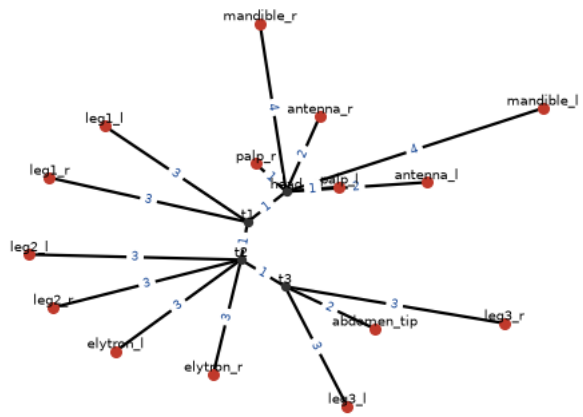

```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q008

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q008/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a tarantula: eight long legs, pedipalps, chelicerae, spinnerets on a big abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0516059 (one unit of tree length -> 0.0516059 of the paper side).

Input `tree`:

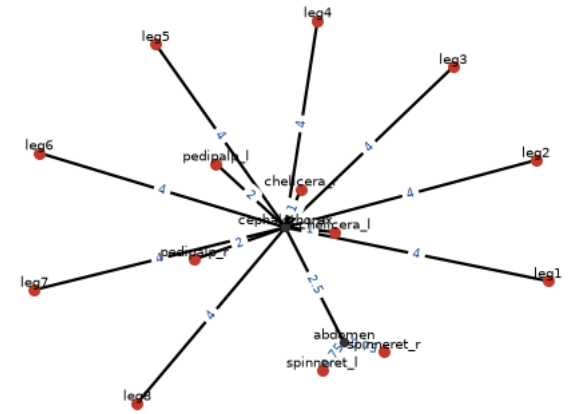

```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q009

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q009/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a crab: two claws, eight legs, eyes on stalks). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0957510 (one unit of tree length -> 0.0957510 of the paper side). In addition, no point of the folded base may have more than 74 layers of paper.

Input `tree`:

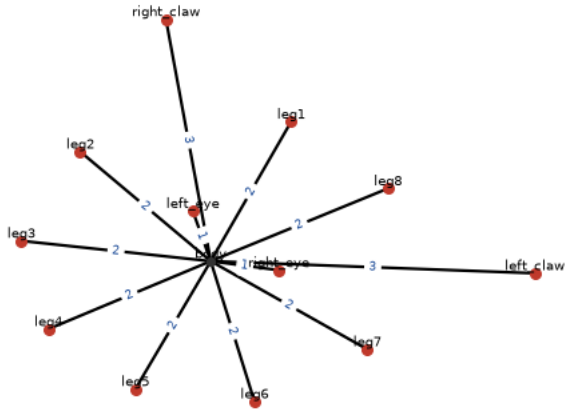

```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["body","leg6",2],["body","leg7",2],["body","leg8",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q010

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q010/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a dragon: head with two horns, two wings, four legs, long tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0687332 (one unit of tree length -> 0.0687332 of the paper side). In addition, no point of the folded base may have more than 36 layers of paper.

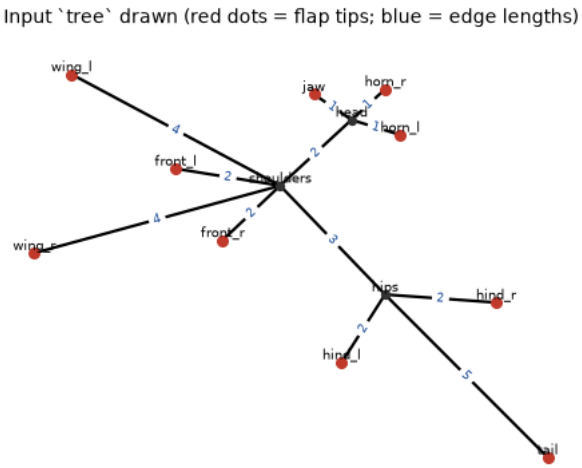
Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulders","wing_l",4],["shoulders","wing_r",4],["shoulder
s","front_l",2],["shoulders","front_r",2],["shoulders","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q011

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q011/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a lobster: two big claws, long antennae, antennules, eight walking legs, a three-lobed tail fan). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0453081 (one unit of tree length -> 0.0453081 of the paper side). In addition, no point of the folded base may have more than 100 layers of paper.

Input `tree`:

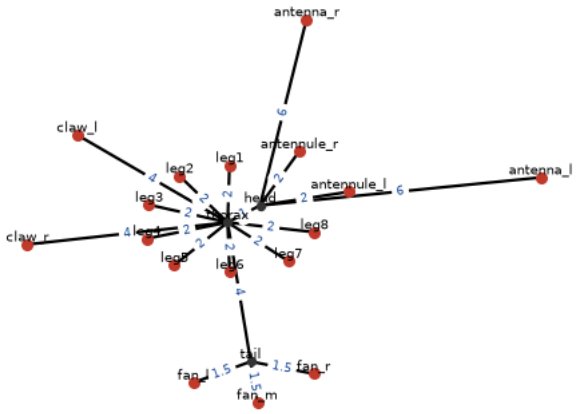

```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thorax","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thorax","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tail","fan_r",1.5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q012

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q012/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a praying mantis: antennae, raptorial forelegs, four walking legs, four wings, abdomen with two cerci). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0443844 (one unit of tree length -> 0.0443844 of the paper side). In addition, no point of the folded base may have more than 60 layers of paper.

Input `tree`:

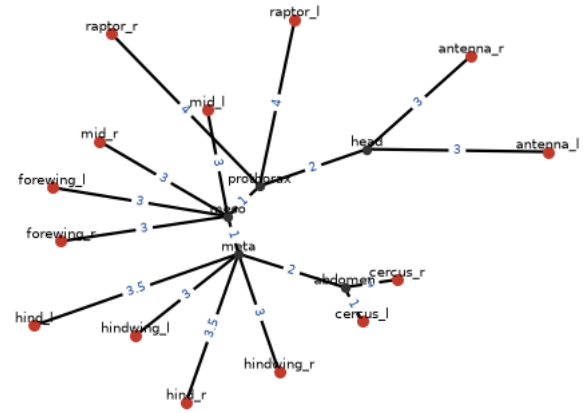

```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["meso","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],["abdomen","cercus_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q013

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q013/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a pegasus: head with two ears and a muzzle, wings with three primary feathers each, four legs, a tail). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0432756 (one unit of tree length -> 0.0432756 of the paper side). In addition, no point of the folded base may have more than 62 layers of paper.

Input `tree`:

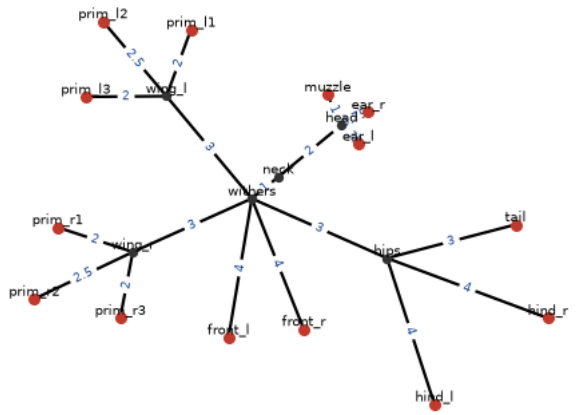

```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["neck","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","prim_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hips","hind_r",4],["hips","tail",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q014

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q014/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a wingless wyrm: a head with two horns, two forelegs, and twelve dorsal scales along a long body ending in a tail fin). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0609262 (one unit of tree length -> 0.0609262 of the paper side). In addition, no point of the folded base may have more than 54 layers of paper.

Input `tree`:

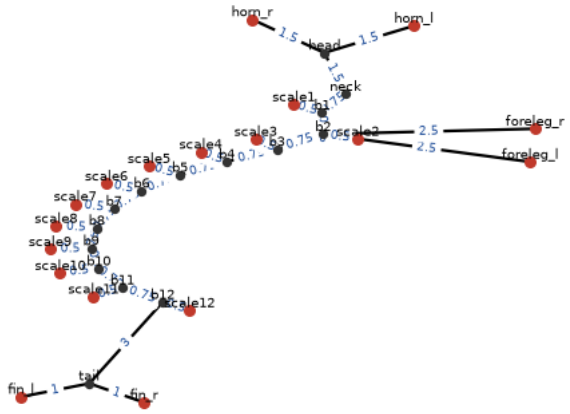

```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale1",0.5],["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale9",0.5],["b10","scale10",0.5],["b11","scale11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tail","fin_l",1],["tail","fin_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q015

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q015/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a stag beetle: long mandibles, antennae, palps, six legs, two elytra (wings), abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0454431 (one unit of tree length -> 0.0454431 of the paper side). In addition, no point of the folded base may have more than 62 layers of paper.

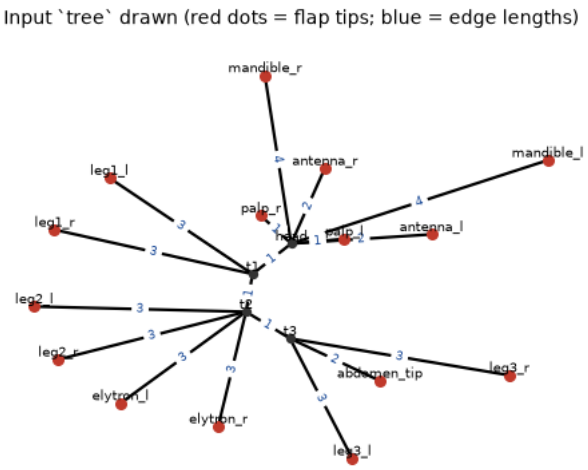
Input `tree`:


```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q016

split: expert · family: design · conventions: advanced (front matter) · files: inputs/Q016/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below (the stick figure of a tarantula: eight long legs, pedipalps, chelicerae, spinnerets on a big abdomen). This is graded pass/fail: the base must fold flat and realize the tree at a scale of at least 0.0516059 (one unit of tree length -> 0.0516059 of the paper side). In addition, no point of the folded base may have more than 76 layers of paper.

Input `tree`:

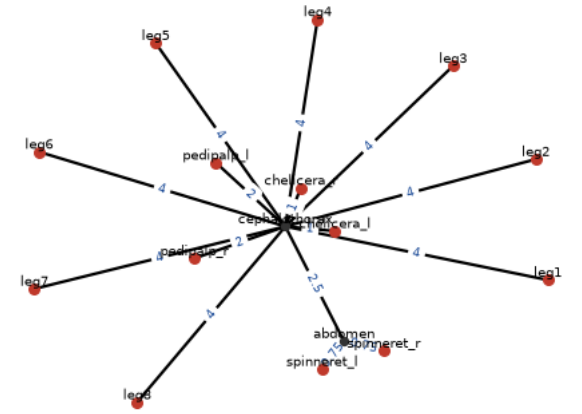

```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q017

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q017/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.09384, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",1],["body","leg1",2],["body","leg2",2],["body","leg3",
2],["body","leg4",2],["body","leg5",2],["body","leg6",2],["body","leg7",2],["body","leg8",2]]}
```

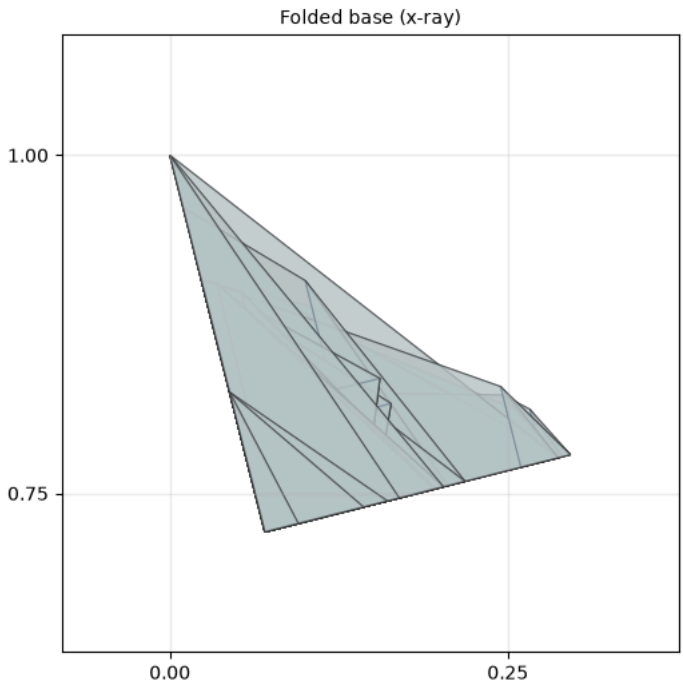

Input `folded_outline`:


```
[[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.
0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.
0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],
[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.
0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[0.0,1.0],[1e-06,
1.0],[1e-06,1.0],[1e-06,1.0],[1e-06,1.0],[0.198822,0.84513],[0.198822,0.84513],[0.198822,0.84513],[0.198822,0.84513],[0.200861,0.844409],[0.244648,0.82892
2],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,0.828922],[0.244648,
0.828922],[0.25669,0.81718],[0.25669,0.81718],[0.25669,0.81718],[0.26625,0.812132],[0.26625,0.812132],[0.26 ...
[truncated: the full input is in inputs/Q017/]
```

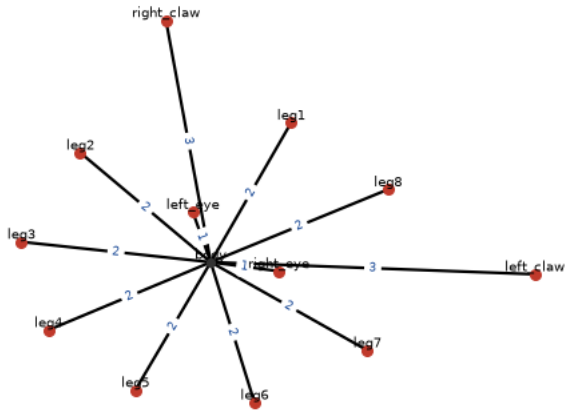

Answer format: the complete crease pattern of the unit square.


```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```


Paper edges may be omitted (they are added automatically).



Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q018

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q018/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.06736, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulders","wing_l",4],["shoulders","wing_r",4],["shoulder
s","front_l",2],["shoulders","front_r",2],["shoulders","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```

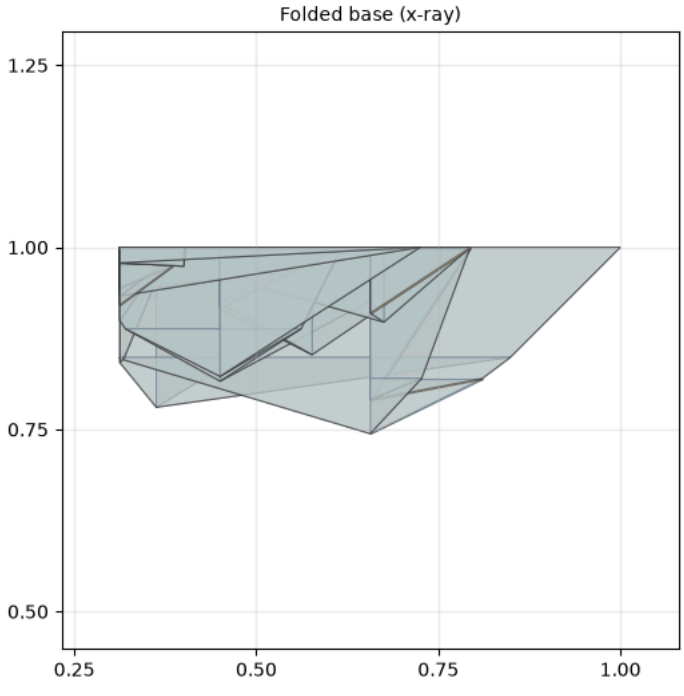

Input `folded_outline`:

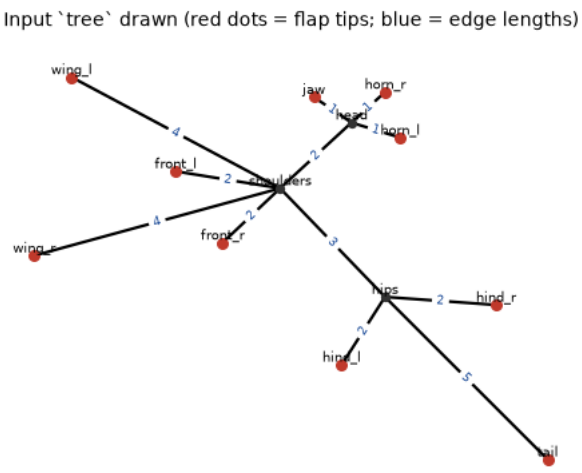

```
[[0.312664,1.0],[0.312664,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.
312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.3126
65,1.0],[0.312665,1.0],[0.312665,1.0],[0.312665,1.0],[0.32145,1.0],[0.362533,1.0],[0.362533,1.0],[0.362533,1.0],[0.362533,1.0],[0.362533,1.0],[0.368039,1.
0],[0.450132,1.0],[0.673527,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],
[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.7
93799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.793799,1.0],[0.79379
9,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7938,1.0],[0.7
938,1.0],[0.7938,1.0],[1.0,1.0],[1.0,1.0],[1.0,1.0],[0.848854,0.848854],[0.848854,0.848854],[0.848854,0.848854],[0.848854,0 ...
[truncated: the full input is in inputs/Q018/]
```


Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q019

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q019/

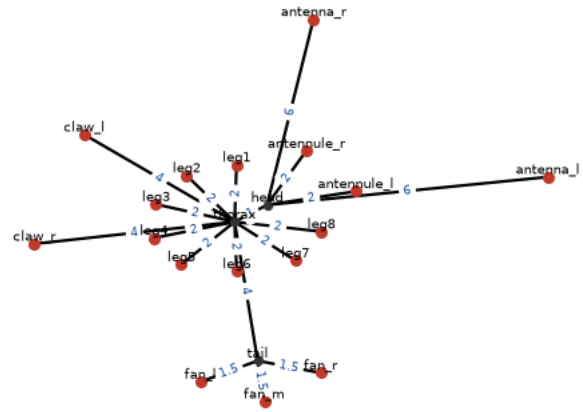
Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04440, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thorax","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thorax","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tail","fan_r",1.5]]}
```


Input `folded_outline`:


```
[[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.257938,0.648069],[0.343705,0.676699],[0.504243,0.730289],[0.504243,0.730289],[0.504243,0.730289],[0.504242,0.730288],[0.504242,0.730288],[0.504242,0.730288
```



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q020

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q020/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04350, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_l",4],["prothorax","raptor_r",4],["prothorax","meso",
1],["meso","mid_l",3],["meso","mid_r",3],["meso","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",3.5],["m
eta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],["abdomen","cercus_r",1]]}
```

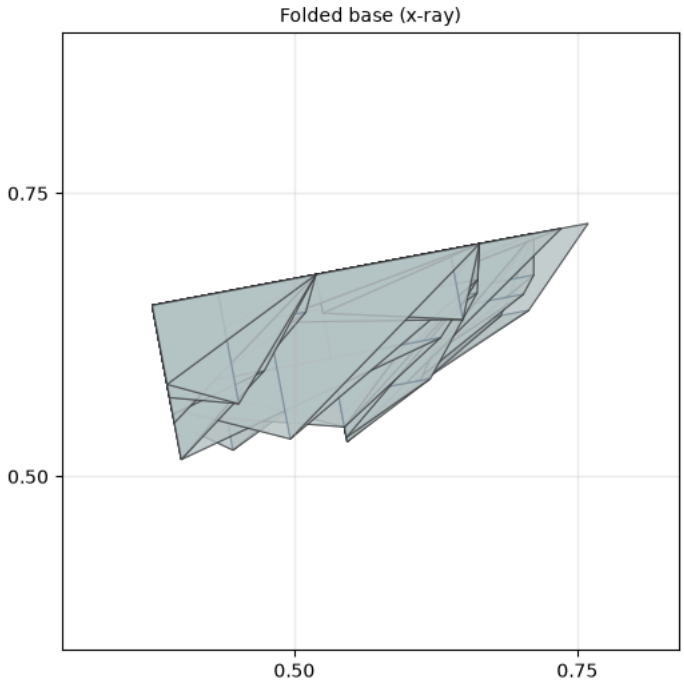

Input `folded_outline`:

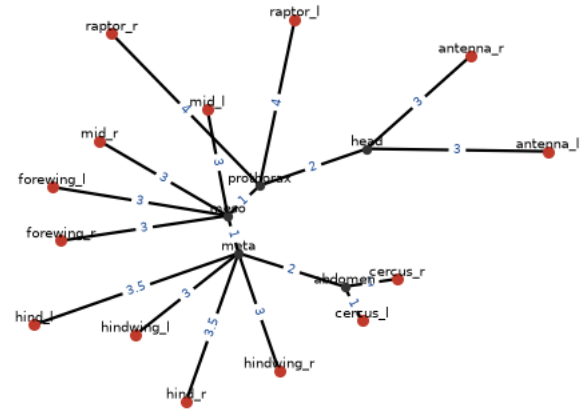

```
[[[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.
651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374
575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],
[0.374575,0.651021],[0.374575,0.651021],[0.374575,0.651021],[0.374776,0.651058],[0.519028,0.678177],[0.519028,0.678177],[0.519028,0.678177],[0.519027,0.67
8177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.51902
7,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],
[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.67
8177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519027,0.678177],[0.519028,0.678177 ...
[truncated: the full input is in inputs/Q020/]
```


Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q021

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q021/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04241, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["neck","withers",1],["withers","wing_l",3],["wing_l","prim_l
1",2],["wing_l","prim_l2",2.5],["wing_l","prim_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",2],["with
ers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hips","hind_r",4],["hips","tail",3]]}
```

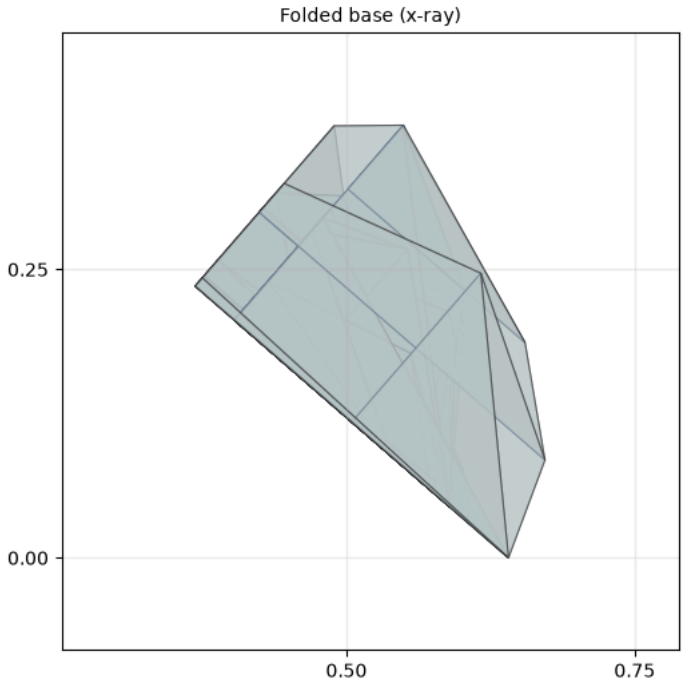

Input `folded_outline`:

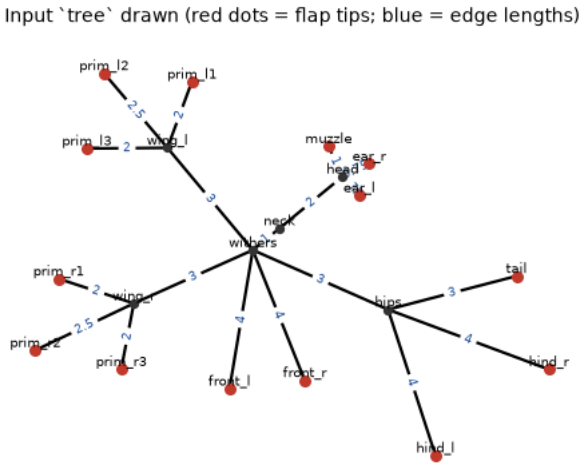

```
[[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.235491],[0.369035,0.2
35491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.3690
36,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],
[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.235491],[0.369036,0.23
5491],[0.369036,0.235491],[0.370136,0.23676],[0.376846,0.244497],[0.376846,0.244497],[0.376846,0.244497],[0.376846,0.244497],[0.376846,0.244497],[0.38890
2,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.388902,0.258398],[0.424627,0.299591],
[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.32
4407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407],[0.446149,0.324407], ...
[truncated: the full input is in inputs/Q021/]
```


Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q022

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q022/

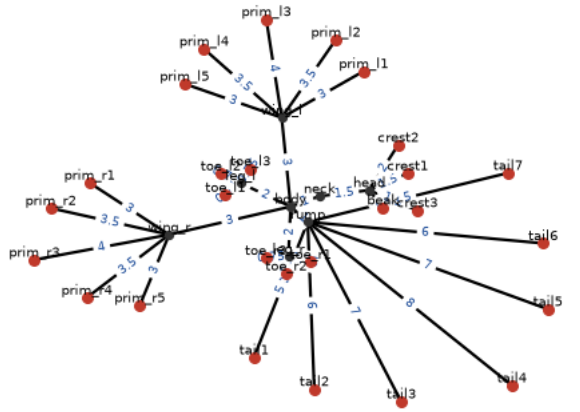
Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.01470, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",1.5],["head","beak",1],["head","crest1",1.5],["head","crest2",2],["head","crest3",1.5],["neck","body",1],["body","wing_l",3],["body","wing_r",3],["wing_l","prim_l1",3],["wing_l","prim_l2",3.5],["wing_l","prim_l3",4],["wing_l","prim_l4",3.5],["wing_l","prim_l5",3],["wing_r","prim_r1",3],["wing_r","prim_r2",3.5],["wing_r","prim_r3",4],["wing_r","prim_r4",3.5],["wing_r","prim_r5",3],["body","leg_l",2],["body","leg_r",2],["leg_l","toe_l1",0.75],["leg_l","toe_l2",0.75],["leg_l","toe_l3",0.75],["leg_r","toe_r1",0.75],["leg_r","toe_r2",0.75],["leg_r","toe_r3",0.75],["body","rump",1],["rump","tail1",5],["rump","tail2",6],["rump","tail3",7],["rump","tail4",8],["rump","tail5",7],["rump","tail6",6],["rump","tail7",5]]}
```


Input `folded_outline`:


```
[[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781281,0.117908],[0.781284,0.117909],[0.781284,0.117909],[0.781284,0.117909
```



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q023

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q023/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.05971, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["neck","head",1.5],["head","horn_l",1.5],["head","horn_r",1.5],["neck","b1",0.75],["b1","b2",0.75],["b2","b3",0.75],["b3","b4",0.75],["b4","b
5",0.75],["b5","b6",0.75],["b6","b7",0.75],["b7","b8",0.75],["b8","b9",0.75],["b9","b10",0.75],["b10","b11",0.75],["b11","b12",0.75],["b1","scale1",0.5],
["b2","scale2",0.5],["b3","scale3",0.5],["b4","scale4",0.5],["b5","scale5",0.5],["b6","scale6",0.5],["b7","scale7",0.5],["b8","scale8",0.5],["b9","scale
9",0.5],["b10","scale10",0.5],["b11","scale11",0.5],["b12","scale12",0.5],["b2","foreleg_l",2.5],["b2","foreleg_r",2.5],["b12","tail",3],["tail","fin_l",
1],["tail","fin_r",1]]}
```

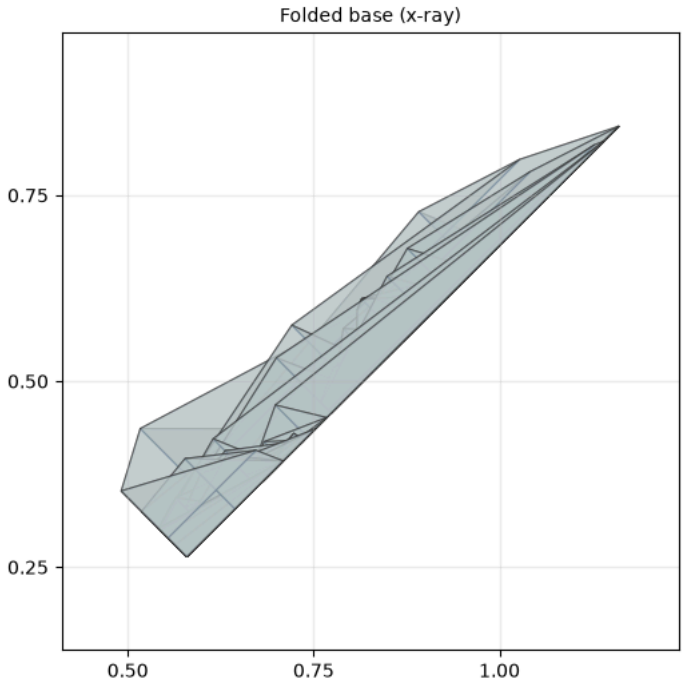

Input `folded_outline`:

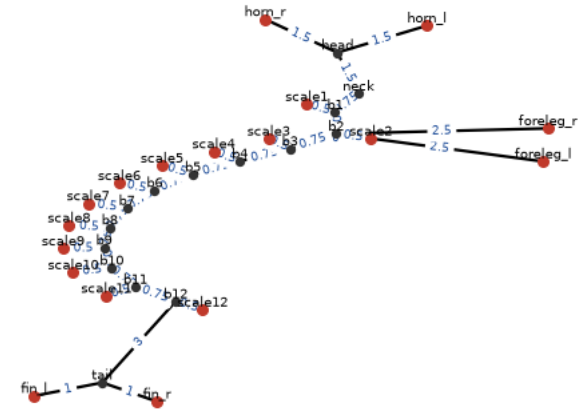

```
[[0.491096,0.352392],[0.491096,0.352392],[0.491096,0.352392],[0.491096,0.352392],[0.506259,0.401948],[0.506259,0.401948],[0.506259,0.401948],[0.516767,0.4
36292],[0.516767,0.436292],[0.516767,0.436292],[0.529715,0.443173],[0.5719,0.465591],[0.614085,0.48801],[0.656269,0.510428],[0.688812,0.527722],[0.688812,
0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.688812,0.527722],[0.694732,0.536584],[0.720632,0.575356],[0.720632,0.575356],[0.7
20632,0.575356],[0.722699,0.576857],[0.758062,0.60255],[0.795491,0.629743],[0.830138,0.654915],[0.830138,0.654915],[0.830138,0.654915],[0.832333,0.65752
6],[0.861865,0.692651],[0.861865,0.692651],[0.879428,0.71354],[0.891397,0.727776],[0.891397,0.727776],[0.891397,0.727776],[0.891397,0.727776],[0.891397,0.
727776],[0.891399,0.727777],[1.027361,0.798203],[1.027361,0.798203],[1.027361,0.798203],[1.027363,0.798204],[1.027363,0.798204],[1.027363,0.798204],[1.066
317,0.811241],[1.160926,0.842904],[1.160926,0.842904],[1.160926,0.842904],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],
[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.160925,0.842903],[1.1 ...
[truncated: the full input is in inputs/Q023/]
```


Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q024

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q024/

Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.04453, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",2],["head","palp_l",1],["head","palp_r",1],["head","t
1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",1],["t
3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```

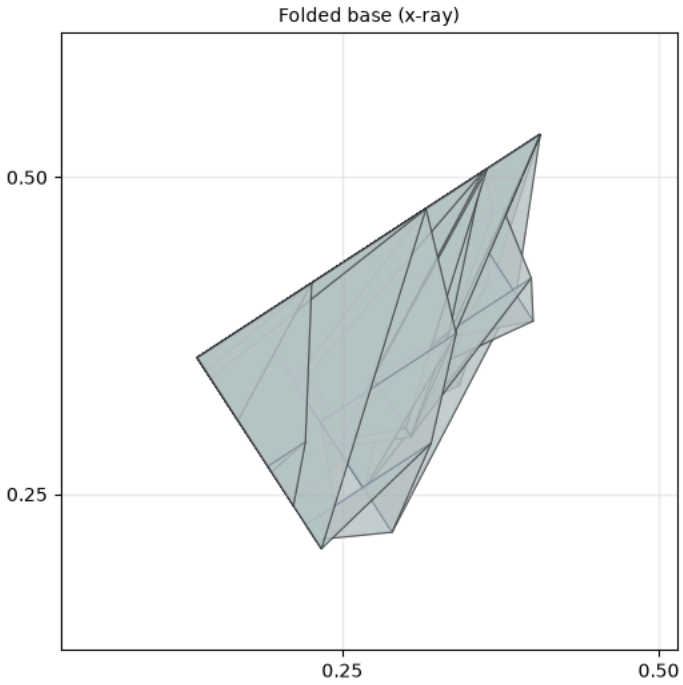

Input `folded_outline`:

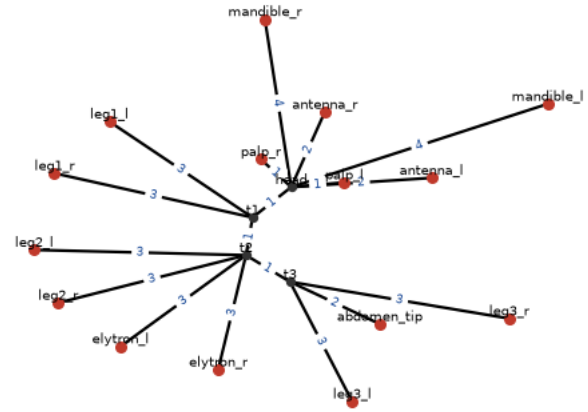

```
[[0.135546,0.357646],[0.135546,0.357646],[0.196486,0.397289],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.4
75114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316
19,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],
[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.475114],[0.316119,0.47
5114],[0.316119,0.475114],[0.406405,0.533848],[0.406405,0.533848],[0.406405,0.533848],[0.406405,0.533848],[0.406405,0.533847],[0.406405,0.533847],[0.40640
5,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],
[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.53
3847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847],[0.406405,0.533847] ...
[truncated: the full input is in inputs/Q024/]
```


Answer format: the complete crease pattern of the unit square.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Paper edges may be omitted (they are added automatically).





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q025

split: expert · family: inverse · conventions: advanced (front matter) · files: inputs/Q025/

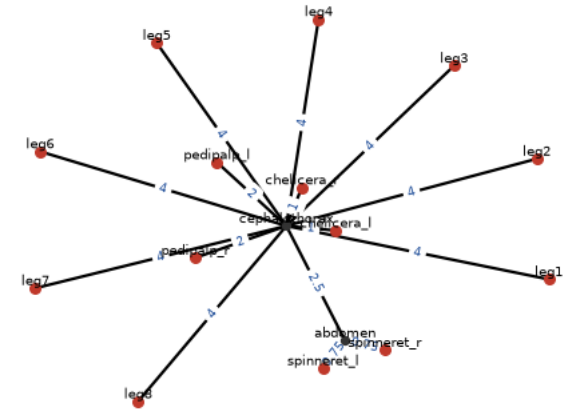
Task (design, hard):
Below is a flat-folded origami base, folded from ONE uncut unit square: its outline (`folded_outline`, coordinates in units of the paper side), an x-ray image of the folded state (all layers drawn translucently), and the stick-figure tree it realizes. Reconstruct the crease pattern. Pass/fail: your pattern must fold flat, realize the tree at a scale of at least 0.05057, and its folded outline must match the given one (IoU >= 0.97 after the best rotation/reflection/translation).

Input `tree`:


```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Input `folded_outline`:


```
[[[0.446872,-1.1e-05],[0.446872,-1.1e-05],[0.500762,0.053882],[0.500762,0.053882],[0.500762,0.053882],[0.500762,0.053882],[0.500785,0.053892],[0.500785,0.053892],[0.500785,0.053892
```



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q026

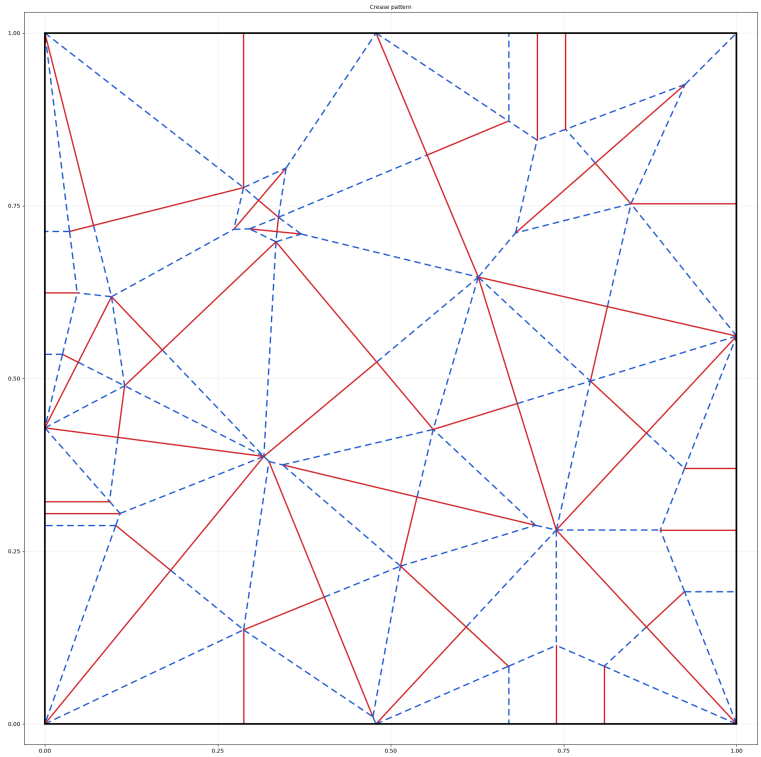
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q026/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q027

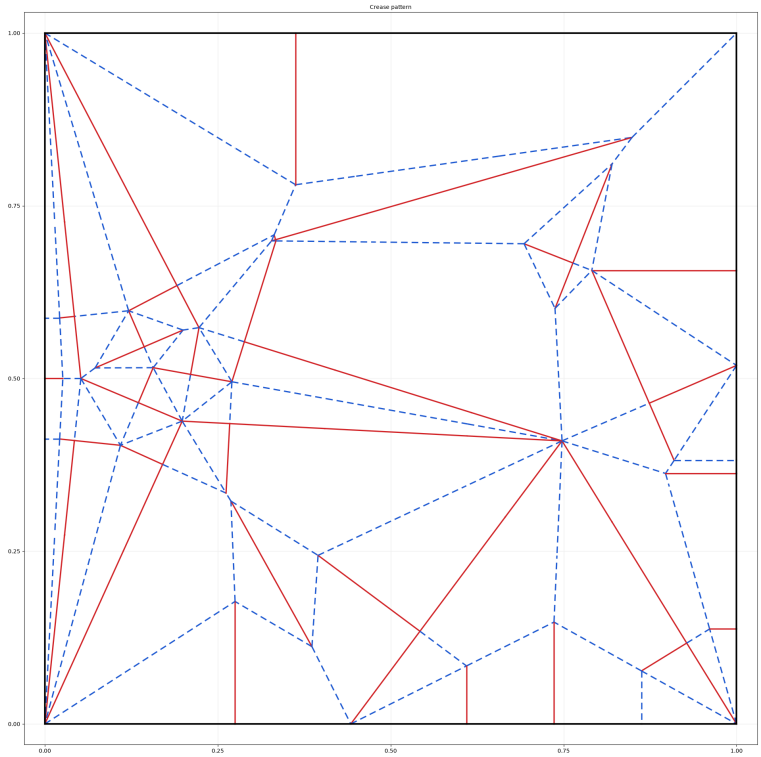
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q027/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q028

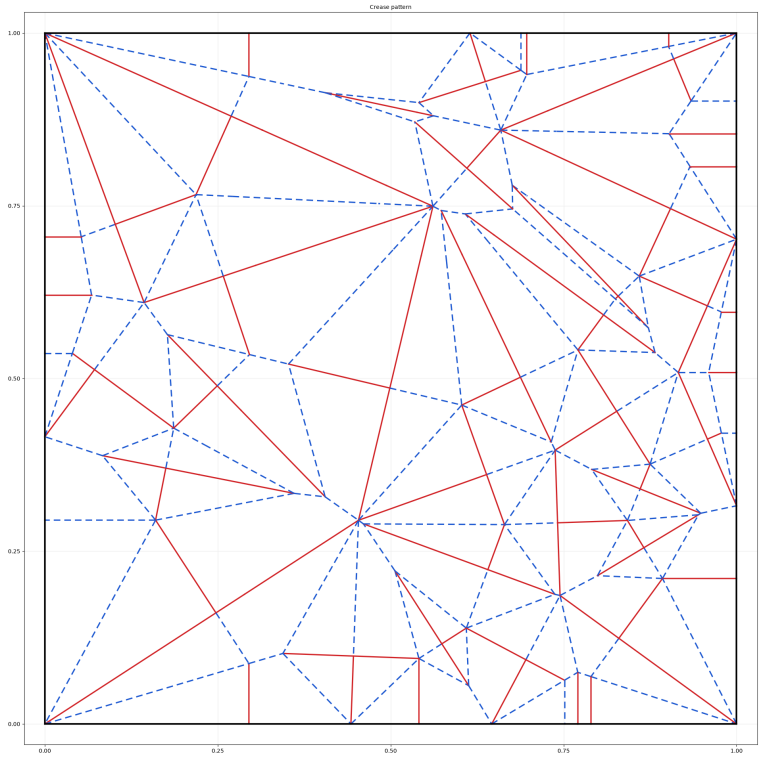
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q028/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q029

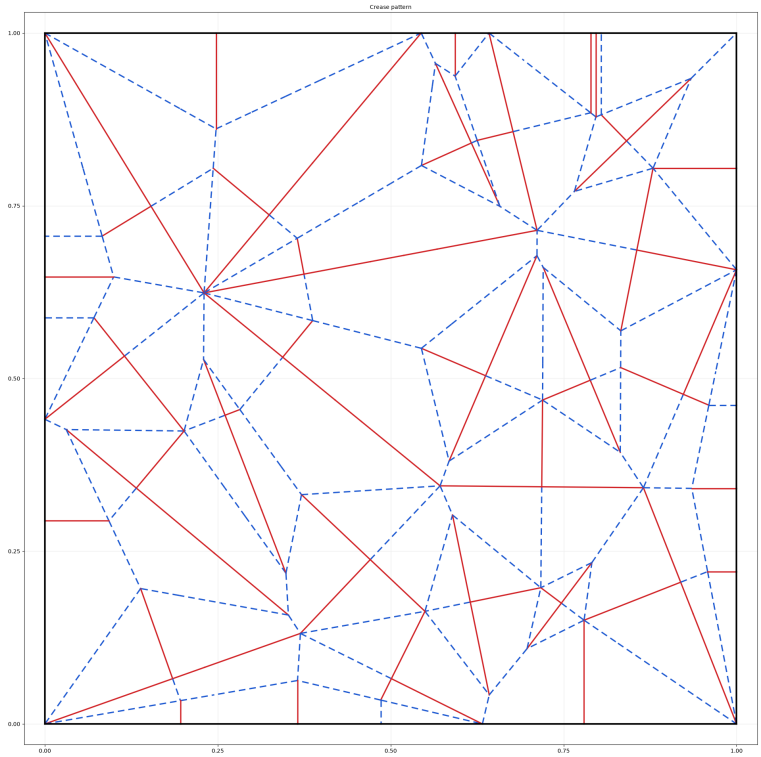
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q029/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q030

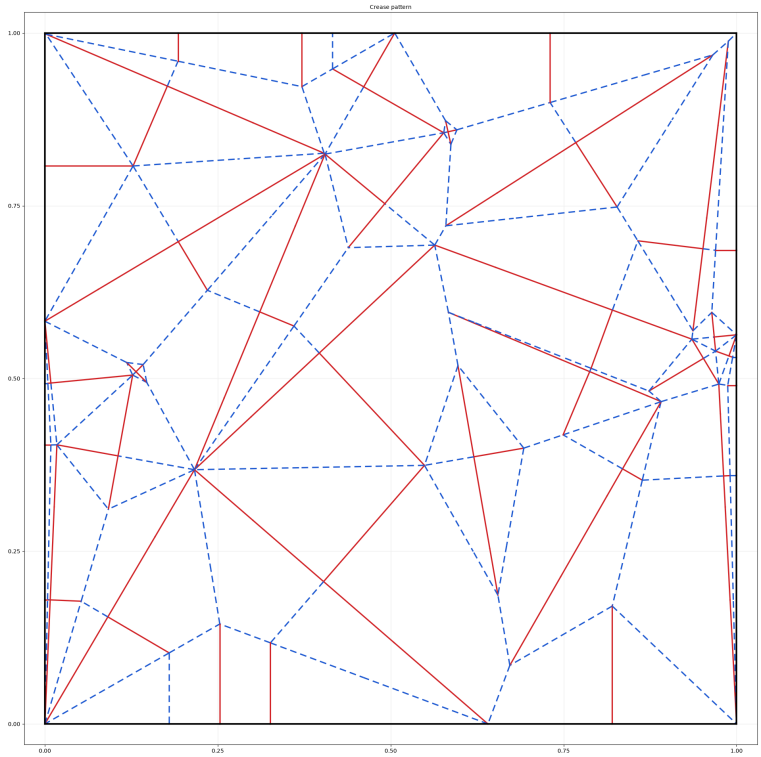
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q030/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q031

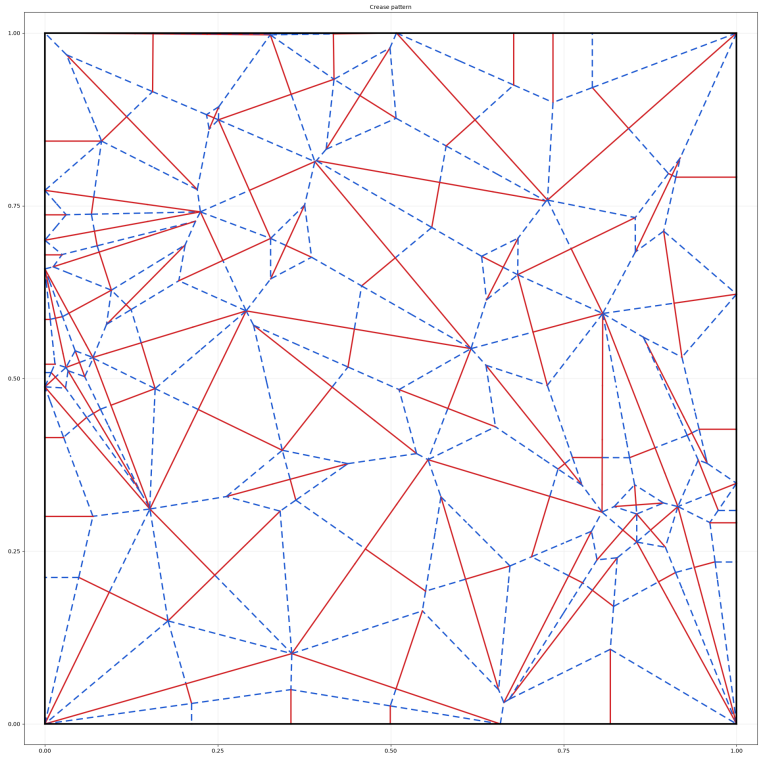
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q031/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q032

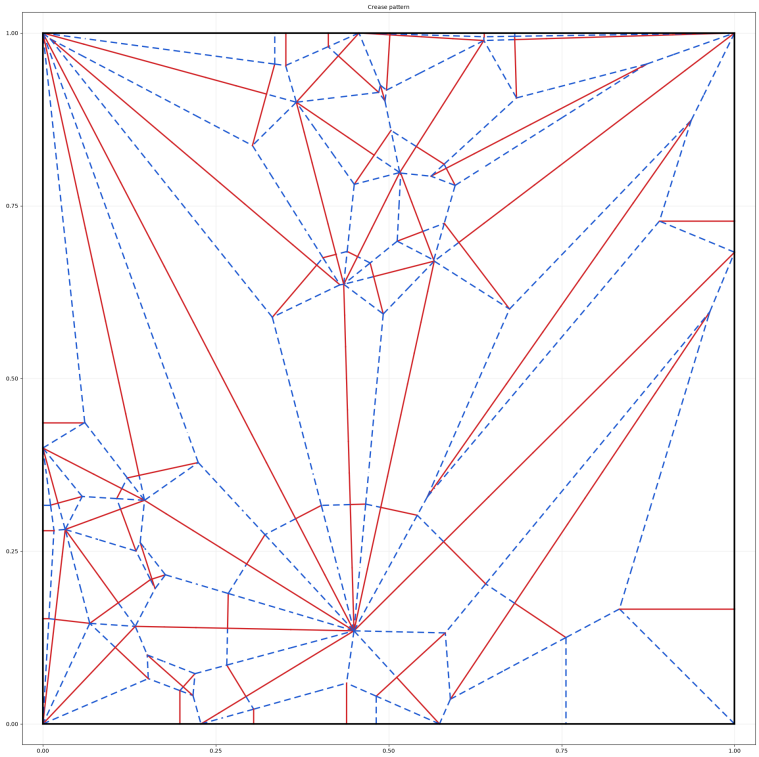
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q032/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q033

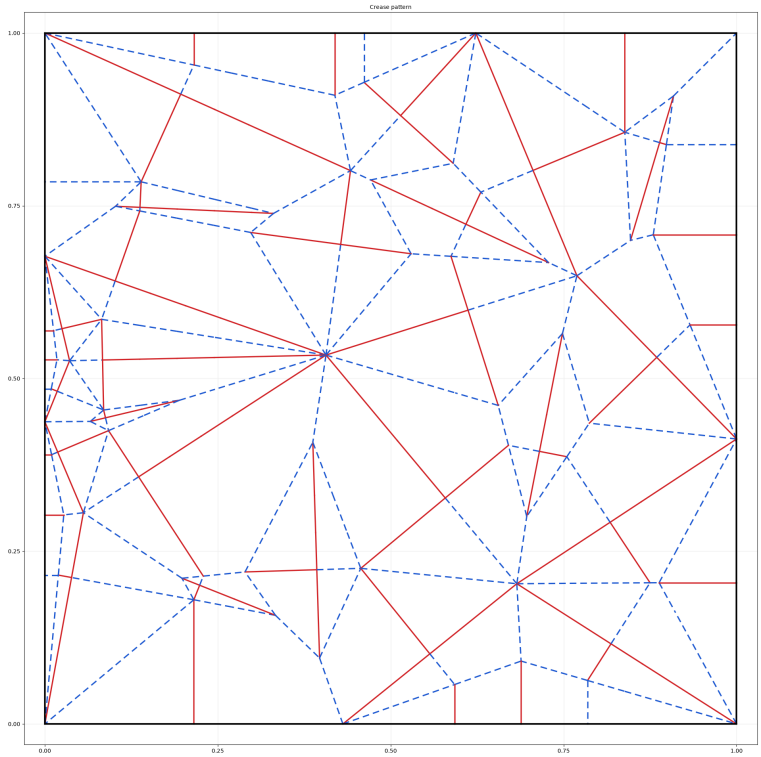
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q033/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q034

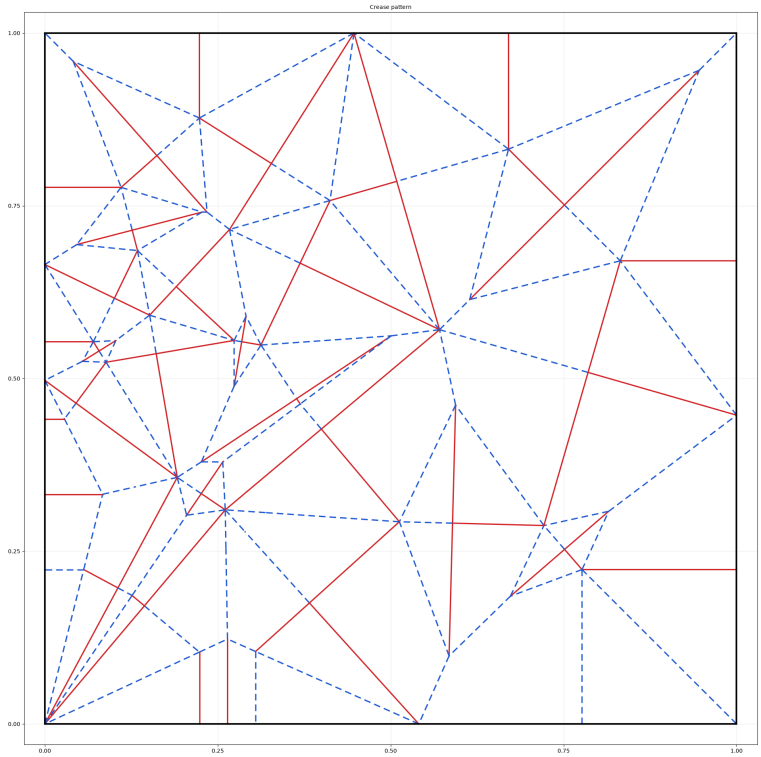
split: expert · family: read-tree · conventions: advanced (front matter) · files: inputs/Q034/

Task (read, hard):
The image `cp.png` shows the crease pattern of a flat-foldable origami base (one uncut unit square; red solid = mountain, blue dashed = valley, axes in units of the paper side). No coordinates are given: read the pattern from the image. Report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment (in units of the paper side). Pass/fail: exact tree, every segment length within 0.01 of the paper side (or 2%, whichever is larger).

Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q035

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q035/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 12 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, K, L, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.07030663,0.72148322],[0.03569616,0.71274642],[0.0481769,0.5234063],[0.02546801,0.53493564],[0.09635379,0.61829849],[0.04674418,0.62384102],[0.0,1.0],[0.0,0.4285141],[0.0,0.71274642],[0.0,0.62384102],[0.0,0.53493564],[0.30889424,0.75782987],[0.28725364,0.77624759],[0.27318569,0.71587295],[0.34875421,0.80466463],[0.33556711,0.71252174],[0.33409025,0.69780028],[0.37053504,0.70901377],[0.29583321,0.71650782],[0.33765665,0.73335083],[0.16976004,0.54124614],[0.47977888,0.52376454],[0.5526887,0.82334459],[0.2872537,1.0],[0.31657255,0.38714143],[0.62662133,0.64668915],[0.47875609,1.0],[0.11527581,0.48934008],[0.10552417,0.41472321],[0.18183805,0.22237254],[0.10249402,0.28725358],[0.09341833,0.32209259],[0.10870942,0.30467308],[0.0,0.0],[0.0,0.32209259],[0.0,0.30467308],[0.0,0.28725358],[0.40379295,0.1835696],[0.28725358,0.13617235],[0.47423699,0.01036327],[0.32397538,0.37982337],[0.53806533,0.32846288],[0.5141629,0.22845772],[0.70933356,0.28752773],[0.34358464,0.37494608],[0.5614315,0.42622445],[0.28725358,0.0],[0.60906683,0.14032908],[0.6829995,0.46367366],[0.47875597,0.0],[0.73937768,0.28065814],[0.8696887,0.14032923],[0.80849719,0.0835061],[0.67025837,0.08350605], ...
[truncated: the full input is in inputs/Q035/]
```


Input `tree`:


```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["body","leg6",2],["body","leg7",2],["body","leg8",2]]}
```

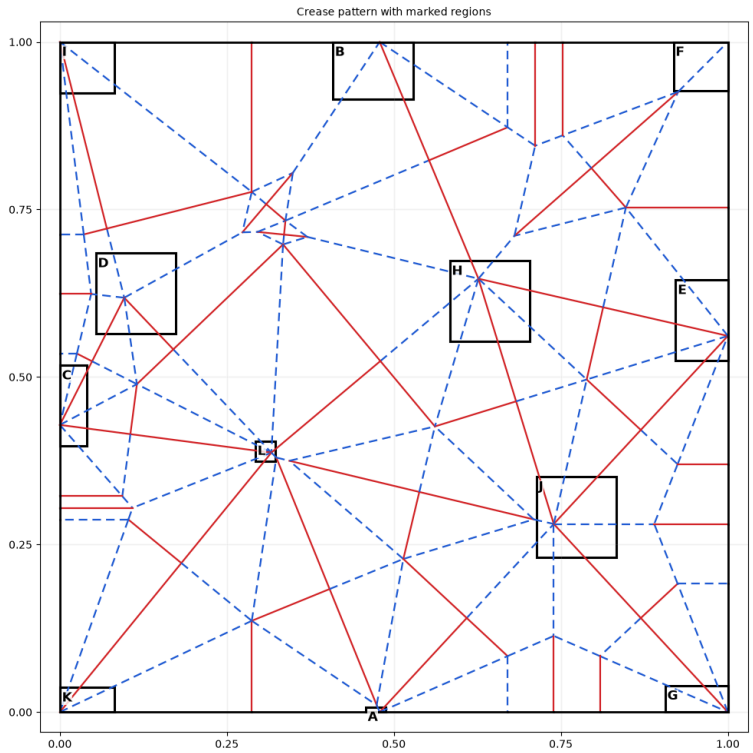

Input `regions`:

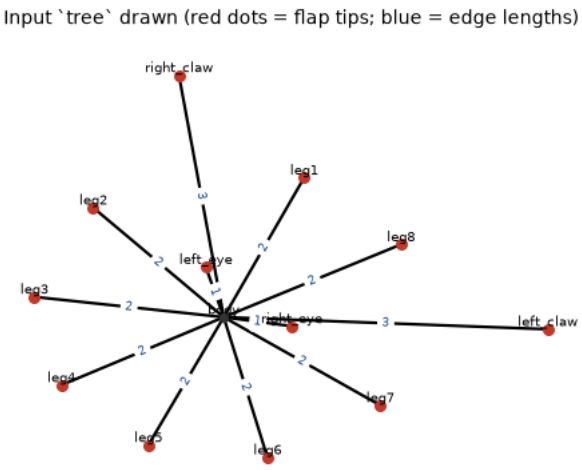

```
{"A":[0.4577,0.0,0.4877,0.0075],"B":[0.4086,0.9144,0.5286,1.0],"C":[0.0,0.3971,0.0407,0.5171],"D":[0.0537,0.5646,0.1737,0.6846],"E":[0.9219,0.5246,1.0,0.6446],"F":[0.9186,0.9271,1.0,1.0],"G":[0.9062,0.0,1.0,0.0393],"H":[0.5836,0.5532,0.7036,0.6732],"I":[0.0,0.9235,0.0811,1.0],"J":[0.7134,0.2314,0.8334,0.3514],"K":[0.0,0.0,0.0816,0.0365],"L":[0.2927,0.3745,0.3227,0.4045]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q036

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q036/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 10 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.21068246,0.50590869],[0.27034906,0.49521506],[0.26719778,0.43467973],[0.28841725,0.55307693],[0.22280777,0.57356426],[0.19855715,0.4382531],[0.40672998,0.47077249],[0.61130116,0.43410867],[0.74768195,0.40966613],[0.61040076,0.41681287],[0.40447897,0.42753299],[0.61646347,0.45064064],[0.41963574,0.51210242],[0.1435487,0.54582664],[0.15656378,0.5156079],[0.19974391,0.57002968],[0.07261342,0.51527512],[0.12109135,0.59796864],[0.19097808,0.63448367],[0.04266327,0.58981143],[0.13521614,0.4649405],[0.08072756,0.7319791],[0.0,1.0],[0.12731872,0.75632245],[0.12731874,0.75632241],[0.05200446,0.5],[0.02844173,0.72654528],[0.02844218,0.72654096],[0.02844241,0.72653879],[0.10923045,0.40326499],[0.04266327,0.41018857],[0.17019184,0.37564551],[0.0728203,0.26884332],[0.0,0.0],[0.02844173,0.27345472],[0.02844218,0.27345904],[0.02844241,0.27346121],[0.11346123,0.25043034],[0.33350835,0.70080671],[0.33147326,0.70789076],[0.32772645,0.6991457],[0.36253292,0.7803841],[0.45013163,0.79271726],[0.65633227,0.82174858],[0.84885403,0.84885403],[0.66381635,0.7956968 3],[0.46563155,0.73876276],[0.7634463,0.66708616],[0.69275545,0.69496167],[0.73753346,0.60137223],[0.82003289,0.81058752], ...
[truncated: the full input is in inputs/Q036/]
```


Input `tree`:


```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulders","wing_l",4],["shoulders","wing_r",4],["shoulder s","front_l",2],["shoulders","front_r",2],["shoulders","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```

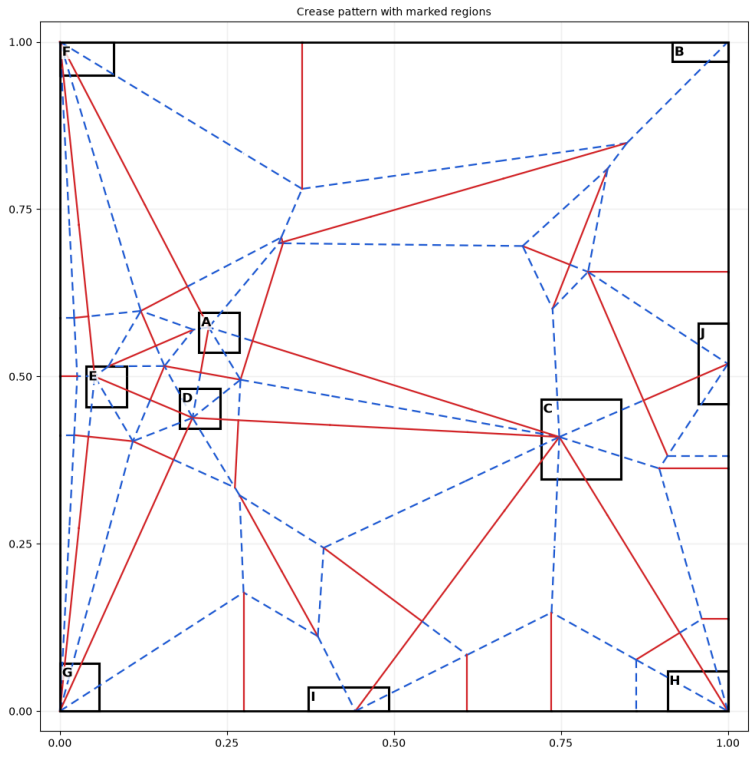

Input `regions`:

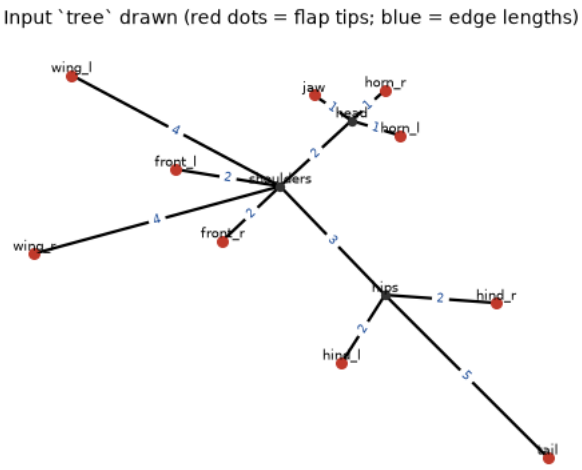

```
{"A":[0.2083,0.5358,0.2683,0.5958],"B":[0.9169,0.9706,1.0,1.0],"C":[0.7203,0.3465,0.8403,0.4665],"D":[0.1796,0.4223,0.2396,0.4823],"E":[0.0396,0.4549,0.0996,0.5149],"F":[0.0,0.9501,0.0809,1.0],"G":[0.0,0.0,0.0582,0.0719],"H":[0.9093,0.0,1.0,0.06],"I":[0.3724,0.0,0.4924,0.036],"J":[0.9557,0.4592,1.0,0.5792]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q037

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q037/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 17 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.17486932,0.37021691],[0.16006054,0.2949799],[0.36031757,0.33371586],[0.21215044,0.30505571],[0.08337999,0.3882244],[0.22310711,0.36
072246],[0.18627062,0.42814267],[0.23154277,0.40358088],[0.24921097,0.48928984],[0.40524793,0.32867699],[0.17714415,0.56347007],[0.28346872,0.45402746],
[0.29582263,0.53457353],[0.35223762,0.52083727],[0.0,0.29497973],[0.24730653,0.16078782],[0.49872676,0.48613474],[0.25793777,0.64806892],[0.0717222,0.5127
6711],[0.0,0.41568328],[0.0,0.0],[0.45339507,0.29477752],[0.28852423,0.18758576],[0.50410685,0.68695611],[0.56105778,0.74925063],[0.54405842,0.67749181],
[0.54405841,0.67749176],[0.14344441,0.60985094],[0.49110701,0.72590101],[0.30457162,0.66363534],[0.30457162,0.66363534],[0.03975035,0.53638682],[0.1017919
1,0.72314002],[0.05250868,0.70502027],[0.0,1.0],[0.06751771,0.62070354],[0.0,0.70502027],[0.0,0.62070354],[0.0,0.53638682],[0.21854957,0.76606767],[0.2693
0785,0.87964025],[0.27124325,0.76348043],[0.48201745,0.75313149],[0.49373087,0.77934055],[0.31419245,0.85958031],[0.31419229,0.85958039],[0.29498006,0.087
46151],[0.29497992,0.0],[0.44611154,0.09825914],[0.3441432,0.10203837],[0.44246979,0.0],[0.57387376,0.11597489],[0.54079664,0 ...
[truncated: the full input is in inputs/Q037/]
```


Input `tree`:


```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_r",2],["head","thorax",1],["thorax","claw_l",4],["thor
ax","claw_r",4],["thorax","leg1",2],["thorax","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thorax","leg7",
2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tail","fan_r",1.5]]}
```

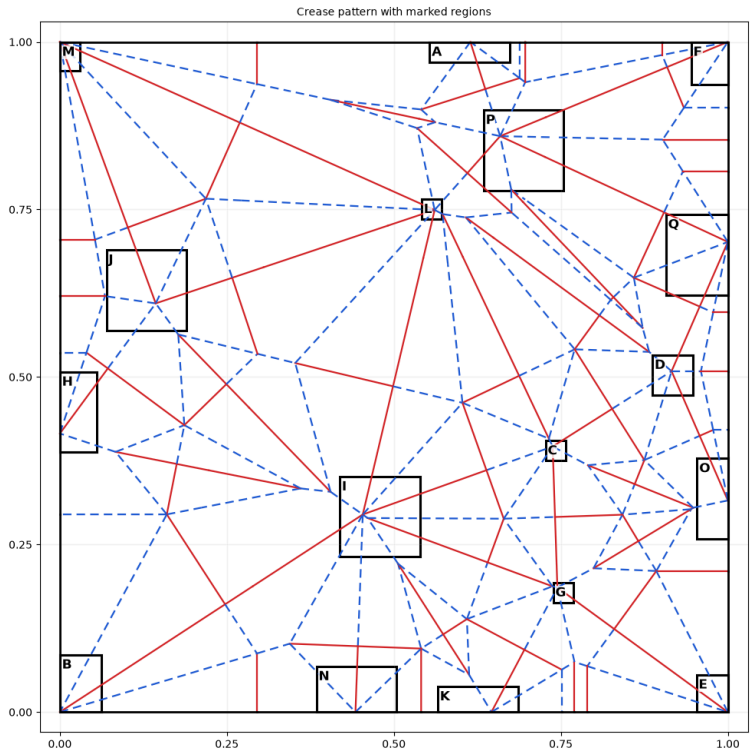

Input `regions`:

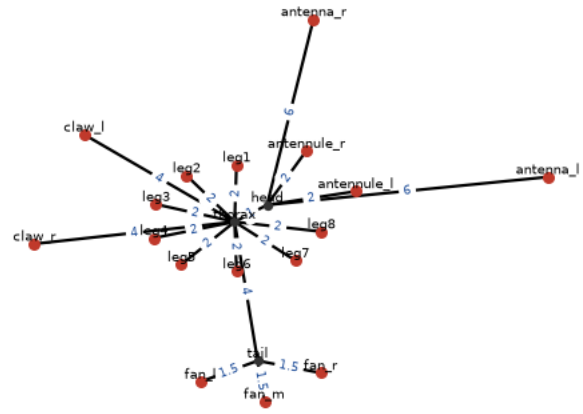

```
{"A":[0.5535,0.9699,0.6735,1.0],"B":[0.0,0.0,0.0617,0.0857],"C":[0.7271,0.375,0.7571,0.405],"D":[0.8872,0.4729,0.9472,0.5329],"E":[0.953,0.0,1.0,0.055
9],"F":[0.9451,0.936,1.0,1.0],"G":[0.7385,0.1633,0.7685,0.1933],"H":[0.0,0.3878,0.0551,0.5078],"I":[0.419,0.2315,0.539,0.3515],"J":[0.0696,0.5698,0.1896,
0.6898],"K":[0.5657,0.0,0.6857,0.0382],"L":[0.5416,0.7355,0.5716,0.7655],"M":[0.0,0.957,0.0303,1.0],"N":[0.3841,0.0,0.5041,0.0683],"O":[0.954,0.2586,1.0,
0.3786],"P":[0.6344,0.7784,0.7544,0.8984],"Q":[0.9074,0.6225,1.0,0.7425]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q038

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q038/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 14 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, K, L, M, N, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.11506158,0.53240593],[0.07092753,0.58792981],[0.1534154,0.74924308],[0.08279224,0.70603518],[0.0,0.44094739],[0.09942385,0.6469824
4],[0.23012319,0.62386449],[0.05519483,0.80402345],[0.0,1.0],[0.10227694,0.83282872],[0.0,0.70603518],[0.0,0.64698244],[0.0,0.58792981],[0.26051423,0.4467
9328],[0.20124782,0.42397732],[0.34854015,0.21813819],[0.2896077,0.30049553],[0.2601544,0.34165613],[0.22944009,0.52751112],[0.31332051,0.30962431],[0.295
71842,0.3553473],[0.28188021,0.45501859],[0.34129011,0.37289115],[0.3709951,0.33182738],[0.13221137,0.34135964],[0.03084452,0.42606313],[0.35209105,0.1576
2529],[0.24499983,0.24711212],[0.20740367,0.27852796],[0.09260777,0.29396498],[0.1939432,0.18601134],[0.13842274,0.19597674],[0.0,0.29396482],[0.18470022,
0.06551889],[0.47039184,0.23782999],[0.34387673,0.53078355],[0.0,0.0],[0.36940022,0.13103771],[0.23087522,0.0818986],[0.57138329,0.3446221],[0.45763001,0.
43770283],[0.41971225,0.46872973],[0.2433591,0.80427147],[0.32430854,0.73670534],[0.19597655,0.89015501],[0.24753973,0.86125382],[0.29910312,0.88537996],
[0.34809726,0.90830397],[0.3970914,0.93122798],[0.49507967,0.97707599],[0.54407381,1.0],[0.51267877,0.96238648],[0.44988869, ...
[truncated: the full input is in inputs/Q038/]
```


Input `tree`:


```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_l",4],["prothorax","raptor_r",4],["prothorax","meso",
1],["meso","mid_l",3],["meso","mid_r",3],["meso","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",3.5],["m
eta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],["abdomen","cercus_r",1]]}
```

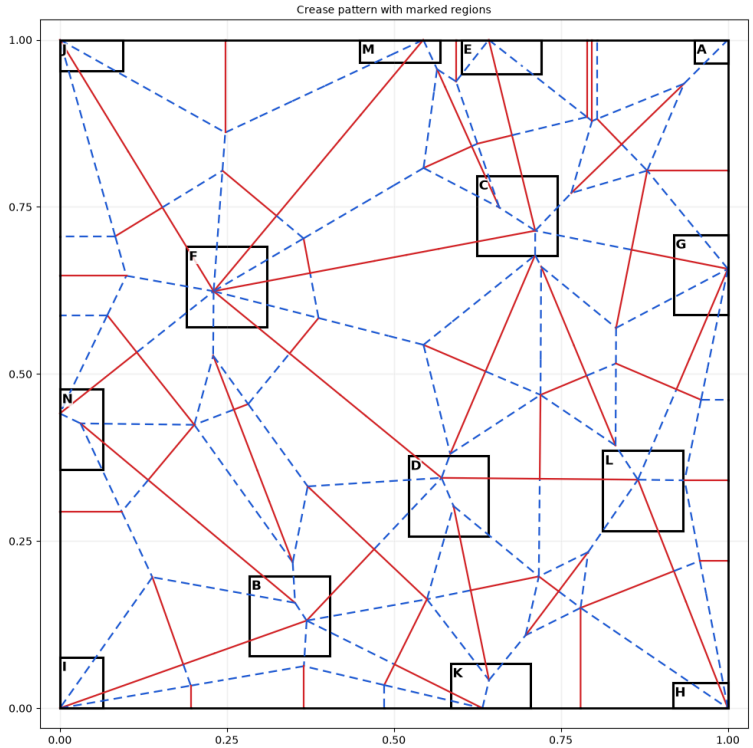

Input `regions`:

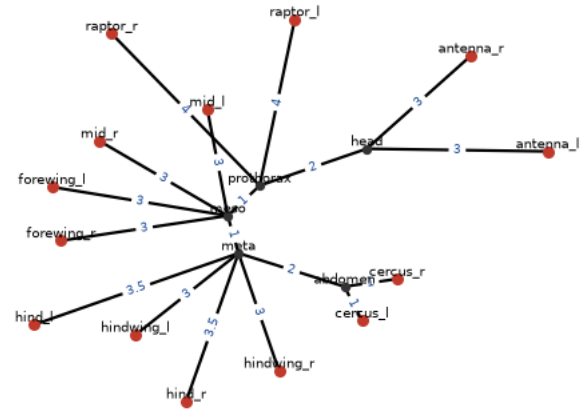

```
{"A":[0.95,0.965,1.0,1.0],"B":[0.2837,0.078,0.4037,0.198],"C":[0.6241,0.6767,0.7441,0.7967],"D":[0.5218,0.2573,0.6418,0.3773],"E":[0.6007,0.9492,0.7207,1.
0],"F":[0.1899,0.5705,0.3099,0.6905],"G":[0.9187,0.5884,1.0,0.7084],"H":[0.9174,0.0,1.0,0.0383],"I":[0.0,0.0,0.0644,0.0765],"J":[0.0,0.9534,0.0937,1.
0],"K":[0.5848,0.0,0.7048,0.067],"L":[0.8129,0.2653,0.9329,0.3853],"M":[0.4488,0.9665,0.5688,1.0],"N":[0.0,0.3571,0.0649,0.4771]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q039

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q039/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 15 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.13749551,0.74764319],[0.13917569,0.78458091],[0.10219977,0.74924885],[0.19250666,0.77183921],[0.24583744,0.75909755],[0.29916822,0.74635588],[0.33053669,0.7388614],[0.29889277,0.74030094],[0.24509368,0.74274836],[0.19129461,0.74519577],[0.13725864,0.74243634],[0.29757289,0.71128467],[0.24413473,0.72166858],[0.19069669,0.73205246],[0.0,0.7845811],[0.10160182,0.64111543],[0.4276765,0.69400543],[0.19638105,0.91145651],[0.0,1.0],[0.0,0.67687164],[0.40640582,0.53384729],[0.25400382,0.58748136],[0.20320315,0.60535938],[0.15240249,0.62323741],[0.44185693,0.80077734],[0.34366658,0.84504901],[0.2945714,0.86718485],[0.24547622,0.88932068],[0.22375315,0.20071629],[0.21541914,0.1797319],[0.33344517,0.15715313],[0.2692741,0.16942927],[0.19814275,0.21088723],[0.27380521,0.18083859],[0.22901458,0.21396457],[0.28934939,0.21997882],[0.39314958,0.2232194],[0.39713951,0.09541625],[0.31796449,0.27475313],[0.34657958,0.32952744],[0.38739181,0.40764918],[0.3897886,0.3308764],[0.39146909,0.2770479],[0.45670857,0.22520365],[0.41624637,0.33170239],[0.43647747,0.27845303],[0.03859543,0.21193324],[0.21541938,0.0],[0.55656763,0.10146547],[0.57883745,0.32702461],[0.13554609,0.35764588],[ ...
[truncated: the full input is in inputs/Q039/]
```


Input `tree`:


```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```

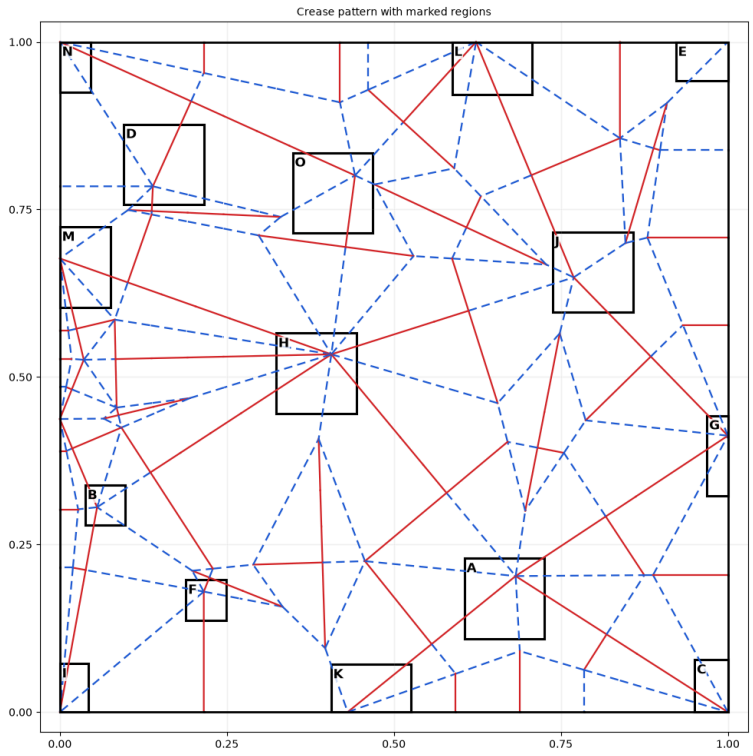

Input `regions`:

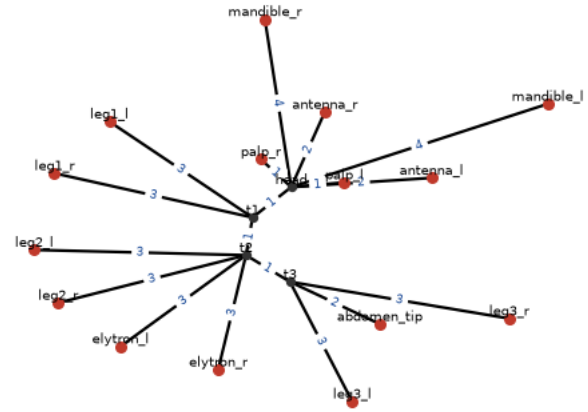

```
{"A":[0.6056,0.1094,0.7256,0.2294],"B":[0.0381,0.2786,0.0981,0.3386],"C":[0.9503,0.0,1.0,0.0781],"D":[0.0956,0.7572,0.2156,0.8772],"E":[0.9222,0.9422,1.0,1.0],"F":[0.1888,0.1372,0.2488,0.1972],"G":[0.9688,0.3225,1.0,0.4425],"H":[0.3238,0.4454,0.4438,0.5654],"I":[0.0,0.0,0.0427,0.0724],"J":[0.7377,0.5964,0.8577,0.7164],"K":[0.406,0.0,0.526,0.0712],"L":[0.587,0.9219,0.707,1.0],"M":[0.0,0.604,0.0758,0.724],"N":[0.0,0.9245,0.0459,1.0],"O":[0.3485,0.7147,0.4685,0.8347]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q040

split: expert-analysis · family: regions · conventions: advanced (front matter) · files: inputs/Q040/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square), the stick-figure tree it realizes (with named flaps), and 14 rectangular regions of the unfolded paper, labelled A, B, C, D, E, F, G, H, I, J, K, L, M, N, also drawn on the image. Each region contains the tip of exactly one flap, and every flap tip is marked. For every region, say which flap of the tree it becomes. Pass/fail: all regions must be right (mirror-image answers, e.g. left/right swapped consistently, are accepted when the tree is symmetric).

Input `crease_pattern`:


```
{"vertices_coords":[[0.75135002,0.25259175],[0.77656726,0.2234321],[0.81511822,0.30774054],[0.67305641,0.18488091],[0.72144941,0.28716498],[0.5893627,0.29
061078],[0.59381905,0.46145556],[0.58435056,0.09845871],[0.51254415,0.29261416],[0.77656799,0.0],[1.0,0.22343346],[0.7853035,0.50873792],[0.3996624,0.4267
3718],[0.38244632,0.17511595],[0.54060344,0.0],[1.0,0.0],[1.0,0.44686547],[0.57060908,0.57061014],[0.3188706,0.30586542],[0.26076869,0.30984079],[0.292821
08,0.33681688],[0.28884814,0.27875045],[0.28884814,0.27875044],[0.30463889,0.10484365],[0.14387256,0.17094686],[0.22343263,0.10398739],[0.0,0.0],[0.264035
76,0.12288441],[0.22343263,0.0],[0.26403576,0.0],[0.30463889,0.0],[0.26173983,0.2542674],[0.23379268,0.27778838],[0.23379266,0.27778835],[0.75108042,0.751
08104],[0.83186361,0.67029827],[0.94610862,0.94610875],[0.6141467,0.61414765],[0.67029782,0.83186438],[1.0,0.67029748],[0.67029741,1.0],[0.50873714,0.7853
0544],[1.0,1.0],[0.4468654,1.0],[0.41226866,0.75750428],[0.32742351,0.81117337],[0.36899693,0.66691384],[0.26699279,0.71563797],[0.16198395,0.82248985],
[0.2234327,0.87695172],[0.04093866,0.95906167],[0.23443418,0.74074641],[0.12626696,0.71456889],[0.110173,0.77656802],[0.0461 ...
[truncated: the full input is in inputs/Q040/]
```


Input `tree`:


```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalo
thorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothora
x","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```

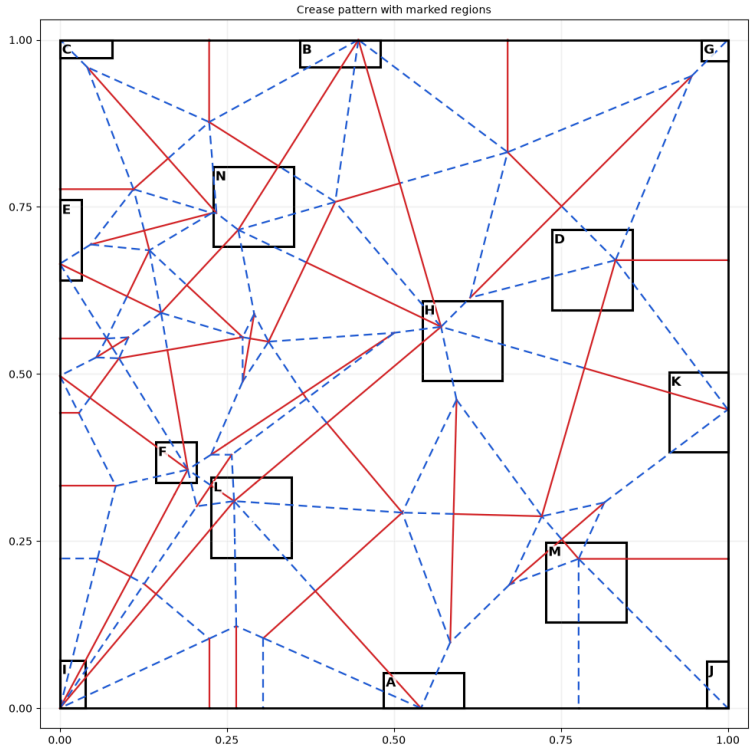

Input `regions`:

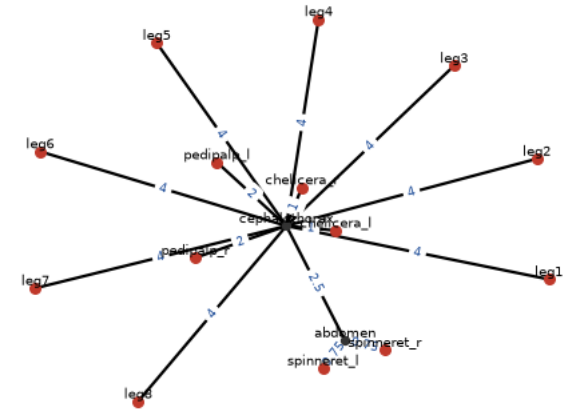

```
{"A":[0.4847,0.0,0.6047,0.0531],"B":[0.3593,0.959,0.4793,1.0],"C":[0.0,0.9733,0.0776,1.0],"D":[0.7368,0.5961,0.8568,0.7161],"E":[0.0,0.6404,0.032,0.760
4],"F":[0.1438,0.3381,0.2038,0.3981],"G":[0.9608,0.9686,1.0,1.0],"H":[0.5422,0.4898,0.6622,0.6098],"I":[0.0,0.0,0.0383,0.0719],"J":[0.9689,0.0,1.0,0.070
4],"K":[0.9117,0.3831,1.0,0.5031],"L":[0.2267,0.2252,0.3467,0.3452],"M":[0.728,0.1286,0.848,0.2486],"N":[0.2296,0.6904,0.3496,0.8104]}
```


Answer format:

```
{"regions": {"A": "flap_name", "B": "flap_name", ...}}
```

Use the flap (leaf) names of the given tree; each region gets a different flap.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q041

split: expert-analysis · family: repair · conventions: advanced (front matter) · files: inputs/Q041/

Task (solve, hard):
The crease pattern below was flat-foldable, but 12 of its mountain/valley assignments have been changed. Every vertex still satisfies Maekawa's and Kawasaki's theorems (and every other local test): the changes only break the GLOBAL layer order, so no flat folded state exists. Repair it: return a full M/V assignment of the SAME creases that folds flat, changing at most 12 assignments relative to the given pattern. Pass/fail.

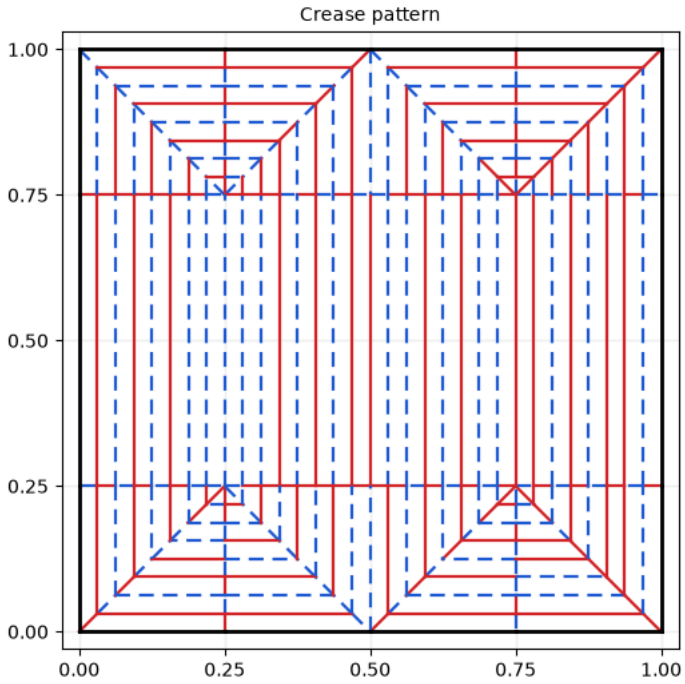
Input `crease_pattern`:


```
{"vertices_coords":[[0.21875,0.25],[0.25,0.25],[0.1875,0.25],[0.15625,0.25],[0.125,0.25],[0.09375,0.25],[0.0625,0.25],[0.03125,0.25],[0.0,0.25],[0.25,0.
0],[0.25,0.03125],[0.25,0.0625],[0.25,0.09375],[0.25,0.125],[0.25,0.15625],[0.25,0.1875],[0.25,0.21875],[0.0,0.0],[0.03125,0.03125],[0.0625,0.0625],[0.093
75,0.09375],[0.125,0.125],[0.15625,0.15625],[0.1875,0.1875],[0.21875,0.21875],[0.0,0.75],[0.03125,0.75],[0.0625,0.75],[0.09375,0.75],[0.125,0.75],[0.1562
5,0.75],[0.1875,0.75],[0.21875,0.75],[0.25,0.75],[0.0,1.0],[0.03125,0.96875],[0.0625,0.9375],[0.09375,0.90625],[0.125,0.875],[0.15625,0.84375],[0.1875,0.8
125],[0.21875,0.78125],[0.25,0.96875],[0.25,1.0],[0.25,0.9375],[0.25,0.90625],[0.25,0.875],[0.25,0.84375],[0.25,0.8125],[0.25,0.78125],[0.46875,0.03125],
[0.5,0.0],[0.4375,0.0625],[0.40625,0.09375],[0.375,0.125],[0.34375,0.15625],[0.3125,0.1875],[0.28125,0.21875],[0.46875,0.96875],[0.5,1.0],[0.4375,0.9375],
[0.40625,0.90625],[0.375,0.875],[0.34375,0.84375],[0.3125,0.8125],[0.28125,0.78125],[0.75,0.21875],[0.75,0.25],[0.75,0.1875],[0.75,0.15625],[0.75,0.125],
[0.75,0.09375],[0.75,0.0625],[0.75,0.03125],[0.75,0.0],[0.71875,0.25],[0.6875,0.25],[0.65625,0.25],[0.625,0.25],[0.59375,0.25], ...
[truncated: the full input is in inputs/Q041/]
```


Answer format: the SAME creases, each assigned M or V.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear exactly once, nothing may be added, and at most the stated number of assignments may differ from the input.



Working / answer

Q042

split: expert-analysis · family: repair · conventions: advanced (front matter) · files: inputs/Q042/

Task (solve, hard):
The crease pattern below was flat-foldable, but 12 of its mountain/valley assignments have been changed. Every vertex still satisfies Maekawa's and Kawasaki's theorems (and every other local test): the changes only break the GLOBAL layer order, so no flat folded state exists. Repair it: return a full M/V assignment of the SAME creases that folds flat, changing at most 12 assignments relative to the given pattern. Pass/fail.

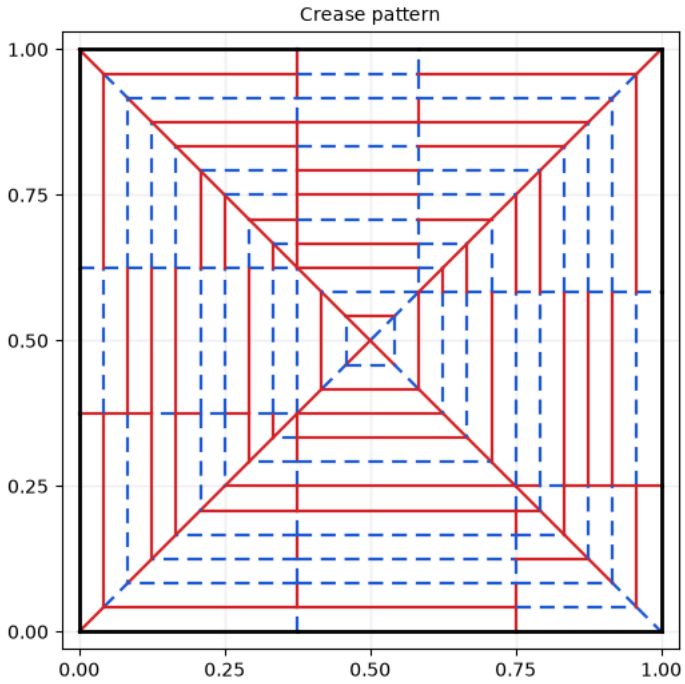
Input `crease_pattern`:


```
{"vertices_coords":[[0.33333333,0.375],[0.375,0.375],[0.29166667,0.375],[0.25,0.375],[0.20833333,0.375],[0.16666667,0.375],[0.125,0.375],[0.08333333,0.375],[0.04166667,0.375],[0.0,0.375],[0.375,0.0],[0.375,0.04166667],[0.375,0.08333333],[0.375,0.125],[0.375,0.16666667],[0.375,0.20833333],[0.375,0.25],[0.375,0.29166667],[0.375,0.33333333],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.125],[0.16666667,0.16666667],[0.20833333,0.20833333],[0.25,0.25],[0.29166667,0.29166667],[0.33333333,0.33333333],[0.0,0.625],[0.04166667,0.625],[0.08333333,0.625],[0.125,0.625],[0.16666667,0.625],[0.20833333,0.625],[0.25,0.625],[0.29166667,0.625],[0.33333333,0.625],[0.375,0.625],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],[0.29166667,0.70833333],[0.33333333,0.66666667],[0.375,0.66666667],[0.375,0.70833333],[0.375,0.75],[0.375,0.79166667],[0.375,0.83333333],[0.375,0.875],[0.375,0.91666667],[0.375,0.95833333],[0.375,1.0],[0.58333333,0.95833333],[0.58333333,1.0],[0.58333333,0.91666667],[0.58333333,0.875],[0.58333333,0.83333333],[0.58333333,0.79166667],[0.58333333,0.75],[0.58333333,0.70833333],[0.583 ...
[truncated: the full input is in inputs/Q042/]
```


Answer format: the SAME creases, each assigned M or V.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear exactly once, nothing may be added, and at most the stated number of assignments may differ from the input.



Working / answer

Q043

split: expert-analysis · family: repair · conventions: advanced (front matter) · files: inputs/Q043/

Task (solve, hard):
The crease pattern below was flat-foldable, but 11 of its mountain/valley assignments have been changed. Every vertex still satisfies Maekawa's and Kawasaki's theorems (and every other local test): the changes only break the GLOBAL layer order, so no flat folded state exists. Repair it: return a full M/V assignment of the SAME creases that folds flat, changing at most 11 assignments relative to the given pattern. Pass/fail.

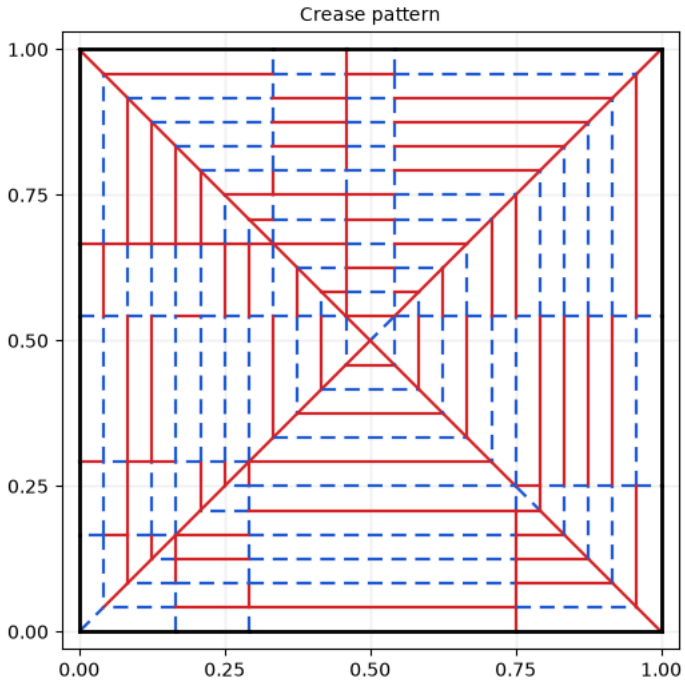
Input `crease_pattern`:


```
{"vertices_coords":[[0.125,0.16666667],[0.16666667,0.16666667],[0.08333333,0.16666667],[0.04166667,0.16666667],[0.0,0.16666667],[0.16666667,0.0],[0.16666667,0.04166667],[0.16666667,0.08333333],[0.16666667,0.125],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.125],[0.29166667,0.0],[0.29166667,0.04166667],[0.29166667,0.08333333],[0.29166667,0.125],[0.29166667,0.16666667],[0.29166667,0.20833333],[0.29166667,0.25],[0.29166667,0.29166667],[0.25,0.29166667],[0.20833333,0.29166667],[0.16666667,0.29166667],[0.125,0.29166667],[0.08333333,0.29166667],[0.04166667,0.29166667],[0.0,0.29166667],[0.25,0.25],[0.20833333,0.20833333],[0.0,0.66666667],[0.04166667,0.66666667],[0.08333333,0.66666667],[0.125,0.66666667],[0.16666667,0.66666667],[0.20833333,0.66666667],[0.25,0.66666667],[0.29166667,0.66666667],[0.33333333,0.66666667],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],[0.29166667,0.70833333],[0.33333333,0.70833333],[0.33333333,0.75],[0.33333333,0.79166667],[0.33333333,0.83333333],[0.33333333,0.875],[0.33333333,0.91666667],[0.33333333,0.95833333],[0.33333333,1.0],[0.0,0.54166667],[0.04166667 ...
[truncated: the full input is in inputs/Q043/]
```


Answer format: the SAME creases, each assigned M or V.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear exactly once, nothing may be added, and at most the stated number of assignments may differ from the input.



Working / answer

Q044

split: expert-analysis · family: repair · conventions: advanced (front matter) · files: inputs/Q044/

Task (solve, hard):
The crease pattern below was flat-foldable, but 12 of its mountain/valley assignments have been changed. Every vertex still satisfies Maekawa's and Kawasaki's theorems (and every other local test): the changes only break the GLOBAL layer order, so no flat folded state exists. Repair it: return a full M/V assignment of the SAME creases that folds flat, changing at most 12 assignments relative to the given pattern. Pass/fail.

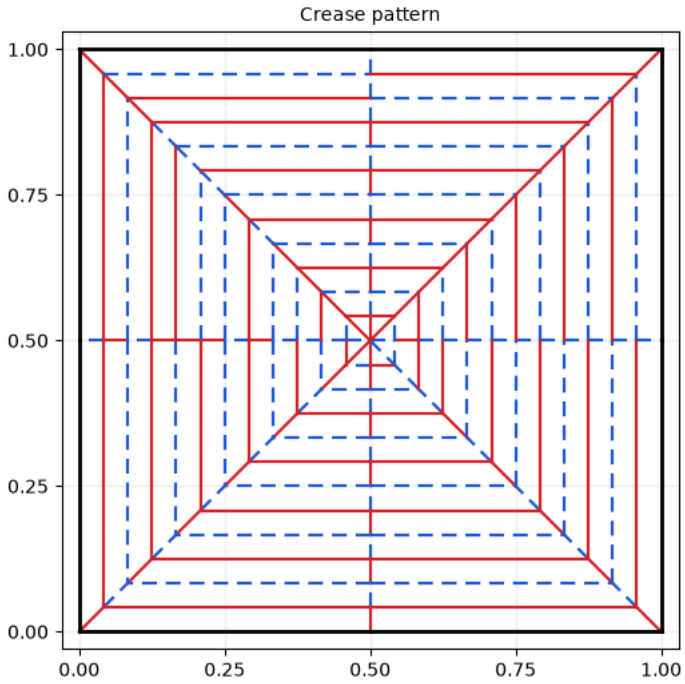
Input `crease_pattern`:


```
{"vertices_coords":[[0.45833333,0.5],[0.5,0.5],[0.41666667,0.5],[0.375,0.5],[0.33333333,0.5],[0.29166667,0.5],[0.25,0.5],[0.20833333,0.5],[0.16666667,0.
5],[0.125,0.5],[0.08333333,0.5],[0.04166667,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.04166667],[0.5,0.08333333],[0.5,0.125],[0.5,0.16666667],[0.5,0.20833333],[0.5,
0.25],[0.5,0.29166667],[0.5,0.33333333],[0.5,0.375],[0.5,0.41666667],[0.5,0.45833333],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.1
25],[0.16666667,0.16666667],[0.20833333,0.20833333],[0.25,0.25],[0.29166667,0.29166667],[0.33333333,0.33333333],[0.375,0.375],[0.41666667,0.41666667],[0.4
5833333,0.45833333],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],
[0.29166667,0.70833333],[0.33333333,0.66666667],[0.375,0.625],[0.41666667,0.58333333],[0.45833333,0.54166667],[0.5,0.54166667],[0.5,0.58333333],[0.5,0.62
5],[0.5,0.66666667],[0.5,0.70833333],[0.5,0.75],[0.5,0.79166667],[0.5,0.83333333],[0.5,0.875],[0.5,0.91666667],[0.5,0.95833333],[0.5,1.0],[0.95833333,0.04
166667],[1.0,0.0],[0.91666667,0.08333333],[0.875,0.125],[0.83333333,0.16666667],[0.79166667,0.20833333],[0.75,0.25],[0.70833333 ...
[truncated: the full input is in inputs/Q044/]
```


Answer format: the SAME creases, each assigned M or V.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear exactly once, nothing may be added, and at most the stated number of assignments may differ from the input.



Working / answer

Q045

split: expert-analysis · family: repair · conventions: advanced (front matter) · files: inputs/Q045/

Task (solve, hard):
The crease pattern below was flat-foldable, but 15 of its mountain/valley assignments have been changed. Every vertex still satisfies Maekawa's and Kawasaki's theorems (and every other local test): the changes only break the GLOBAL layer order, so no flat folded state exists. Repair it: return a full M/V assignment of the SAME creases that folds flat, changing at most 15 assignments relative to the given pattern. Pass/fail.

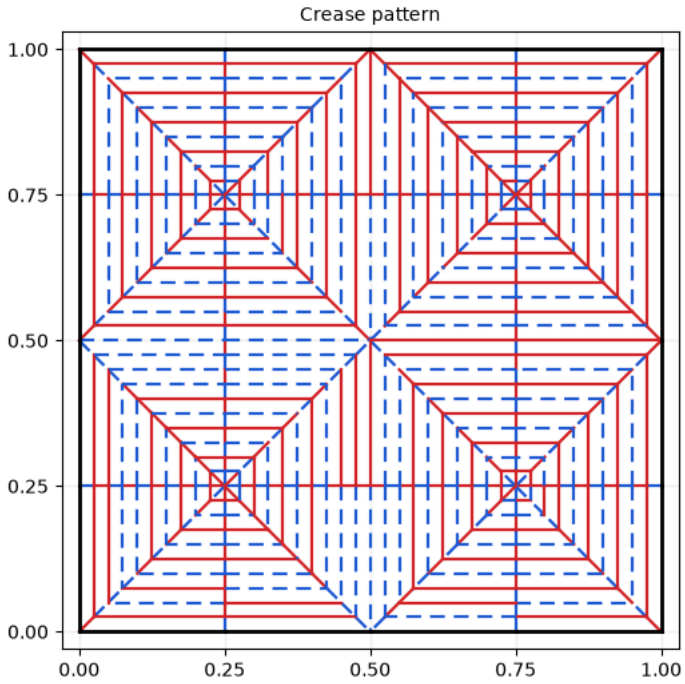
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.25],[0.025,0.25],[0.0,0.75],[0.025,0.75],[1.0,0.25],[0.975,0.25],[1.0,0.75],[0.975,0.75],[0.05,0.25],[0.05,0.75],[0.95,0.25],
[0.95,0.75],[0.075,0.25],[0.075,0.75],[0.925,0.25],[0.925,0.75],[0.1,0.25],[0.1,0.75],[0.9,0.25],[0.9,0.75],[0.125,0.25],[0.125,0.75],[0.875,0.25],[0.875,
0.75],[0.15,0.25],[0.15,0.75],[0.85,0.25],[0.85,0.75],[0.175,0.25],[0.175,0.75],[0.825,0.25],[0.825,0.75],[0.2,0.25],[0.2,0.75],[0.8,0.25],[0.8,0.75],[0.2
25,0.25],[0.225,0.75],[0.775,0.25],[0.775,0.75],[0.25,0.25],[0.25,0.75],[0.75,0.25],[0.75,0.75],[0.0,0.5],[0.025,0.475],[0.025,0.525],[1.0,0.5],[0.975,0.4
75],[0.975,0.525],[0.05,0.45],[0.05,0.55],[0.95,0.45],[0.95,0.55],[0.075,0.425],[0.075,0.575],[0.925,0.425],[0.925,0.575],[0.1,0.4],[0.1,0.6],[0.9,0.4],
[0.9,0.6],[0.125,0.375],[0.125,0.625],[0.875,0.375],[0.875,0.625],[0.15,0.35],[0.15,0.65],[0.85,0.35],[0.85,0.65],[0.175,0.325],[0.175,0.675],[0.825,0.32
5],[0.825,0.675],[0.2,0.3],[0.2,0.7],[0.8,0.3],[0.8,0.7],[0.225,0.275],[0.225,0.725],[0.775,0.275],[0.775,0.725],[0.25,0.0],[0.25,0.025],[0.25,1.0],[0.25,
0.975],[0.75,0.0],[0.75,0.025],[0.75,1.0],[0.75,0.975],[0.25,0.05],[0.25,0.95],[0.75,0.05],[0.75,0.95],[0.25,0.075],[0.25,0.925 ...
[truncated: the full input is in inputs/Q045/]
```


Answer format: the SAME creases, each assigned M or V.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear exactly once, nothing may be added, and at most the stated number of assignments may differ from the input.



Working / answer

Q046

split: hard · family: analytic · conventions: advanced (front matter) · files: inputs/Q046/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

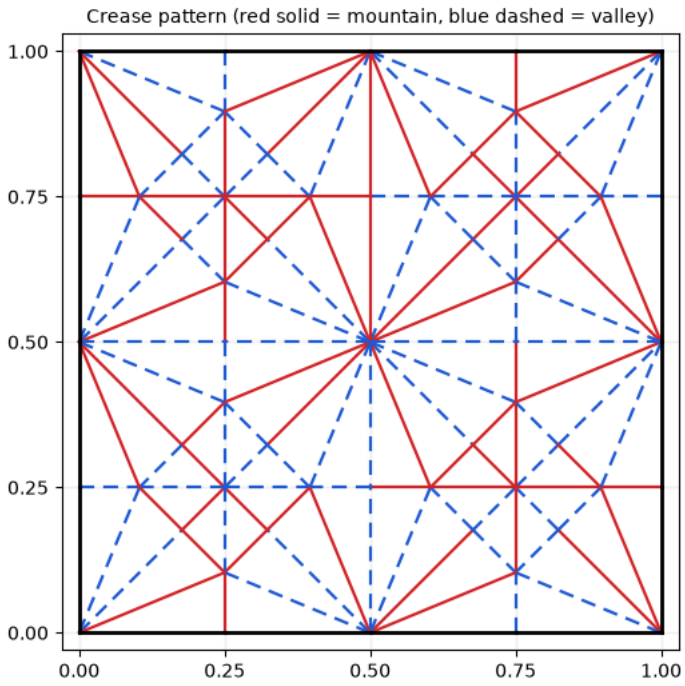
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.176777,0.176777],[0.25,0.25],[0.323223,0.323223],[0.5,0.5],[0.5,0.0],[0.323223,0.176777],[0.176777,0.323223],[0.0,0.5],
[0.25,0.0],[0.25,0.103553],[0.5,0.25],[0.396447,0.25],[0.25,0.5],[0.25,0.396447],[0.0,0.25],[0.103553,0.25],[1.0,0.0],[0.823223,0.176777],[0.75,0.25],[0.6
76777,0.323223],[0.676777,0.176777],[0.823223,0.323223],[1.0,0.5],[0.75,0.0],[0.75,0.103553],[0.603553,0.25],[0.75,0.5],[0.75,0.396447],[1.0,0.25],[0.8964
47,0.25],[0.0,1.0],[0.176777,0.823223],[0.25,0.75],[0.323223,0.676777],[0.5,1.0],[0.323223,0.823223],[0.176777,0.676777],[0.25,1.0],[0.25,0.896447],[0.5,
0.75],[0.396447,0.75],[0.25,0.603553],[0.0,0.75],[0.103553,0.75],[1.0,1.0],[0.823223,0.823223],[0.75,0.75],[0.676777,0.676777],[0.676777,0.823223],[0.8232
23,0.676777],[0.75,1.0],[0.75,0.896447],[0.603553,0.75],[0.75,0.603553],[1.0,0.75],[0.896447,0.75]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[2,6],
[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,16],[0,10],[5,10],[5,1
2],[4,12],[4,14],[8,14],[8,16],[17,18],[18,19],[19,20],[4,20],[5,21],[19,21],[19,22],[22,23],[24,25],[19,25],[11,26],[19,26],[2 ...
[truncated: the full input is in inputs/Q046/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q047

split: hard · family: analytic · conventions: advanced (front matter) · files: inputs/Q047/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

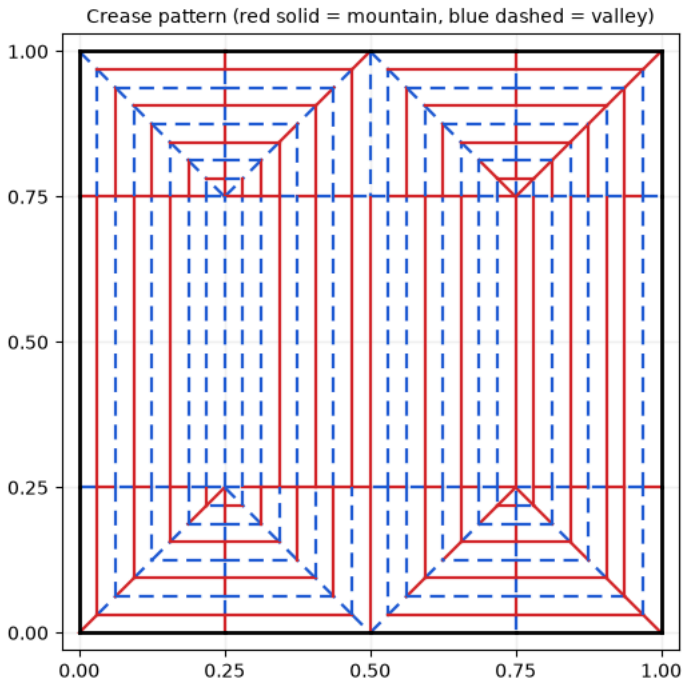
Input `crease_pattern`:


```
{"vertices_coords":[[0.21875,0.25],[0.25,0.25],[0.1875,0.25],[0.15625,0.25],[0.125,0.25],[0.09375,0.25],[0.0625,0.25],[0.03125,0.25],[0.0,0.25],[0.25,0.
0],[0.25,0.03125],[0.25,0.0625],[0.25,0.09375],[0.25,0.125],[0.25,0.15625],[0.25,0.1875],[0.25,0.21875],[0.0,0.0],[0.03125,0.03125],[0.0625,0.0625],[0.093
75,0.09375],[0.125,0.125],[0.15625,0.15625],[0.1875,0.1875],[0.21875,0.21875],[0.0,0.75],[0.03125,0.75],[0.0625,0.75],[0.09375,0.75],[0.125,0.75],[0.1562
5,0.75],[0.1875,0.75],[0.21875,0.75],[0.25,0.75],[0.0,1.0],[0.03125,0.96875],[0.0625,0.9375],[0.09375,0.90625],[0.125,0.875],[0.15625,0.84375],[0.1875,0.8
125],[0.21875,0.78125],[0.25,0.96875],[0.25,1.0],[0.25,0.9375],[0.25,0.90625],[0.25,0.875],[0.25,0.84375],[0.25,0.8125],[0.25,0.78125],[0.46875,0.03125],
[0.5,0.0],[0.4375,0.0625],[0.40625,0.09375],[0.375,0.125],[0.34375,0.15625],[0.3125,0.1875],[0.28125,0.21875],[0.46875,0.96875],[0.5,1.0],[0.4375,0.9375],
[0.40625,0.90625],[0.375,0.875],[0.34375,0.84375],[0.3125,0.8125],[0.28125,0.78125],[0.75,0.21875],[0.75,0.25],[0.75,0.1875],[0.75,0.15625],[0.75,0.125],
[0.75,0.09375],[0.75,0.0625],[0.75,0.03125],[0.75,0.0],[0.71875,0.25],[0.6875,0.25],[0.65625,0.25],[0.625,0.25],[0.59375,0.25], ...
[truncated: the full input is in inputs/Q047/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q048

split: hard · family: analytic · conventions: advanced (front matter) · files: inputs/Q048/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

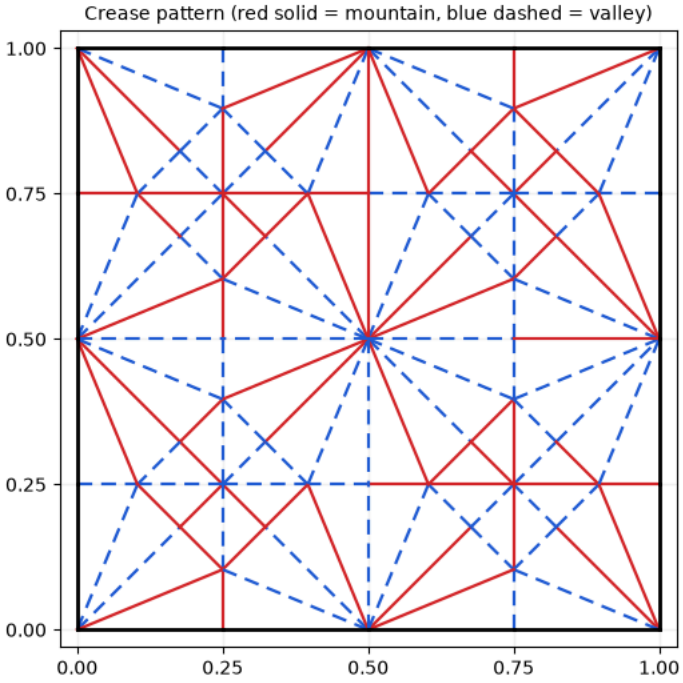
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.176777,0.176777],[0.25,0.25],[0.323223,0.323223],[0.5,0.5],[0.5,0.0],[0.323223,0.176777],[0.176777,0.323223],[0.0,0.5],
[0.25,0.0],[0.25,0.103553],[0.5,0.25],[0.396447,0.25],[0.25,0.5],[0.25,0.396447],[0.0,0.25],[0.103553,0.25],[1.0,0.0],[0.823223,0.176777],[0.75,0.25],[0.6
76777,0.323223],[0.676777,0.176777],[0.823223,0.323223],[1.0,0.5],[0.75,0.0],[0.75,0.103553],[0.603553,0.25],[0.75,0.5],[0.75,0.396447],[1.0,0.25],[0.8964
47,0.25],[0.0,1.0],[0.176777,0.823223],[0.25,0.75],[0.323223,0.676777],[0.5,1.0],[0.323223,0.823223],[0.176777,0.676777],[0.25,1.0],[0.25,0.896447],[0.5,
0.75],[0.396447,0.75],[0.25,0.603553],[0.0,0.75],[0.103553,0.75],[1.0,1.0],[0.823223,0.823223],[0.75,0.75],[0.676777,0.676777],[0.676777,0.823223],[0.8232
23,0.676777],[0.75,1.0],[0.75,0.896447],[0.603553,0.75],[0.75,0.603553],[1.0,0.75],[0.896447,0.75]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[2,6],
[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,16],[0,10],[5,10],[5,1
2],[4,12],[4,14],[8,14],[8,16],[17,18],[18,19],[19,20],[4,20],[5,21],[19,21],[19,22],[22,23],[24,25],[19,25],[11,26],[19,26],[2 ...
[truncated: the full input is in inputs/Q048/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q049

split: hard · family: analytic · conventions: advanced (front matter) · files: inputs/Q049/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

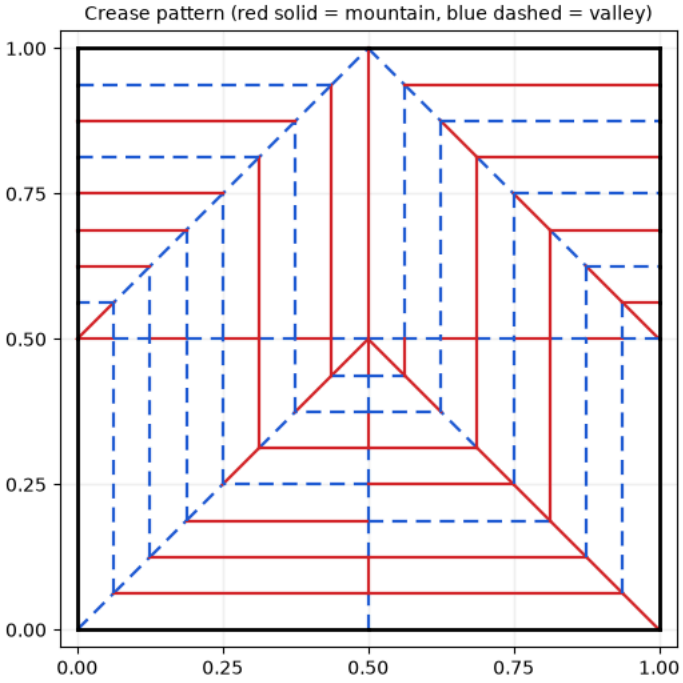
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.0625],[0.5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.5,0.5],[0.0,0.0],[0.0625,0.0625],
[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.0,0.5],[0.0625,0.5],[0.125,0.5],[0.1875,0.5],[0.25,0.5],[0.312
5,0.5],[0.375,0.5],[0.4375,0.5],[0.4375,0.9375],[0.5,1.0],[0.375,0.875],[0.3125,0.8125],[0.25,0.75],[0.1875,0.6875],[0.125,0.625],[0.0625,0.5625],[0.9375,
0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.5625,0.9375],[0.625,0.875],[0.6875,0.8125],
[0.75,0.75],[0.8125,0.6875],[0.875,0.625],[0.9375,0.5625],[1.0,0.5],[0.9375,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.
5],[0.0,0.5625],[0.0,0.625],[0.0,0.6875],[0.0,0.75],[0.0,0.8125],[0.0,0.875],[0.0,0.9375],[1.0,0.5625],[1.0,0.625],[1.0,0.6875],[1.0,0.75],[1.0,0.8125],
[1.0,0.875],[1.0,0.9375],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[1
4,15],[15,16],[8,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[8,24],[25,26],[25,27],[27,28],[28,29],[29,30],[30,31] ...
[truncated: the full input is in inputs/Q049/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q050

split: hard · family: analytic · conventions: advanced (front matter) · files: inputs/Q050/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

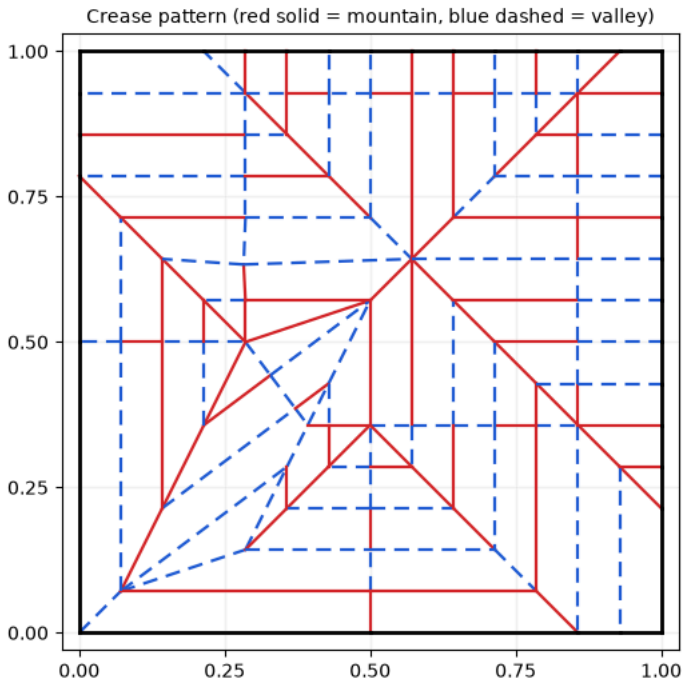
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.071429,0.071429],[0.142857,0.214286],[0.214286,0.357143],[0.285714,0.142857],[0.214286,0.5],[0.285714,0.5],[0.142857,0.
5],[0.071429,0.5],[0.0,0.5],[0.285714,0.857143],[0.285714,0.928571],[0.285714,0.785714],[0.285714,0.714286],[0.282584,0.63336],[0.285714,0.571429],[0.3285
71,0.442857],[0.357143,0.285714],[0.371429,0.385714],[0.392857,0.357143],[0.428571,0.428571],[0.5,0.0],[0.5,0.071429],[0.5,0.142857],[0.5,0.214286],[0.5,
0.285714],[0.5,0.357143],[0.428571,0.357143],[0.357143,0.214286],[0.428571,0.285714],[0.5,0.571429],[0.5,0.714286],[0.571429,0.642857],[0.428571,0.78571
4],[0.357143,0.857143],[0.571429,0.357143],[0.642857,0.357143],[0.714286,0.357143],[0.785714,0.357143],[0.857143,0.357143],[0.642857,0.571429],[0.714286,
0.5],[0.785714,0.428571],[0.785714,0.928571],[0.857143,0.928571],[0.714286,0.928571],[0.642857,0.928571],[0.571429,0.928571],[0.5,0.928571],[0.428571,0.92
8571],[0.357143,0.928571],[0.857143,0.428571],[0.857143,0.5],[0.857143,0.571429],[0.857143,0.642857],[0.857143,0.714286],[0.857143,0.785714],[0.857143,0.8
57143],[0.642857,0.714286],[0.714286,0.785714],[0.785714,0.857143],[0.214286,0.571429],[0.142857,0.642857],[0.071429,0.714286],[ ...
[truncated: the full input is in inputs/Q050/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q051

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q051/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a multi-flap base on two-tone paper). Colour requirement: at the tip of each listed flap the BACK side of the paper must be visible from above: {"n2": {"above": "back"}, "n6": {"above": "back"}, "n7": {"above": "back"}, "n8": {"above": "back"}, "n9": {"above": "back"}}

Input `tree`:

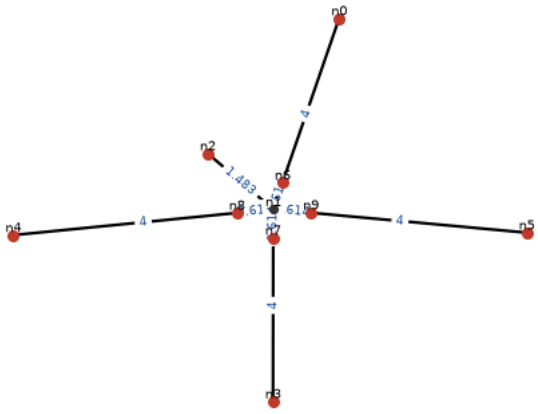

```
{"edges":[["n0","n1",4.0],["n1","n9",0.614],["n1","n6",0.614],["n1","n8",0.614],["n1","n7",0.614],["n1","n2",1.483],["n1","n4",4.0],["n1","n3",4.0],["n
1","n5",4.0]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q052

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q052/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a multi-flap base on two-tone paper). Use only creases at multiples of 22.5 degrees, and at the tip of each listed flap the BACK side must be visible from above: {"n2": {"above": "back"}, "n6": {"above": "back"}, "n7": {"above": "back"}, "n8": {"above": "back"}, "n9": {"above": "back"}}

Input `tree`:

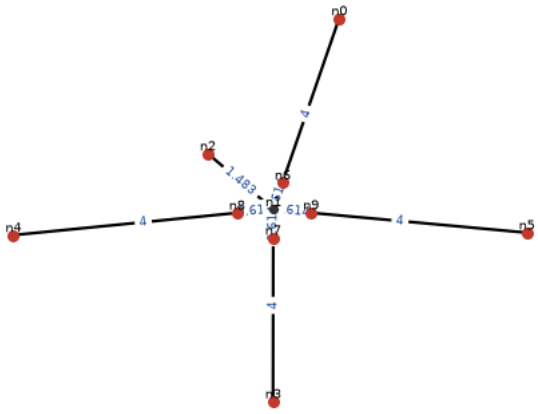

```
{"edges":[["n0","n1",4.0],["n1","n9",0.614],["n1","n6",0.614],["n1","n8",0.614],["n1","n7",0.614],["n1","n2",1.483],["n1","n4",4.0],["n1","n3",4.0],["n
1","n5",4.0]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q053

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q053/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a beetle: head with two antennae, six legs (two per thoracic segment), abdomen).

Input `tree`:

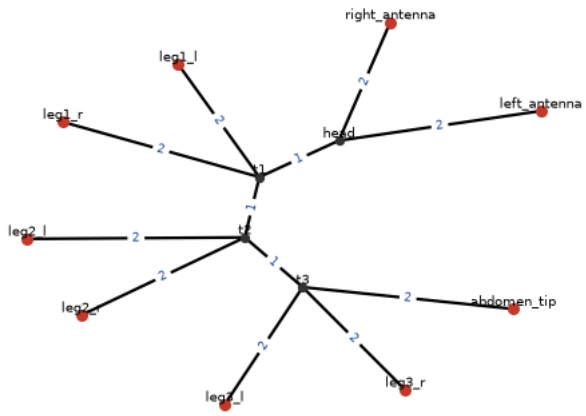

```
{"edges":[["head","left_antenna",2],["head","right_antenna",2],["head","t1",1],["t1","leg1_l",2],["t1","leg1_r",2],["t1","t2",1],["t2","leg2_l",2],["t
2","leg2_r",2],["t2","t3",1],["t3","leg3_l",2],["t3","leg3_r",2],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q054

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q054/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a standing bird: head and beak, two wings, two legs, a tail).

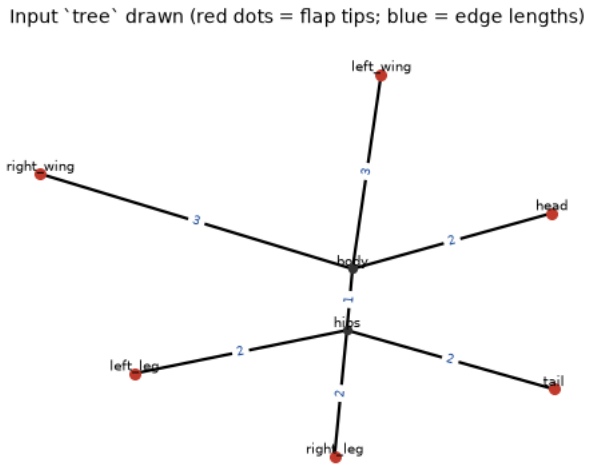
Input `tree`:

```
{"edges":[["body","head",2],["body","left_wing",3],["body","right_wing",3],["body","hips",1],["hips","left_leg",2],["hips","right_leg",2],["hips","tail",2]]}
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q055

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q055/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a centipede: antennae, forcipules, nine pairs of legs along a segmented body, long tail legs).

Input `tree`:

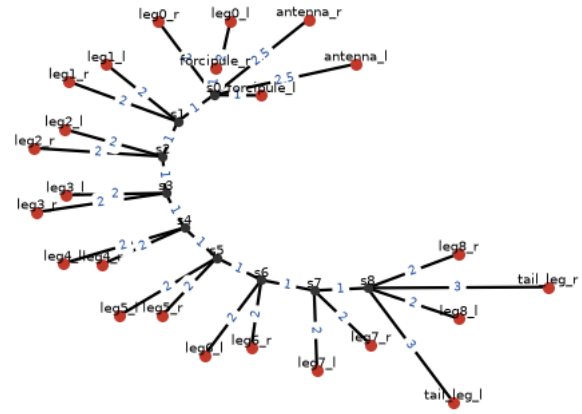

```
{"edges":[["s0","antenna_l",2.5],["s0","antenna_r",2.5],["s0","forcipule_l",1],["s0","forcipule_r",1],["s0","s1",1],["s1","s2",1],["s2","s3",1],["s3","s4",1],["s4","s5",1],["s5","s6",1],["s6","s7",1],["s7","s8",1],["s0","leg0_l",2],["s0","leg0_r",2],["s1","leg1_l",2],["s1","leg1_r",2],["s2","leg2_l",2],["s2","leg2_r",2],["s3","leg3_l",2],["s3","leg3_r",2],["s4","leg4_l",2],["s4","leg4_r",2],["s5","leg5_l",2],["s5","leg5_r",2],["s6","leg6_l",2],["s6","leg6_r",2],["s7","leg7_l",2],["s7","leg7_r",2],["s8","leg8_l",2],["s8","leg8_r",2],["s8","tail_leg_l",3],["s8","tail_leg_r",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q056

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q056/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a crab: two claws, eight legs, eyes on stalks).

Input `tree`:

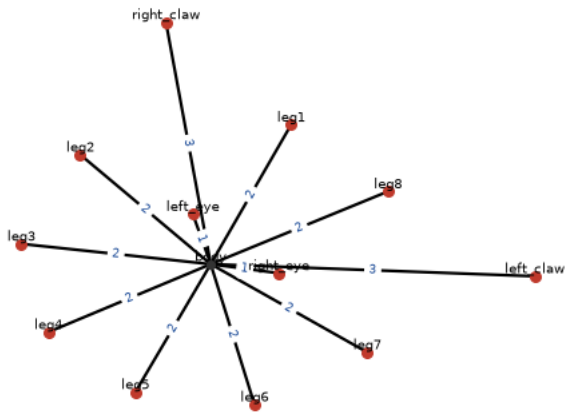

```
{"edges":[["body","left_claw",3],["body","right_claw",3],["body","left_eye",1],["body","right_eye",1],["body","leg1",2],["body","leg2",2],["body","leg3",2],["body","leg4",2],["body","leg5",2],["body","leg6",2],["body","leg7",2],["body","leg8",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q057

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q057/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a deer: branched antlers (two tines each), head, four legs, short tail).

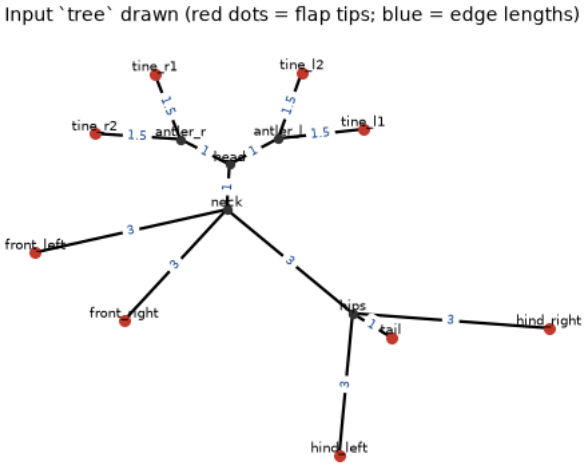
Input `tree`:


```
{"edges":[["neck","head",1],["head","antler_l",1],["antler_l","tine_l1",1.5],["antler_l","tine_l2",1.5],["head","antler_r",1],["antler_r","tine_r1",1.5],
["antler_r","tine_r2",1.5],["neck","front_left",3],["neck","front_right",3],["neck","hips",3],["hips","hind_left",3],["hips","hind_right",3],["hips","tai
l",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q058

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q058/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a dragon: head with two horns, two wings, four legs, long tail).

Input `tree`:

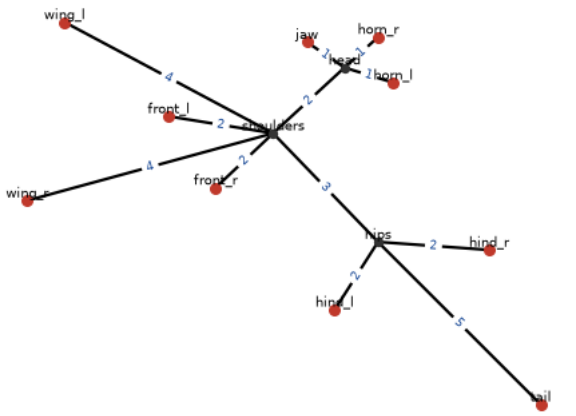

```
{"edges":[["head","horn_l",1],["head","horn_r",1],["head","jaw",1],["head","shoulders",2],["shoulders","wing_l",4],["shoulders","wing_r",4],["shoulder
s","front_l",2],["shoulders","front_r",2],["shoulders","hips",3],["hips","hind_l",2],["hips","hind_r",2],["hips","tail",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q059

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q059/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a generated multi-flap base).

Input `tree`:

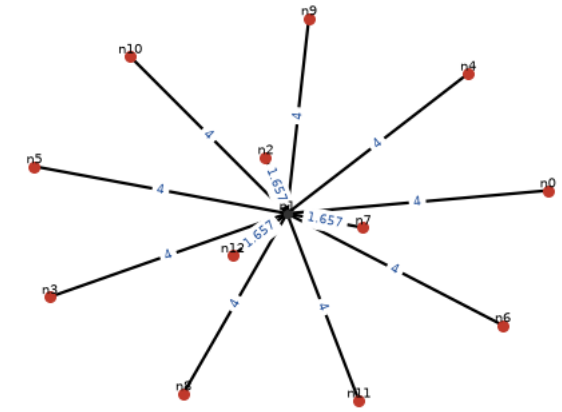

```
{"edges":[["n0","n1",4.0],["n1","n4",4.0],["n1","n9",4.0],["n1","n10",4.0],["n1","n5",4.0],["n1","n2",1.657],["n1","n3",4.0],["n1","n12",1.657],["n1","n7",1.657],["n1","n8",4.0],["n1","n11",4.0],["n1","n6",4.0]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q060

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q060/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a generated multi-flap base).

Input `tree`:

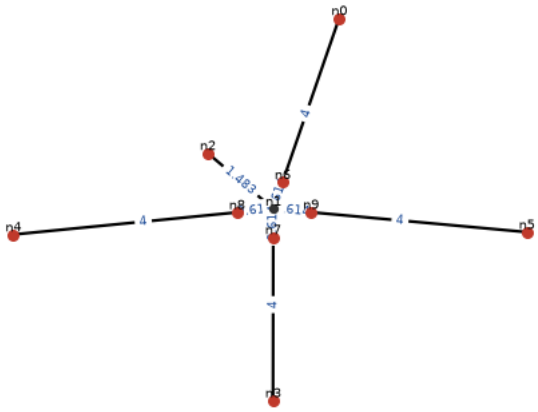

```
{"edges":[["n0","n1",4.0],["n1","n9",0.614],["n1","n6",0.614],["n1","n8",0.614],["n1","n7",0.614],["n1","n2",1.483],["n1","n4",4.0],["n1","n3",4.0],["n
1","n5",4.0]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q061

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q061/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a lizard: head, four legs, a long tail and a body).

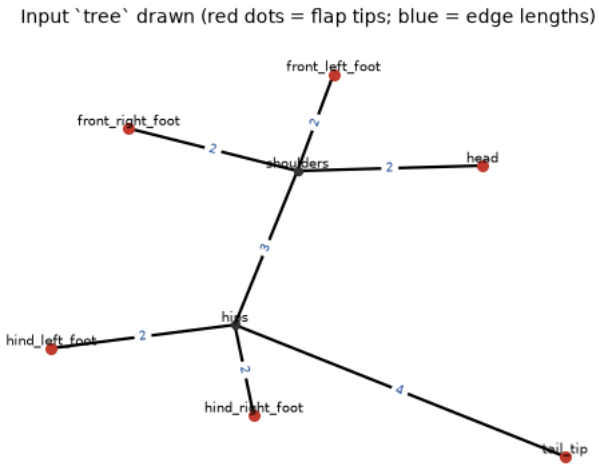
Input `tree`:


```
{"edges":[["shoulders","head",2],["shoulders","front_left_foot",2],["shoulders","front_right_foot",2],["shoulders","hips",3],["hips","hind_left_foot",2],
["hips","hind_right_foot",2],["hips","tail_tip",4]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q062

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q062/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a lobster: two big claws, long antennae, antennules, eight walking legs, a three-lobed tail fan).

Input `tree`:

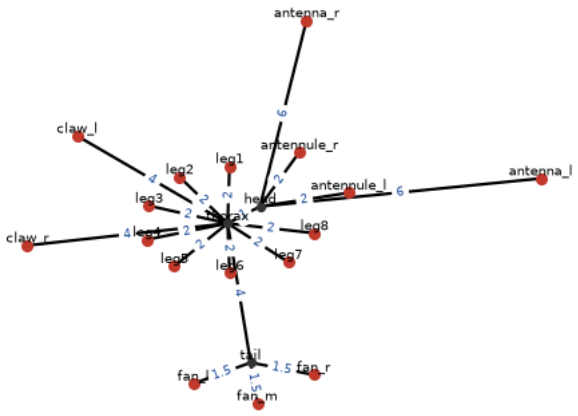

```
{"edges":[["head","antenna_l",6],["head","antenna_r",6],["head","antennule_l",2],["head","antennule_r",2],["head","thorax",1],["thorax","claw_l",4],["thorax","claw_r",4],["thorax","leg1",2],["thorax","leg2",2],["thorax","leg3",2],["thorax","leg4",2],["thorax","leg5",2],["thorax","leg6",2],["thorax","leg7",2],["thorax","leg8",2],["thorax","tail",4],["tail","fan_l",1.5],["tail","fan_m",1.5],["tail","fan_r",1.5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q063

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q063/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a praying mantis: antennae, raptorial forelegs, four walking legs, four wings, abdomen with two cerci).

Input `tree`:

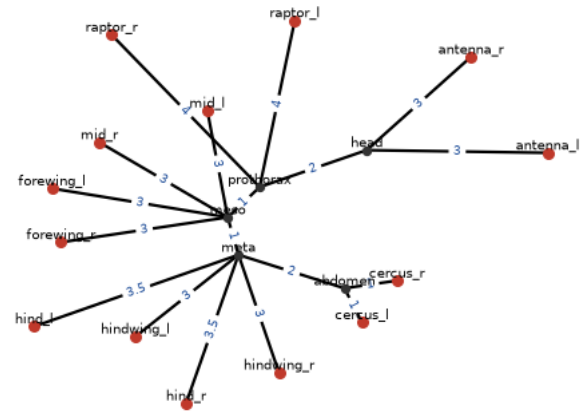

```
{"edges":[["head","antenna_l",3],["head","antenna_r",3],["head","prothorax",2],["prothorax","raptor_l",4],["prothorax","raptor_r",4],["prothorax","meso",1],["meso","mid_l",3],["meso","mid_r",3],["meso","forewing_l",3],["meso","forewing_r",3],["meso","meta",1],["meta","hind_l",3.5],["meta","hind_r",3.5],["meta","hindwing_l",3],["meta","hindwing_r",3],["meta","abdomen",2],["abdomen","cercus_l",1],["abdomen","cercus_r",1]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q064

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q064/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a pegasus: head with two ears and a muzzle, wings with three primary feathers each, four legs, a tail).

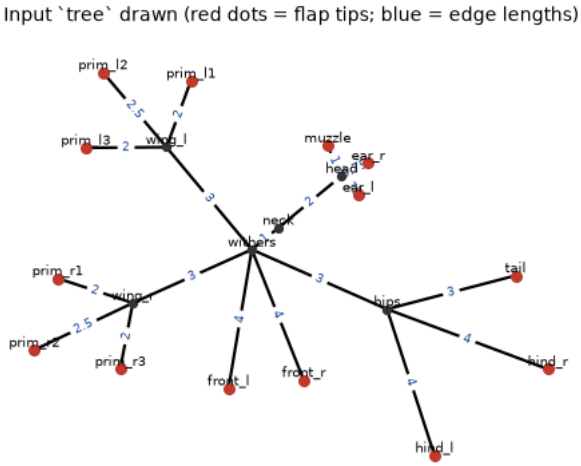
Input `tree`:


```
{"edges":[["neck","head",2],["head","ear_l",0.75],["head","ear_r",0.75],["head","muzzle",1],["neck","withers",1],["withers","wing_l",3],["wing_l","prim_l1",2],["wing_l","prim_l2",2.5],["wing_l","prim_l3",2],["withers","wing_r",3],["wing_r","prim_r1",2],["wing_r","prim_r2",2.5],["wing_r","prim_r3",2],["withers","front_l",4],["withers","front_r",4],["withers","hips",3],["hips","hind_l",4],["hips","hind_r",4],["hips","tail",3]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q065

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q065/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a scorpion: two pincers, eight legs, a long segmented tail with a stinger).

Input `tree`:

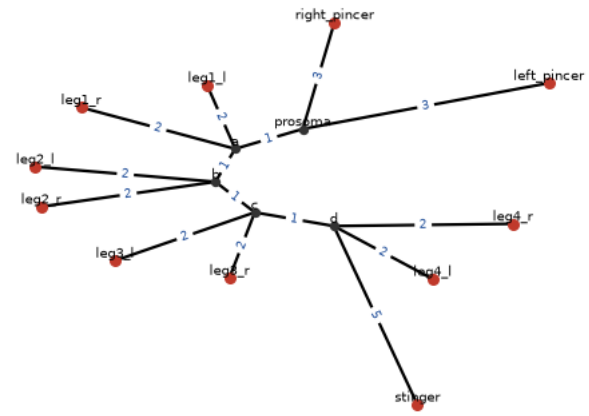

```
{"edges":[["prosoma","left_pincer",3],["prosoma","right_pincer",3],["prosoma","a",1],["a","leg1_l",2],["a","leg1_r",2],["a","b",1],["b","leg2_l",2],
["b","leg2_r",2],["b","c",1],["c","leg3_l",2],["c","leg3_r",2],["c","d",1],["d","leg4_l",2],["d","leg4_r",2],["d","stinger",5]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q066

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q066/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a spider: eight equal legs, a small head and an abdomen).

Input `tree`:

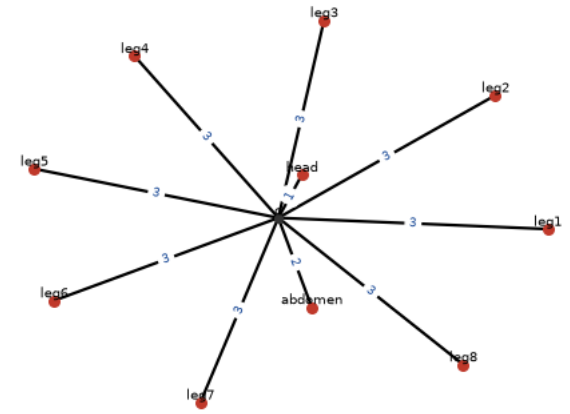

```
{"edges":[["c","leg1",3],["c","leg2",3],["c","leg3",3],["c","leg4",3],["c","leg5",3],["c","leg6",3],["c","leg7",3],["c","leg8",3],["c","head",1],["c","abdomen",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q067

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q067/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a stag beetle: long mandibles, antennae, palps, six legs, two elytra (wings), abdomen).

Input `tree`:

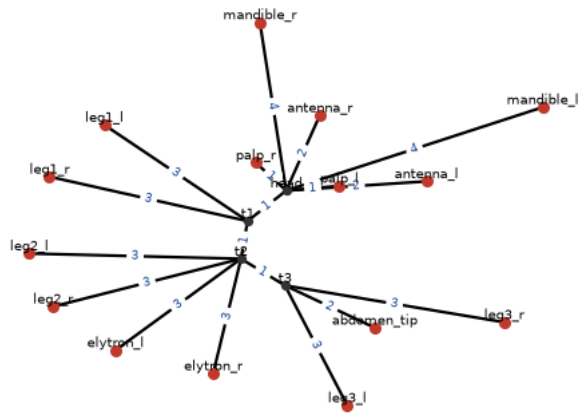

```
{"edges":[["head","mandible_l",4],["head","mandible_r",4],["head","antenna_l",2],["head","antenna_r",2],["head","palp_l",1],["head","palp_r",1],["head","t1",1],["t1","leg1_l",3],["t1","leg1_r",3],["t1","t2",1],["t2","leg2_l",3],["t2","leg2_r",3],["t2","elytron_l",3],["t2","elytron_r",3],["t2","t3",1],["t3","leg3_l",3],["t3","leg3_r",3],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q068

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q068/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a tarantula: eight long legs, pedipalps, chelicerae, spinnerets on a big abdomen).

Input `tree`:

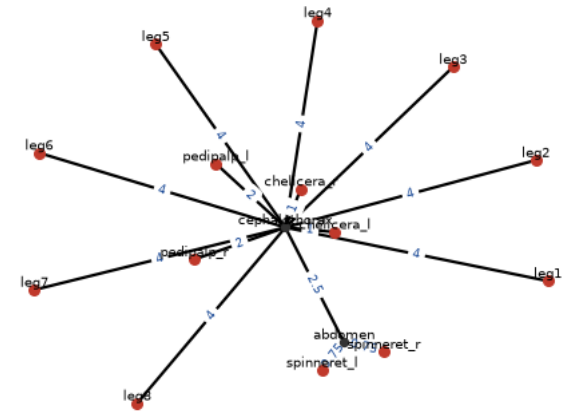

```
{"edges":[["cephalothorax","leg1",4],["cephalothorax","leg2",4],["cephalothorax","leg3",4],["cephalothorax","leg4",4],["cephalothorax","leg5",4],["cephalothorax","leg6",4],["cephalothorax","leg7",4],["cephalothorax","leg8",4],["cephalothorax","pedipalp_l",2],["cephalothorax","pedipalp_r",2],["cephalothorax","chelicera_l",1],["cephalothorax","chelicera_r",1],["cephalothorax","abdomen",2.5],["abdomen","spinneret_l",0.75],["abdomen","spinneret_r",0.75]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q069

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q069/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a bird base enlarged by border grafting). Start from the given bird-base crease pattern: shrink it by 1/(1+2*0.1) and graft a border strip of width 0.1 (before shrinking) all around it, so that at least 90% of the original creases are kept (same positions after the shrink, assignments may all be swapped). The border may be folded.

Input `tree`:


```
{"edges":[["n0","n1",4.0],["n1","n9",0.614],["n1","n6",0.614],["n1","n8",0.614],["n1","n7",0.614],["n1","n2",1.483],["n1","n4",4.0],["n1","n3",4.0],["n
1","n5",4.0]]}
```

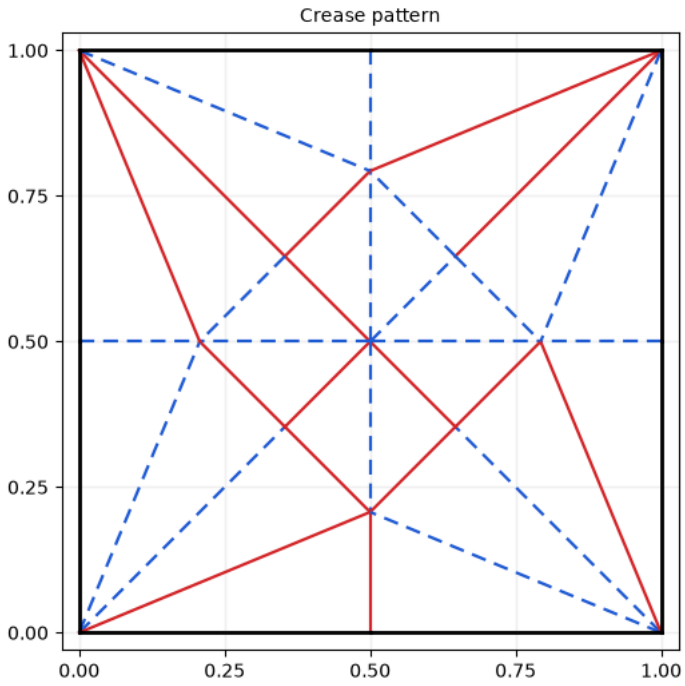

Input `base_crease_pattern`:

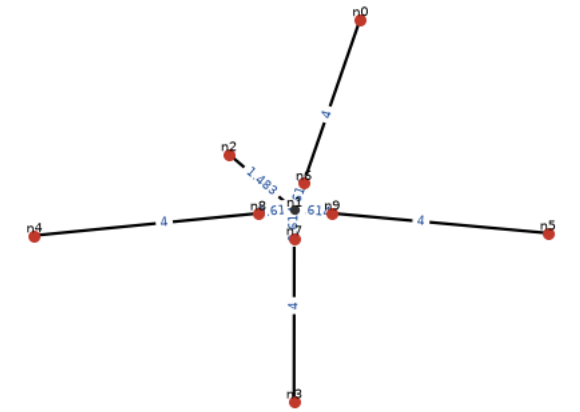

```
{"vertices_coords":[[0.0,0.0],[0.35355339,0.35355339],[0.5,0.5],[0.64644661,0.64644661],[1.0,1.0],[1.0,0.0],[0.64644661,0.35355339],[0.35355339,0.6464466
1],[0.0,1.0],[0.5,0.0],[0.5,0.20710678],[1.0,0.5],[0.79289322,0.5],[0.5,1.0],[0.5,0.79289322],[0.0,0.5],[0.20710678,0.5]],"edges_vertices":[[0,1],[1,2],
[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,
16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[0,9],[5,9],[5,11],[4,11],[4,13],[8,13],[8,15],[0,15]],"edges_assignment":
["V","M","V","M","V","M","M","M","M","V","V","V","V","V","V","V","M","M","V","V","M","V","M","V","M","M","V","M","V","M","V","M","V","M","V","M","B","B","B","B","B","B","B",
aces_vertices":[[0,1,16],[1,0,10],[1,2,16],[2,1,10],[2,3,14],[3,2,12],[3,4,14],[4,3,12],[5,6,10],[6,5,12],[2,6,12],[6,2,10],[2,7,16],[7,2,14],[7,8,16],[8,
7,14],[9,10,0],[10,9,5],[11,12,5],[12,11,4],[13,14,4],[14,13,8],[15,16,8],[16,15,0]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.





Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q070

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q070/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a lizard: head, four legs, a long tail and a body). The crease pattern must be box-pleated on an N x N grid with 8 <= N <= 32.

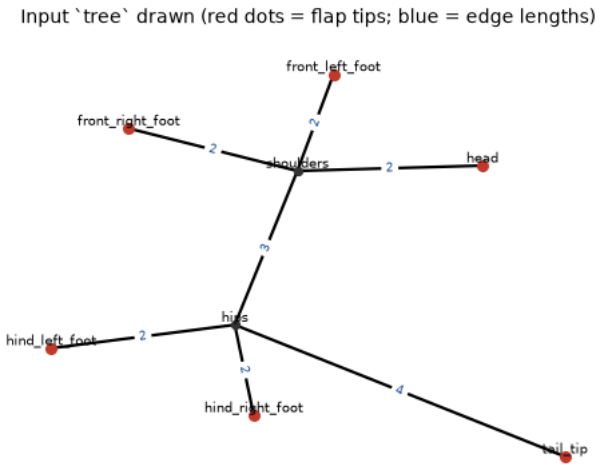
Input `tree`:


```
{"edges":[["shoulders","head",2],["shoulders","front_left_foot",2],["shoulders","front_right_foot",2],["shoulders","hips",3],["hips","hind_left_foot",2],
["hips","hind_right_foot",2],["hips","tail_tip",4]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q071

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q071/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a box-pleated base (tree lengths in grid units)). The crease pattern must be box-pleated on an N x N grid with 16 <= N <= 32.

Input `tree`:

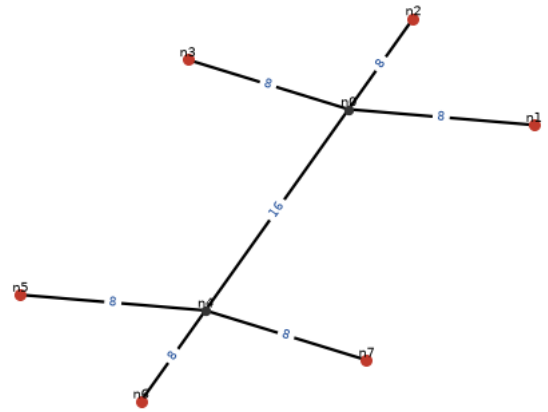
```
{"edges":[["n0","n1",8.0],["n0","n2",8.0],["n0","n3",8.0],["n0","n4",16.0],["n4","n5",8.0],["n4","n6",8.0],["n4","n7",8.0]]}
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q072

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q072/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a beetle: head with two antennae, six legs (two per thoracic segment), abdomen). Use box pleating with at least one Pythagorean stretch (grid at most 48).

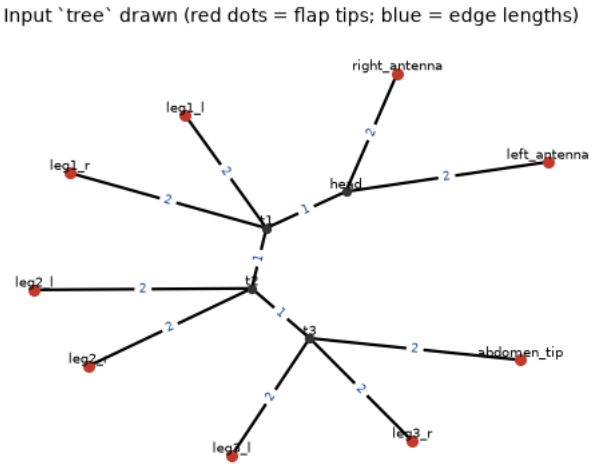
Input `tree`:


```
{"edges":[["head","left_antenna",2],["head","right_antenna",2],["head","t1",1],["t1","leg1_l",2],["t1","leg1_r",2],["t1","t2",1],["t2","leg2_l",2],["t
2","leg2_r",2],["t2","t3",1],["t3","leg3_l",2],["t3","leg3_r",2],["t3","abdomen_tip",2]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q073

split: hard · family: design · conventions: advanced (front matter) · files: inputs/Q073/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a lizard: head, four legs, a long tail and a body). Use box pleating with at least one Pythagorean stretch (grid at most 48).

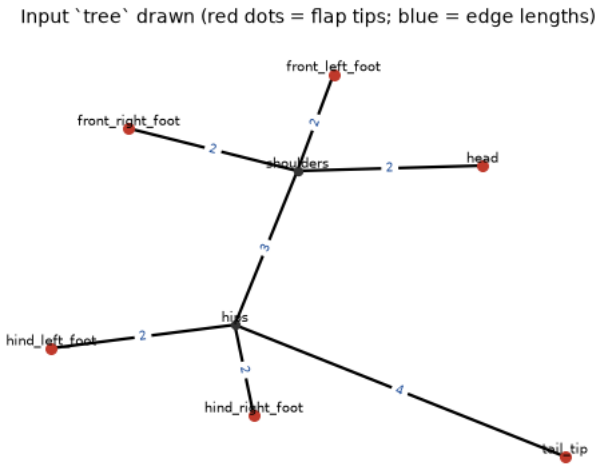
Input `tree`:


```
{"edges":[["shoulders","head",2],["shoulders","front_left_foot",2],["shoulders","front_right_foot",2],["shoulders","hips",3],["hips","hind_left_foot",2],
["hips","hind_right_foot",2],["hips","tail_tip",4]]}
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q074

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q074/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

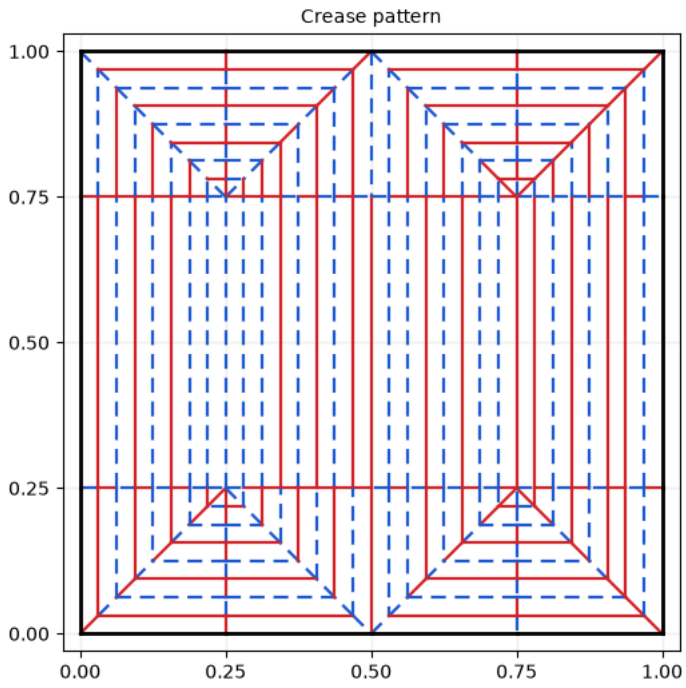
Input `crease_pattern`:


```
{"vertices_coords":[[0.21875,0.25],[0.25,0.25],[0.1875,0.25],[0.15625,0.25],[0.125,0.25],[0.09375,0.25],[0.0625,0.25],[0.03125,0.25],[0.0,0.25],[0.25,0.
0],[0.25,0.03125],[0.25,0.0625],[0.25,0.09375],[0.25,0.125],[0.25,0.15625],[0.25,0.1875],[0.25,0.21875],[0.0,0.0],[0.03125,0.03125],[0.0625,0.0625],[0.093
75,0.09375],[0.125,0.125],[0.15625,0.15625],[0.1875,0.1875],[0.21875,0.21875],[0.0,0.75],[0.03125,0.75],[0.0625,0.75],[0.09375,0.75],[0.125,0.75],[0.1562
5,0.75],[0.1875,0.75],[0.21875,0.75],[0.25,0.75],[0.0,1.0],[0.03125,0.96875],[0.0625,0.9375],[0.09375,0.90625],[0.125,0.875],[0.15625,0.84375],[0.1875,0.8
125],[0.21875,0.78125],[0.25,0.96875],[0.25,1.0],[0.25,0.9375],[0.25,0.90625],[0.25,0.875],[0.25,0.84375],[0.25,0.8125],[0.25,0.78125],[0.46875,0.03125],
[0.5,0.0],[0.4375,0.0625],[0.40625,0.09375],[0.375,0.125],[0.34375,0.15625],[0.3125,0.1875],[0.28125,0.21875],[0.46875,0.96875],[0.5,1.0],[0.4375,0.9375],
[0.40625,0.90625],[0.375,0.875],[0.34375,0.84375],[0.3125,0.8125],[0.28125,0.78125],[0.75,0.21875],[0.75,0.25],[0.75,0.1875],[0.75,0.15625],[0.75,0.125],
[0.75,0.09375],[0.75,0.0625],[0.75,0.03125],[0.75,0.0],[0.71875,0.25],[0.6875,0.25],[0.65625,0.25],[0.625,0.25],[0.59375,0.25], ...
[truncated: the full input is in inputs/Q074/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q075

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q075/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

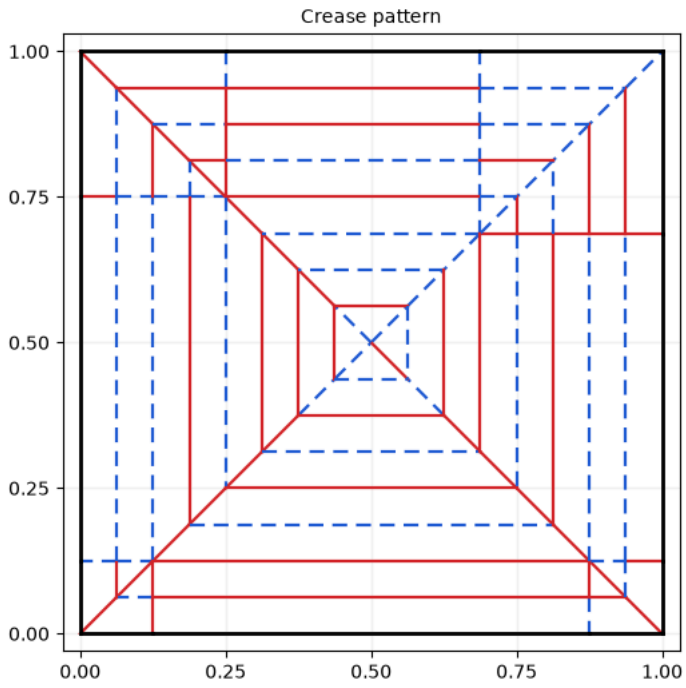
Input `crease_pattern`:


```
{"vertices_coords":[[0.0625,0.125],[0.125,0.125],[0.0,0.125],[0.125,0.0],[0.125,0.0625],[0.0,0.0],[0.0625,0.0625],[0.0,0.75],[0.0625,0.75],[0.125,0.75],
[0.1875,0.75],[0.25,0.75],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.8125],[0.25,0.875],[0.25,0.9375],[0.25,1.0],[0.6875,0.9375],[0.6
875,1.0],[0.6875,0.875],[0.6875,0.8125],[0.6875,0.75],[0.6875,0.6875],[0.875,0.0625],[0.875,0.125],[0.875,0.0],[0.9375,0.0625],[1.0,0.0],[0.9375,0.125],
[1.0,0.125],[0.75,0.6875],[0.8125,0.6875],[0.875,0.6875],[0.9375,0.6875],[1.0,0.6875],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],
[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.625,0.625],[0.8125,0.1875],[0.75,0.25],[0.6875,0.31
25],[0.625,0.375],[0.5625,0.4375],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[3,4],[1,4],[5,6],[1,6],[7,8],[8,9],[9,10],
[10,11],[12,13],[13,14],[14,15],[11,15],[11,16],[16,17],[17,18],[18,19],[20,21],[20,22],[22,23],[23,24],[24,25],[26,27],[26,28],[29,30],[27,29],[31,32],[2
7,31],[25,33],[33,34],[34,35],[35,36],[36,37],[38,39],[38,40],[40,41],[41,42],[25,42],[1,43],[43,44],[44,45],[45,46],[46,47],[ ...
[truncated: the full input is in inputs/Q075/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q076

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q076/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

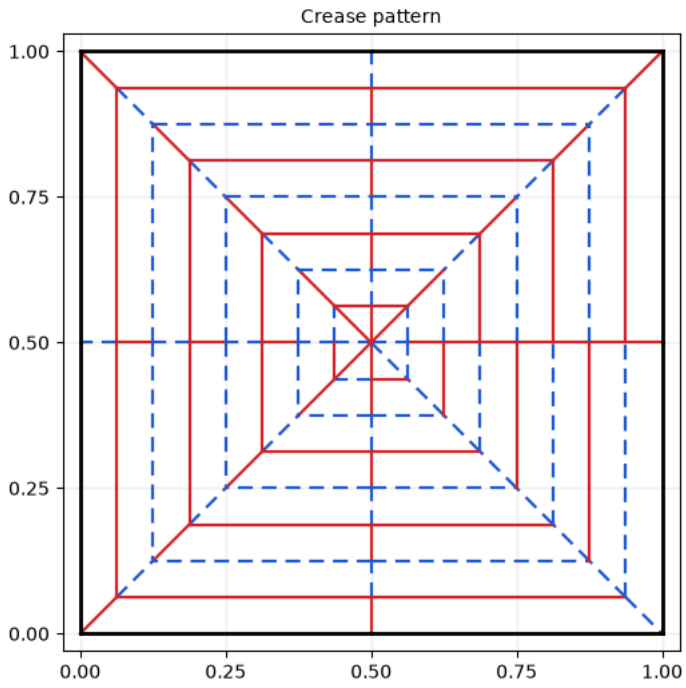
Input `crease_pattern`:


```
{"vertices_coords":[[0.4375,0.5],[0.5,0.5],[0.375,0.5],[0.3125,0.5],[0.25,0.5],[0.1875,0.5],[0.125,0.5],[0.0625,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.0625],[0.
5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.312
5],[0.375,0.375],[0.4375,0.4375],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625],[0.5,0.
5625],[0.5,0.625],[0.5,0.6875],[0.5,0.75],[0.5,0.8125],[0.5,0.875],[0.5,0.9375],[0.5,1.0],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.
25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.9375,0.5],[1.0,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.5],[0.937
5,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.625,0.625],[0.5625,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,
5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[14,15],[15,16],[1,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[1,24],[25,26],
[26,27],[27,28],[28,29],[29,30],[30,31],[31,32],[1,32],[1,33],[33,34],[34,35],[35,36],[36,37],[37,38],[38,39],[39,40],[41,42], ...
[truncated: the full input is in inputs/Q076/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q077

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q077/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

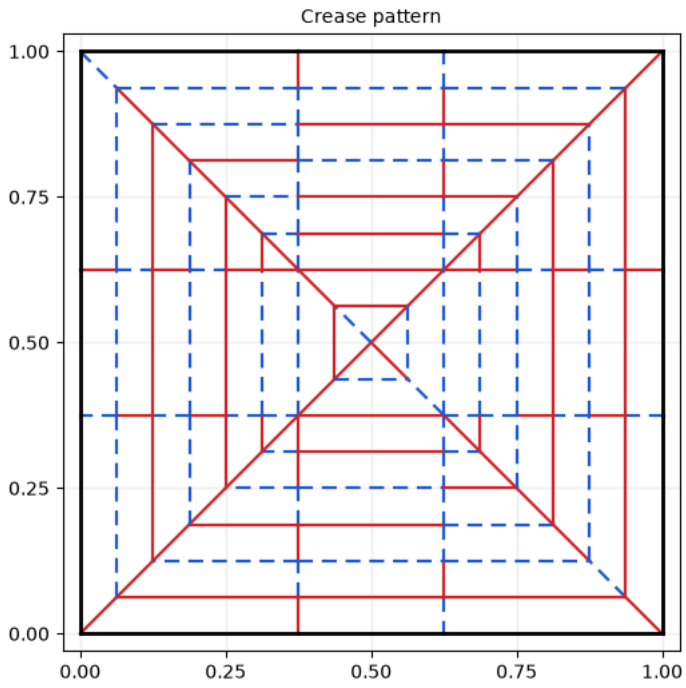
Input `crease_pattern`:


```
{"vertices_coords":[[0.3125,0.375],[0.375,0.375],[0.25,0.375],[0.1875,0.375],[0.125,0.375],[0.0625,0.375],[0.0,0.375],[0.375,0.0],[0.375,0.0625],[0.375,0.125],[0.375,0.1875],[0.375,0.25],[0.375,0.3125],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.0,0.625],[0.0625,0.625],[0.125,0.625],[0.1875,0.625],[0.25,0.625],[0.3125,0.625],[0.375,0.625],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.6875],[0.375,0.75],[0.375,0.8125],[0.375,0.875],[0.375,0.9375],[0.375,1.0],[0.625,0.3125],[0.625,0.375],[0.625,0.25],[0.625,0.1875],[0.625,0.125],[0.625,0.0625],[0.625,0.0],[0.625,0.9375],[0.625,1.0],[0.625,0.875],[0.625,0.8125],[0.625,0.75],[0.625,0.6875],[0.625,0.625],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.9375,0.375],[1.0,0.375],[0.875,0.375],[0.8125,0.375],[0.75,0.375],[0.6875,0.375],[0.6875,0.625],[0.75,0.625],[0.8125,0.625],[0.875,0.625],[0.9375,0.625],[1.0,0.625],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.5625,0.4375],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4] ...
[truncated: the full input is in inputs/Q077/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q078

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q078/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

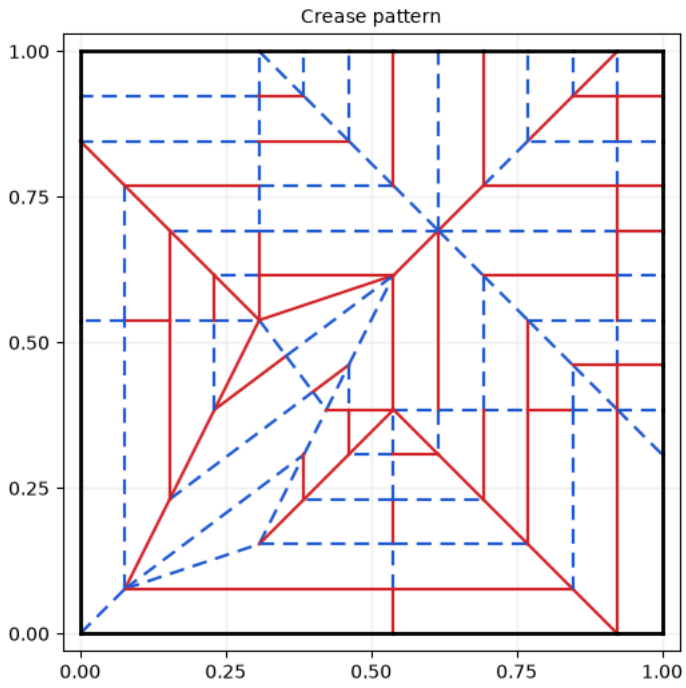
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.07692308],[0.15384615,0.23076923],[0.23076923,0.38461538],[0.30769231,0.15384615],[0.23076923,0.53846154],[0.3
0769231,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.92307692],[0.30769231,1.0],[0.30769231,0.84615385],[0.3
0769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.35384615,0.47692308],[0.38461538,0.30769231],[0.4,0.41538462],[0.42307692,0.3846153
8],[0.46153846,0.46153846],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.
38461538],[0.46153846,0.38461538],[0.38461538,0.23076923],[0.46153846,0.30769231],[0.53846154,0.61538462],[0.53846154,0.76923077],[0.61538462,0.69230769],
[0.46153846,0.84615385],[0.38461538,0.92307692],[0.61538462,0.38461538],[0.69230769,0.38461538],[0.76923077,0.38461538],[0.84615385,0.38461538],[0.9230769
2,0.38461538],[0.69230769,0.61538462],[0.76923077,0.53846154],[0.84615385,0.46153846],[0.92307692,0.46153846],[0.92307692,0.53846154],[0.92307692,0.615384
62],[0.92307692,0.69230769],[0.92307692,0.76923077],[0.92307692,0.84615385],[0.92307692,0.92307692],[0.92307692,1.0],[0.69 ...
[truncated: the full input is in inputs/Q078/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q079

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q079/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

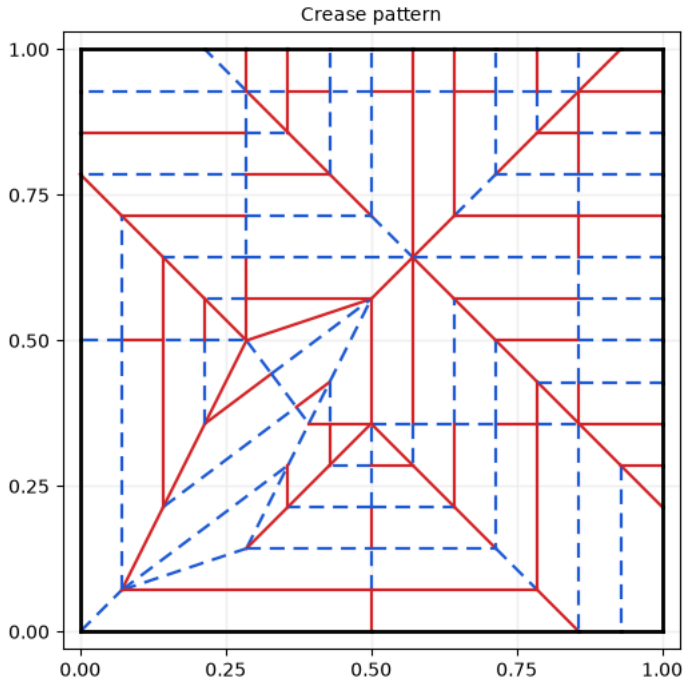
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07142857,0.07142857],[0.14285714,0.21428571],[0.21428571,0.35714286],[0.28571429,0.14285714],[0.21428571,0.5],[0.28571429,0.5],[0.14285714,0.5],[0.07142857,0.5],[0.0,0.5],[0.28571429,0.85714286],[0.28571429,0.92857143],[0.28571429,0.78571429],[0.28571429,0.71428571],[0.28571429,0.64285714],[0.28571429,0.57142857],[0.32857143,0.44285714],[0.35714286,0.28571429],[0.37142857,0.38571429],[0.39285714,0.35714286],[0.42857143,0.42857143],[0.5,0.0],[0.5,0.07142857],[0.5,0.14285714],[0.5,0.21428571],[0.5,0.28571429],[0.5,0.35714286],[0.42857143,0.35714286],[0.35714286,0.21428571],[0.42857143,0.28571429],[0.5,0.57142857],[0.5,0.71428571],[0.57142857,0.64285714],[0.42857143,0.78571429],[0.35714286,0.85714286],[0.57142857,0.35714286],[0.64285714,0.35714286],[0.71428571,0.35714286],[0.78571429,0.35714286],[0.85714286,0.35714286],[0.64285714,0.57142857],[0.71428571,0.5],[0.78571429,0.42857143],[0.78571429,0.92857143],[0.85714286,0.92857143],[0.71428571,0.92857143],[0.64285714,0.92857143],[0.57142857,0.92857143],[0.5,0.92857143],[0.42857143,0.92857143],[0.35714286,0.92857143],[0.85714286,0.42857143],[0.85714286,0.5],[0.85714286,0.57142857],[0.85714286,0.64285714], ...
[truncated: the full input is in inputs/Q079/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q080

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q080/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

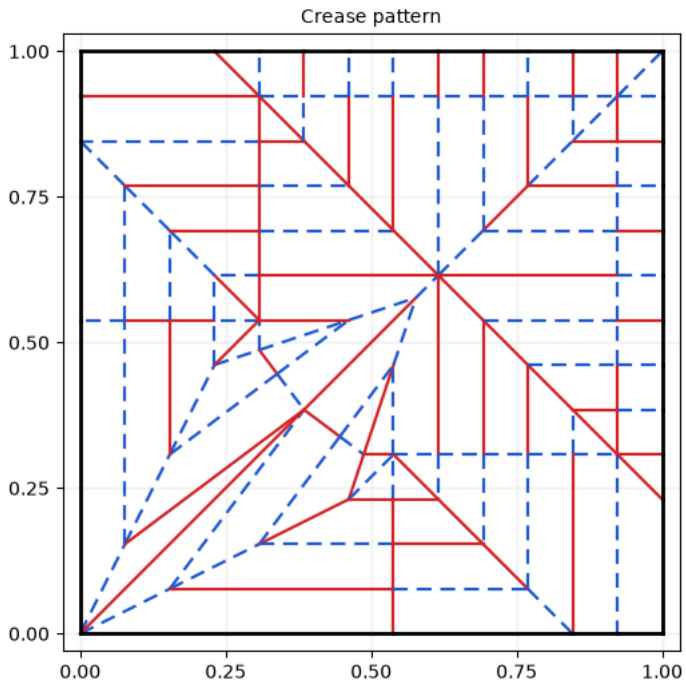
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.15384615],[0.15384615,0.07692308],[0.15384615,0.30769231],[0.23076923,0.46153846],[0.30769231,0.15384615],[0.3
0769231,0.48717949],[0.30769231,0.53846154],[0.23076923,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.8461538
5],[0.30769231,0.92307692],[0.30769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.33846154,0.44615385],[0.38461538,0.38461538],[0.4461
5385,0.33846154],[0.46153846,0.23076923],[0.46153846,0.53846154],[0.48717949,0.30769231],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],
[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.46153846],[0.57692308,0.57692308],[0.53846154,0.69230769],[0.61538462,0.61538462],[0.4615384
6,0.76923077],[0.38461538,0.84615385],[0.61538462,0.30769231],[0.69230769,0.30769231],[0.76923077,0.30769231],[0.84615385,0.30769231],[0.92307692,0.307692
31],[0.69230769,0.53846154],[0.76923077,0.46153846],[0.84615385,0.38461538],[0.84615385,0.92307692],[0.92307692,0.92307692],[0.76923077,0.92307692],[0.692
30769,0.92307692],[0.61538462,0.92307692],[0.53846154,0.92307692],[0.46153846,0.92307692],[0.38461538,0.92307692],[0.92307 ...
[truncated: the full input is in inputs/Q080/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q081

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q081/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

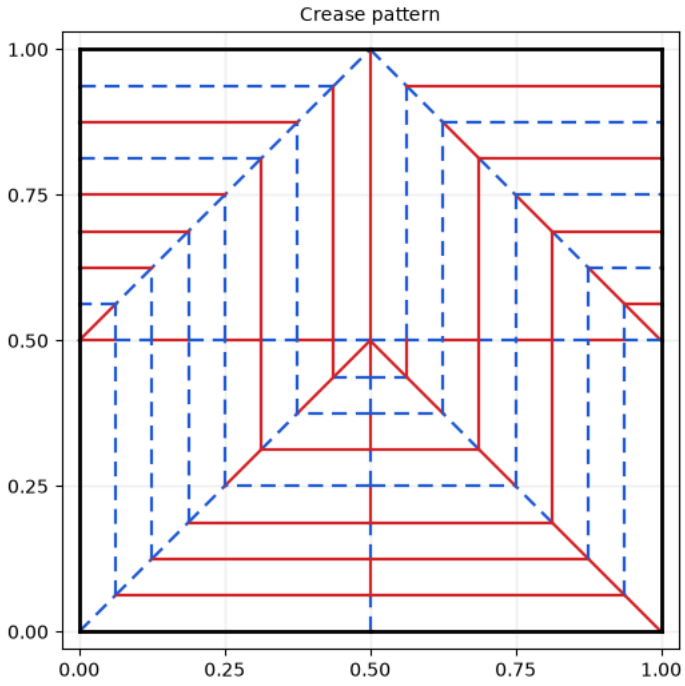
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.0625],[0.5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.5,0.5],[0.0,0.0],[0.0625,0.0625],
[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.0,0.5],[0.0625,0.5],[0.125,0.5],[0.1875,0.5],[0.25,0.5],[0.312
5,0.5],[0.375,0.5],[0.4375,0.5],[0.4375,0.9375],[0.5,1.0],[0.375,0.875],[0.3125,0.8125],[0.25,0.75],[0.1875,0.6875],[0.125,0.625],[0.0625,0.5625],[0.9375,
0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.5625,0.9375],[0.625,0.875],[0.6875,0.8125],
[0.75,0.75],[0.8125,0.6875],[0.875,0.625],[0.9375,0.5625],[1.0,0.5],[0.9375,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.
5],[0.0,0.5625],[0.0,0.625],[0.0,0.6875],[0.0,0.75],[0.0,0.8125],[0.0,0.875],[0.0,0.9375],[1.0,0.5625],[1.0,0.625],[1.0,0.6875],[1.0,0.75],[1.0,0.8125],
[1.0,0.875],[1.0,0.9375],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[1
4,15],[15,16],[8,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[8,24],[25,26],[25,27],[27,28],[28,29],[29,30],[30,31] ...
[truncated: the full input is in inputs/Q081/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q082

split: hard · family: read-tree · conventions: advanced (front matter) · files: inputs/Q082/

Task (read, hard):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

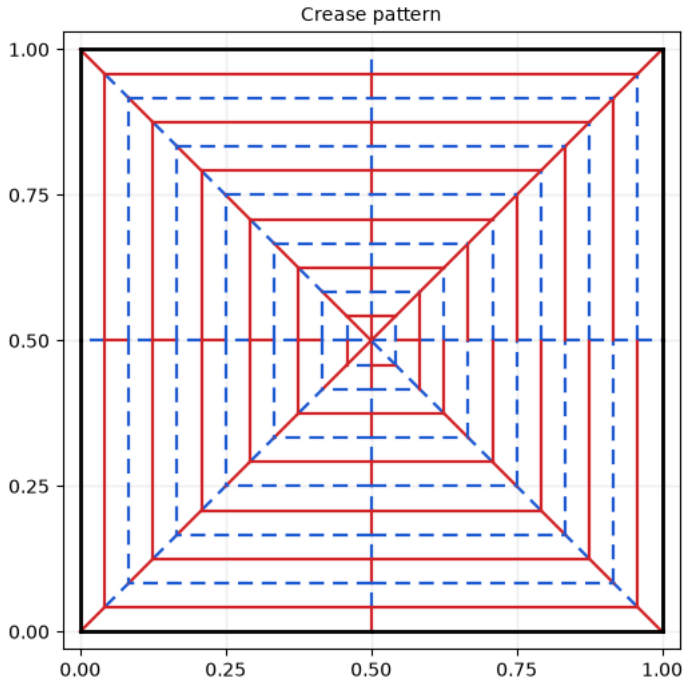
Input `crease_pattern`:


```
{"vertices_coords":[[0.45833333,0.5],[0.5,0.5],[0.41666667,0.5],[0.375,0.5],[0.33333333,0.5],[0.29166667,0.5],[0.25,0.5],[0.20833333,0.5],[0.16666667,0.
5],[0.125,0.5],[0.08333333,0.5],[0.04166667,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.04166667],[0.5,0.08333333],[0.5,0.125],[0.5,0.16666667],[0.5,0.20833333],[0.5,
0.25],[0.5,0.29166667],[0.5,0.33333333],[0.5,0.375],[0.5,0.41666667],[0.5,0.45833333],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.1
25],[0.16666667,0.16666667],[0.20833333,0.20833333],[0.25,0.25],[0.29166667,0.29166667],[0.33333333,0.33333333],[0.375,0.375],[0.41666667,0.41666667],[0.4
5833333,0.45833333],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],
[0.29166667,0.70833333],[0.33333333,0.66666667],[0.375,0.625],[0.41666667,0.58333333],[0.45833333,0.54166667],[0.5,0.54166667],[0.5,0.58333333],[0.5,0.62
5],[0.5,0.66666667],[0.5,0.70833333],[0.5,0.75],[0.5,0.79166667],[0.5,0.83333333],[0.5,0.875],[0.5,0.91666667],[0.5,0.95833333],[0.5,1.0],[0.95833333,0.04
166667],[1.0,0.0],[0.91666667,0.08333333],[0.875,0.125],[0.83333333,0.16666667],[0.79166667,0.20833333],[0.75,0.25],[0.70833333 ...
[truncated: the full input is in inputs/Q082/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q083

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q083/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

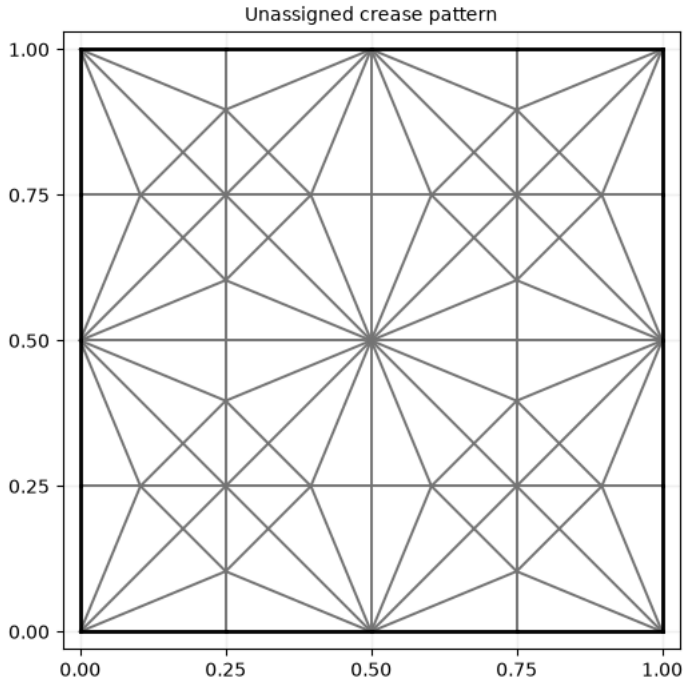
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1767767,0.1767767],[0.25,0.25],[0.3232233,0.3232233],[0.5,0.5],[0.5,0.0],[0.3232233,0.1767767],[0.1767767,0.3232233],[0.
0,0.5],[0.25,0.0],[0.25,0.10355339],[0.5,0.25],[0.39644661,0.25],[0.25,0.5],[0.25,0.39644661],[0.0,0.25],[0.10355339,0.25],[1.0,0.0],[0.8232233,0.176776
7],[0.75,0.25],[0.6767767,0.3232233],[0.6767767,0.1767767],[0.8232233,0.3232233],[1.0,0.5],[0.75,0.0],[0.75,0.10355339],[0.60355339,0.25],[0.75,0.5],[0.7
5,0.39644661],[1.0,0.25],[0.89644661,0.25],[0.0,1.0],[0.1767767,0.8232233],[0.25,0.75],[0.3232233,0.6767767],[0.5,1.0],[0.3232233,0.8232233],[0.1767767,0.
6767767],[0.25,1.0],[0.25,0.89644661],[0.5,0.75],[0.39644661,0.75],[0.25,0.60355339],[0.0,0.75],[0.10355339,0.75],[1.0,1.0],[0.8232233,0.8232233],[0.75,0.
75],[0.6767767,0.6767767],[0.6767767,0.8232233],[0.8232233,0.6767767],[0.75,1.0],[0.75,0.89644661],[0.60355339,0.75],[0.75,0.60355339],[1.0,0.75],[0.89644
661,0.75]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,1
2],[3,14],[7,14],[7,16],[1,16],[1,10],[0,16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[17,18],[18,19],[19,20],[4,20],[ ...
[truncated: the full input is in inputs/Q083/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q084

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q084/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

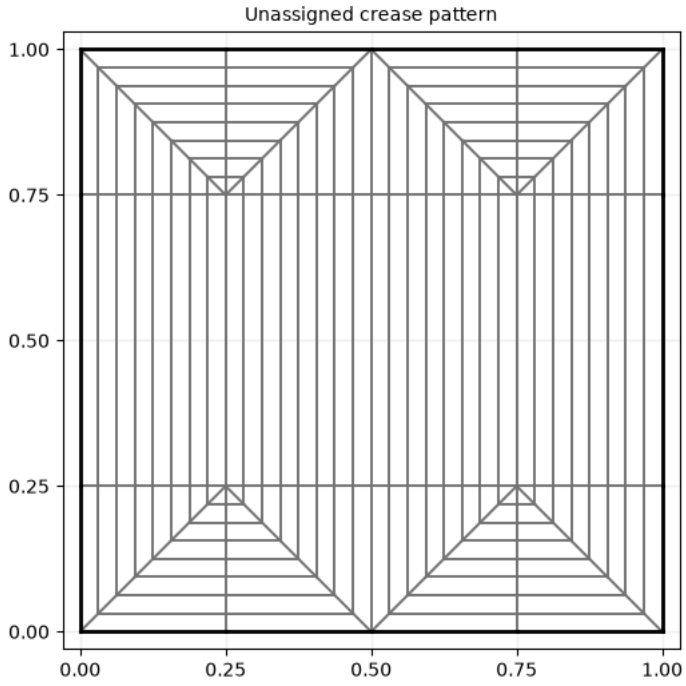
Input `crease_pattern`:


```
{"vertices_coords":[[0.21875,0.25],[0.25,0.25],[0.1875,0.25],[0.15625,0.25],[0.125,0.25],[0.09375,0.25],[0.0625,0.25],[0.03125,0.25],[0.0,0.25],[0.25,0.
0],[0.25,0.03125],[0.25,0.0625],[0.25,0.09375],[0.25,0.125],[0.25,0.15625],[0.25,0.1875],[0.25,0.21875],[0.0,0.0],[0.03125,0.03125],[0.0625,0.0625],[0.093
75,0.09375],[0.125,0.125],[0.15625,0.15625],[0.1875,0.1875],[0.21875,0.21875],[0.0,0.75],[0.03125,0.75],[0.0625,0.75],[0.09375,0.75],[0.125,0.75],[0.1562
5,0.75],[0.1875,0.75],[0.21875,0.75],[0.25,0.75],[0.0,1.0],[0.03125,0.96875],[0.0625,0.9375],[0.09375,0.90625],[0.125,0.875],[0.15625,0.84375],[0.1875,0.8
125],[0.21875,0.78125],[0.25,0.96875],[0.25,1.0],[0.25,0.9375],[0.25,0.90625],[0.25,0.875],[0.25,0.84375],[0.25,0.8125],[0.25,0.78125],[0.46875,0.03125],
[0.5,0.0],[0.4375,0.0625],[0.40625,0.09375],[0.375,0.125],[0.34375,0.15625],[0.3125,0.1875],[0.28125,0.21875],[0.46875,0.96875],[0.5,1.0],[0.4375,0.9375],
[0.40625,0.90625],[0.375,0.875],[0.34375,0.84375],[0.3125,0.8125],[0.28125,0.78125],[0.75,0.21875],[0.75,0.25],[0.75,0.1875],[0.75,0.15625],[0.75,0.125],
[0.75,0.09375],[0.75,0.0625],[0.75,0.03125],[0.75,0.0],[0.71875,0.25],[0.6875,0.25],[0.65625,0.25],[0.625,0.25],[0.59375,0.25], ...
[truncated: the full input is in inputs/Q084/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q085

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q085/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

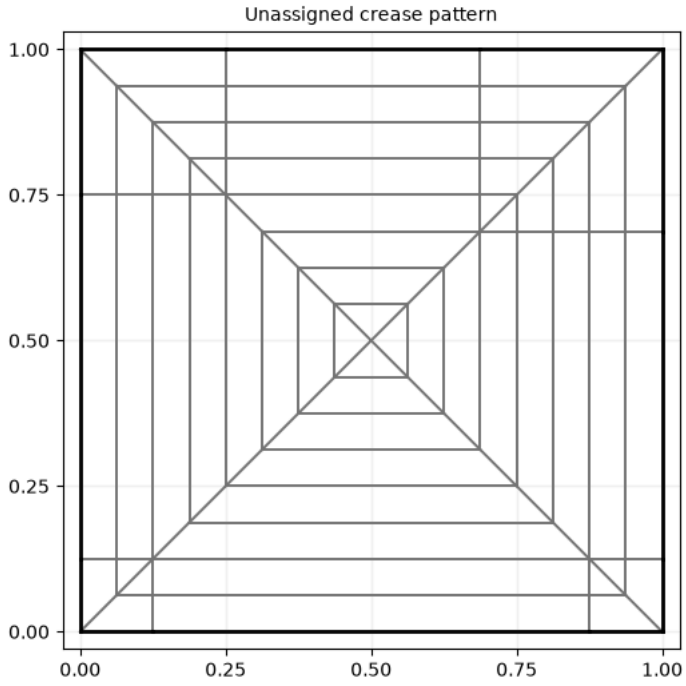
Input `crease_pattern`:


```
{"vertices_coords":[[0.0625,0.125],[0.125,0.125],[0.0,0.125],[0.125,0.0],[0.125,0.0625],[0.0,0.0],[0.0625,0.0625],[0.0,0.75],[0.0625,0.75],[0.125,0.75],
[0.1875,0.75],[0.25,0.75],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.8125],[0.25,0.875],[0.25,0.9375],[0.25,1.0],[0.6875,0.9375],[0.6
875,1.0],[0.6875,0.875],[0.6875,0.8125],[0.6875,0.75],[0.6875,0.6875],[0.875,0.0625],[0.875,0.125],[0.875,0.0],[0.9375,0.0625],[1.0,0.0],[0.9375,0.125],
[1.0,0.125],[0.75,0.6875],[0.8125,0.6875],[0.875,0.6875],[0.9375,0.6875],[1.0,0.6875],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],
[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.625,0.625],[0.8125,0.1875],[0.75,0.25],[0.6875,0.31
25],[0.625,0.375],[0.5625,0.4375],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[3,4],[1,4],[5,6],[1,6],[7,8],[8,9],[9,10],
[10,11],[12,13],[13,14],[14,15],[11,15],[11,16],[16,17],[17,18],[18,19],[20,21],[20,22],[22,23],[23,24],[24,25],[26,27],[26,28],[29,30],[27,29],[31,32],[2
7,31],[25,33],[33,34],[34,35],[35,36],[36,37],[38,39],[38,40],[40,41],[41,42],[25,42],[1,43],[43,44],[44,45],[45,46],[46,47],[ ...
[truncated: the full input is in inputs/Q085/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q086

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q086/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

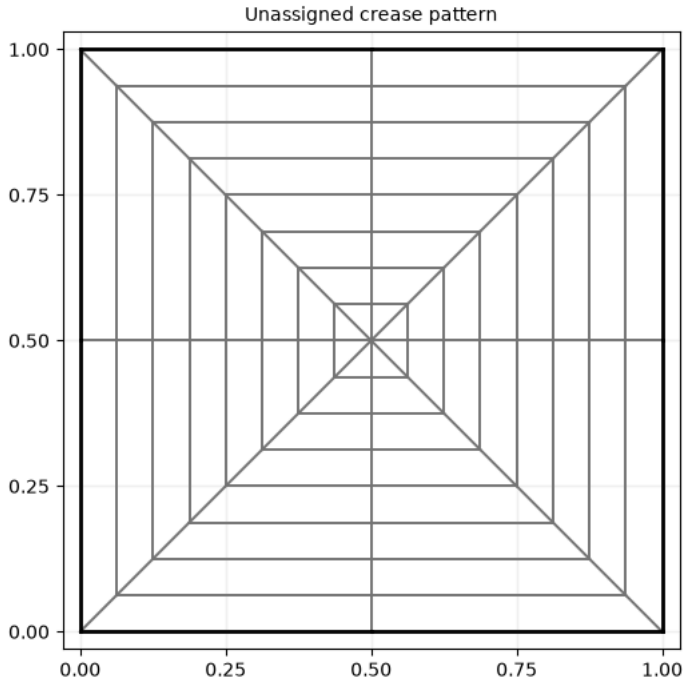
Input `crease_pattern`:


```
{"vertices_coords":[[0.4375,0.5],[0.5,0.5],[0.375,0.5],[0.3125,0.5],[0.25,0.5],[0.1875,0.5],[0.125,0.5],[0.0625,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.0625],[0.
5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.312
5],[0.375,0.375],[0.4375,0.4375],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625],[0.5,0.
5625],[0.5,0.625],[0.5,0.6875],[0.5,0.75],[0.5,0.8125],[0.5,0.875],[0.5,0.9375],[0.5,1.0],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.
25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.9375,0.5],[1.0,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.5],[0.937
5,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.625,0.625],[0.5625,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,
5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[14,15],[15,16],[1,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[1,24],[25,26],
[26,27],[27,28],[28,29],[29,30],[30,31],[31,32],[1,32],[1,33],[33,34],[34,35],[35,36],[36,37],[37,38],[38,39],[39,40],[41,42], ...
[truncated: the full input is in inputs/Q086/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q087

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q087/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

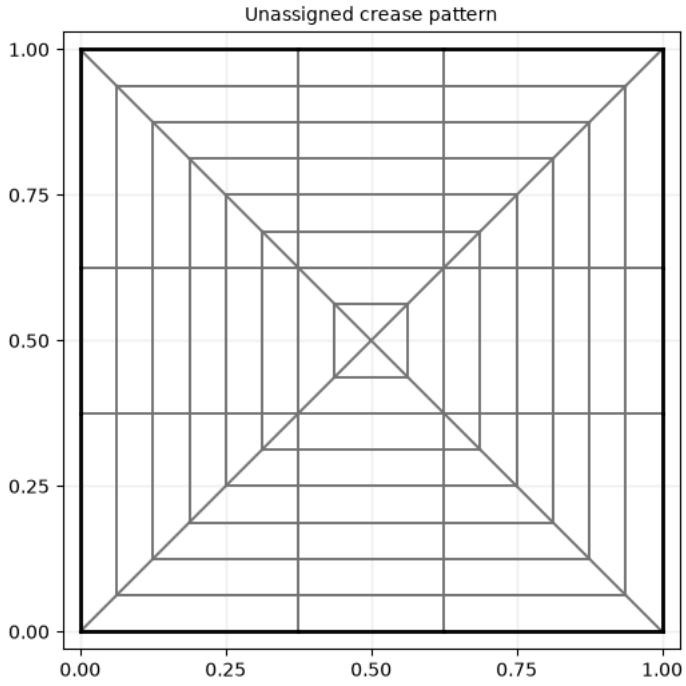
Input `crease_pattern`:


```
{"vertices_coords":[[0.3125,0.375],[0.375,0.375],[0.25,0.375],[0.1875,0.375],[0.125,0.375],[0.0625,0.375],[0.0,0.375],[0.375,0.0],[0.375,0.0625],[0.375,0.125],[0.375,0.1875],[0.375,0.25],[0.375,0.3125],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.0,0.625],[0.0625,0.625],[0.125,0.625],[0.1875,0.625],[0.25,0.625],[0.3125,0.625],[0.375,0.625],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.6875],[0.375,0.75],[0.375,0.8125],[0.375,0.875],[0.375,0.9375],[0.375,1.0],[0.625,0.3125],[0.625,0.375],[0.625,0.25],[0.625,0.1875],[0.625,0.125],[0.625,0.0625],[0.625,0.0],[0.625,0.9375],[0.625,1.0],[0.625,0.875],[0.625,0.8125],[0.625,0.75],[0.625,0.6875],[0.625,0.625],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.9375,0.375],[1.0,0.375],[0.875,0.375],[0.8125,0.375],[0.75,0.375],[0.6875,0.375],[0.6875,0.625],[0.75,0.625],[0.8125,0.625],[0.875,0.625],[0.9375,0.625],[1.0,0.625],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.5625,0.4375],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4] ...
[truncated: the full input is in inputs/Q087/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q088

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q088/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

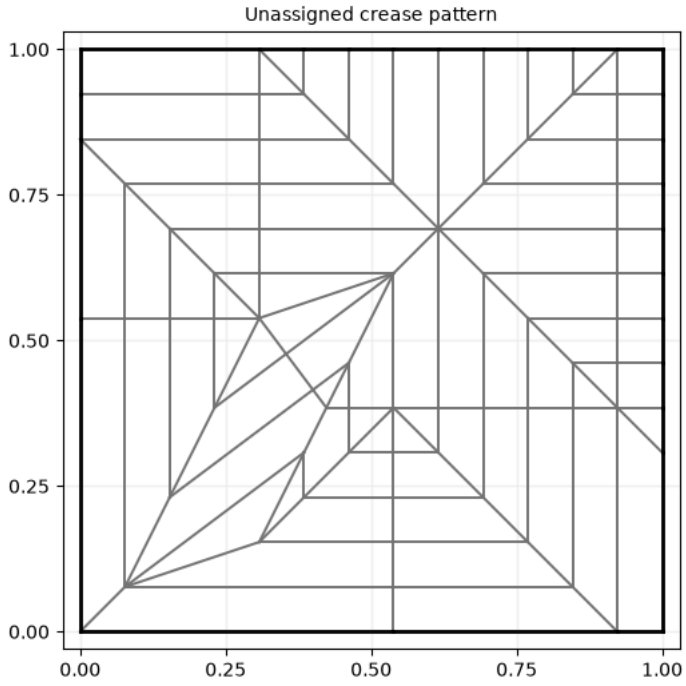
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.07692308],[0.15384615,0.23076923],[0.23076923,0.38461538],[0.30769231,0.15384615],[0.23076923,0.53846154],[0.3
0769231,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.92307692],[0.30769231,1.0],[0.30769231,0.84615385],[0.3
0769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.35384615,0.47692308],[0.38461538,0.30769231],[0.4,0.41538462],[0.42307692,0.3846153
8],[0.46153846,0.46153846],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.
38461538],[0.46153846,0.38461538],[0.38461538,0.23076923],[0.46153846,0.30769231],[0.53846154,0.61538462],[0.53846154,0.76923077],[0.61538462,0.69230769],
[0.46153846,0.84615385],[0.38461538,0.92307692],[0.61538462,0.38461538],[0.69230769,0.38461538],[0.76923077,0.38461538],[0.84615385,0.38461538],[0.9230769
2,0.38461538],[0.69230769,0.61538462],[0.76923077,0.53846154],[0.84615385,0.46153846],[0.92307692,0.46153846],[0.92307692,0.53846154],[0.92307692,0.615384
62],[0.92307692,0.69230769],[0.92307692,0.76923077],[0.92307692,0.84615385],[0.92307692,0.92307692],[0.92307692,1.0],[0.69 ...
[truncated: the full input is in inputs/Q088/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q089

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q089/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

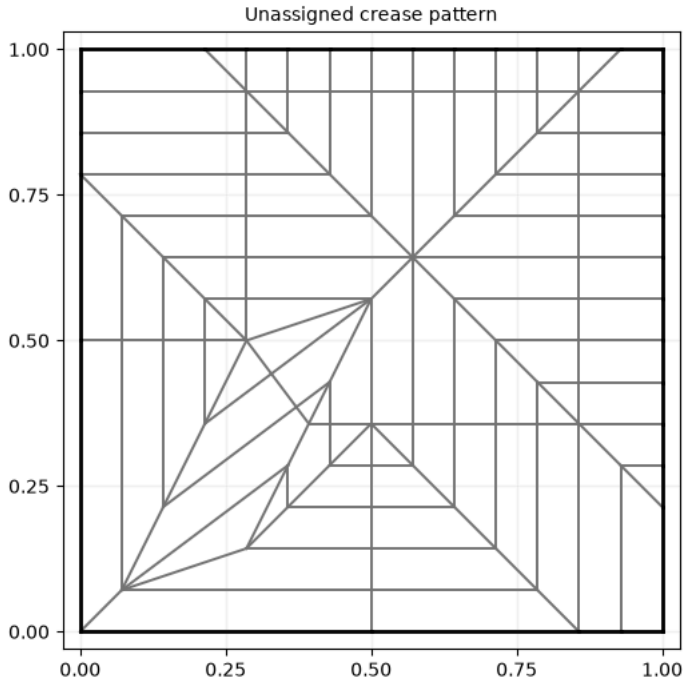
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07142857,0.07142857],[0.14285714,0.21428571],[0.21428571,0.35714286],[0.28571429,0.14285714],[0.21428571,0.5],[0.28571429,0.5],[0.14285714,0.5],[0.07142857,0.5],[0.0,0.5],[0.28571429,0.85714286],[0.28571429,0.92857143],[0.28571429,0.78571429],[0.28571429,0.71428571],[0.28571429,0.64285714],[0.28571429,0.57142857],[0.32857143,0.44285714],[0.35714286,0.28571429],[0.37142857,0.38571429],[0.39285714,0.35714286],[0.42857143,0.42857143],[0.5,0.0],[0.5,0.07142857],[0.5,0.14285714],[0.5,0.21428571],[0.5,0.28571429],[0.5,0.35714286],[0.42857143,0.35714286],[0.35714286,0.21428571],[0.42857143,0.28571429],[0.5,0.57142857],[0.5,0.71428571],[0.57142857,0.64285714],[0.42857143,0.78571429],[0.35714286,0.85714286],[0.57142857,0.35714286],[0.64285714,0.35714286],[0.71428571,0.35714286],[0.78571429,0.35714286],[0.85714286,0.35714286],[0.64285714,0.57142857],[0.71428571,0.5],[0.78571429,0.42857143],[0.78571429,0.92857143],[0.85714286,0.92857143],[0.71428571,0.92857143],[0.64285714,0.92857143],[0.57142857,0.92857143],[0.5,0.92857143],[0.42857143,0.92857143],[0.35714286,0.92857143],[0.85714286,0.42857143],[0.85714286,0.5],[0.85714286,0.57142857],[0.85714286,0.64285714], ...
[truncated: the full input is in inputs/Q089/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q090

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q090/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

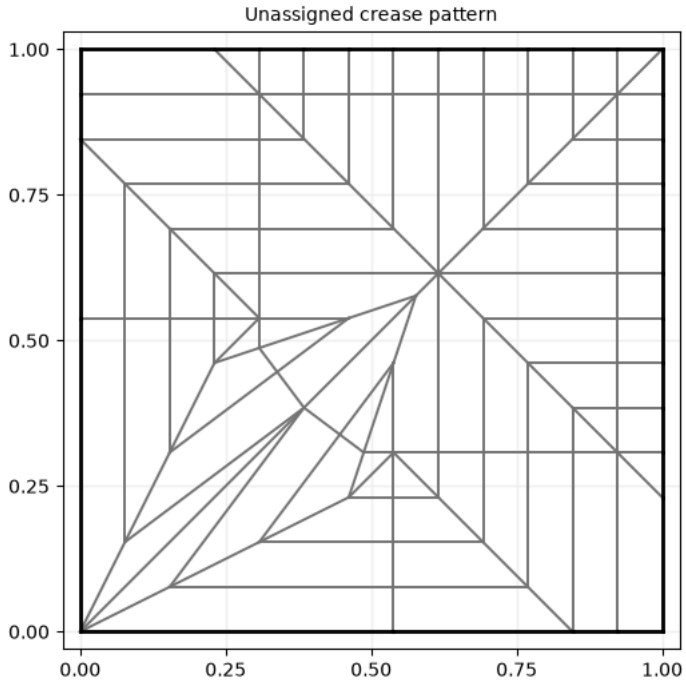
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.15384615],[0.15384615,0.07692308],[0.15384615,0.30769231],[0.23076923,0.46153846],[0.30769231,0.15384615],[0.3
0769231,0.48717949],[0.30769231,0.53846154],[0.23076923,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.8461538
5],[0.30769231,0.92307692],[0.30769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.33846154,0.44615385],[0.38461538,0.38461538],[0.4461
5385,0.33846154],[0.46153846,0.23076923],[0.46153846,0.53846154],[0.48717949,0.30769231],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],
[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.46153846],[0.57692308,0.57692308],[0.53846154,0.69230769],[0.61538462,0.61538462],[0.4615384
6,0.76923077],[0.38461538,0.84615385],[0.61538462,0.30769231],[0.69230769,0.30769231],[0.76923077,0.30769231],[0.84615385,0.30769231],[0.92307692,0.307692
31],[0.69230769,0.53846154],[0.76923077,0.46153846],[0.84615385,0.38461538],[0.84615385,0.92307692],[0.92307692,0.92307692],[0.76923077,0.92307692],[0.692
30769,0.92307692],[0.61538462,0.92307692],[0.53846154,0.92307692],[0.46153846,0.92307692],[0.38461538,0.92307692],[0.92307 ...
[truncated: the full input is in inputs/Q090/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q091

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q091/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

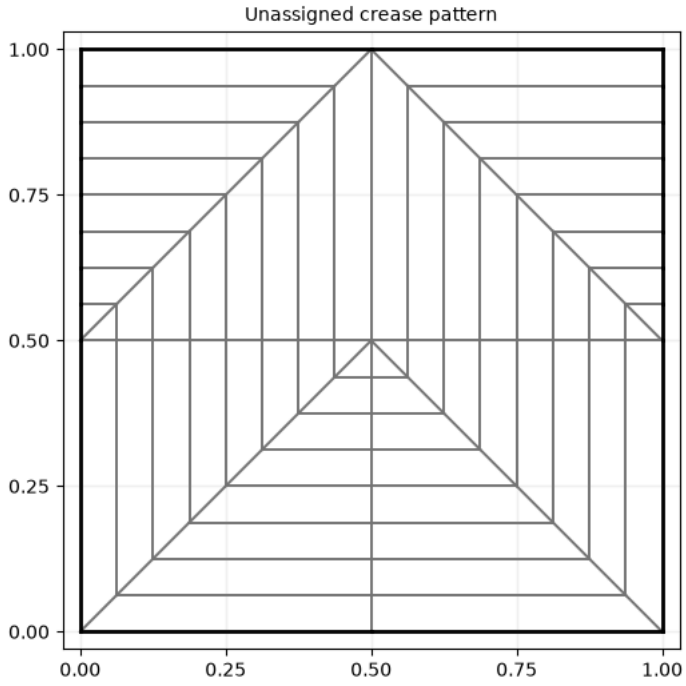
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.0625],[0.5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.5,0.5],[0.0,0.0],[0.0625,0.0625],
[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.0,0.5],[0.0625,0.5],[0.125,0.5],[0.1875,0.5],[0.25,0.5],[0.312
5,0.5],[0.375,0.5],[0.4375,0.5],[0.4375,0.9375],[0.5,1.0],[0.375,0.875],[0.3125,0.8125],[0.25,0.75],[0.1875,0.6875],[0.125,0.625],[0.0625,0.5625],[0.9375,
0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.5625,0.9375],[0.625,0.875],[0.6875,0.8125],
[0.75,0.75],[0.8125,0.6875],[0.875,0.625],[0.9375,0.5625],[1.0,0.5],[0.9375,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.
5],[0.0,0.5625],[0.0,0.625],[0.0,0.6875],[0.0,0.75],[0.0,0.8125],[0.0,0.875],[0.0,0.9375],[1.0,0.5625],[1.0,0.625],[1.0,0.6875],[1.0,0.75],[1.0,0.8125],
[1.0,0.875],[1.0,0.9375],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[1
4,15],[15,16],[8,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[8,24],[25,26],[25,27],[27,28],[28,29],[29,30],[30,31] ...
[truncated: the full input is in inputs/Q091/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q092

split: hard · family: solve · conventions: advanced (front matter) · files: inputs/Q092/

Task (solve, hard):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

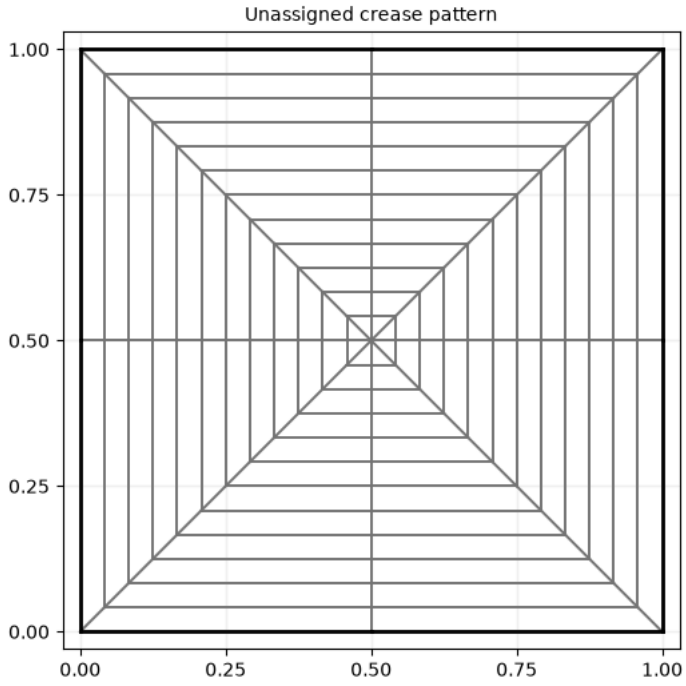
Input `crease_pattern`:


```
{"vertices_coords":[[0.45833333,0.5],[0.5,0.5],[0.41666667,0.5],[0.375,0.5],[0.33333333,0.5],[0.29166667,0.5],[0.25,0.5],[0.20833333,0.5],[0.16666667,0.
5],[0.125,0.5],[0.08333333,0.5],[0.04166667,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.04166667],[0.5,0.08333333],[0.5,0.125],[0.5,0.16666667],[0.5,0.20833333],[0.5,
0.25],[0.5,0.29166667],[0.5,0.33333333],[0.5,0.375],[0.5,0.41666667],[0.5,0.45833333],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.1
25],[0.16666667,0.16666667],[0.20833333,0.20833333],[0.25,0.25],[0.29166667,0.29166667],[0.33333333,0.33333333],[0.375,0.375],[0.41666667,0.41666667],[0.4
5833333,0.45833333],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],
[0.29166667,0.70833333],[0.33333333,0.66666667],[0.375,0.625],[0.41666667,0.58333333],[0.45833333,0.54166667],[0.5,0.54166667],[0.5,0.58333333],[0.5,0.62
5],[0.5,0.66666667],[0.5,0.70833333],[0.5,0.75],[0.5,0.79166667],[0.5,0.83333333],[0.5,0.875],[0.5,0.91666667],[0.5,0.95833333],[0.5,1.0],[0.95833333,0.04
166667],[1.0,0.0],[0.91666667,0.08333333],[0.875,0.125],[0.83333333,0.16666667],[0.79166667,0.20833333],[0.75,0.25],[0.70833333 ...
[truncated: the full input is in inputs/Q092/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q093

split: core · family: analytic · conventions: advanced (front matter) · files: inputs/Q093/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

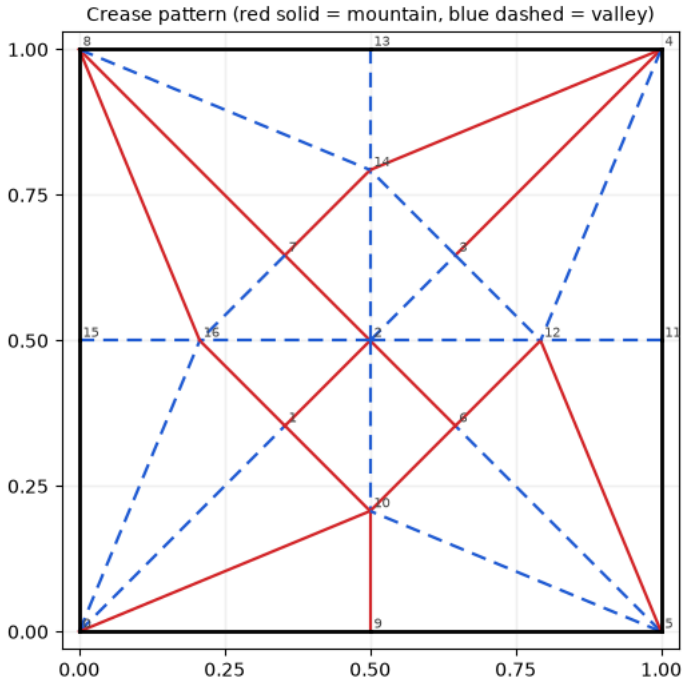
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.353553,0.353553],[0.5,0.5],[0.646447,0.646447],[1.0,1.0],[1.0,0.0],[0.646447,0.353553],[0.353553,0.646447],[0.0,1.0],[0.5,0.0],[0.5,0.207107],[1.0,0.5],[0.792893,0.5],[0.5,1.0],[0.5,0.792893],[0.0,0.5],[0.207107,0.5]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[0,9],[5,9],[5,11],[4,11],[4,13],[8,13],[8,15],[0,15]],"edges_assignment":
["V","M","V","M","V","M","M","M","M","V","V","V","V","V","V","V","M","M","V","V","M","V","M","V","M","M","V","M","V","M","V","M","V","M","V","M","B","B","B","B","B","B","B",
aces_vertices":[[0,1,16],[1,0,10],[1,2,16],[2,1,10],[2,3,14],[3,2,12],[3,4,14],[4,3,12],[5,6,10],[6,5,12],[2,6,12],[6,2,10],[2,7,16],[7,2,14],[7,8,16],[8,7,14],[9,10,0],[10,9,5],[11,12,5],[12,11,4],[13,14,4],[14,13,8],[15,16,8],[16,15,0]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q094

split: core · family: analytic · conventions: advanced (front matter) · files: inputs/Q094/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

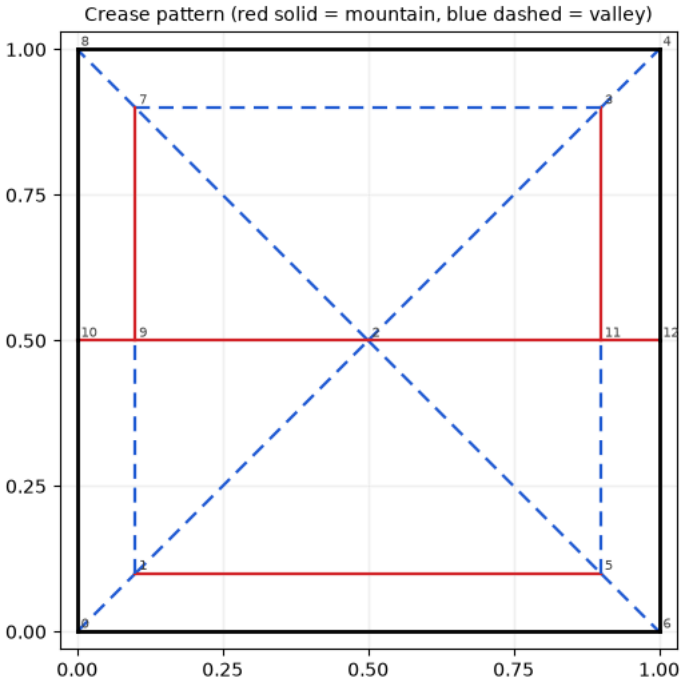
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1,0.1],[0.5,0.5],[0.9,0.9],[1.0,1.0],[0.9,0.1],[1.0,0.0],[0.1,0.9],[0.0,1.0],[0.1,0.5],[0.0,0.5],[0.9,0.5],[1.0,0.5]],"ed
ges_vertices":[[0,1],[1,2],[2,3],[3,4],[2,5],[5,6],[2,7],[7,8],[2,9],[9,10],[2,11],[11,12],[1,5],[5,11],[3,11],[3,7],[7,9],[1,9],[0,6],[6,12],[4,12],[4,
8],[8,10],[0,10]],"edges_assignment":["V","V","V","V","V","V","V","V","M","M","M","M","M","V","M","V","M","V","B","B","B","B","B","B"],"faces_vertices":
[[0,1,9,10],[1,0,6,5],[1,2,9],[2,1,5],[2,3,7],[3,2,11],[3,4,8,7],[4,3,11,12],[2,5,11],[5,6,12,11],[2,7,9],[7,8,10,9]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q095

split: core · family: analytic · conventions: advanced (front matter) · files: inputs/Q095/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

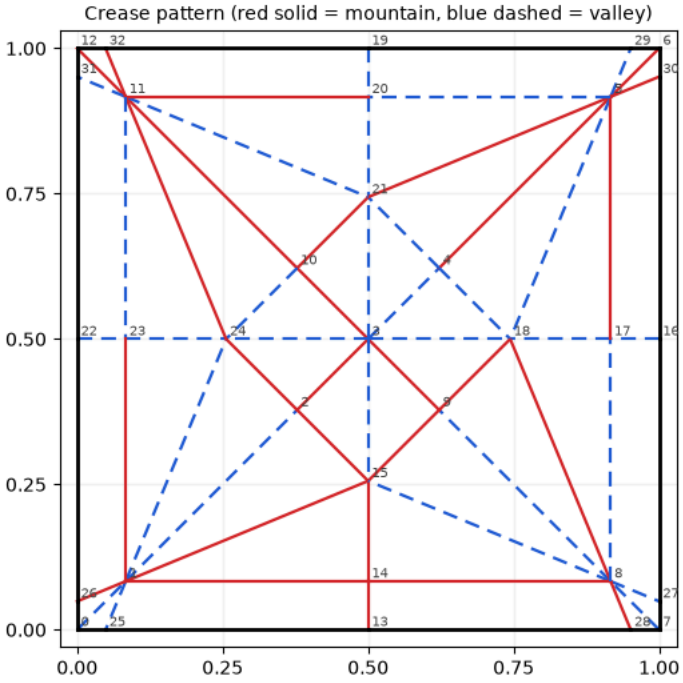
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.083333,0.083333],[0.377961,0.377961],[0.5,0.5],[0.622039,0.622039],[0.916667,0.916667],[1.0,1.0],[1.0,0.0],[0.916667,0.083333],[0.622039,0.377961],[0.377961,0.622039],[0.083333,0.916667],[0.0,1.0],[0.5,0.0],[0.5,0.083333],[0.5,0.255922],[1.0,0.5],[0.916667,0.5],[0.744078,0.5],[0.5,1.0],[0.5,0.916667],[0.5,0.744078],[0.0,0.5],[0.083333,0.5],[0.255922,0.5],[0.048816,0.0],[0.0,0.048816],[1.0,0.048816],[0.951184,0.0],[0.951184,1.0],[1.0,0.951184],[0.0,0.951184],[0.048816,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[7,8],[8,9],[3,9],[3,10],[10,11],[11,12],[13,14],[14,15],[3,15],[16,17],[17,18],[3,18],[19,20],[20,21],[3,21],[22,23],[23,24],[3,24],[9,15],[9,18],[4,18],[4,21],[10,21],[10,24],[2,24],[2,15],[1,25],[1,24],[1,26],[1,15],[8,27],[8,15],[8,28],[8,18],[5,29],[5,18],[5,30],[5,21],[11,31],[11,21],[11,32],[11,24],[1,14],[8,14],[8,17],[5,17],[5,20],[11,20],[11,23],[1,23],[0,25],[13,25],[13,28],[7,28],[7,27],[16,27],[16,30],[6,30],[6,29],[19,29],[19,32],[12,32],[12,31],[22,31],[22,26],[0,26]],"edges_assignment":["V","V","M","V","M","M","V","V","M","M","M","M","M","M","V","V","V","V","V","V","V","V","V","V","M","M","V","V","M","V","M","M" ...
[truncated: the full input is in inputs/Q095/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q096

split: core · family: design · conventions: advanced (front matter) · files: inputs/Q096/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a four-flap star). Every crease must lie at a multiple of 22.5 degrees.

Input `tree`:

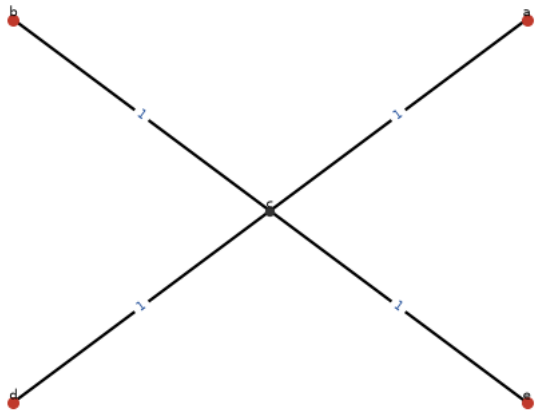
```
{"edges":[["c","a",1],["c","b",1],["c","d",1],["c","e",1]]}
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q097

split: core · family: design · conventions: advanced (front matter) · files: inputs/Q097/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a box-pleated base (tree lengths in grid units)). The crease pattern must be box-pleated on an N x N grid with 8 <= N <= 16.

Input `tree`:

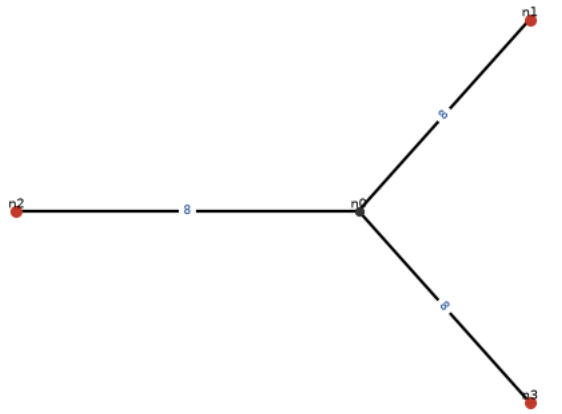
```
{"edges":[["n0","n1",8.0],["n0","n2",8.0],["n0","n3",8.0]]}
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q098

split: core · family: design · conventions: advanced (front matter) · files: inputs/Q098/

Task (design, hard):
Design a crease pattern for ONE uncut unit square whose flat-folded base realizes the target tree below at the largest scale you can (the stick figure of a box-pleated base (tree lengths in grid units)). The crease pattern must be box-pleated on an N x N grid with 12 <= N <= 24.

Input `tree`:

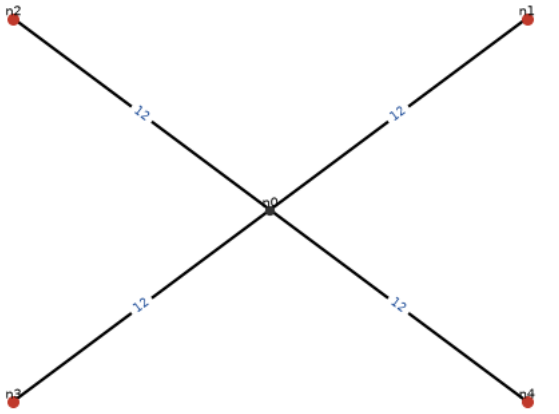
```
{"edges":[["n0","n1",12.0],["n0","n2",12.0],["n0","n3",12.0],["n0","n4",12.0]]}
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q099

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q099/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. Distances are Euclidean (circle packing).

Input `tree`:

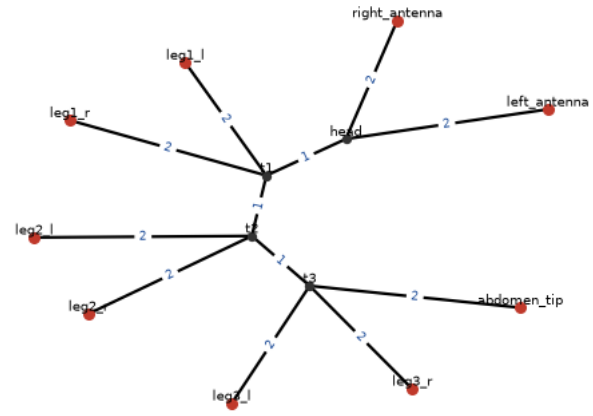

```
{"edges":[["head","left_antenna",2],["head","right_antenna",2],["head","t1",1],["t1","leg1_l",2],["t1","leg1_r",2],["t1","t2",1],["t2","leg2_l",2],["t2","leg2_r",2],["t2","t3",1],["t3","leg3_l",2],["t3","leg3_r",2],["t3","abdomen_tip",2]]}
```


Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q100

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q100/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. Distances are Euclidean (circle packing).

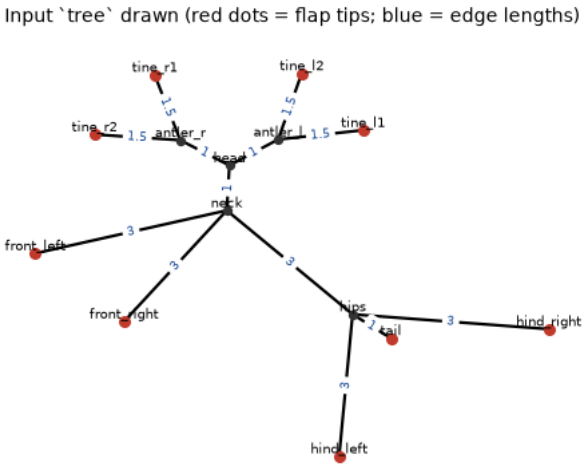
Input `tree`:


```
{"edges":[["neck","head",1],["head","antler_l",1],["antler_l","tine_l1",1.5],["antler_l","tine_l2",1.5],["head","antler_r",1],["antler_r","tine_r1",1.5],["antler_r","tine_r2",1.5],["neck","front_left",3],["neck","front_right",3],["neck","hips",3],["hips","hind_left",3],["hips","hind_right",3],["hips","tail",1]]}
```


Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q101

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q101/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. Distances are Euclidean (circle packing).

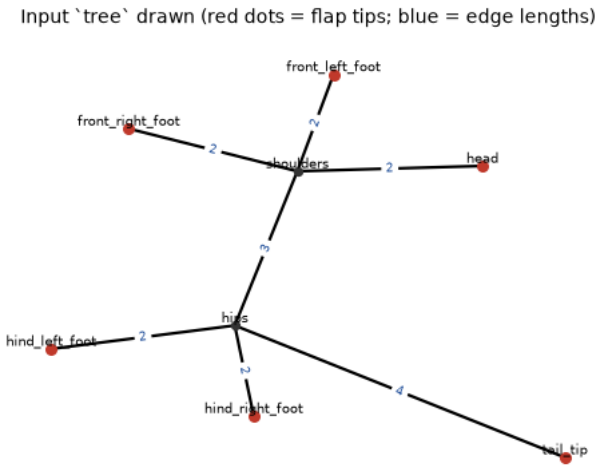
Input `tree`:

```
{"edges":[["shoulders","head",2],["shoulders","front_left_foot",2],["shoulders","front_right_foot",2],["shoulders","hips",3],["hips","hind_left_foot",2],["hips","hind_right_foot",2],["hips","tail_tip",4]]}
```

Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q102

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q102/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. Distances are Euclidean (circle packing).

Input `tree`:

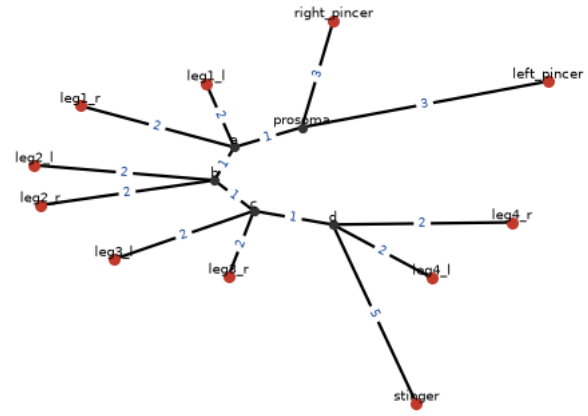

```
{"edges":[["prosoma","left_pincer",3],["prosoma","right_pincer",3],["prosoma","a",1],["a","leg1_l",2],["a","leg1_r",2],["a","b",1],["b","leg2_l",2],
["b","leg2_r",2],["b","c",1],["c","leg3_l",2],["c","leg3_r",2],["c","d",1],["d","leg4_l",2],["d","leg4_r",2],["d","stinger",5]]}
```


Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q103

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q103/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. This is BOX PLEATING: distances use the L-infinity norm max(|dx|, |dy|) (square packing).

Input `tree`:

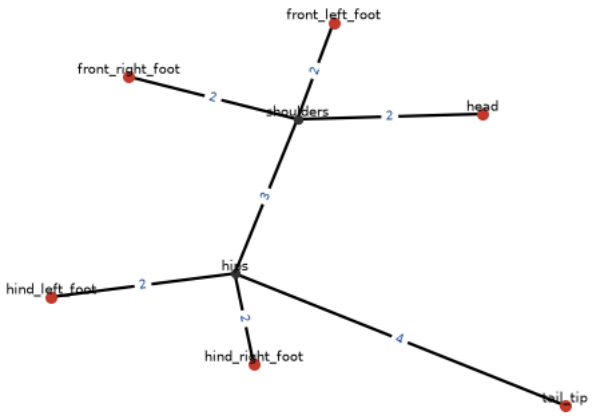

```
{"edges":[["shoulders","head",2],["shoulders","front_left_foot",2],["shoulders","front_right_foot",2],["shoulders","hips",3],["hips","hind_left_foot",2],["hips","hind_right_foot",2],["hips","tail_tip",4]]}
```


Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.

Input `tree` drawn (red dots = flap tips; blue = edge lengths)



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q104

split: core · family: pack · conventions: advanced (front matter) · files: inputs/Q104/

Task (pack, medium):
Target tree for an origami base (the stick figure of the model). Choose where each flap tip lies on the unit-square sheet so that the scale m is as large as possible, where m is the largest value with dist(u_i, u_j) >= m * d_tree(i, j) for every pair of leaves. This is BOX PLEATING: distances use the L-infinity norm max(|dx|, |dy|) (square packing).

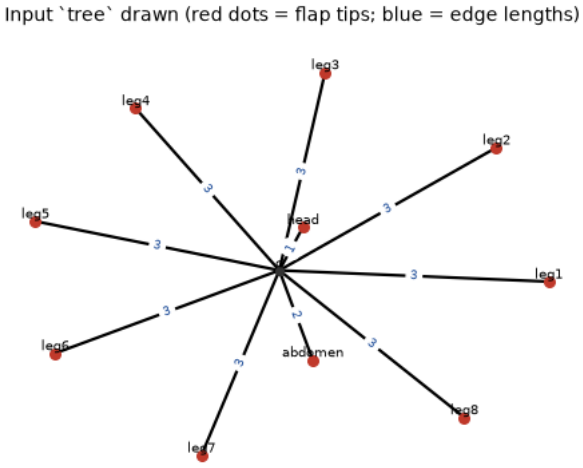
Input `tree`:


```
{"edges":[["c","leg1",3],["c","leg2",3],["c","leg3",3],["c","leg4",3],["c","leg5",3],["c","leg6",3],["c","leg7",3],["c","leg8",3],["c","head",1],["c","abdomen",2]]}
```


Answer format:

```
{"positions": {"leaf_name": [x, y], ...}, "scale": m}
```

Give a point in the unit square for every leaf of the tree; graded on the scale m they allow.



Drawn for this booklet; layout is arbitrary, only the labelled lengths matter.

Working / answer

Q105

split: core · family: read-color · conventions: advanced (front matter) · files: inputs/Q105/

Task (read, hard):
The crease pattern below is folded from two-tone paper. The side facing you in the drawing is 'front', the other side 'back'. For each listed flap tip (given by its position on the unfolded paper), say which side of the paper is visible at that tip in the folded base, seen from above and from below (the folded model is viewed with the paper of the largest face, near its crease-pattern position, still showing its front side upward). Where layer order is a choice, the listed tips are ones whose colour does not depend on it.

Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.35355339,0.35355339],[0.5,0.5],[0.64644661,0.64644661],[1.0,1.0],[1.0,0.0],[0.64644661,0.35355339],[0.35355339,0.6464466
1],[0.0,1.0],[0.5,0.0],[0.5,0.20710678],[1.0,0.5],[0.79289322,0.5],[0.5,1.0],[0.5,0.79289322],[0.0,0.5],[0.20710678,0.5]],"edges_vertices":[[0,1],[1,2],
[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,
16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[0,9],[5,9],[5,11],[4,11],[4,13],[8,13],[8,15],[0,15]],"edges_assignment":
["V","M","V","M","V","M","M","M","M","V","V","V","V","V","V","V","M","M","V","V","M","V","M","M","V","M","V","M","V","M","V","M","B","B","B","B","B","B","B",
aces_vertices":[[0,1,16],[1,0,10],[1,2,16],[2,1,10],[2,3,14],[3,2,12],[3,4,14],[4,3,12],[5,6,10],[6,5,12],[2,6,12],[6,2,10],[2,7,16],[7,2,14],[7,8,16],[8,
7,14],[9,10,0],[10,9,5],[11,12,5],[12,11,4],[13,14,4],[14,13,8],[15,16,8],[16,15,0]]}
```

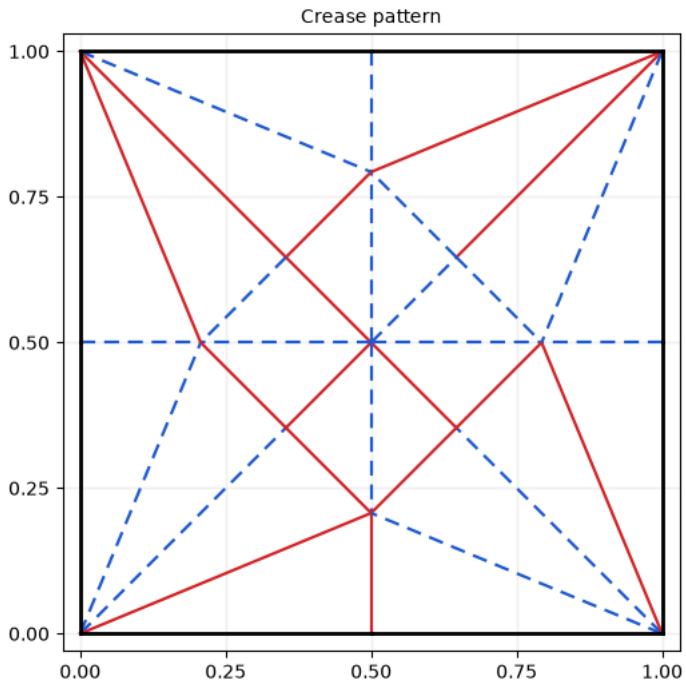

Input `flap_tips`:


```
{"tip0":[0.0,0.0],"tip1":[0.5,0.5],"tip2":[1.0,1.0],"tip3":[1.0,0.0],"tip4":[0.0,1.0],"tip5":[0.5,0.207107],"tip6":[0.792893,0.5],"tip7":[0.5,0.792893],"t
ip8":[0.207107,0.5]}
```


Answer format:

```
{"colors": {"tip_id": {"above": "front" | "back", "below": "front" | "back"}, ...}}
```

"front" is the side of the paper facing you in the crease-pattern drawing.



Working / answer

Q106

split: core · family: read-color · conventions: advanced (front matter) · files: inputs/Q106/

Task (read, hard):
The crease pattern below is folded from two-tone paper. The side facing you in the drawing is 'front', the other side 'back'. For each listed flap tip (given by its position on the unfolded paper), say which side of the paper is visible at that tip in the folded base, seen from above and from below (the folded model is viewed with the paper of the largest face, near its crease-pattern position, still showing its front side upward). Where layer order is a choice, the listed tips are ones whose colour does not depend on it.

Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.08333333,0.08333333],[0.37796116,0.37796116],[0.5,0.5],[0.62203884,0.62203884],[0.91666667,0.91666667],[1.0,1.0],[1.0,0.
0],[0.91666667,0.08333333],[0.62203884,0.37796116],[0.37796116,0.62203884],[0.08333333,0.91666667],[0.0,1.0],[0.5,0.0],[0.5,0.08333333],[0.5,0.25592232],
[1.0,0.5],[0.91666667,0.5],[0.74407768,0.5],[0.5,1.0],[0.5,0.91666667],[0.5,0.74407768],[0.0,0.5],[0.08333333,0.5],[0.25592232,0.5],[0.04881554,0.0],[0.0,
0.04881554],[1.0,0.04881554],[0.95118446,0.0],[0.95118446,1.0],[1.0,0.95118446],[0.0,0.95118446],[0.04881554,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,
4],[4,5],[5,6],[7,8],[8,9],[3,9],[3,10],[10,11],[11,12],[13,14],[14,15],[3,15],[16,17],[17,18],[3,18],[19,20],[20,21],[3,21],[22,23],[23,24],[3,24],[9,1
5],[9,18],[4,18],[4,21],[10,21],[10,24],[2,24],[2,15],[1,25],[1,24],[1,26],[1,15],[8,27],[8,15],[8,28],[8,18],[5,29],[5,18],[5,30],[5,21],[11,31],[11,21],
[11,32],[11,24],[1,14],[8,14],[8,17],[5,17],[5,20],[11,20],[11,23],[1,23],[0,25],[13,25],[13,28],[7,28],[7,27],[16,27],[16,30],[6,30],[6,29],[19,29],[19,3
2],[12,32],[12,31],[22,31],[22,26],[0,26]],"edges_assignment":["V","V","M","V","M","M","V","V","M","M","M","M","M","M","V","V" ...
[truncated: the full input is in inputs/Q106/]
```

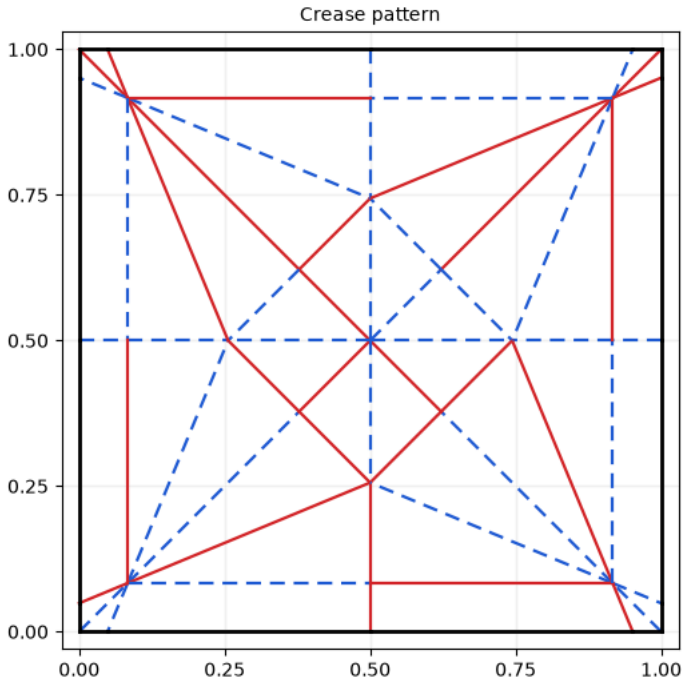

Input `flap_tips`:


```
{"tip1":[0.5,0.5],"tip5":[0.5,0.0],"tip6":[0.5,0.255922],"tip7":[1.0,0.5],"tip8":[0.744078,0.5],"tip9":[0.5,1.0],"tip10":[0.5,0.744078],"tip11":[0.0,0.
5],"tip12":[0.255922,0.5]}
```


Answer format:

```
{"colors": {"tip_id": {"above": "front" | "back", "below": "front" | "back"}, ...}}
```

"front" is the side of the paper facing you in the crease-pattern drawing.



Working / answer

Q107

split: core · family: read-color · conventions: advanced (front matter) · files: inputs/Q107/

Task (read, hard):
The crease pattern below is folded from two-tone paper. The side facing you in the drawing is 'front', the other side 'back'. For each listed flap tip (given by its position on the unfolded paper), say which side of the paper is visible at that tip in the folded base, seen from above and from below (the folded model is viewed with the paper of the largest face, near its crease-pattern position, still showing its front side upward). Where layer order is a choice, the listed tips are ones whose colour does not depend on it.

Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1,0.1],[0.5,0.5],[0.9,0.9],[1.0,1.0],[0.9,0.1],[1.0,0.0],[0.1,0.9],[0.0,1.0],[0.1,0.5],[0.0,0.5],[0.9,0.5],[1.0,0.5]],"ed
ges_vertices":[[0,1],[1,2],[2,3],[3,4],[2,5],[5,6],[2,7],[7,8],[2,9],[9,10],[2,11],[11,12],[1,5],[5,11],[3,11],[3,7],[7,9],[1,9],[0,6],[6,12],[4,12],[4,
8],[8,10],[0,10]],"edges_assignment":["V","V","V","V","V","V","V","V","M","M","M","M","M","V","M","V","M","V","B","B","B","B","B","B"],"faces_vertices":
[[0,1,9,10],[1,0,6,5],[1,2,9],[2,1,5],[2,3,7],[3,2,11],[3,4,8,7],[4,3,11,12],[2,5,11],[5,6,12,11],[2,7,9],[7,8,10,9]]}
```

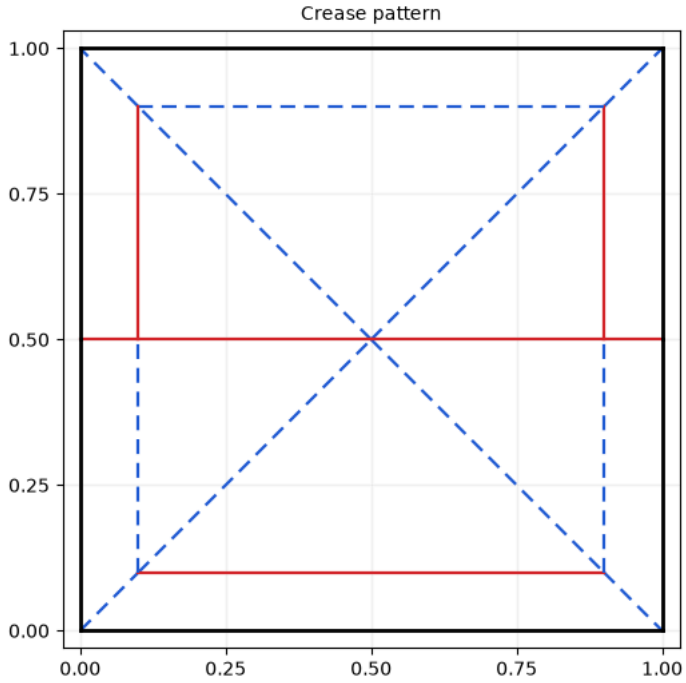

Input `flap_tips`:


```
{"tip0":[0.0,0.0],"tip1":[1.0,1.0],"tip2":[1.0,0.0],"tip3":[0.0,1.0]}
```


Answer format:

```
{"colors": {"tip_id": {"above": "front" | "back", "below": "front" | "back"}, ...}}
```

"front" is the side of the paper facing you in the crease-pattern drawing.



Working / answer

Q108

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q108/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

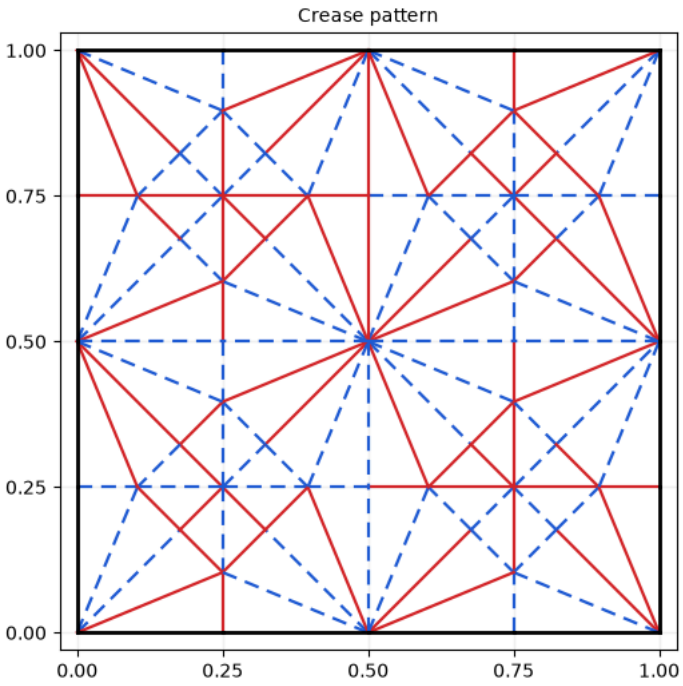
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1767767,0.1767767],[0.25,0.25],[0.3232233,0.3232233],[0.5,0.5],[0.5,0.0],[0.3232233,0.1767767],[0.1767767,0.3232233],[0.
0,0.5],[0.25,0.0],[0.25,0.10355339],[0.5,0.25],[0.39644661,0.25],[0.25,0.5],[0.25,0.39644661],[0.0,0.25],[0.10355339,0.25],[1.0,0.0],[0.8232233,0.176776
7],[0.75,0.25],[0.6767767,0.3232233],[0.6767767,0.1767767],[0.8232233,0.3232233],[1.0,0.5],[0.75,0.0],[0.75,0.10355339],[0.60355339,0.25],[0.75,0.5],[0.7
5,0.39644661],[1.0,0.25],[0.89644661,0.25],[0.0,1.0],[0.1767767,0.8232233],[0.25,0.75],[0.3232233,0.6767767],[0.5,1.0],[0.3232233,0.8232233],[0.1767767,0.
6767767],[0.25,1.0],[0.25,0.89644661],[0.5,0.75],[0.39644661,0.75],[0.25,0.60355339],[0.0,0.75],[0.10355339,0.75],[1.0,1.0],[0.8232233,0.8232233],[0.75,0.
75],[0.6767767,0.6767767],[0.6767767,0.8232233],[0.8232233,0.6767767],[0.75,1.0],[0.75,0.89644661],[0.60355339,0.75],[0.75,0.60355339],[1.0,0.75],[0.89644
661,0.75]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,1
2],[3,14],[7,14],[7,16],[1,16],[1,10],[0,16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[17,18],[18,19],[19,20],[4,20],[ ...
[truncated: the full input is in inputs/Q108/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q109

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q109/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

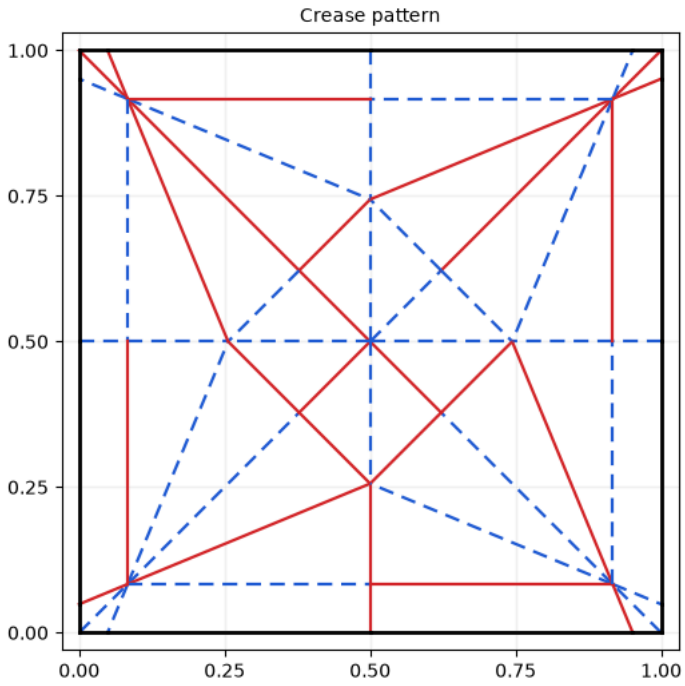
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.08333333,0.08333333],[0.37796116,0.37796116],[0.5,0.5],[0.62203884,0.62203884],[0.91666667,0.91666667],[1.0,1.0],[1.0,0.
0],[0.91666667,0.08333333],[0.62203884,0.37796116],[0.37796116,0.62203884],[0.08333333,0.91666667],[0.0,1.0],[0.5,0.0],[0.5,0.08333333],[0.5,0.25592232],
[1.0,0.5],[0.91666667,0.5],[0.74407768,0.5],[0.5,1.0],[0.5,0.91666667],[0.5,0.74407768],[0.0,0.5],[0.08333333,0.5],[0.25592232,0.5],[0.04881554,0.0],[0.0,
0.04881554],[1.0,0.04881554],[0.95118446,0.0],[0.95118446,1.0],[1.0,0.95118446],[0.0,0.95118446],[0.04881554,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,
4],[4,5],[5,6],[7,8],[8,9],[3,9],[3,10],[10,11],[11,12],[13,14],[14,15],[3,15],[16,17],[17,18],[3,18],[19,20],[20,21],[3,21],[22,23],[23,24],[3,24],[9,1
5],[9,18],[4,18],[4,21],[10,21],[10,24],[2,24],[2,15],[1,25],[1,24],[1,26],[1,15],[8,27],[8,15],[8,28],[8,18],[5,29],[5,18],[5,30],[5,21],[11,31],[11,21],
[11,32],[11,24],[1,14],[8,14],[8,17],[5,17],[5,20],[11,20],[11,23],[1,23],[0,25],[13,25],[13,28],[7,28],[7,27],[16,27],[16,30],[6,30],[6,29],[19,29],[19,3
2],[12,32],[12,31],[22,31],[22,26],[0,26]],"edges_assignment":["V","V","M","V","M","M","V","V","M","M","M","M","M","M","V","V" ...
[truncated: the full input is in inputs/Q109/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q110

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q110/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

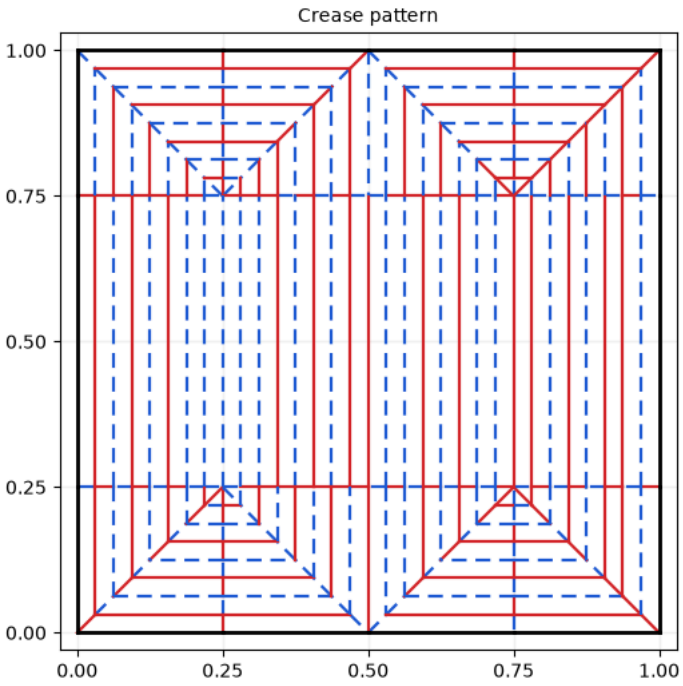
Input `crease_pattern`:


```
{"vertices_coords":[[0.21875,0.25],[0.25,0.25],[0.1875,0.25],[0.15625,0.25],[0.125,0.25],[0.09375,0.25],[0.0625,0.25],[0.03125,0.25],[0.0,0.25],[0.25,0.
0],[0.25,0.03125],[0.25,0.0625],[0.25,0.09375],[0.25,0.125],[0.25,0.15625],[0.25,0.1875],[0.25,0.21875],[0.0,0.0],[0.03125,0.03125],[0.0625,0.0625],[0.093
75,0.09375],[0.125,0.125],[0.15625,0.15625],[0.1875,0.1875],[0.21875,0.21875],[0.0,0.75],[0.03125,0.75],[0.0625,0.75],[0.09375,0.75],[0.125,0.75],[0.1562
5,0.75],[0.1875,0.75],[0.21875,0.75],[0.25,0.75],[0.0,1.0],[0.03125,0.96875],[0.0625,0.9375],[0.09375,0.90625],[0.125,0.875],[0.15625,0.84375],[0.1875,0.8
125],[0.21875,0.78125],[0.25,0.96875],[0.25,1.0],[0.25,0.9375],[0.25,0.90625],[0.25,0.875],[0.25,0.84375],[0.25,0.8125],[0.25,0.78125],[0.46875,0.03125],
[0.5,0.0],[0.4375,0.0625],[0.40625,0.09375],[0.375,0.125],[0.34375,0.15625],[0.3125,0.1875],[0.28125,0.21875],[0.46875,0.96875],[0.5,1.0],[0.4375,0.9375],
[0.40625,0.90625],[0.375,0.875],[0.34375,0.84375],[0.3125,0.8125],[0.28125,0.78125],[0.75,0.21875],[0.75,0.25],[0.75,0.1875],[0.75,0.15625],[0.75,0.125],
[0.75,0.09375],[0.75,0.0625],[0.75,0.03125],[0.75,0.0],[0.71875,0.25],[0.6875,0.25],[0.65625,0.25],[0.625,0.25],[0.59375,0.25], ...
[truncated: the full input is in inputs/Q110/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q111

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q111/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

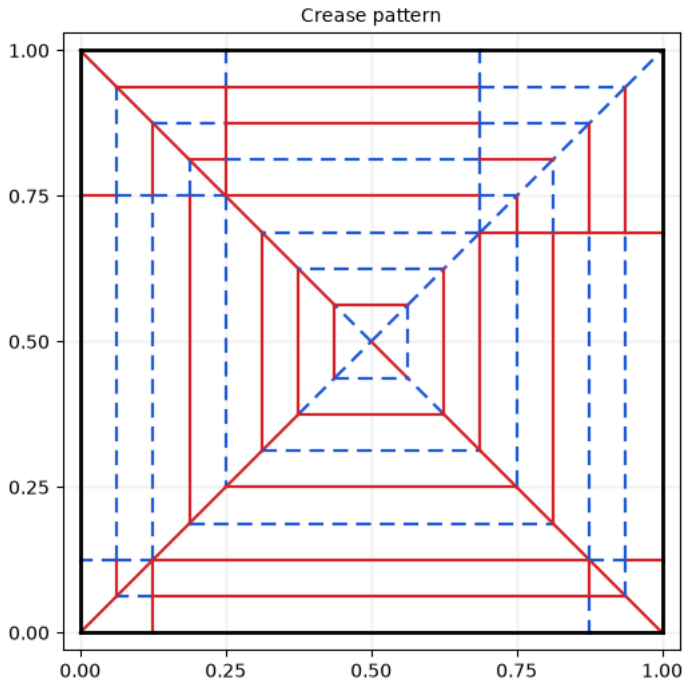
Input `crease_pattern`:


```
{"vertices_coords":[[0.0625,0.125],[0.125,0.125],[0.0,0.125],[0.125,0.0],[0.125,0.0625],[0.0,0.0],[0.0625,0.0625],[0.0,0.75],[0.0625,0.75],[0.125,0.75],
[0.1875,0.75],[0.25,0.75],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.8125],[0.25,0.875],[0.25,0.9375],[0.25,1.0],[0.6875,0.9375],[0.6
875,1.0],[0.6875,0.875],[0.6875,0.8125],[0.6875,0.75],[0.6875,0.6875],[0.875,0.0625],[0.875,0.125],[0.875,0.0],[0.9375,0.0625],[1.0,0.0],[0.9375,0.125],
[1.0,0.125],[0.75,0.6875],[0.8125,0.6875],[0.875,0.6875],[0.9375,0.6875],[1.0,0.6875],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],
[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.625,0.625],[0.8125,0.1875],[0.75,0.25],[0.6875,0.31
25],[0.625,0.375],[0.5625,0.4375],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[3,4],[1,4],[5,6],[1,6],[7,8],[8,9],[9,10],
[10,11],[12,13],[13,14],[14,15],[11,15],[11,16],[16,17],[17,18],[18,19],[20,21],[20,22],[22,23],[23,24],[24,25],[26,27],[26,28],[29,30],[27,29],[31,32],[2
7,31],[25,33],[33,34],[34,35],[35,36],[36,37],[38,39],[38,40],[40,41],[41,42],[25,42],[1,43],[43,44],[44,45],[45,46],[46,47],[ ...
[truncated: the full input is in inputs/Q111/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q112

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q112/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

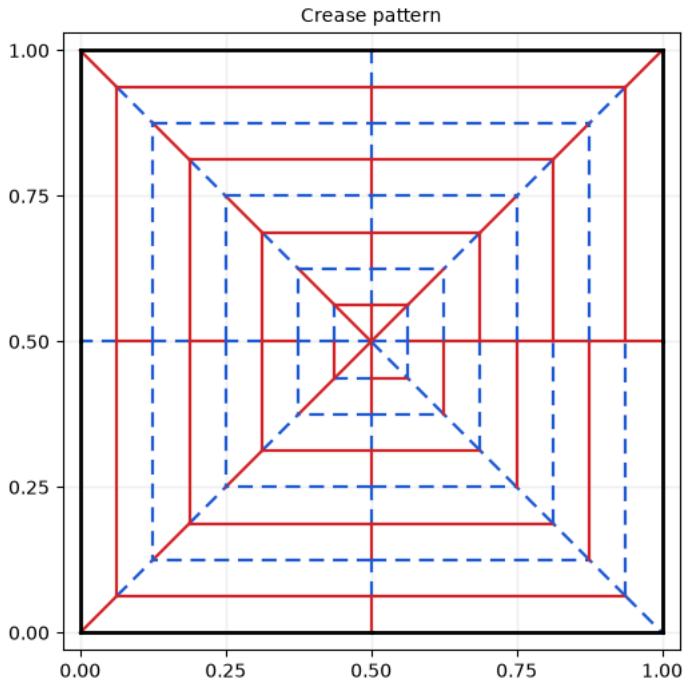
Input `crease_pattern`:


```
{"vertices_coords":[[0.4375,0.5],[0.5,0.5],[0.375,0.5],[0.3125,0.5],[0.25,0.5],[0.1875,0.5],[0.125,0.5],[0.0625,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.0625],[0.
5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.312
5],[0.375,0.375],[0.4375,0.4375],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.625],[0.4375,0.5625],[0.5,0.
5625],[0.5,0.625],[0.5,0.6875],[0.5,0.75],[0.5,0.8125],[0.5,0.875],[0.5,0.9375],[0.5,1.0],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.
25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.9375,0.5],[1.0,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.5],[0.937
5,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.625,0.625],[0.5625,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,
5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[14,15],[15,16],[1,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[1,24],[25,26],
[26,27],[27,28],[28,29],[29,30],[30,31],[31,32],[1,32],[1,33],[33,34],[34,35],[35,36],[36,37],[37,38],[38,39],[39,40],[41,42], ...
[truncated: the full input is in inputs/Q112/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q113

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q113/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

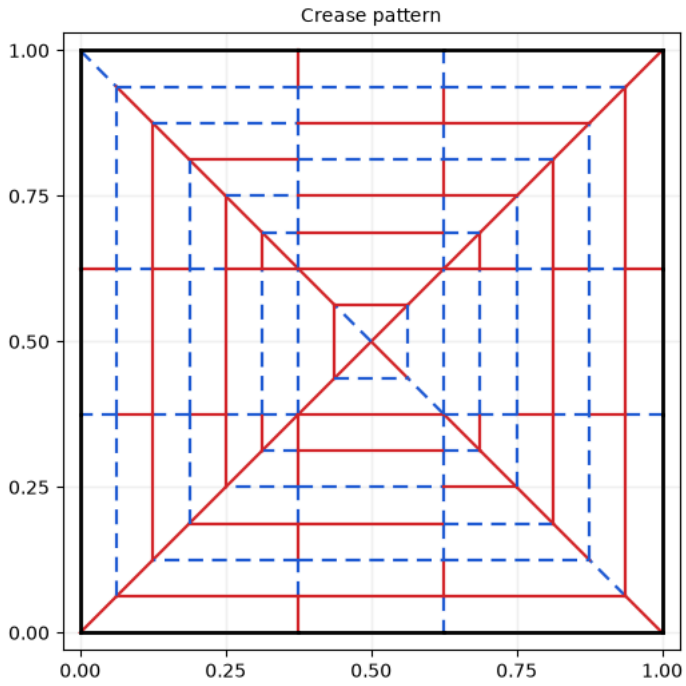
Input `crease_pattern`:


```
{"vertices_coords":[[0.3125,0.375],[0.375,0.375],[0.25,0.375],[0.1875,0.375],[0.125,0.375],[0.0625,0.375],[0.0,0.375],[0.375,0.0],[0.375,0.0625],[0.375,0.125],[0.375,0.1875],[0.375,0.25],[0.375,0.3125],[0.0,0.0],[0.0625,0.0625],[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.0,0.625],[0.0625,0.625],[0.125,0.625],[0.1875,0.625],[0.25,0.625],[0.3125,0.625],[0.375,0.625],[0.0,1.0],[0.0625,0.9375],[0.125,0.875],[0.1875,0.8125],[0.25,0.75],[0.3125,0.6875],[0.375,0.6875],[0.375,0.75],[0.375,0.8125],[0.375,0.875],[0.375,0.9375],[0.375,1.0],[0.625,0.3125],[0.625,0.375],[0.625,0.25],[0.625,0.1875],[0.625,0.125],[0.625,0.0625],[0.625,0.0],[0.625,0.9375],[0.625,1.0],[0.625,0.875],[0.625,0.8125],[0.625,0.75],[0.625,0.6875],[0.625,0.625],[0.9375,0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.9375,0.375],[1.0,0.375],[0.875,0.375],[0.8125,0.375],[0.75,0.375],[0.6875,0.375],[0.6875,0.625],[0.75,0.625],[0.8125,0.625],[0.875,0.625],[0.9375,0.625],[1.0,0.625],[0.9375,0.9375],[1.0,1.0],[0.875,0.875],[0.8125,0.8125],[0.75,0.75],[0.6875,0.6875],[0.4375,0.4375],[0.5,0.5],[0.5625,0.5625],[0.5625,0.4375],[0.4375,0.5625]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4] ...
[truncated: the full input is in inputs/Q113/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q114

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q114/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

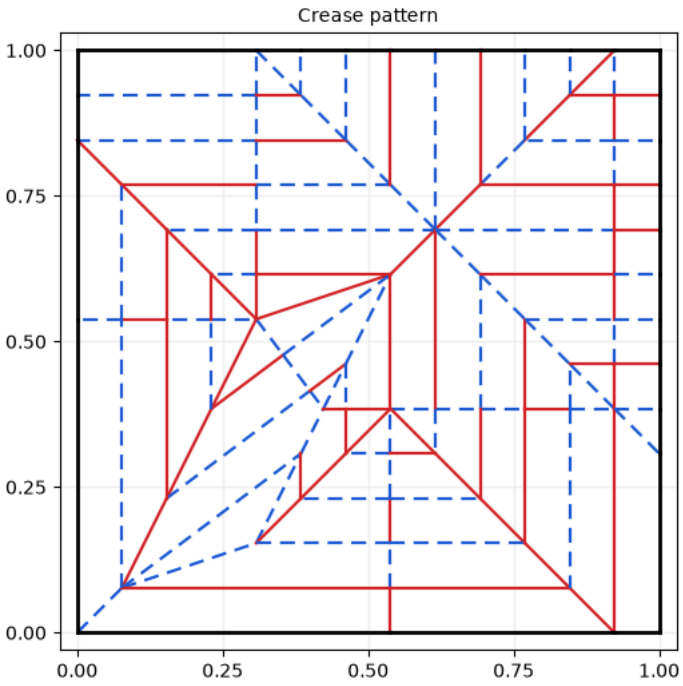
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.07692308],[0.15384615,0.23076923],[0.23076923,0.38461538],[0.30769231,0.15384615],[0.23076923,0.53846154],[0.3
0769231,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.92307692],[0.30769231,1.0],[0.30769231,0.84615385],[0.3
0769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.35384615,0.47692308],[0.38461538,0.30769231],[0.4,0.41538462],[0.42307692,0.3846153
8],[0.46153846,0.46153846],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.
38461538],[0.46153846,0.38461538],[0.38461538,0.23076923],[0.46153846,0.30769231],[0.53846154,0.61538462],[0.53846154,0.76923077],[0.61538462,0.69230769],
[0.46153846,0.84615385],[0.38461538,0.92307692],[0.61538462,0.38461538],[0.69230769,0.38461538],[0.76923077,0.38461538],[0.84615385,0.38461538],[0.9230769
2,0.38461538],[0.69230769,0.61538462],[0.76923077,0.53846154],[0.84615385,0.46153846],[0.92307692,0.46153846],[0.92307692,0.53846154],[0.92307692,0.615384
62],[0.92307692,0.69230769],[0.92307692,0.76923077],[0.92307692,0.84615385],[0.92307692,0.92307692],[0.92307692,1.0],[0.69 ...
[truncated: the full input is in inputs/Q114/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q115

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q115/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

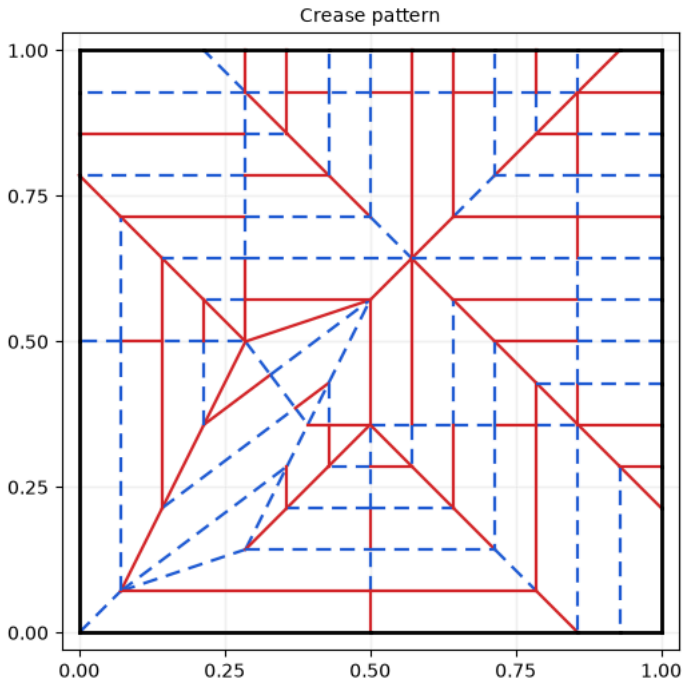
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07142857,0.07142857],[0.14285714,0.21428571],[0.21428571,0.35714286],[0.28571429,0.14285714],[0.21428571,0.5],[0.28571429,0.5],[0.14285714,0.5],[0.07142857,0.5],[0.0,0.5],[0.28571429,0.85714286],[0.28571429,0.92857143],[0.28571429,0.78571429],[0.28571429,0.71428571],[0.28571429,0.64285714],[0.28571429,0.57142857],[0.32857143,0.44285714],[0.35714286,0.28571429],[0.37142857,0.38571429],[0.39285714,0.35714286],[0.42857143,0.42857143],[0.5,0.0],[0.5,0.07142857],[0.5,0.14285714],[0.5,0.21428571],[0.5,0.28571429],[0.5,0.35714286],[0.42857143,0.35714286],[0.35714286,0.21428571],[0.42857143,0.28571429],[0.5,0.57142857],[0.5,0.71428571],[0.57142857,0.64285714],[0.42857143,0.78571429],[0.35714286,0.85714286],[0.57142857,0.35714286],[0.64285714,0.35714286],[0.71428571,0.35714286],[0.78571429,0.35714286],[0.85714286,0.35714286],[0.64285714,0.57142857],[0.71428571,0.5],[0.78571429,0.42857143],[0.78571429,0.92857143],[0.85714286,0.92857143],[0.71428571,0.92857143],[0.64285714,0.92857143],[0.57142857,0.92857143],[0.5,0.92857143],[0.42857143,0.92857143],[0.35714286,0.92857143],[0.85714286,0.42857143],[0.85714286,0.5],[0.85714286,0.57142857],[0.85714286,0.64285714], ...
[truncated: the full input is in inputs/Q115/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q116

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q116/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

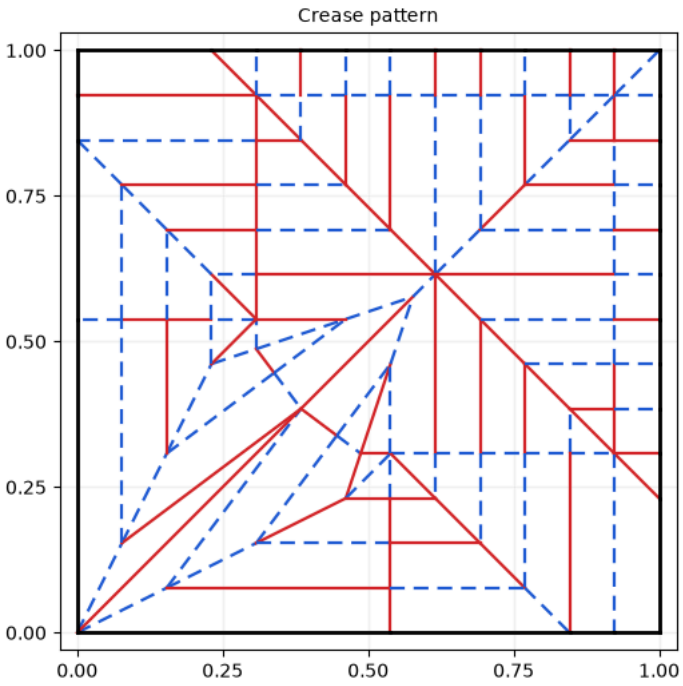
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.07692308,0.15384615],[0.15384615,0.07692308],[0.15384615,0.30769231],[0.23076923,0.46153846],[0.30769231,0.15384615],[0.30769231,0.48717949],[0.30769231,0.53846154],[0.23076923,0.53846154],[0.15384615,0.53846154],[0.07692308,0.53846154],[0.0,0.53846154],[0.30769231,0.84615385],[0.30769231,0.92307692],[0.30769231,0.76923077],[0.30769231,0.69230769],[0.30769231,0.61538462],[0.33846154,0.44615385],[0.38461538,0.38461538],[0.44615385,0.33846154],[0.46153846,0.23076923],[0.46153846,0.53846154],[0.48717949,0.30769231],[0.53846154,0.0],[0.53846154,0.07692308],[0.53846154,0.15384615],[0.53846154,0.23076923],[0.53846154,0.30769231],[0.53846154,0.46153846],[0.57692308,0.57692308],[0.53846154,0.69230769],[0.61538462,0.61538462],[0.46153846,0.76923077],[0.38461538,0.84615385],[0.61538462,0.30769231],[0.69230769,0.30769231],[0.76923077,0.30769231],[0.84615385,0.30769231],[0.92307692,0.30769231],[0.69230769,0.53846154],[0.76923077,0.46153846],[0.84615385,0.38461538],[0.84615385,0.92307692],[0.92307692,0.92307692],[0.76923077,0.92307692],[0.69230769,0.92307692],[0.61538462,0.92307692],[0.53846154,0.92307692],[0.46153846,0.92307692],[0.38461538,0.92307692],[0.92307 ...
[truncated: the full input is in inputs/Q116/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q117

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q117/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

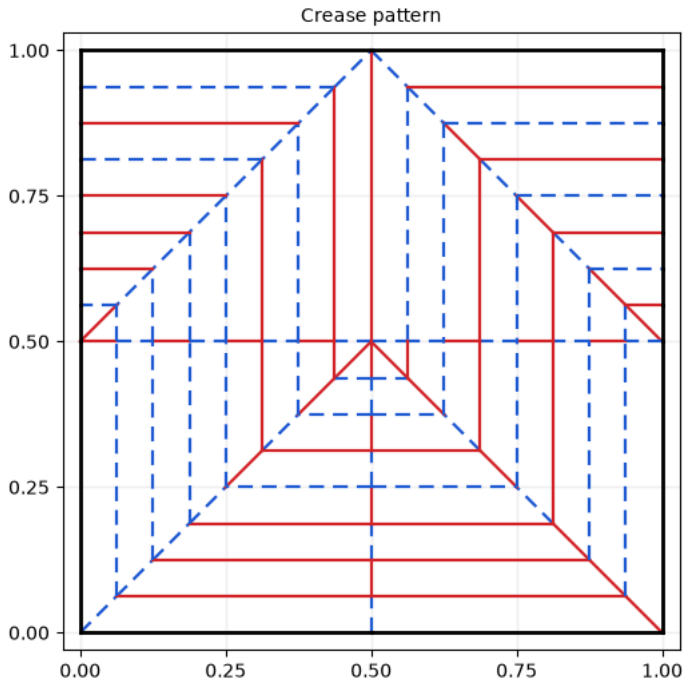
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.0625],[0.5,0.125],[0.5,0.1875],[0.5,0.25],[0.5,0.3125],[0.5,0.375],[0.5,0.4375],[0.5,0.5],[0.0,0.0],[0.0625,0.0625],
[0.125,0.125],[0.1875,0.1875],[0.25,0.25],[0.3125,0.3125],[0.375,0.375],[0.4375,0.4375],[0.0,0.5],[0.0625,0.5],[0.125,0.5],[0.1875,0.5],[0.25,0.5],[0.312
5,0.5],[0.375,0.5],[0.4375,0.5],[0.4375,0.9375],[0.5,1.0],[0.375,0.875],[0.3125,0.8125],[0.25,0.75],[0.1875,0.6875],[0.125,0.625],[0.0625,0.5625],[0.9375,
0.0625],[1.0,0.0],[0.875,0.125],[0.8125,0.1875],[0.75,0.25],[0.6875,0.3125],[0.625,0.375],[0.5625,0.4375],[0.5625,0.9375],[0.625,0.875],[0.6875,0.8125],
[0.75,0.75],[0.8125,0.6875],[0.875,0.625],[0.9375,0.5625],[1.0,0.5],[0.9375,0.5],[0.875,0.5],[0.8125,0.5],[0.75,0.5],[0.6875,0.5],[0.625,0.5],[0.5625,0.
5],[0.0,0.5625],[0.0,0.625],[0.0,0.6875],[0.0,0.75],[0.0,0.8125],[0.0,0.875],[0.0,0.9375],[1.0,0.5625],[1.0,0.625],[1.0,0.6875],[1.0,0.75],[1.0,0.8125],
[1.0,0.875],[1.0,0.9375],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[7,8],[9,10],[10,11],[11,12],[12,13],[13,14],[1
4,15],[15,16],[8,16],[17,18],[18,19],[19,20],[20,21],[21,22],[22,23],[23,24],[8,24],[25,26],[25,27],[27,28],[28,29],[29,30],[30,31] ...
[truncated: the full input is in inputs/Q117/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q118

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q118/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

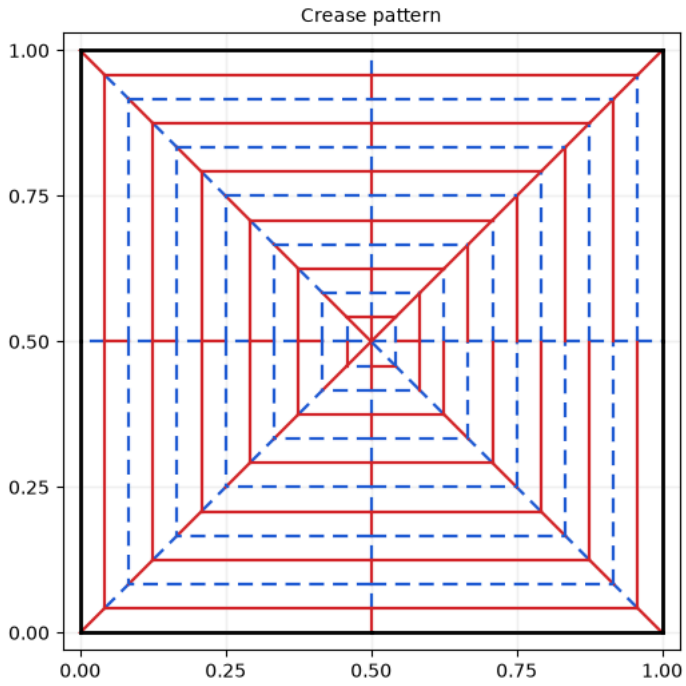
Input `crease_pattern`:


```
{"vertices_coords":[[0.45833333,0.5],[0.5,0.5],[0.41666667,0.5],[0.375,0.5],[0.33333333,0.5],[0.29166667,0.5],[0.25,0.5],[0.20833333,0.5],[0.16666667,0.
5],[0.125,0.5],[0.08333333,0.5],[0.04166667,0.5],[0.0,0.5],[0.5,0.0],[0.5,0.04166667],[0.5,0.08333333],[0.5,0.125],[0.5,0.16666667],[0.5,0.20833333],[0.5,
0.25],[0.5,0.29166667],[0.5,0.33333333],[0.5,0.375],[0.5,0.41666667],[0.5,0.45833333],[0.0,0.0],[0.04166667,0.04166667],[0.08333333,0.08333333],[0.125,0.1
25],[0.16666667,0.16666667],[0.20833333,0.20833333],[0.25,0.25],[0.29166667,0.29166667],[0.33333333,0.33333333],[0.375,0.375],[0.41666667,0.41666667],[0.4
5833333,0.45833333],[0.0,1.0],[0.04166667,0.95833333],[0.08333333,0.91666667],[0.125,0.875],[0.16666667,0.83333333],[0.20833333,0.79166667],[0.25,0.75],
[0.29166667,0.70833333],[0.33333333,0.66666667],[0.375,0.625],[0.41666667,0.58333333],[0.45833333,0.54166667],[0.5,0.54166667],[0.5,0.58333333],[0.5,0.62
5],[0.5,0.66666667],[0.5,0.70833333],[0.5,0.75],[0.5,0.79166667],[0.5,0.83333333],[0.5,0.875],[0.5,0.91666667],[0.5,0.95833333],[0.5,1.0],[0.95833333,0.04
166667],[1.0,0.0],[0.91666667,0.08333333],[0.875,0.125],[0.83333333,0.16666667],[0.79166667,0.20833333],[0.75,0.25],[0.70833333 ...
[truncated: the full input is in inputs/Q118/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q119

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q119/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

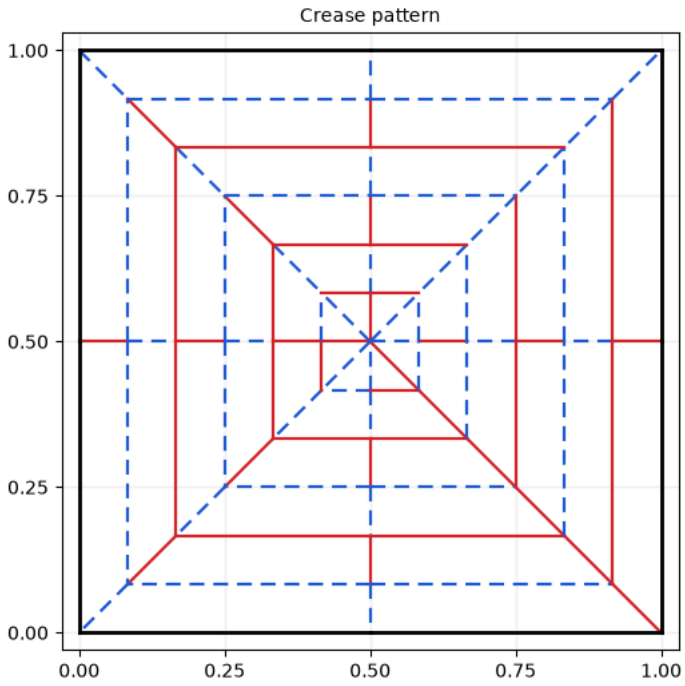
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.41666667],[0.5,0.5],[0.5,0.33333333],[0.5,0.25],[0.5,0.16666667],[0.5,0.08333333],[0.5,0.0],[0.0,0.5],[0.08333333,0.5],[0.16666667,0.5],[0.25,0.5],[0.33333333,0.5],[0.41666667,0.5],[0.0,1.0],[0.08333333,0.91666667],[0.16666667,0.83333333],[0.25,0.75],[0.33333333,0.66666667],[0.41666667,0.58333333],[0.5,0.58333333],[0.5,0.66666667],[0.5,0.75],[0.5,0.83333333],[0.5,0.91666667],[0.5,1.0],[0.91666667,0.08333333],[1.0,0.0],[0.83333333,0.16666667],[0.75,0.25],[0.66666667,0.33333333],[0.58333333,0.41666667],[0.91666667,0.5],[1.0,0.5],[0.83333333,0.5],[0.75,0.5],[0.66666667,0.5],[0.58333333,0.5],[0.91666667,0.91666667],[1.0,1.0],[0.83333333,0.83333333],[0.75,0.75],[0.66666667,0.66666667],[0.58333333,0.58333333],[0.41666667,0.41666667],[0.33333333,0.33333333],[0.25,0.25],[0.16666667,0.16666667],[0.08333333,0.08333333],[0.0,0.0]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,5],[5,6],[7,8],[8,9],[9,10],[10,11],[11,12],[1,12],[13,14],[14,15],[15,16],[16,17],[17,18],[1,18],[1,19],[19,20],[20,21],[21,22],[22,23],[23,24],[25,26],[25,27],[27,28],[28,29],[29,30],[1,30],[31,32],[31,33],[33,34],[34,35],[35,36],[1,36],[37,38],[37,39],[39,40],[40,41],[41,42],[1,42],[1,43],[43 ...
[truncated: the full input is in inputs/Q119/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q120

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q120/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

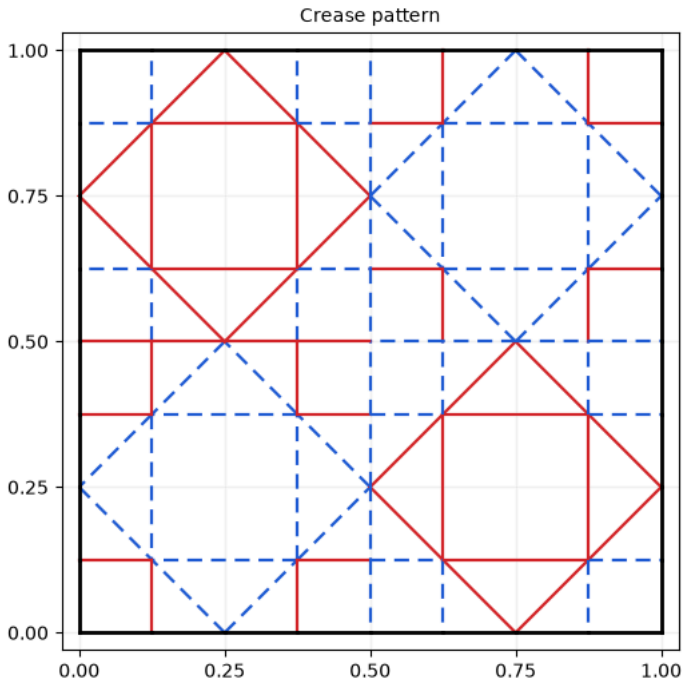
Input `crease_pattern`:


```
{"vertices_coords":[[0.25,0.0],[0.375,0.125],[0.5,0.25],[0.375,0.375],[0.25,0.5],[0.125,0.375],[0.0,0.25],[0.125,0.125],[0.0,0.375],[0.0,0.125],[0.375,0.
0],[0.125,0.0],[0.5,0.125],[0.5,0.375],[0.375,0.5],[0.125,0.5],[0.75,0.0],[0.625,0.125],[0.625,0.375],[0.75,0.5],[0.875,0.375],[1.0,0.25],[0.875,0.125],
[1.0,0.375],[1.0,0.125],[0.625,0.0],[0.875,0.0],[0.625,0.5],[0.875,0.5],[0.25,1.0],[0.375,0.875],[0.5,0.75],[0.375,0.625],[0.125,0.625],[0.0,0.75],[0.125,
0.875],[0.0,0.625],[0.0,0.875],[0.375,1.0],[0.125,1.0],[0.5,0.875],[0.5,0.625],[0.75,1.0],[0.625,0.875],[0.625,0.625],[0.875,0.625],[1.0,0.75],[0.875,0.87
5],[1.0,0.625],[1.0,0.875],[0.625,1.0],[0.875,1.0],[0.5,0.0],[0.5,0.5],[0.5,1.0],[0.0,0.5],[1.0,0.5],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_verti
ces":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[0,7],[5,7],[5,8],[7,9],[1,7],[1,10],[7,11],[1,3],[1,12],[3,13],[3,5],[3,14],[5,15],[16,17],[2,17],[2,18],
[18,19],[19,20],[20,21],[21,22],[16,22],[20,22],[20,23],[22,24],[17,22],[17,25],[22,26],[17,18],[12,17],[13,18],[18,20],[18,27],[20,28],[29,30],[30,31],[3
1,32],[4,32],[4,33],[33,34],[34,35],[29,35],[33,35],[33,36],[35,37],[30,35],[30,38],[35,39],[30,32],[30,40],[32,41],[32,33],[ ...
[truncated: the full input is in inputs/Q120/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q121

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q121/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

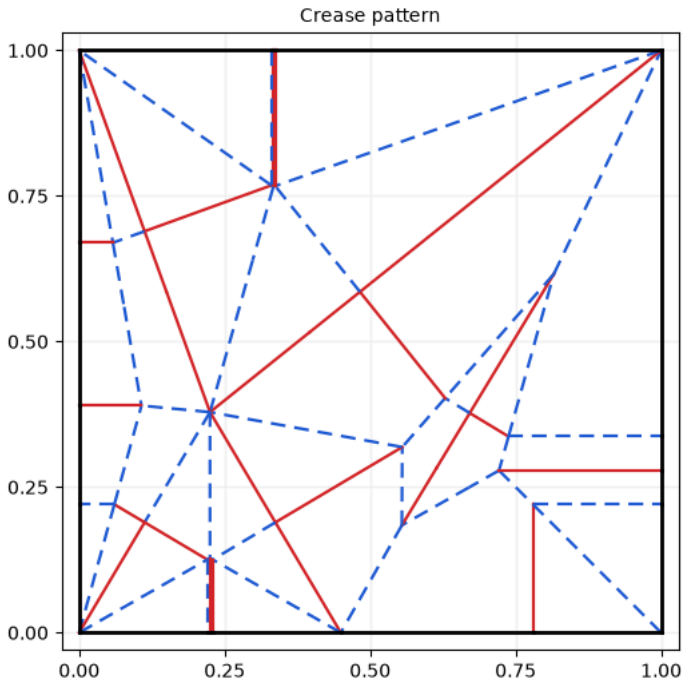
Input `crease_pattern`:


```
{"vertices_coords":[[0.67114,0.3771613],[0.73774143,0.33686288],[0.8155856,0.61588735],[0.72146791,0.27853209],[0.66969919,0.24934811],[0.55451494,0.1844144],[0.61414361,0.28296304],[0.62879913,0.40278043],[0.55509556,0.31869115],[0.48252704,0.58530424],[0.3371409,0.18937529],[0.7797987,0.0],[0.7797987,0.2202013],[1.0,0.0],[1.0,1.0],[1.0,0.2202013],[1.0,0.27853209],[1.0,0.33686288],[0.22478162,0.37875058],[0.3966119,0.51645302],[0.44950018,0.0],[0.11240229,0.68934356],[0.33036601,0.7682076],[0.33686288,0.76706945],[0.33361444,0.76592842],[0.0,1.0],[0.33686288,1.0],[0.33361444,1.0],[0.33036601,1.0],[0.29805749,0.63943291],[0.14986207,0.58581257],[0.11239902,0.18938913],[0.06043101,0.2202312],[0.05792917,0.66963399],[0.07723494,0.55953448],[0.1069831,0.38988284],[0.0,0.0],[0.0,0.66963399],[0.0,0.38988284],[0.0,0.2202312],[0.2202312,0.12539268],[0.22930116,0.12539231],[0.22476618,0.12797475],[0.2202312,0.0],[0.22476618,0.0],[0.22930116,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[5,6],[0,6],[0,2],[0,7],[5,8],[7,8],[2,7],[7,9],[8,10],[11,12],[12,13],[3,12],[2,14],[12,15],[3,16],[1,17],[8,18],[18,19],[9,19],[9,14],[5,20],[10,20],[10,18],[21,22],[9,23],[14,23],[23,24],[22,24],[ ...
[truncated: the full input is in inputs/Q121/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q122

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q122/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

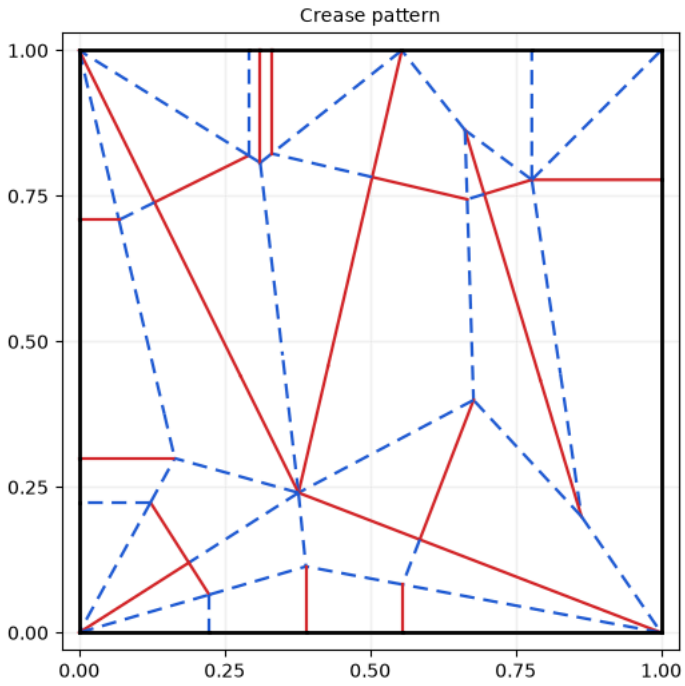
Input `crease_pattern`:


```
{"vertices_coords":[[0.69610678,0.75314504],[0.77777833,0.77777824],[0.6629485,0.86308373],[0.82695228,0.44444444],[0.86308387,0.19952043],[0.79235982,0.43401112],[0.66671561,0.74428046],[0.67765092,0.39941419],[0.77777778,1.0],[0.50349674,0.78241699],[0.58530412,0.1598693],[1.0,0.77777778],[1.0,1.0],[0.55433539,1.0],[0.37681679,0.2402432],[0.42765547,0.45782615],[1.0,0.0],[0.12942802,0.73904078],[0.29129245,0.81932073],[0.33089215,0.82274639],[0.3110923,0.80703953],[0.0,1.0],[0.33089221,1.0],[0.31109236,1.0],[0.29129251,1.0],[0.34946794,0.47609482],[0.27753575,0.44041859],[0.18840843,0.12012159],[0.12253467,0.22344329],[0.06826854,0.70870749],[0.14638995,0.37537417],[0.16419353,0.29940873],[0.0,0.0],[0.0,0.70870749],[0.0,0.29940872],[0.0,0.22344328],[0.22344337,0.06516998],[0.55555556,0.08270247],[0.38949944,0.11360228],[0.22344337,0.0],[0.38949944,0.0],[0.55555556,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[0,5],[0,2],[0,6],[4,7],[6,7],[2,6],[1,8],[6,9],[7,10],[1,11],[1,12],[2,13],[7,14],[14,15],[9,15],[9,13],[4,16],[10,16],[10,14],[17,18],[9,19],[13,19],[19,20],[18,20],[18,21],[19,22],[20,23],[18,24],[20,25],[14,25],[14,26],[17,26],[17,21],[27,28],[17,29],[21,29],[29, ...
[truncated: the full input is in inputs/Q122/]
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q123

split: core · family: read-tech · conventions: advanced (front matter) · files: inputs/Q123/

Task (read, medium):
Below is an origami crease pattern. Identify the design system it uses: is it box-pleated (and on which grid), is it a 22.5-degree design, and does it contain Pythagorean stretches (which triples)?

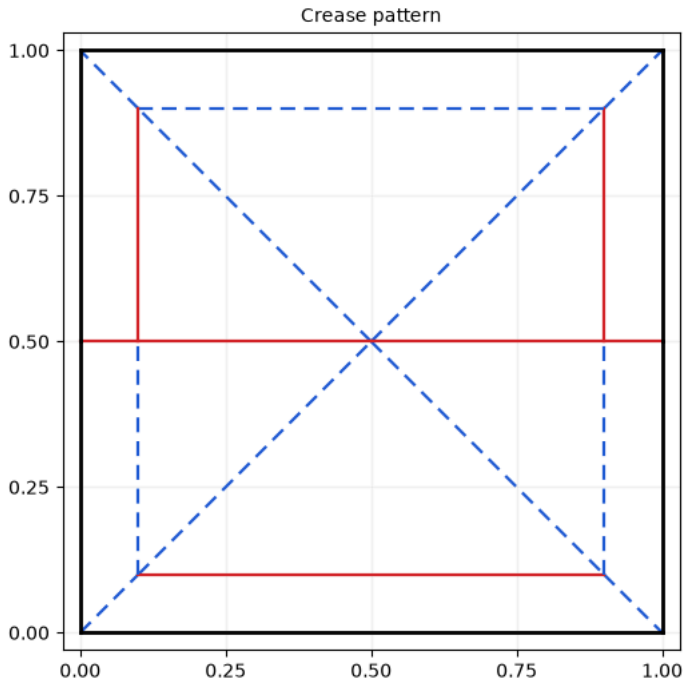
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1,0.1],[0.5,0.5],[0.9,0.9],[1.0,1.0],[0.9,0.1],[1.0,0.0],[0.1,0.9],[0.0,1.0],[0.1,0.5],[0.0,0.5],[0.9,0.5],[1.0,0.5]],"ed
ges_vertices":[[0,1],[1,2],[2,3],[3,4],[2,5],[5,6],[2,7],[7,8],[2,9],[9,10],[2,11],[11,12],[1,5],[5,11],[3,11],[3,7],[7,9],[1,9],[0,6],[6,12],[4,12],[4,
8],[8,10],[0,10]],"edges_assignment":["V","V","V","V","V","V","V","V","M","M","M","M","M","V","M","V","M","V","B","B","B","B","B","B"],"faces_vertices":
[[0,1,9,10],[1,0,6,5],[1,2,9],[2,1,5],[2,3,7],[3,2,11],[3,4,8,7],[4,3,11,12],[2,5,11],[5,6,12,11],[2,7,9],[7,8,10,9]]}
```


Answer format:

```
{"box_pleat": true, "grid": 16, "angles_22_5": false, "pythagorean_stretch": false, "stretch_triples": []}
```

`box_pleat` is true only if every crease is at 0/45/90/135 degrees (a pattern with stretch gadgets is not);
`grid` is the smallest N such that all vertices lie on the 1/N lattice (null if none <= 512);
`stretch_triples` lists the Pythagorean triples [a, b, c] whose directions occur.



Working / answer

Q124

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q124/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

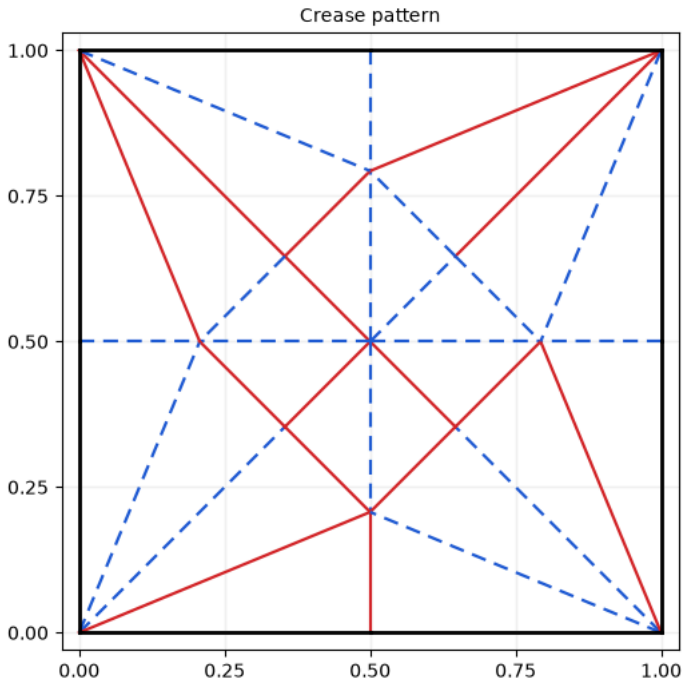
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.35355339,0.35355339],[0.5,0.5],[0.64644661,0.64644661],[1.0,1.0],[1.0,0.0],[0.64644661,0.35355339],[0.35355339,0.6464466
1],[0.0,1.0],[0.5,0.0],[0.5,0.20710678],[1.0,0.5],[0.79289322,0.5],[0.5,1.0],[0.5,0.79289322],[0.0,0.5],[0.20710678,0.5]],"edges_vertices":[[0,1],[1,2],
[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,
16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[0,9],[5,9],[5,11],[4,11],[4,13],[8,13],[8,15],[0,15]],"edges_assignment":
["V","M","V","M","V","M","M","M","M","V","V","V","V","V","V","V","M","M","V","V","M","V","M","M","V","M","V","M","V","M","V","M","B","B","B","B","B","B","B",
aces_vertices":[[0,1,16],[1,0,10],[1,2,16],[2,1,10],[2,3,14],[3,2,12],[3,4,14],[4,3,12],[5,6,10],[6,5,12],[2,6,12],[6,2,10],[2,7,16],[7,2,14],[7,8,16],[8,
7,14],[9,10,0],[10,9,5],[11,12,5],[12,11,4],[13,14,4],[14,13,8],[15,16,8],[16,15,0]]}
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q125

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q125/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

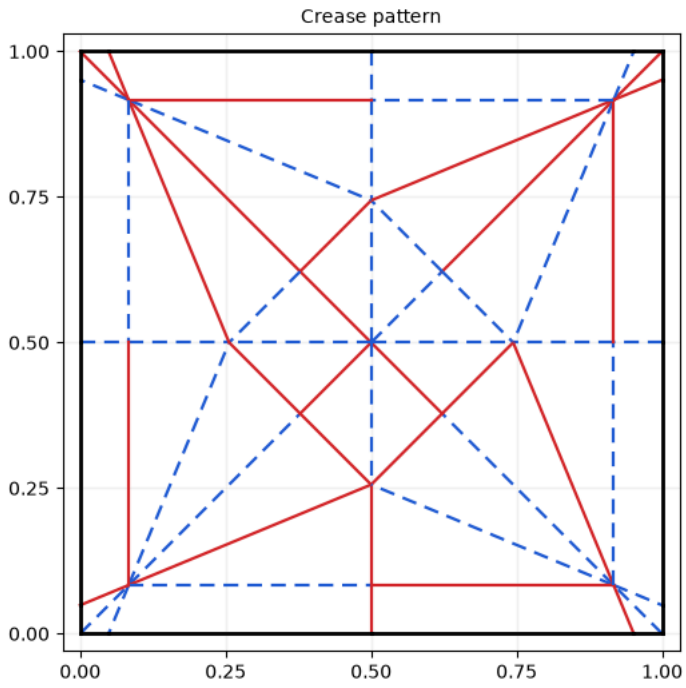
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.08333333,0.08333333],[0.37796116,0.37796116],[0.5,0.5],[0.62203884,0.62203884],[0.91666667,0.91666667],[1.0,1.0],[1.0,0.
0],[0.91666667,0.08333333],[0.62203884,0.37796116],[0.37796116,0.62203884],[0.08333333,0.91666667],[0.0,1.0],[0.5,0.0],[0.5,0.08333333],[0.5,0.25592232],
[1.0,0.5],[0.91666667,0.5],[0.74407768,0.5],[0.5,1.0],[0.5,0.91666667],[0.5,0.74407768],[0.0,0.5],[0.08333333,0.5],[0.25592232,0.5],[0.04881554,0.0],[0.0,
0.04881554],[1.0,0.04881554],[0.95118446,0.0],[0.95118446,1.0],[1.0,0.95118446],[0.0,0.95118446],[0.04881554,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,
4],[4,5],[5,6],[7,8],[8,9],[3,9],[3,10],[10,11],[11,12],[13,14],[14,15],[3,15],[16,17],[17,18],[3,18],[19,20],[20,21],[3,21],[22,23],[23,24],[3,24],[9,1
5],[9,18],[4,18],[4,21],[10,21],[10,24],[2,24],[2,15],[1,25],[1,24],[1,26],[1,15],[8,27],[8,15],[8,28],[8,18],[5,29],[5,18],[5,30],[5,21],[11,31],[11,21],
[11,32],[11,24],[1,14],[8,14],[8,17],[5,17],[5,20],[11,20],[11,23],[1,23],[0,25],[13,25],[13,28],[7,28],[7,27],[16,27],[16,30],[6,30],[6,29],[19,29],[19,3
2],[12,32],[12,31],[22,31],[22,26],[0,26]],"edges_assignment":["V","V","M","V","M","M","V","V","M","M","M","M","M","M","V","V" ...
[truncated: the full input is in inputs/Q125/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q126

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q126/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

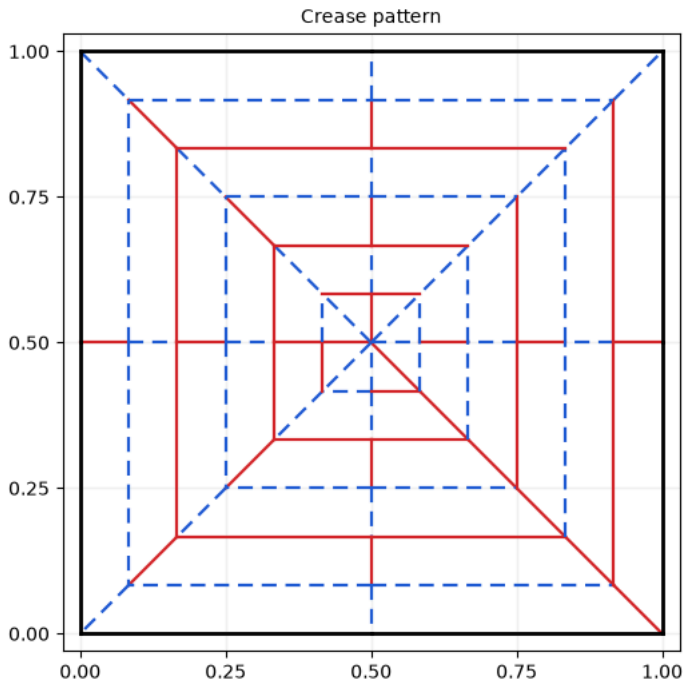
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.41666667],[0.5,0.5],[0.5,0.33333333],[0.5,0.25],[0.5,0.16666667],[0.5,0.08333333],[0.5,0.0],[0.0,0.5],[0.08333333,0.5],[0.16666667,0.5],[0.25,0.5],[0.33333333,0.5],[0.41666667,0.5],[0.0,1.0],[0.08333333,0.91666667],[0.16666667,0.83333333],[0.25,0.75],[0.33333333,0.66666667],[0.41666667,0.58333333],[0.5,0.58333333],[0.5,0.66666667],[0.5,0.75],[0.5,0.83333333],[0.5,0.91666667],[0.5,1.0],[0.91666667,0.08333333],[1.0,0.0],[0.83333333,0.16666667],[0.75,0.25],[0.66666667,0.33333333],[0.58333333,0.41666667],[0.91666667,0.5],[1.0,0.5],[0.83333333,0.5],[0.75,0.5],[0.66666667,0.5],[0.58333333,0.5],[0.91666667,0.91666667],[1.0,1.0],[0.83333333,0.83333333],[0.75,0.75],[0.66666667,0.66666667],[0.58333333,0.58333333],[0.41666667,0.41666667],[0.33333333,0.33333333],[0.25,0.25],[0.16666667,0.16666667],[0.08333333,0.08333333],[0.0,0.0]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,5],[5,6],[7,8],[8,9],[9,10],[10,11],[11,12],[1,12],[13,14],[14,15],[15,16],[16,17],[17,18],[1,18],[1,19],[19,20],[20,21],[21,22],[22,23],[23,24],[25,26],[25,27],[27,28],[28,29],[29,30],[1,30],[31,32],[31,33],[33,34],[34,35],[35,36],[1,36],[37,38],[37,39],[39,40],[40,41],[41,42],[1,42],[1,43],[43 ...
[truncated: the full input is in inputs/Q126/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q127

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q127/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

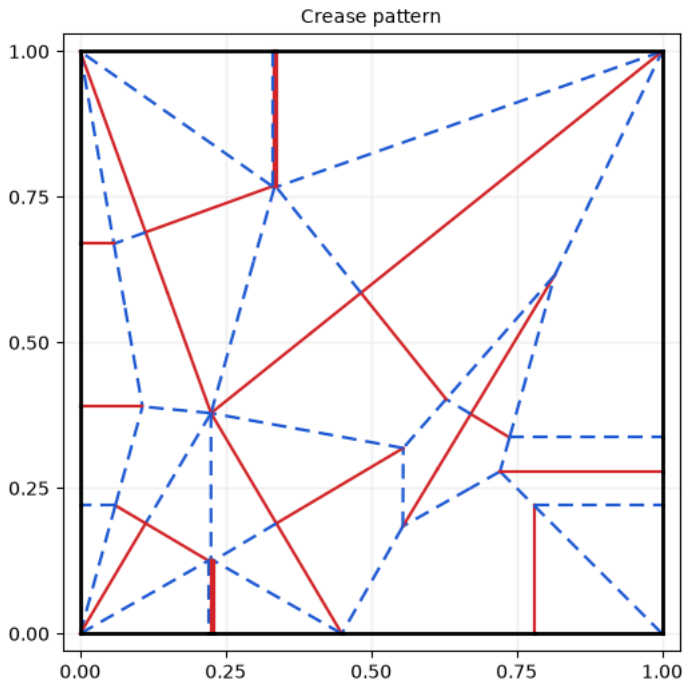
Input `crease_pattern`:


```
{"vertices_coords":[[0.67114,0.3771613],[0.73774143,0.33686288],[0.8155856,0.61588735],[0.72146791,0.27853209],[0.66969919,0.24934811],[0.55451494,0.18441
44],[0.61414361,0.28296304],[0.62879913,0.40278043],[0.55509556,0.31869115],[0.48252704,0.58530424],[0.3371409,0.18937529],[0.7797987,0.0],[0.7797987,0.22
02013],[1.0,0.0],[1.0,1.0],[1.0,0.2202013],[1.0,0.27853209],[1.0,0.33686288],[0.22478162,0.37875058],[0.3966119,0.51645302],[0.44950018,0.0],[0.11240229,
0.68934356],[0.33036601,0.7682076],[0.33686288,0.76706945],[0.33361444,0.76592842],[0.0,1.0],[0.33686288,1.0],[0.33361444,1.0],[0.33036601,1.0],[0.2980574
9,0.63943291],[0.14986207,0.58581257],[0.11239902,0.18938913],[0.06043101,0.2202312],[0.05792917,0.66963399],[0.07723494,0.55953448],[0.1069831,0.3898828
4],[0.0,0.0],[0.0,0.66963399],[0.0,0.38988284],[0.0,0.2202312],[0.2202312,0.12539268],[0.22930116,0.12539231],[0.22476618,0.12797475],[0.2202312,0.0],[0.2
2476618,0.0],[0.22930116,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[5,6],[0,6],[0,2],[0,7],[5,8],[7,8],[2,7],[7,9],[8,10],[11,12],[12,13],[3,1
2],[2,14],[12,15],[3,16],[1,17],[8,18],[18,19],[9,19],[9,14],[5,20],[10,20],[10,18],[21,22],[9,23],[14,23],[23,24],[22,24],[ ...
[truncated: the full input is in inputs/Q127/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q128

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q128/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

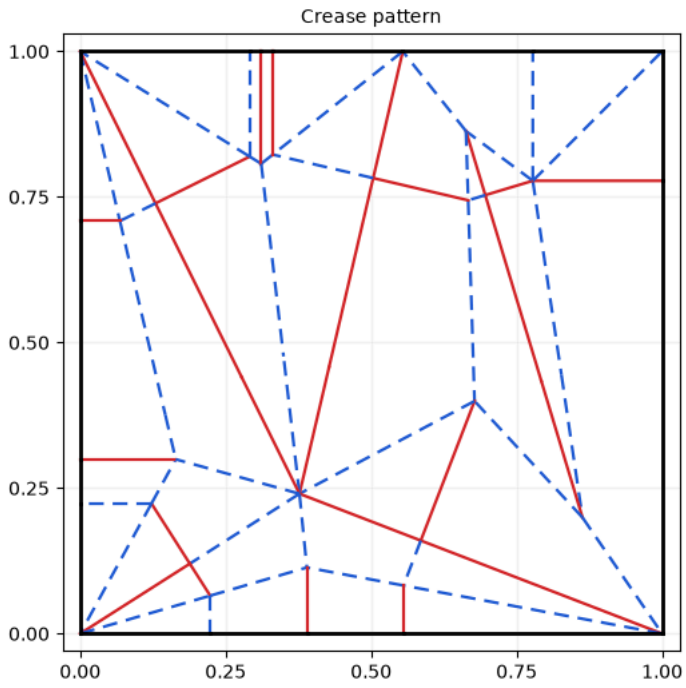
Input `crease_pattern`:


```
{"vertices_coords":[[0.69610678,0.75314504],[0.77777833,0.77777824],[0.6629485,0.86308373],[0.82695228,0.44444444],[0.86308387,0.19952043],[0.79235982,0.43401112],[0.66671561,0.74428046],[0.67765092,0.39941419],[0.77777778,1.0],[0.50349674,0.78241699],[0.58530412,0.1598693],[1.0,0.77777778],[1.0,1.0],[0.55433539,1.0],[0.37681679,0.2402432],[0.42765547,0.45782615],[1.0,0.0],[0.12942802,0.73904078],[0.29129245,0.81932073],[0.33089215,0.82274639],[0.3110923,0.80703953],[0.0,1.0],[0.33089221,1.0],[0.31109236,1.0],[0.29129251,1.0],[0.34946794,0.47609482],[0.27753575,0.44041859],[0.18840843,0.12012159],[0.12253467,0.22344329],[0.06826854,0.70870749],[0.14638995,0.37537417],[0.16419353,0.29940873],[0.0,0.0],[0.0,0.70870749],[0.0,0.29940872],[0.0,0.22344328],[0.22344337,0.06516998],[0.55555556,0.08270247],[0.38949944,0.11360228],[0.22344337,0.0],[0.38949944,0.0],[0.55555556,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[0,5],[0,2],[0,6],[4,7],[6,7],[2,6],[1,8],[6,9],[7,10],[1,11],[1,12],[2,13],[7,14],[14,15],[9,15],[9,13],[4,16],[10,16],[10,14],[17,18],[9,19],[13,19],[19,20],[18,20],[18,21],[19,22],[20,23],[18,24],[20,25],[14,25],[14,26],[17,26],[17,21],[27,28],[17,29],[21,29],[29, ...
[truncated: the full input is in inputs/Q128/]
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q129

split: core · family: read-tree · conventions: advanced (front matter) · files: inputs/Q129/

Task (read, medium):
Below is the crease pattern of a flat-foldable origami base (one uncut square; also shown as an image). Fold it in your head (or with code) and report the stick-figure tree of the folded base: its flaps, branch points and the length of every segment.

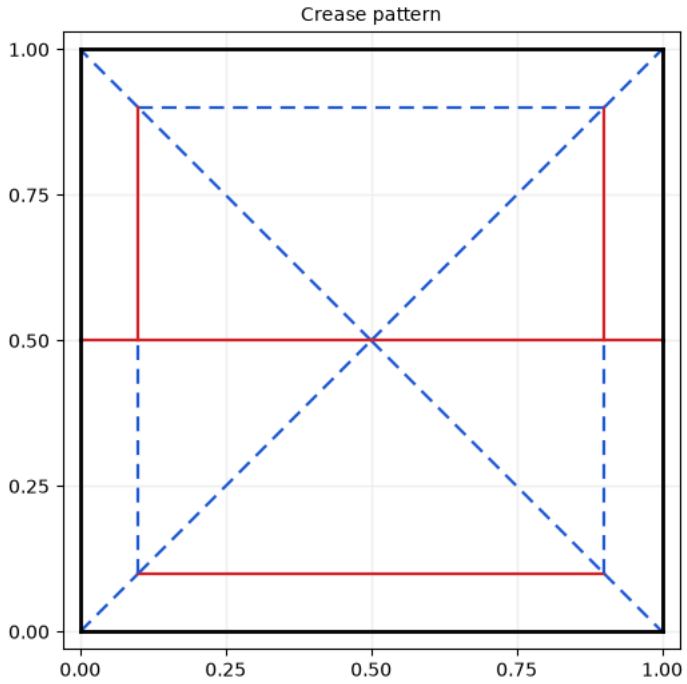
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1,0.1],[0.5,0.5],[0.9,0.9],[1.0,1.0],[0.9,0.1],[1.0,0.0],[0.1,0.9],[0.0,1.0],[0.1,0.5],[0.0,0.5],[0.9,0.5],[1.0,0.5]],"ed
ges_vertices":[[0,1],[1,2],[2,3],[3,4],[2,5],[5,6],[2,7],[7,8],[2,9],[9,10],[2,11],[11,12],[1,5],[5,11],[3,11],[3,7],[7,9],[1,9],[0,6],[6,12],[4,12],[4,
8],[8,10],[0,10]],"edges_assignment":["V","V","V","V","V","V","V","V","M","M","M","M","M","V","M","V","M","V","B","B","B","B","B","B"],"faces_vertices":
[[0,1,9,10],[1,0,6,5],[1,2,9],[2,1,5],[2,3,7],[3,2,11],[3,4,8,7],[4,3,11,12],[2,5,11],[5,6,12,11],[2,7,9],[7,8,10,9]]}
```


Answer format:

```
{"tree": {"edges": [["a", "b", 0.25], ...]}}
```

Report the tree of the folded base (all leaves, including short ones; lengths to 3 decimals).



Working / answer

Q130

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q130/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

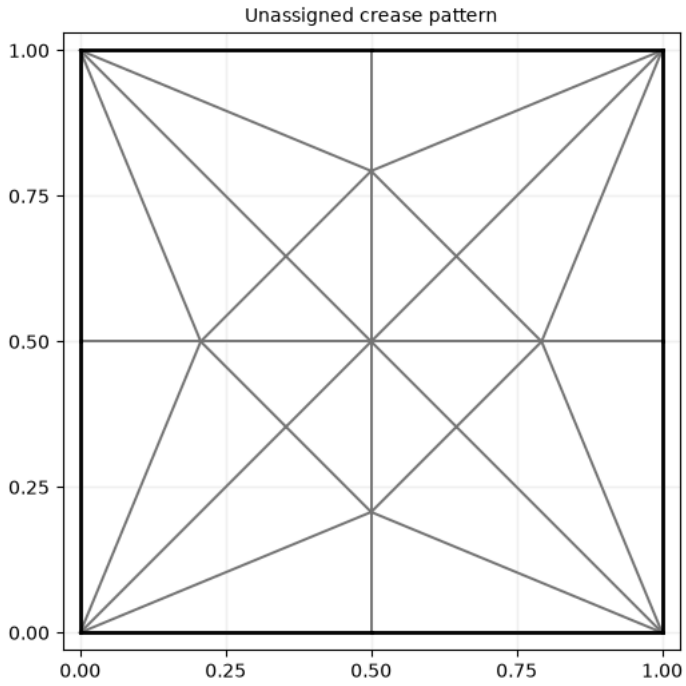
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.35355339,0.35355339],[0.5,0.5],[0.64644661,0.64644661],[1.0,1.0],[1.0,0.0],[0.64644661,0.35355339],[0.35355339,0.6464466
1],[0.0,1.0],[0.5,0.0],[0.5,0.20710678],[1.0,0.5],[0.79289322,0.5],[0.5,1.0],[0.5,0.79289322],[0.0,0.5],[0.20710678,0.5]],"edges_vertices":[[0,1],[1,2],
[2,3],[3,4],[5,6],[2,6],[2,7],[7,8],[9,10],[2,10],[11,12],[2,12],[13,14],[2,14],[15,16],[2,16],[6,10],[6,12],[3,12],[3,14],[7,14],[7,16],[1,16],[1,10],[0,
16],[0,10],[5,10],[5,12],[4,12],[4,14],[8,14],[8,16],[0,9],[5,9],[5,11],[4,11],[4,13],[8,13],[8,15],[0,15]],"edges_assignment":
["U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","B","B","B","B","B","B",
aces_vertices":[[0,1,16],[1,0,10],[1,2,16],[2,1,10],[2,3,14],[3,2,12],[3,4,14],[4,3,12],[5,6,10],[6,5,12],[2,6,12],[6,2,10],[2,7,16],[7,2,14],[7,8,16],[8,
7,14],[9,10,0],[10,9,5],[11,12,5],[12,11,4],[13,14,4],[14,13,8],[15,16,8],[16,15,0]]}
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q131

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q131/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

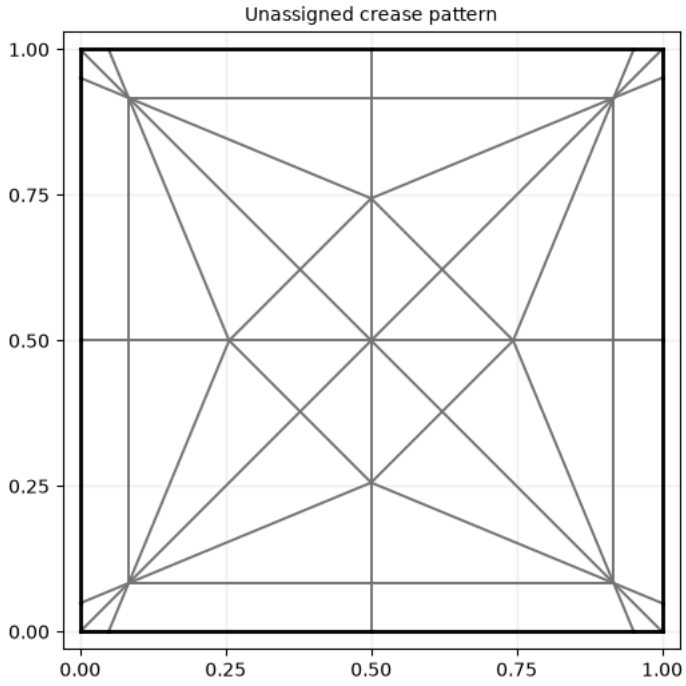
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.08333333,0.08333333],[0.37796116,0.37796116],[0.5,0.5],[0.62203884,0.62203884],[0.91666667,0.91666667],[1.0,1.0],[1.0,0.
0],[0.91666667,0.08333333],[0.62203884,0.37796116],[0.37796116,0.62203884],[0.08333333,0.91666667],[0.0,1.0],[0.5,0.0],[0.5,0.08333333],[0.5,0.25592232],
[1.0,0.5],[0.91666667,0.5],[0.74407768,0.5],[0.5,1.0],[0.5,0.91666667],[0.5,0.74407768],[0.0,0.5],[0.08333333,0.5],[0.25592232,0.5],[0.04881554,0.0],[0.0,
0.04881554],[1.0,0.04881554],[0.95118446,0.0],[0.95118446,1.0],[1.0,0.95118446],[0.0,0.95118446],[0.04881554,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,
4],[4,5],[5,6],[7,8],[8,9],[3,9],[3,10],[10,11],[11,12],[13,14],[14,15],[3,15],[16,17],[17,18],[3,18],[19,20],[20,21],[3,21],[22,23],[23,24],[3,24],[9,1
5],[9,18],[4,18],[4,21],[10,21],[10,24],[2,24],[2,15],[1,25],[1,24],[1,26],[1,15],[8,27],[8,15],[8,28],[8,18],[5,29],[5,18],[5,30],[5,21],[11,31],[11,21],
[11,32],[11,24],[1,14],[8,14],[8,17],[5,17],[5,20],[11,20],[11,23],[1,23],[0,25],[13,25],[13,28],[7,28],[7,27],[16,27],[16,30],[6,30],[6,29],[19,29],[19,3
2],[12,32],[12,31],[22,31],[22,26],[0,26]],"edges_assignment":["U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U" ...
[truncated: the full input is in inputs/Q131/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q132

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q132/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

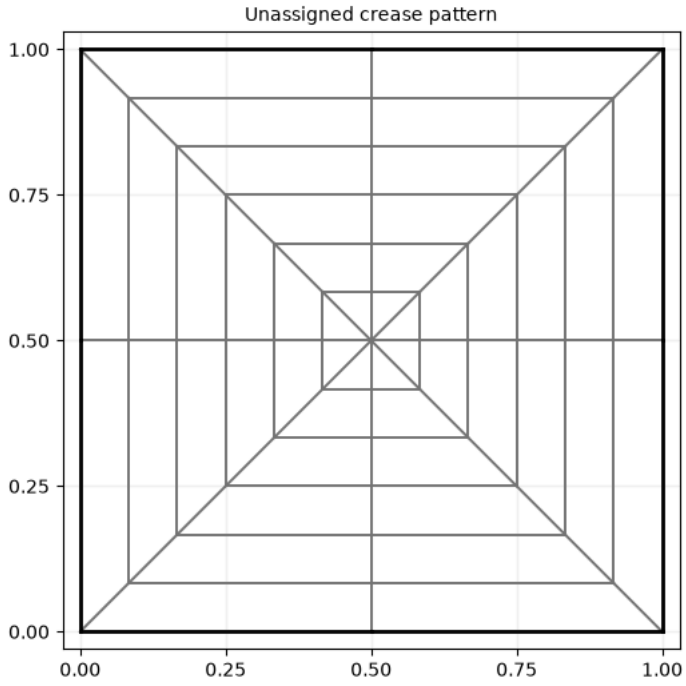
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.41666667],[0.5,0.5],[0.5,0.33333333],[0.5,0.25],[0.5,0.16666667],[0.5,0.08333333],[0.5,0.0],[0.0,0.5],[0.08333333,0.5],[0.16666667,0.5],[0.25,0.5],[0.33333333,0.5],[0.41666667,0.5],[0.0,1.0],[0.08333333,0.91666667],[0.16666667,0.83333333],[0.25,0.75],[0.33333333,0.66666667],[0.41666667,0.58333333],[0.5,0.58333333],[0.5,0.66666667],[0.5,0.75],[0.5,0.83333333],[0.5,0.91666667],[0.5,1.0],[0.91666667,0.08333333],[1.0,0.0],[0.83333333,0.16666667],[0.75,0.25],[0.66666667,0.33333333],[0.58333333,0.41666667],[0.91666667,0.5],[1.0,0.5],[0.83333333,0.5],[0.75,0.5],[0.66666667,0.5],[0.58333333,0.5],[0.91666667,0.91666667],[1.0,1.0],[0.83333333,0.83333333],[0.75,0.75],[0.66666667,0.66666667],[0.58333333,0.58333333],[0.41666667,0.41666667],[0.33333333,0.33333333],[0.25,0.25],[0.16666667,0.16666667],[0.08333333,0.08333333],[0.0,0.0]],"edges_vertices":[[0,1],[0,2],[2,3],[3,4],[4,5],[5,6],[7,8],[8,9],[9,10],[10,11],[11,12],[1,12],[13,14],[14,15],[15,16],[16,17],[17,18],[1,18],[1,19],[19,20],[20,21],[21,22],[22,23],[23,24],[25,26],[25,27],[27,28],[28,29],[29,30],[1,30],[31,32],[31,33],[33,34],[34,35],[35,36],[1,36],[37,38],[37,39],[39,40],[40,41],[41,42],[1,42],[1,43],[43 ...
[truncated: the full input is in inputs/Q132/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q133

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q133/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

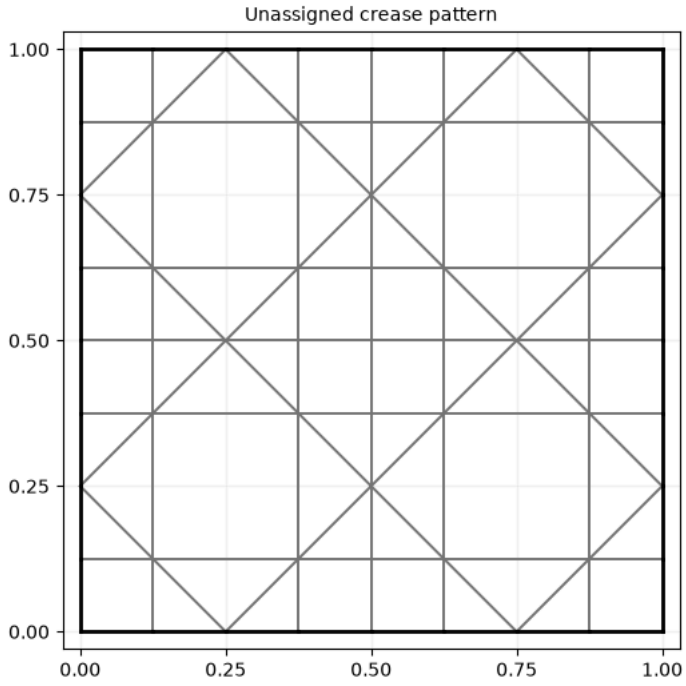
Input `crease_pattern`:


```
{"vertices_coords":[[0.25,0.0],[0.375,0.125],[0.5,0.25],[0.375,0.375],[0.25,0.5],[0.125,0.375],[0.0,0.25],[0.125,0.125],[0.0,0.375],[0.0,0.125],[0.375,0.
0],[0.125,0.0],[0.5,0.125],[0.5,0.375],[0.375,0.5],[0.125,0.5],[0.75,0.0],[0.625,0.125],[0.625,0.375],[0.75,0.5],[0.875,0.375],[1.0,0.25],[0.875,0.125],
[1.0,0.375],[1.0,0.125],[0.625,0.0],[0.875,0.0],[0.625,0.5],[0.875,0.5],[0.25,1.0],[0.375,0.875],[0.5,0.75],[0.375,0.625],[0.125,0.625],[0.0,0.75],[0.125,
0.875],[0.0,0.625],[0.0,0.875],[0.375,1.0],[0.125,1.0],[0.5,0.875],[0.5,0.625],[0.75,1.0],[0.625,0.875],[0.625,0.625],[0.875,0.625],[1.0,0.75],[0.875,0.87
5],[1.0,0.625],[1.0,0.875],[0.625,1.0],[0.875,1.0],[0.5,0.0],[0.5,0.5],[0.5,1.0],[0.0,0.5],[1.0,0.5],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_verti
ces":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[0,7],[5,7],[5,8],[7,9],[1,7],[1,10],[7,11],[1,3],[1,12],[3,13],[3,5],[3,14],[5,15],[16,17],[2,17],[2,18],
[18,19],[19,20],[20,21],[21,22],[16,22],[20,22],[20,23],[22,24],[17,22],[17,25],[22,26],[17,18],[12,17],[13,18],[18,20],[18,27],[20,28],[29,30],[30,31],[3
1,32],[4,32],[4,33],[33,34],[34,35],[29,35],[33,35],[33,36],[35,37],[30,35],[30,38],[35,39],[30,32],[30,40],[32,41],[32,33],[ ...
[truncated: the full input is in inputs/Q133/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q134

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q134/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

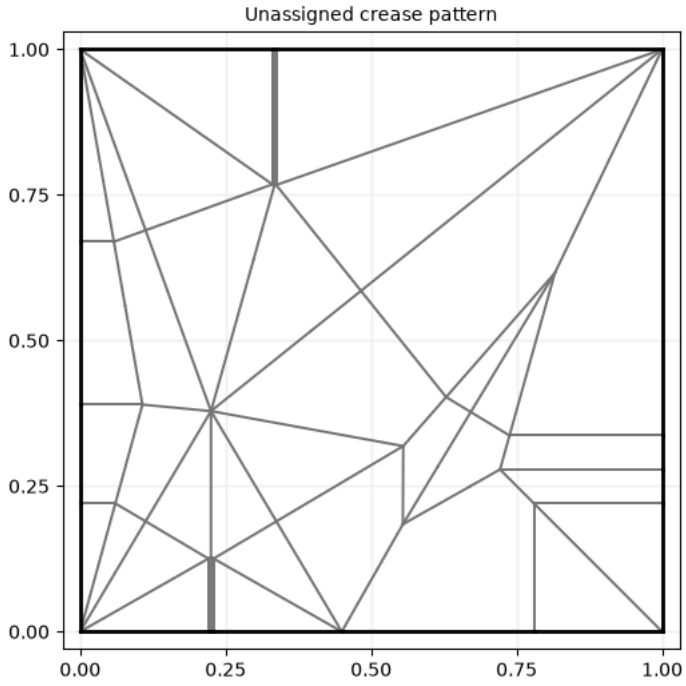
Input `crease_pattern`:


```
{"vertices_coords":[[0.67114,0.3771613],[0.73774143,0.33686288],[0.8155856,0.61588735],[0.72146791,0.27853209],[0.66969919,0.24934811],[0.55451494,0.1844144],[0.61414361,0.28296304],[0.62879913,0.40278043],[0.55509556,0.31869115],[0.48252704,0.58530424],[0.3371409,0.18937529],[0.7797987,0.0],[0.7797987,0.2202013],[1.0,0.0],[1.0,1.0],[1.0,0.2202013],[1.0,0.27853209],[1.0,0.33686288],[0.22478162,0.37875058],[0.3966119,0.51645302],[0.44950018,0.0],[0.11240229,0.68934356],[0.33036601,0.7682076],[0.33686288,0.76706945],[0.33361444,0.76592842],[0.0,1.0],[0.33686288,1.0],[0.33361444,1.0],[0.33036601,1.0],[0.29805749,0.63943291],[0.14986207,0.58581257],[0.11239902,0.18938913],[0.06043101,0.2202312],[0.05792917,0.66963399],[0.07723494,0.55953448],[0.1069831,0.38988284],[0.0,0.0],[0.0,0.66963399],[0.0,0.38988284],[0.0,0.2202312],[0.2202312,0.12539268],[0.22930116,0.12539231],[0.22476618,0.12797475],[0.2202312,0.0],[0.22476618,0.0],[0.22930116,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[5,6],[0,6],[0,2],[0,7],[5,8],[7,8],[2,7],[7,9],[8,10],[11,12],[12,13],[3,12],[2,14],[12,15],[3,16],[1,17],[8,18],[18,19],[9,19],[9,14],[5,20],[10,20],[10,18],[21,22],[9,23],[14,23],[23,24],[22,24],[ ...
[truncated: the full input is in inputs/Q134/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q135

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q135/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

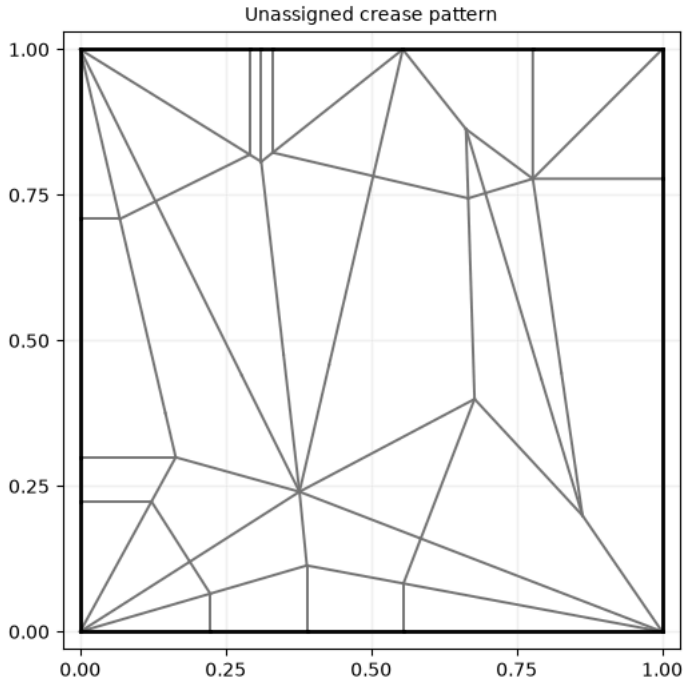
Input `crease_pattern`:


```
{"vertices_coords":[[0.69610678,0.75314504],[0.77777833,0.77777824],[0.6629485,0.86308373],[0.82695228,0.44444444],[0.86308387,0.19952043],[0.79235982,0.43401112],[0.66671561,0.74428046],[0.67765092,0.39941419],[0.77777778,1.0],[0.50349674,0.78241699],[0.58530412,0.1598693],[1.0,0.77777778],[1.0,1.0],[0.55433539,1.0],[0.37681679,0.2402432],[0.42765547,0.45782615],[1.0,0.0],[0.12942802,0.73904078],[0.29129245,0.81932073],[0.33089215,0.82274639],[0.3110923,0.80703953],[0.0,1.0],[0.33089221,1.0],[0.31109236,1.0],[0.29129251,1.0],[0.34946794,0.47609482],[0.27753575,0.44041859],[0.18840843,0.12012159],[0.12253467,0.22344329],[0.06826854,0.70870749],[0.14638995,0.37537417],[0.16419353,0.29940873],[0.0,0.0],[0.0,0.70870749],[0.0,0.29940872],[0.0,0.22344328],[0.22344337,0.06516998],[0.55555556,0.08270247],[0.38949944,0.11360228],[0.22344337,0.0],[0.38949944,0.0],[0.55555556,0.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[4,5],[0,5],[0,2],[0,6],[4,7],[6,7],[2,6],[1,8],[6,9],[7,10],[1,11],[1,12],[2,13],[7,14],[14,15],[9,15],[9,13],[4,16],[10,16],[10,14],[17,18],[9,19],[13,19],[19,20],[18,20],[18,21],[19,22],[20,23],[18,24],[20,25],[14,25],[14,26],[17,26],[17,21],[27,28],[17,29],[21,29],[29, ...
[truncated: the full input is in inputs/Q135/]
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q136

split: core · family: solve · conventions: advanced (front matter) · files: inputs/Q136/

Task (solve, medium):
Below is the geometry of a flat-foldable crease pattern with every crease unassigned ('U'). Assign mountain or valley to every crease so that the whole pattern folds flat (Maekawa and Kawasaki at every vertex, and a valid global layer order).

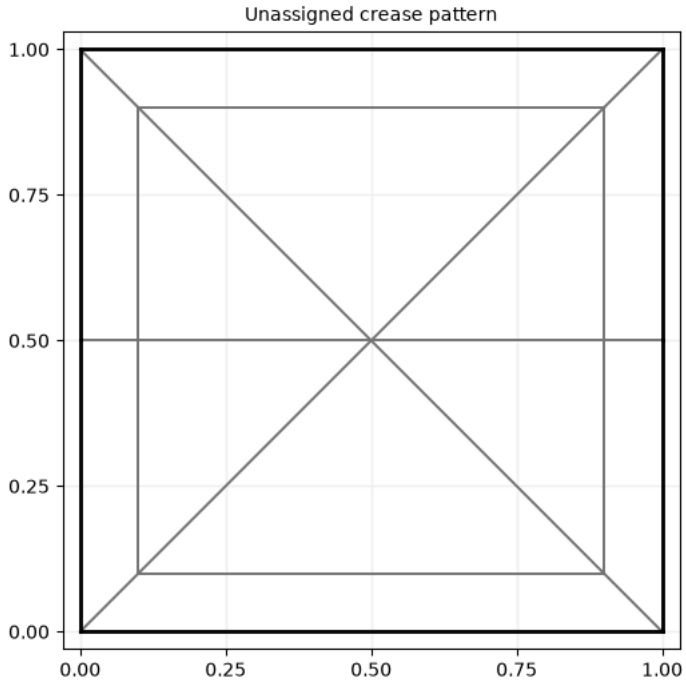
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.1,0.1],[0.5,0.5],[0.9,0.9],[1.0,1.0],[0.9,0.1],[1.0,0.0],[0.1,0.9],[0.0,1.0],[0.1,0.5],[0.0,0.5],[0.9,0.5],[1.0,0.5]],"ed
ges_vertices":[[0,1],[1,2],[2,3],[3,4],[2,5],[5,6],[2,7],[7,8],[2,9],[9,10],[2,11],[11,12],[1,5],[5,11],[3,11],[3,7],[7,9],[1,9],[0,6],[6,12],[4,12],[4,
8],[8,10],[0,10]],"edges_assignment":["U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","U","B","B","B","B","B","B"],"faces_vertices":
[[0,1,9,10],[1,0,6,5],[1,2,9],[2,1,5],[2,3,7],[3,2,11],[3,4,8,7],[4,3,11,12],[2,5,11],[5,6,12,11],[2,7,9],[7,8,10,9]]}
```


Answer format: the SAME creases, each assigned.

```
{"creases": [[x1, y1, x2, y2, "M" | "V"], ...]}
```

Every crease of the input must appear with an assignment, and nothing else may be added.



Working / answer

Q137

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q137/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

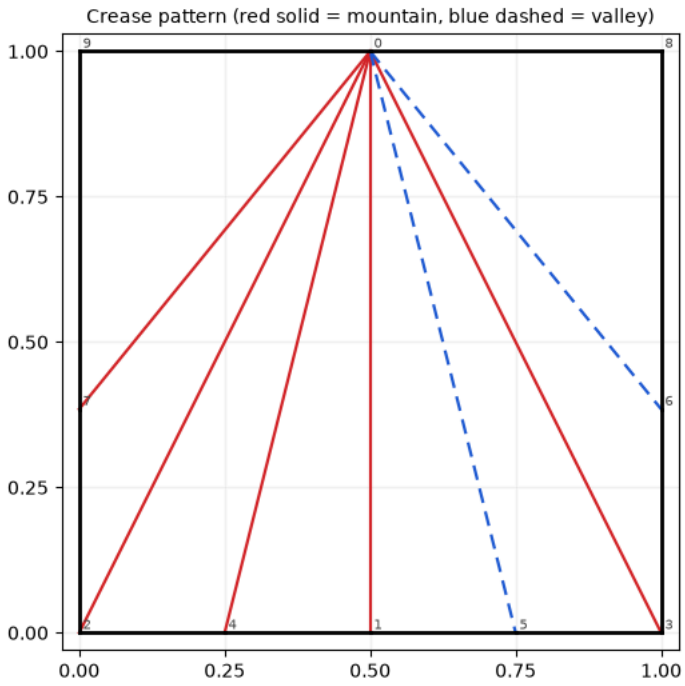
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,1.0],[0.5,0.0],[0.0,0.0],[1.0,0.0],[0.25,0.0],[0.75,0.0],[1.0,0.384615],[0.0,0.384615],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[0,2],[0,3],[0,4],[0,5],[0,6],[0,7],[2,4],[1,4],[1,5],[3,5],[3,6],[6,8],[0,8],[0,9],[7,9],[2,7]],"edges_assignment":["M","M","M","M","V","V","M","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,5],[1,0,4],[0,2,4],[2,0,7],[0,3,6],[3,0,5],[0,6,8],[7,0,9]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q138

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q138/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

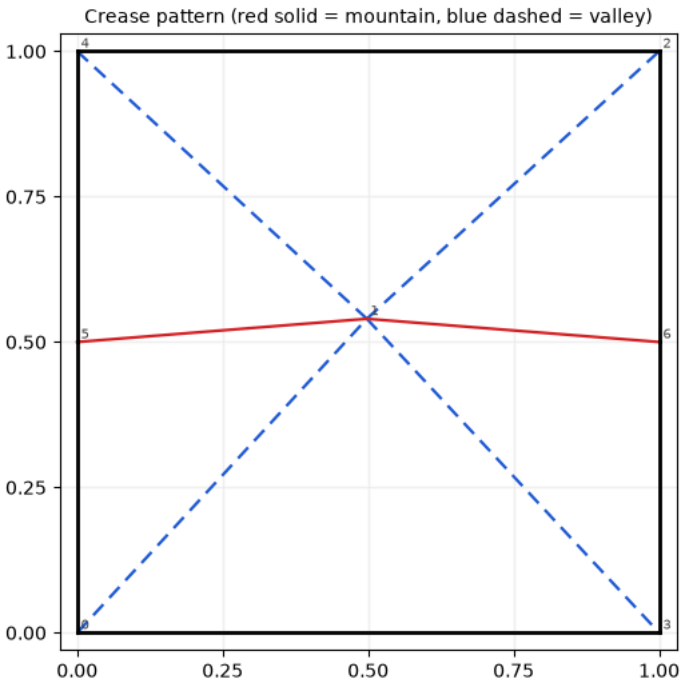
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.49776,0.539937],[1.0,1.0],[1.0,0.0],[0.0,1.0],[0.0,0.5],[1.0,0.5]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[1,5],[1,6],[0,3],[3,6],[2,6],[2,4],[4,5],[0,5]],"edges_assignment":["V","V","V","V","M","M","B","B","B","B","B","B"],"faces_vertices":[[0,1,5],[1,0,3],[1,2,4],[2,1,6],[1,3,6],[1,4,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q139

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q139/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

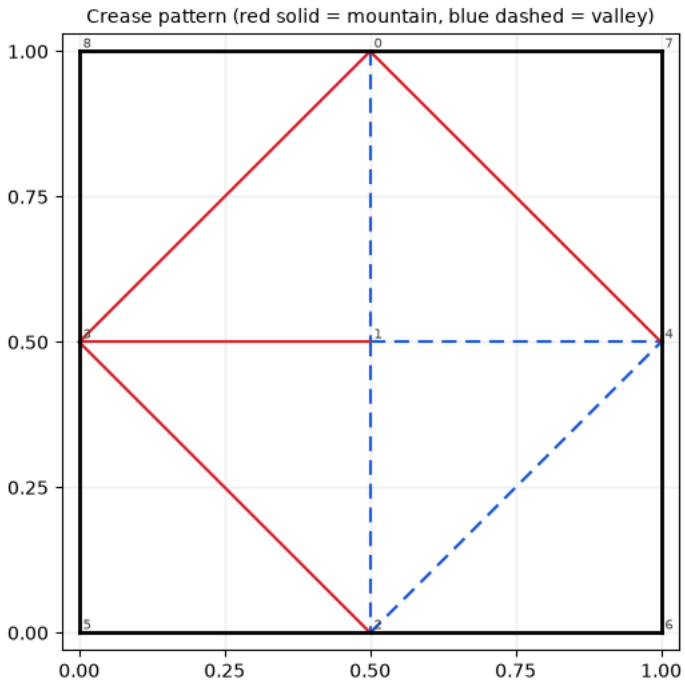
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,1.0],[0.5,0.5],[0.5,0.0],[0.0,0.5],[1.0,0.5],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],
[2,4],[2,3],[0,3],[0,4],[2,5],[2,6],[4,6],[4,7],[0,7],[0,8],[3,8],[3,5]],"edges_assignment":
["V","V","M","V","V","M","M","M","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,4],[1,0,3],[1,2,4],[2,1,3],[4,2,6],[2,3,5],[3,0,8],[0,4,7]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q140

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q140/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

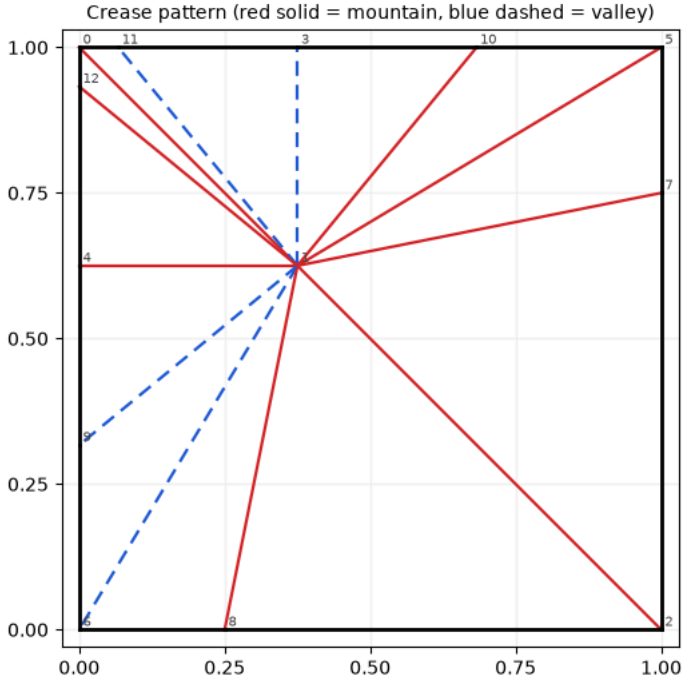
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,1.0],[0.374997,0.625003],[1.0,0.0],[0.374997,1.0],[0.0,0.625003],[1.0,1.0],[0.0,0.0],[1.0,0.750005],[0.249995,0.0],[0.0,0.31716
2],[0.682838,1.0],[0.067155,1.0],[0.0,0.932845]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[1,5],[1,6],[1,7],[1,8],[1,9],[1,10],[1,11],[1,12],[6,8],[2,8],
[2,7],[5,7],[5,10],[3,10],[3,11],[0,11],[0,12],[4,12],[4,9],[6,9]],"edges_assignment":
["M","M","V","M","M","V","M","M","V","M","V","M","B","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,11],[1,0,12],[1,2,7],[2,1,8],[1,
3,11],[3,1,10],[1,4,9],[4,1,12],[1,5,10],[5,1,7],[1,6,8],[6,1,9]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q141

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q141/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

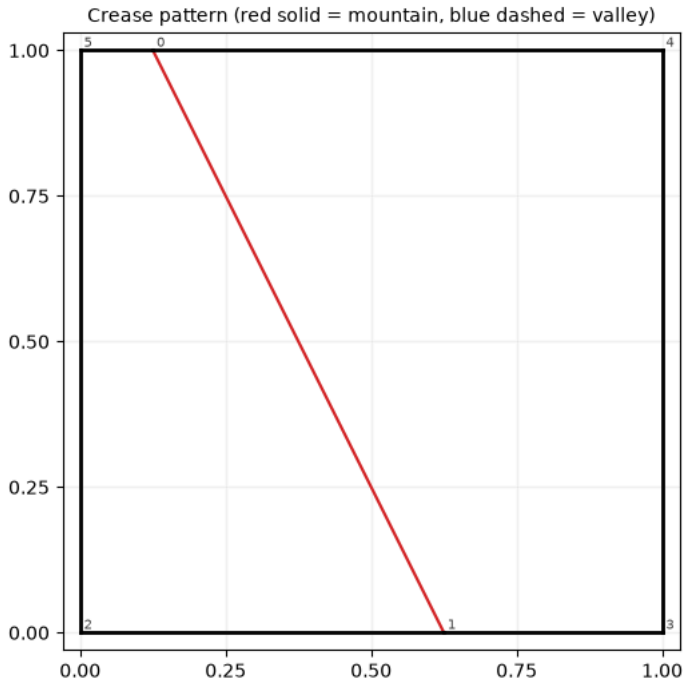
Input `crease_pattern`:


```
{"vertices_coords":[[0.125,1.0],[0.625,0.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[3,4],[0,4],[0,5],[2,5]],"edges_assignment":["M","B","B","B","B","B","B"],"faces_vertices":[[0,1,3,4],[1,0,5,2]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q142

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q142/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

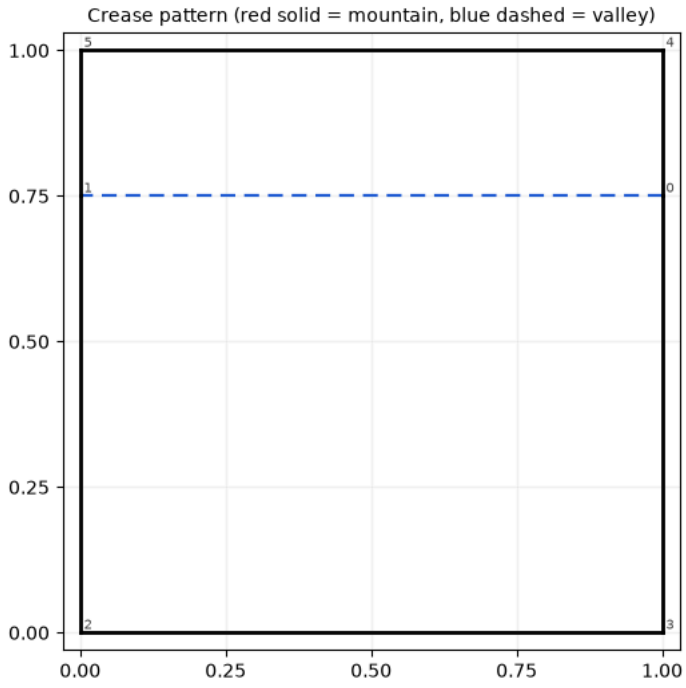
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,0.75],[0.0,0.75],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[2,3],[0,3],[0,4],[4,5],[1,5],[1,2]],"edges_assignment":["V","B","B","B","B","B"],"faces_vertices":[[0,1,2,3],[1,0,4,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q143

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q143/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

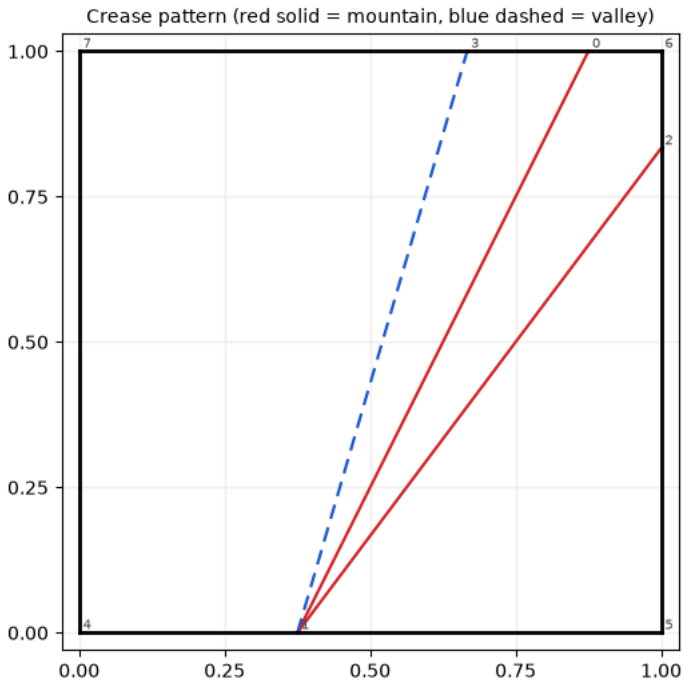
Input `crease_pattern`:


```
{"vertices_coords":[[0.875,1.0],[0.375,0.0],[1.0,0.833333],[0.666667,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[1,5],[2,5],[2,6],[0,6],[0,3],[3,7],[4,7]],"edges_assignment":["M","M","V","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,2,6],[1,0,3],[2,1,5],[1,3,7,4]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q144

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q144/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

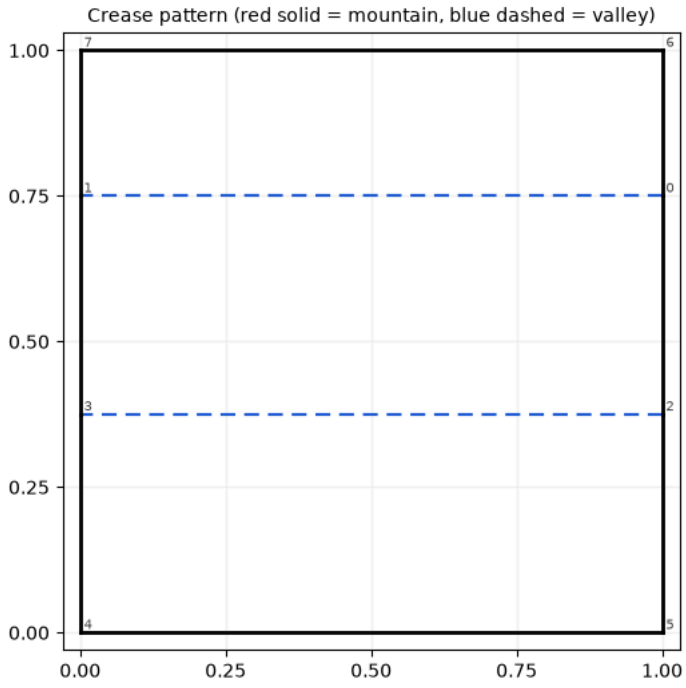
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,0.75],[0.0,0.75],[1.0,0.375],[0.0,0.375],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[2,3],[4,5],[2,5],[0,2],[0,6],[6,7],[1,7],[1,3],[3,4]],"edges_assignment":["V","V","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,3,2],[1,0,6,7],[2,3,4,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q145

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q145/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

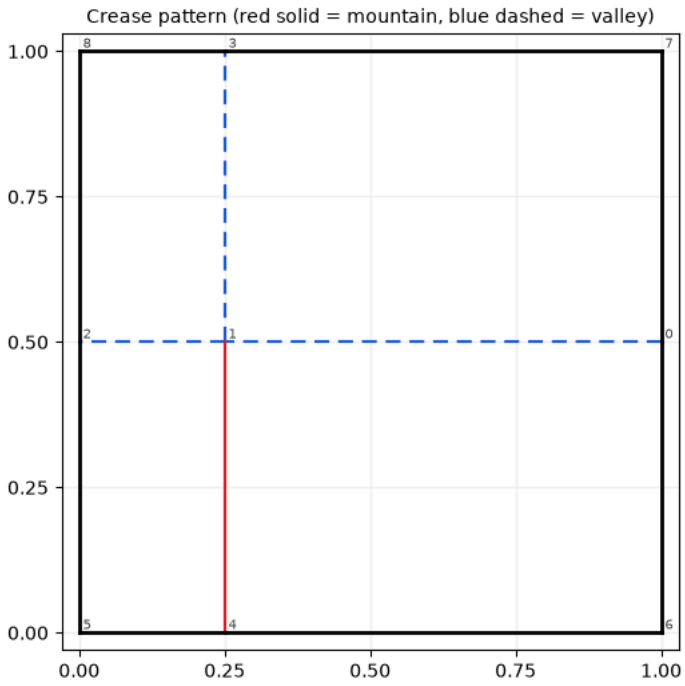
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,0.5],[0.25,0.5],[0.0,0.5],[0.25,1.0],[0.25,0.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[4,5],[4,6],[0,6],[0,7],[3,7],[3,8],[2,8],[2,5]],"edges_assignment":["V","V","V","M","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,4,6],[1,0,7,3],[1,2,5,4],[2,1,3,8]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q146

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q146/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

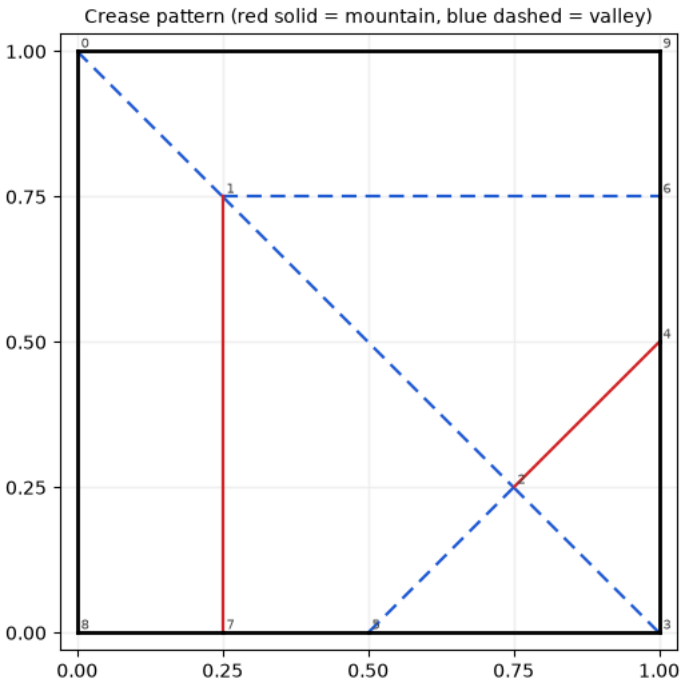
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,1.0],[0.25,0.75],[0.75,0.25],[1.0,0.0],[1.0,0.5],[0.5,0.0],[1.0,0.75],[0.25,0.0],[0.0,0.0],[1.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[2,4],[2,5],[1,6],[1,7],[7,8],[5,7],[3,5],[3,4],[4,6],[6,9],[0,9],[0,8]],"edges_assignment":["V","V","V","M","V","V","M","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,6,9],[1,0,8,7],[1,2,4,6],[2,1,7,5],[2,3,4],[3,2,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q147

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q147/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

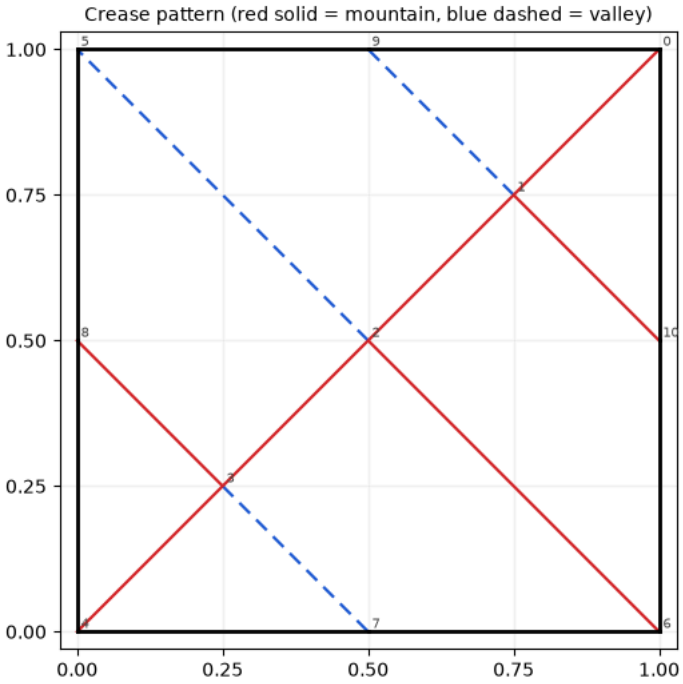
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,1.0],[0.75,0.75],[0.5,0.5],[0.25,0.25],[0.0,0.0],[0.0,1.0],[1.0,0.0],[0.5,0.0],[0.0,0.5],[0.5,1.0],[1.0,0.5]],"edges_vertices":
[[0,1],[1,2],[2,3],[3,4],[2,5],[2,6],[3,7],[3,8],[1,9],[1,10],[4,7],[6,7],[6,10],[0,10],[0,9],[5,9],[5,8],[4,8]],"edges_assignment":
["M","M","M","M","V","M","V","M","V","M","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,10],[1,0,9],[1,2,6,10],[2,1,9,5],[2,3,7,6],[3,2,5,8],[3,
4,7],[4,3,8]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q148

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q148/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

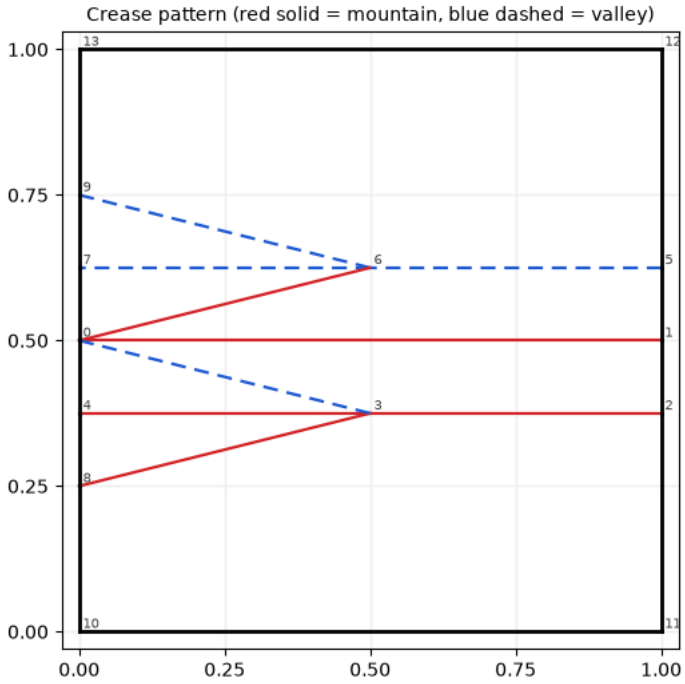
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.5],[1.0,0.5],[1.0,0.375],[0.5,0.375],[0.0,0.375],[1.0,0.625],[0.5,0.625],[0.0,0.625],[0.0,0.25],[0.0,0.75],[0.0,0.0],[1.0,0.0],
[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[2,3],[3,4],[5,6],[6,7],[3,8],[6,9],[0,6],[0,3],[10,11],[2,11],[1,2],[1,5],[5,12],[12,13],[9,13],[7,9],[0,7],
[0,4],[4,8],[8,10]],"edges_assignment":["M","M","M","V","V","M","V","M","V","B","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,5,6],
[1,0,3,2],[2,3,8,10,11],[3,4,8],[4,3,0],[6,5,12,13,9],[6,7,0],[7,6,9]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q149

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q149/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

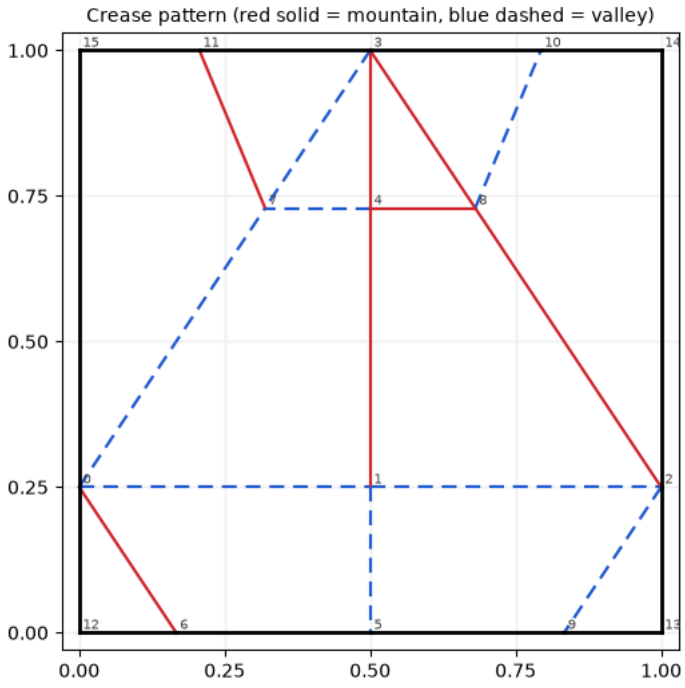
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.25],[0.5,0.25],[1.0,0.25],[0.5,1.0],[0.5,0.729167],[0.5,0.0],[0.166667,0.0],[0.319444,0.729167],[0.680556,0.729167],[0.833333,
0.0],[0.793403,1.0],[0.206597,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[3,4],[1,4],[1,5],[0,6],[0,7],[3,7],[2,8],[3,8],
[2,9],[8,10],[7,11],[4,7],[4,8],[6,12],[5,6],[5,9],[9,13],[2,13],[2,14],[10,14],[3,10],[3,11],[11,15],[0,15],[0,12]],"edges_assignment":
["V","V","M","M","V","M","V","V","M","M","V","V","M","V","M","B","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,4,7],[1,0,6,5],[1,2,
8,4],[2,1,5,9],[3,4,8],[4,3,7],[6,0,12],[0,7,11,15],[7,3,11],[8,2,14,10],[3,8,10],[2,9,13]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q150

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q150/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

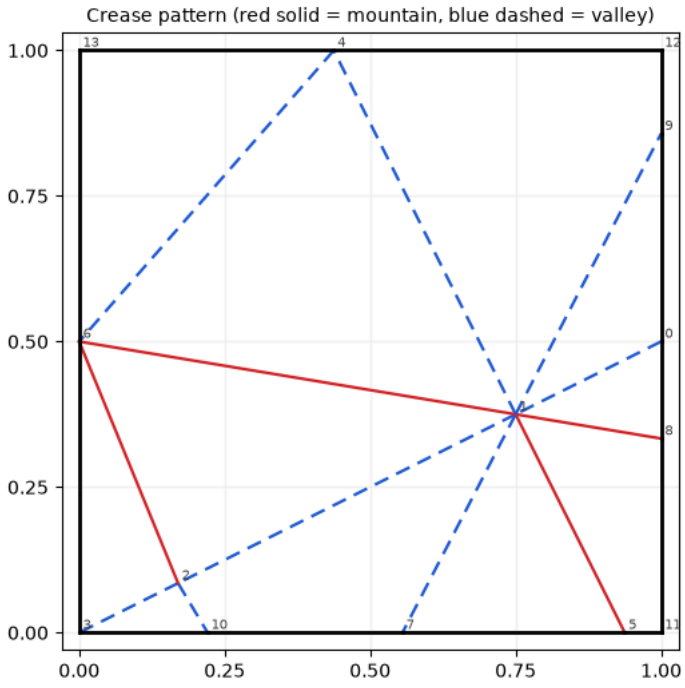
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,0.5],[0.75,0.375],[0.169897,0.084949],[0.0,0.0],[0.4375,1.0],[0.9375,0.0],[0.0,0.5],[0.555556,0.0],[1.0,0.333333],[1.0,0.857143],
[0.220675,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[1,4],[1,5],[1,6],[1,7],[1,8],[1,9],[2,10],[2,6],[4,6],[3,10],[7,10],[5,
7],[5,11],[8,11],[0,8],[0,9],[9,12],[4,12],[4,13],[6,13],[3,6]],"edges_assignment":
["V","V","V","V","M","M","V","M","V","V","M","V","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,8],[1,0,9],[1,2,10,7],[2,1,6],[2,
3,10],[3,2,6],[1,4,6],[4,1,9,12],[1,5,11,8],[5,1,7],[6,4,13]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q151

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q151/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

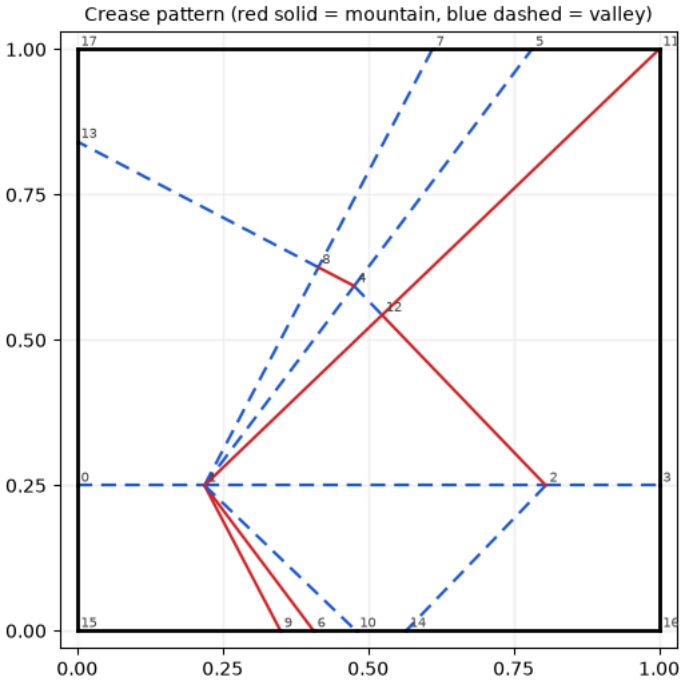
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.25],[0.21875,0.25],[0.805176,0.25],[1.0,0.25],[0.475954,0.592939],[0.78125,1.0],[0.40625,0.0],[0.610352,1.0],[0.414551,0.625],
[0.349284,0.0],[0.479167,0.0],[1.0,1.0],[0.523926,0.542969],[0.0,0.841452],[0.565176,0.0],[0.0,0.0],[1.0,0.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,
3],[1,4],[4,5],[1,6],[7,8],[1,8],[1,9],[1,10],[11,12],[1,12],[8,13],[2,14],[2,12],[4,12],[4,8],[9,15],[6,9],[6,10],[10,14],[14,16],[3,16],[3,11],[5,11],
[5,7],[7,17],[13,17],[0,13],[0,15]],"edges_assignment":
["V","V","V","V","V","M","V","V","M","V","M","M","V","V","M","V","M","B","B","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,8,13],[1,
0,15,9],[1,2,12],[2,1,10,14],[2,3,11,12],[3,2,14,16],[1,4,8],[4,1,12],[4,5,7,8],[5,4,12,11],[1,6,10],[6,1,9],[8,7,17,13]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q152

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q152/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

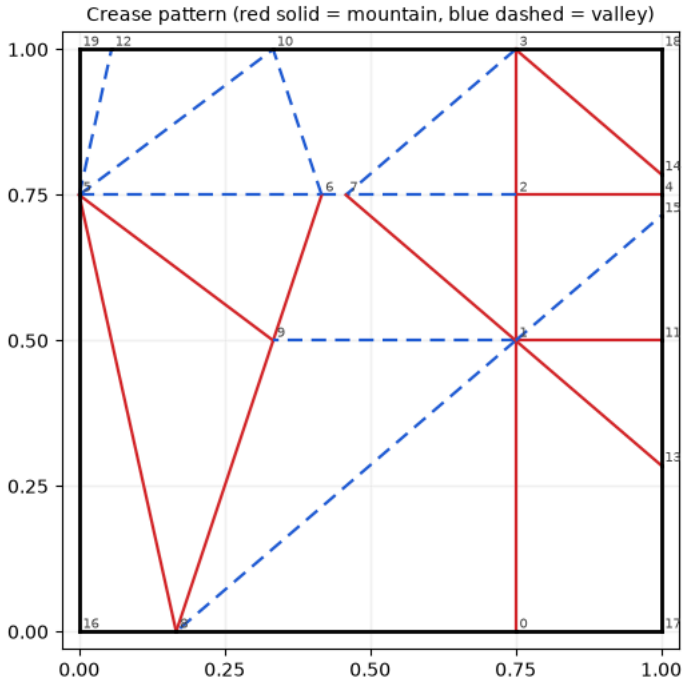
Input `crease_pattern`:


```
{"vertices_coords":[[0.75,0.0],[0.75,0.5],[0.75,0.75],[0.75,1.0],[1.0,0.75],[0.0,0.75],[0.416667,0.75],[0.458333,0.75],[0.166667,0.0],[0.333333,0.5],[0.333333,1.0],[1.0,0.5],[0.055556,1.0],[1.0,0.285714],[1.0,0.785714],[1.0,0.714286],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[2,4],[5,6],[6,7],[2,7],[8,9],[6,9],[6,10],[5,10],[5,9],[1,11],[1,9],[5,12],[5,8],[1,13],[1,8],[3,14],[3,7],[1,7],[1,15],[8,16],[0,8],[0,17],[13,17],[11,13],[11,15],[4,15],[4,14],[14,18],[3,18],[3,10],[10,12],[12,19],[5,19],[5,16]],"edges_assignment":
["M","M","M","M","V","V","V","M","M","V","V","M","M","V","V","M","M","V","M","V","M","V","B","B","B","B","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,8],[1,0,17,13],[1,2,7],[2,1,15,4],[2,3,7],[3,2,4,14],[5,6,10],[6,5,9],[6,7,3,10],[7,6,9,1],[8,9,5],[9,8,1],[5,10,12],[1,11,15],[11,1,13],[5,12,19],[8,5,16],[3,14,18]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q153

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q153/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

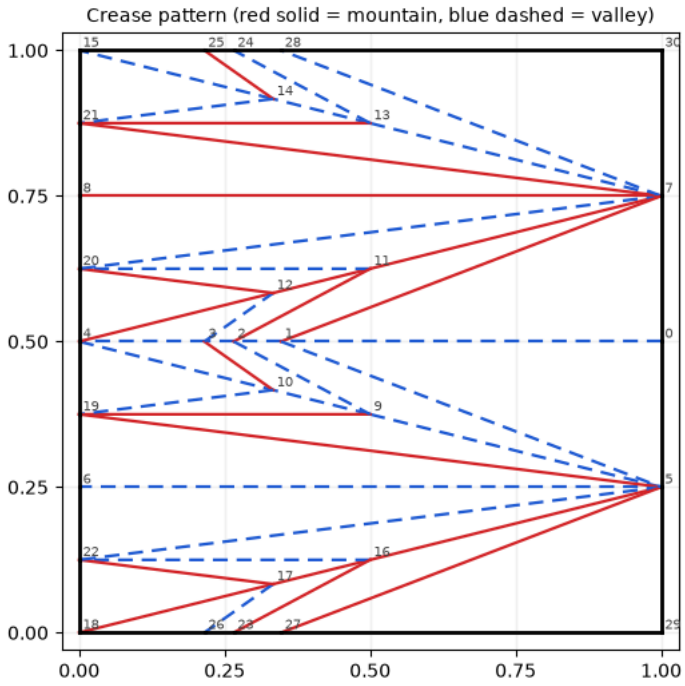
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,0.5],[0.346939,0.5],[0.265623,0.5],[0.215187,0.5],[0.0,0.5],[1.0,0.25],[0.0,0.25],[1.0,0.75],[0.0,0.75],[0.499991,0.375002],[0.33
3325,0.416669],[0.499991,0.624998],[0.333325,0.583331],[0.499991,0.875002],[0.333325,0.916669],[0.0,1.0],[0.499991,0.124998],[0.333325,0.083331],[0.0,0.
0],[0.0,0.375],[0.0,0.625],[0.0,0.875],[0.0,0.125],[0.265623,0.0],[0.265623,1.0],[0.215187,1.0],[0.215187,0.0],[0.346939,0.0],[0.346939,1.0],[1.0,0.0],[1.
0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[5,6],[7,8],[5,9],[9,10],[4,10],[7,11],[11,12],[4,12],[7,13],[13,14],[14,15],[5,16],[16,17],[17,18],[9,1
9],[11,20],[13,21],[16,22],[16,23],[13,24],[2,11],[2,9],[3,10],[3,12],[14,25],[17,26],[17,22],[14,21],[12,20],[10,19],[5,19],[7,20],[7,21],[5,22],[5,27],
[7,28],[1,7],[1,5],[18,26],[23,26],[23,27],[27,29],[5,29],[0,5],[0,7],[7,30],[28,30],[24,28],[24,25],[15,25],[15,21],[8,21],[8,20],[4,20],[4,19],[6,19],
[6,22],[18,22]],"edges_assignment":
["V","V","V","V","V","M","V","V","V","M","M","M","V","V","V","M","M","M","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M",
...
[truncated: the full input is in inputs/Q153/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q154

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q154/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

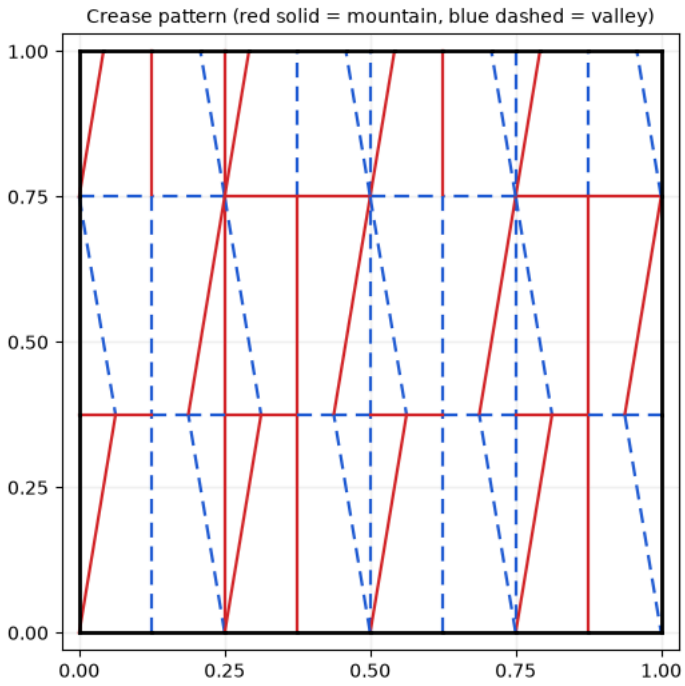
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.375],[0.5,0.75],[0.5,1.0],[0.25,1.0],[0.25,0.75],[0.25,0.375],[0.25,0.0],[0.75,1.0],[0.75,0.75],[0.75,0.375],[0.75,0.0],[1.0,0.75],[0.875,0.75],[0.0,0.75],[0.125,0.75],[0.375,0.75],[0.625,0.75],[0.625,1.0],[0.375,1.0],[0.125,1.0],[0.875,1.0],[0.875,0.0],[0.875,0.375],[0.125,0.0],[0.125,0.375],[0.375,0.0],[0.375,0.375],[0.625,0.0],[0.625,0.375],[0.6875,0.375],[0.3125,0.375],[0.1875,0.375],[0.8125,0.375],[0.9375,0.375],[1.0,0.375],[0.0625,0.375],[0.0,0.375],[0.4375,0.375],[0.5625,0.375],[0.791667,1.0],[0.208333,1.0],[0.291667,1.0],[0.708333,1.0],[0.541667,1.0],[0.458333,1.0],[0.041667,1.0],[0.958333,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[4,5],[5,6],[6,7],[8,9],[9,10],[10,11],[12,13],[9,13],[14,15],[5,15],[2,16],[5,16],[2,17],[9,17],[17,18],[16,19],[15,20],[13,21],[22,23],[13,23],[24,25],[15,25],[26,27],[16,27],[28,29],[17,29],[29,30],[10,30],[27,31],[6,31],[25,32],[6,32],[23,33],[10,33],[23,34],[34,35],[25,36],[36,37],[27,38],[1,38],[29,39],[1,39],[9,30],[5,31],[5,32],[9,33],[9,40],[5,41],[5,42],[9,43],[2,44],[2,45],[14,46],[12,47],[12,34],[14,36],[2,38],[2,39],[0,39],[0,38],[36,48],[34,49],[11,33],[7,32 ...
[truncated: the full input is in inputs/Q154/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q155

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q155/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

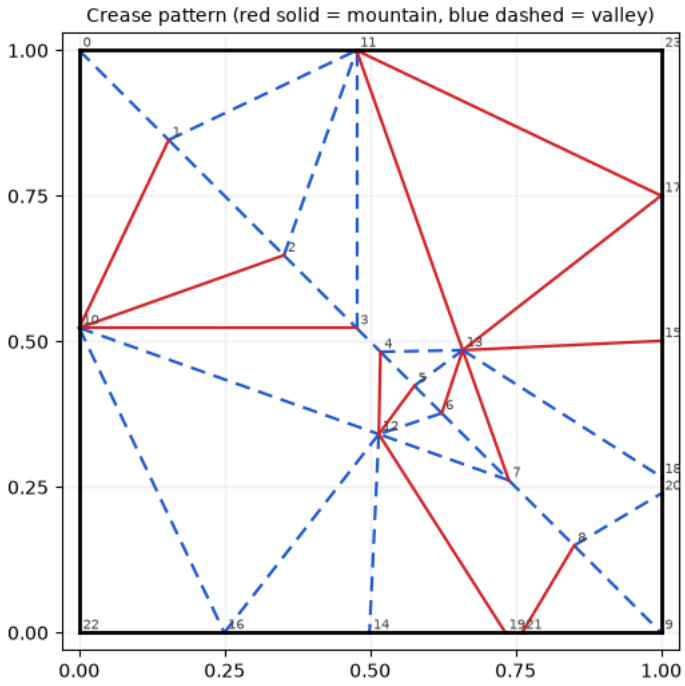
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,1.0],[0.153817,0.846183],[0.351848,0.648152],[0.476574,0.523426],[0.517674,0.482326],[0.576019,0.423981],[0.623054,0.376946],[0.738287,0.261713],[0.850483,0.149517],[1.0,0.0],[0.0,0.523426],[0.476574,1.0],[0.514875,0.34091],[0.65909,0.485125],[0.498902,0.0],[1.0,0.501098],[0.249451,0.0],[1.0,0.750549],[1.0,0.26762],[0.73238,0.0],[1.0,0.23881],[0.76119,0.0],[0.0,0.0],[1.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[7,8],[8,9],[3,10],[3,11],[2,11],[2,10],[10,12],[7,12],[11,13],[7,13],[12,14],[13,15],[5,13],[5,12],[12,16],[13,17],[4,13],[4,12],[6,12],[6,13],[13,18],[12,19],[1,11],[1,10],[10,16],[11,17],[8,20],[8,21],[16,22],[14,16],[14,19],[19,21],[9,21],[9,20],[18,20],[15,18],[15,17],[17,23],[11,23],[0,11],[0,10],[10,22]],"edges_assignment":
["V","V","V","V","V","V","V","V","V","M","V","V","M","V","V","M","M","V","M","V","M","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","V","M","B","B","B","B",
aces_vertices":[[0,1,11],[1,0,10],[1,2,11],[2,1,10],[2,3,11],[3,2,10],[3,4,13,11],[4,3,10,12],[4,5,13],[5,4,12],[5,6,13],[6,5,12],[6,7,13],[7,6,12],[7,8,20,18,13],[8,7,12,19,21],[8,9,20],[9,8,21],[12,10, ...
[truncated: the full input is in inputs/Q155/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q156

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q156/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

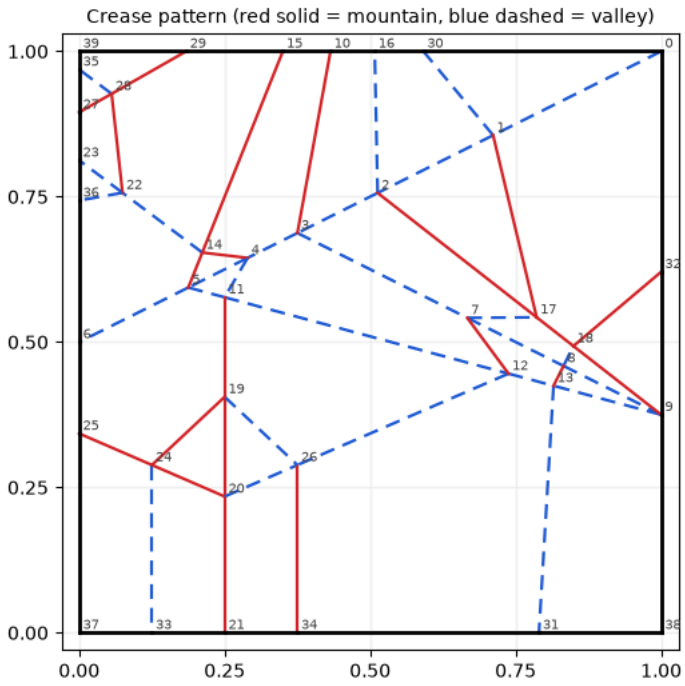
Input `crease_pattern`:


```
{"vertices_coords":[[1.0,1.0],[0.711268,0.855634],[0.512868,0.756434],[0.375,0.6875],[0.289326,0.644663],[0.1875,0.59375],[0.0,0.5],[0.666854,0.541573],
[0.832314,0.458843],[1.0,0.375],[0.431818,1.0],[0.25,0.576923],[0.737682,0.445624],[0.815134,0.424772],[0.211538,0.653846],[0.35,1.0],[0.507801,1.0],[0.78
611,0.54248],[0.849263,0.49303],[0.25,0.405609],[0.25,0.234295],[0.25,0.0],[0.074487,0.756635],[0.0,0.8125],[0.125,0.288462],[0.0,0.342628],[0.375,0.28846
2],[0.0,0.895111],[0.055769,0.926923],[0.183879,1.0],[0.591461,1.0],[0.790058,0.0],[1.0,0.620631],[0.125,0.0],[0.375,0.0],[0.0,0.96875],[0.0,0.743517],[0.
0,0.0],[1.0,0.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[3,7],[7,8],[8,9],[3,10],[5,11],[11,12],[12,13],[9,13],[5,14],[14,15],
[2,16],[2,17],[17,18],[9,18],[11,19],[19,20],[20,21],[14,22],[22,23],[4,14],[4,11],[20,24],[24,25],[20,26],[12,26],[27,28],[28,29],[1,30],[1,17],[7,17],
[7,12],[13,31],[18,32],[8,18],[8,13],[24,33],[26,34],[28,35],[22,28],[19,26],[19,24],[22,36],[33,37],[21,33],[21,34],[31,34],[31,38],[9,38],[9,32],[0,32],
[0,30],[16,30],[10,16],[10,15],[15,29],[29,39],[35,39],[27,35],[23,27],[23,36],[6,36],[6,25],[25,37]],"edges_assignment":["V","V ...
[truncated: the full input is in inputs/Q156/]
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q157

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q157/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

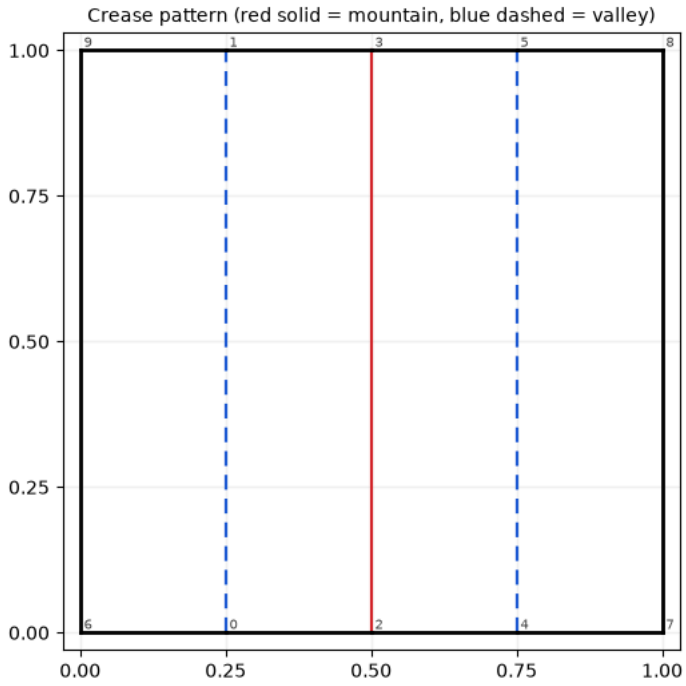
Input `crease_pattern`:


```
{"vertices_coords":[[0.25,0.0],[0.25,1.0],[0.5,0.0],[0.5,1.0],[0.75,0.0],[0.75,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[2,3],[4,5],[0,6],[0,2],[2,4],[4,7],[7,8],[5,8],[3,5],[1,3],[1,9],[6,9]],"edges_assignment":["V","M","V","B","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,9,6],[1,0,2,3],[3,2,4,5],[5,4,7,8]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q158

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q158/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

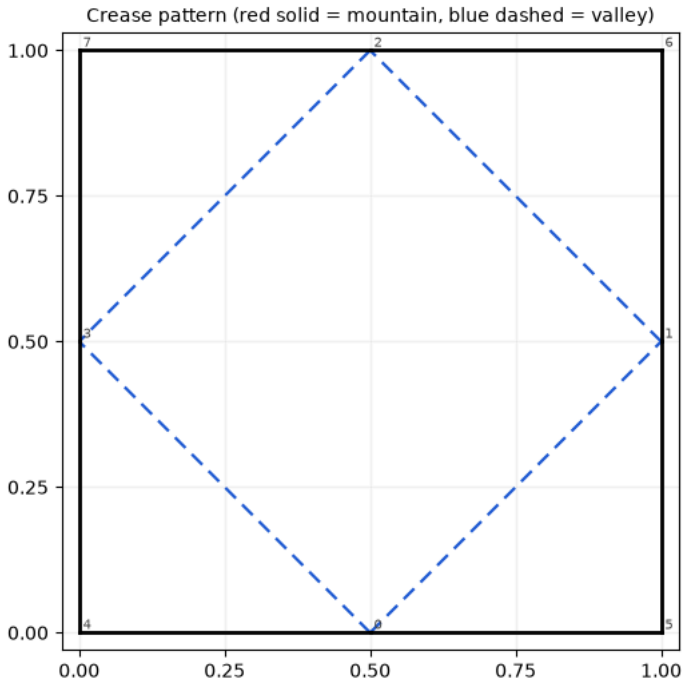
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[1.0,0.5],[0.5,1.0],[0.0,0.5],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[0,3],[0,4],[0,5],[1,5],[1,6],[2,6],[2,7],[3,7],[3,4]],"edges_assignment":["V","V","V","V","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,2,3],[1,0,5],[2,1,6],[3,2,7],[0,3,4]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q159

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q159/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

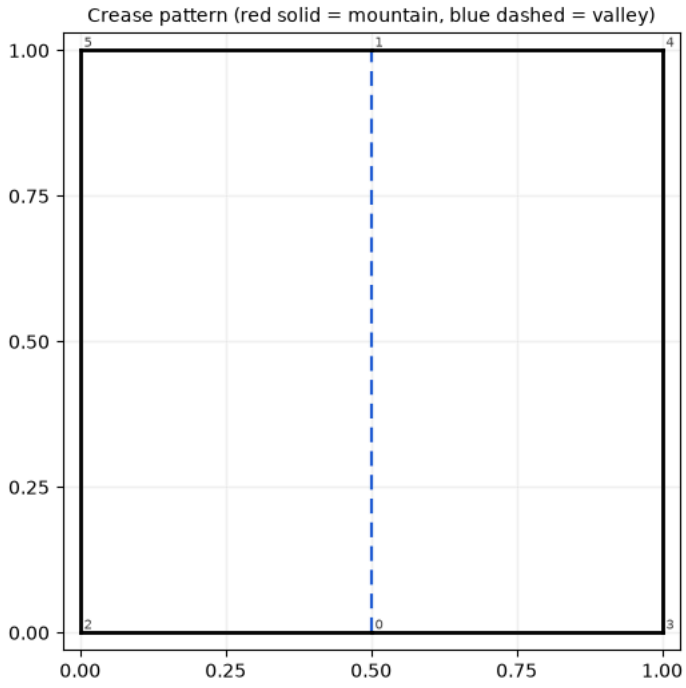
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[0,2],[0,3],[3,4],[1,4],[1,5],[2,5]],"edges_assignment":["V","B","B","B","B","B","B"],"faces_vertices":[[0,1,5,2],[1,0,3,4]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q160

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q160/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

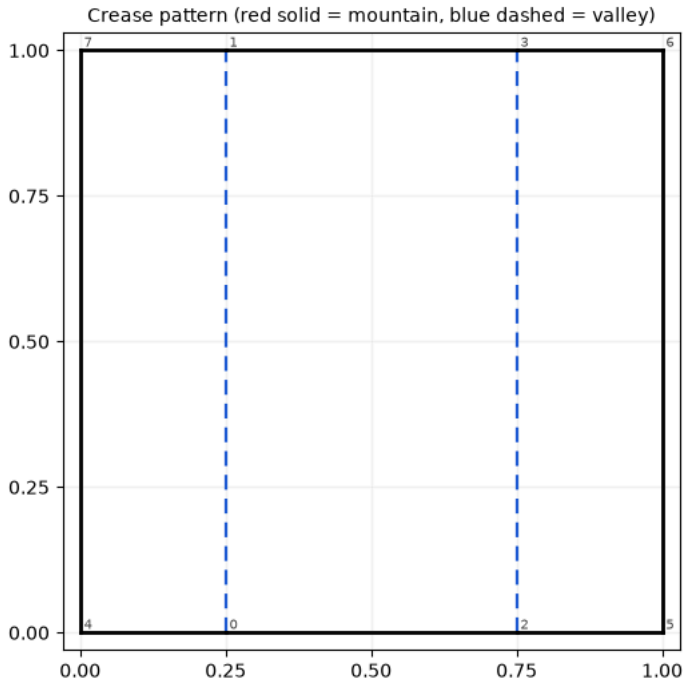
Input `crease_pattern`:


```
{"vertices_coords":[[0.25,0.0],[0.25,1.0],[0.75,0.0],[0.75,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[2,3],[0,4],[0,2],[2,5],
[5,6],[3,6],[1,3],[1,7],[4,7]],"edges_assignment":["V","V","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,7,4],[1,0,2,3],[3,2,5,6]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q161

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q161/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

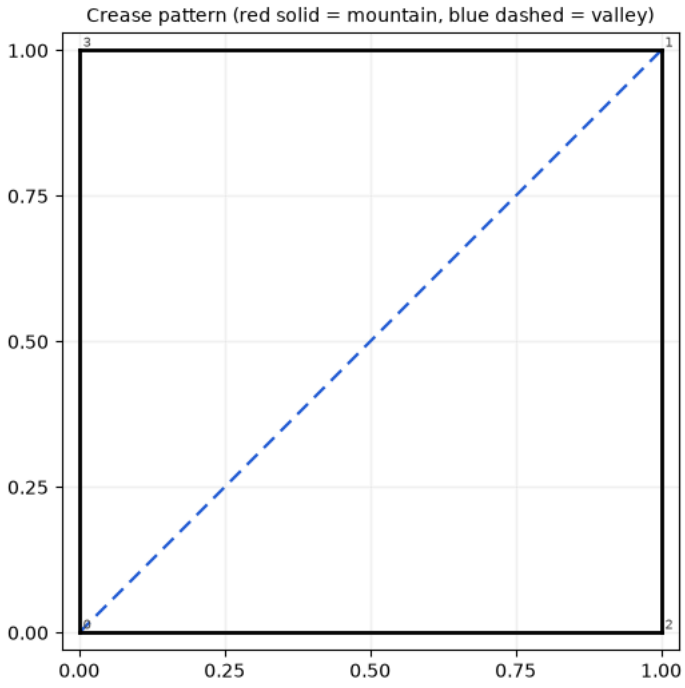
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[1.0,1.0],[1.0,0.0],[0.0,1.0]],"edges_vertices":[[0,1],[0,2],[1,2],[1,3],[0,3]],"edges_assignment":["V","B","B","B","B"],"faces_vertices":[[0,1,3],[1,0,2]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q162

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q162/

Task (analytic, hard):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

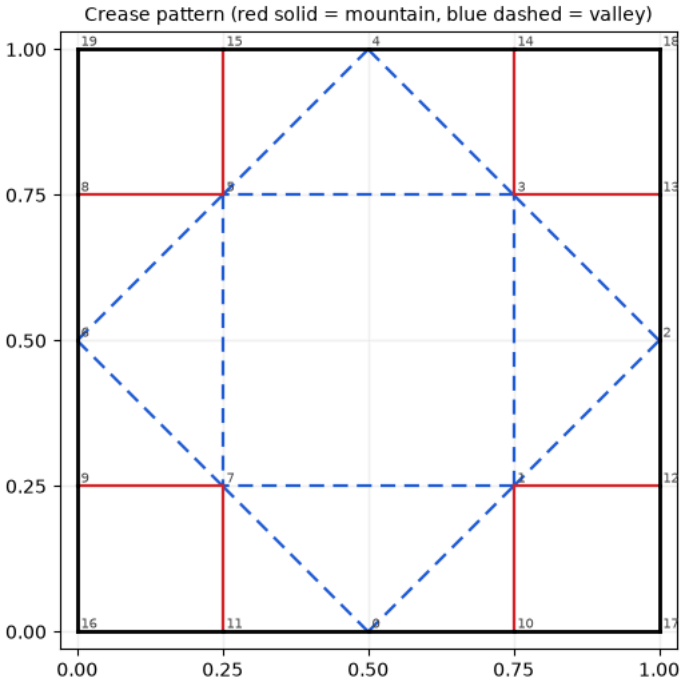
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.75,0.25],[1.0,0.5],[0.75,0.75],[0.5,1.0],[0.25,0.75],[0.0,0.5],[0.25,0.25],[0.0,0.75],[0.0,0.25],[0.75,0.0],[0.25,0.0],
[1.0,0.25],[1.0,0.75],[0.75,1.0],[0.25,1.0],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[2,3],[3,4],[4,5],[5,6],[6,7],[0,7],[5,
7],[5,8],[7,9],[1,7],[1,10],[7,11],[1,3],[1,12],[3,13],[3,5],[3,14],[5,15],[11,16],[0,11],[0,10],[10,17],[12,17],[2,12],[2,13],[13,18],[14,18],[4,14],[4,1
5],[15,19],[8,19],[6,8],[6,9],[9,16]],"edges_assignment":
["V","V","V","V","V","V","V","V","V","M","M","V","M","M","V","M","M","V","M","M","B","B","B","B","B","B","B","B","B","B","B","B","B","B","B","B"],"faces_v
ertices":[[0,1,7],[1,0,10],[1,2,3],[2,1,12],[3,2,13],[3,4,5],[4,3,14],[5,4,15],[5,6,7],[6,5,8],[7,6,9],[0,7,11],[5,7,1,3],[8,5,15,19],[7,9,16,11],[1,10,1
7,12],[3,13,18,14]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q163

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q163/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

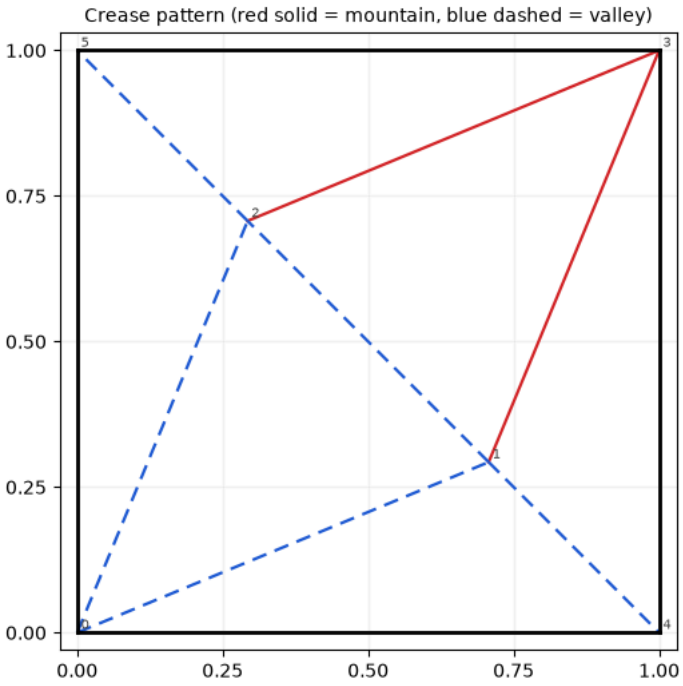
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.707107,0.292893],[0.292893,0.707107],[1.0,1.0],[1.0,0.0],[0.0,1.0]],"edges_vertices":[[0,1],[0,2],[2,3],[1,3],[1,4],[1,2],[2,5],[0,4],[3,4],[3,5],[0,5]],"edges_assignment":["V","V","M","M","V","V","V","B","B","B","B"],"faces_vertices":[[0,1,2],[1,0,4],[0,2,5],[2,3,5],[3,2,1],[3,1,4]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q164

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q164/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

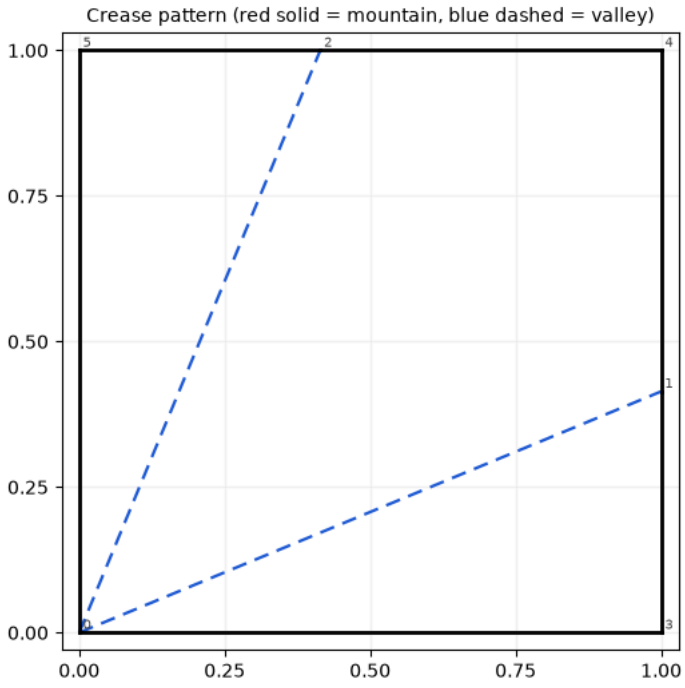
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[1.0,0.414214],[0.414214,1.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[0,2],[0,3],[1,3],[1,4],[2,4],[2,5],[0,5]],"edges_assignment":["V","V","B","B","B","B","B"],"faces_vertices":[[0,1,4,2],[1,0,3],[0,2,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q165

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q165/

Task (analytic, easy):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

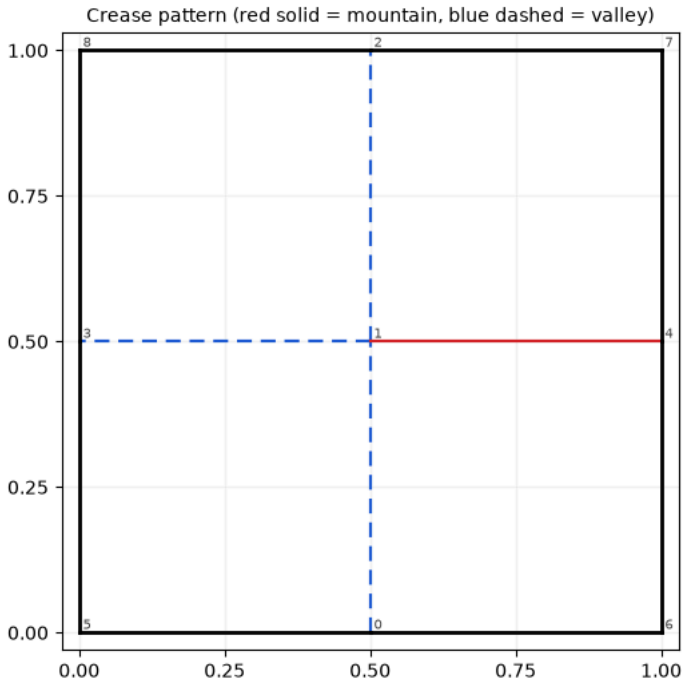
Input `crease_pattern`:


```
{"vertices_coords":[[0.5,0.0],[0.5,0.5],[0.5,1.0],[0.0,0.5],[1.0,0.5],[0.0,0.0],[1.0,0.0],[1.0,1.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[0,5],[0,6],[4,6],[4,7],[2,7],[2,8],[3,8],[3,5]],"edges_assignment":["V","V","V","M","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,3,5],[1,0,6,4],[1,2,8,3],[2,1,4,7]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q166

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q166/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

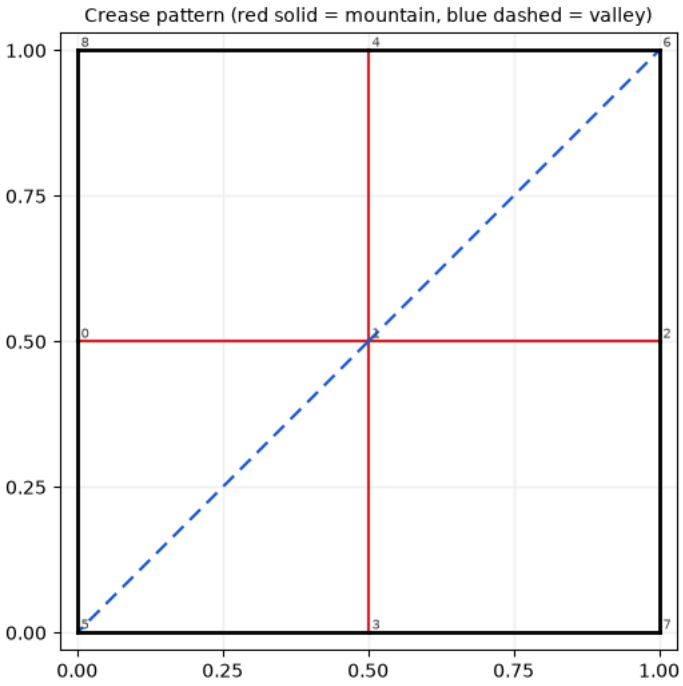
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.5],[0.5,0.5],[1.0,0.5],[0.5,0.0],[0.5,1.0],[0.0,0.0],[1.0,1.0],[1.0,0.0],[0.0,1.0]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[1,5],[1,6],[3,5],[3,7],[2,7],[2,6],[4,6],[4,8],[0,8],[0,5]],"edges_assignment":["M","M","M","M","V","V","B","B","B","B","B","B","B","B"],"faces_vertices":[[0,1,4,8],[1,0,5],[1,2,6],[2,1,3,7],[3,1,5],[4,1,6]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q167

split: warmup · family: analytic · conventions: basic (front matter) · files: inputs/Q167/

Task (analytic, medium):
Below is a crease pattern (also shown as an image). Fold every crease flat and report the folded model: the folded position of every vertex and, if you can, the layer order of overlapping faces. If the pattern cannot be folded flat, say so instead.

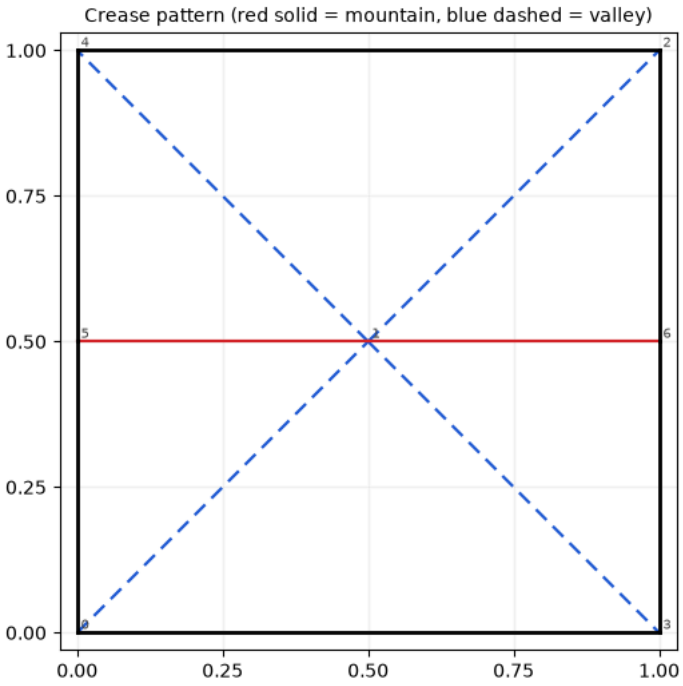
Input `crease_pattern`:


```
{"vertices_coords":[[0.0,0.0],[0.5,0.5],[1.0,1.0],[1.0,0.0],[0.0,1.0],[0.0,0.5],[1.0,0.5]],"edges_vertices":[[0,1],[1,2],[1,3],[1,4],[1,5],[1,6],[0,3],[3,6],[2,6],[2,4],[4,5],[0,5]],"edges_assignment":["V","V","V","V","M","M","B","B","B","B","B","B"],"faces_vertices":[[0,1,5],[1,0,3],[1,2,4],[2,1,6],[1,3,6],[1,4,5]]}
```


Answer format:

```
{
  "foldable": true,
  "folded_vertices": [[x, y], ...],
  "face_above": [[f, g], ...],
  "silhouette": [[x, y], ...]
}
```

- `folded_vertices`: folded position of every CP vertex, in the same order as `vertices_coords` (main score).
- `face_above`: optional layer order for overlapping face pairs (bonus; full credit requires every overlapping pair).
- `silhouette`: optional outline polygon of the folded model; only used if `folded_vertices` is missing (worth at most 75% of the shape score).
- If the CP CANNOT be flat-folded, answer `{"foldable": false, "violation_vertices": [i, ...]}` listing the vertices where a local flat-foldability condition fails (empty list if the failure is global).



Working / answer

Q168

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q168/

Task (design_image, easy):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

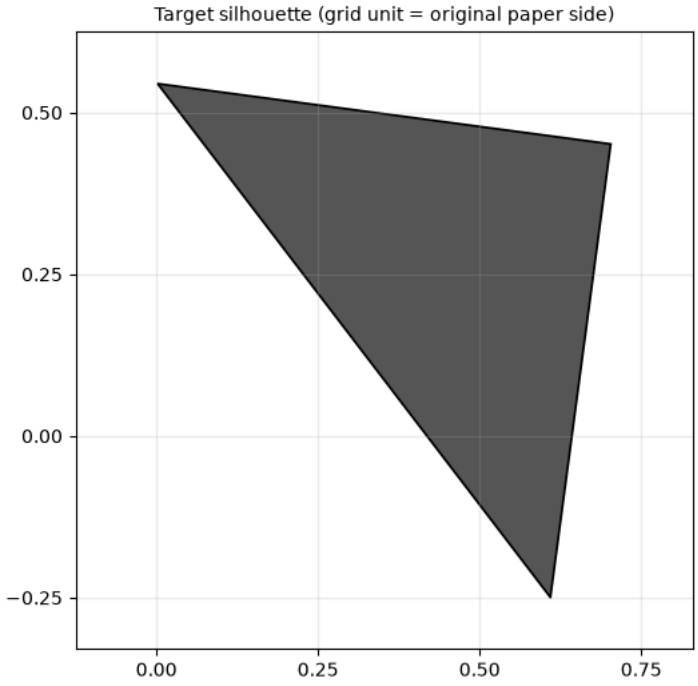
Input `target_silhouette`:

```
[[0.001836,0.54465],[0.001836,0.54465],[0.70276,0.451348],[0.70276,0.451348],[0.70276,0.451348],[0.609457,-0.249576],[0.609457,-0.249576],[0.609457,-0.249576],[0.609457,-0.249576],[0.609457,-0.249576],[0.00184,0.544645],[0.001836,0.54465]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q169

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q169/

Task (design_image, medium):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

Input `target_silhouette`:


```
[[0.220157,0.288229],[0.220157,0.288229],[0.220157,0.288229],[0.220157,0.288229],[0.220157,0.288229],[0.220157,0.288229],[0.220157,0.288229],[0.351155,1.067869],[0.351155,1.067869],[0.351155,1.067869],[0.351155,1.067869],[0.351155,1.067869],[0.351155,1.067869],[0.689211,1.171398],[0.689211,1.171398],[0.689211,1.171398],[0.689211,1.171398],[0.689211,1.171398],[0.555196,0.919064],[0.555196,0.919064],[0.555196,0.919064
```

Q170

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q170/

Task (design_image, hard):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

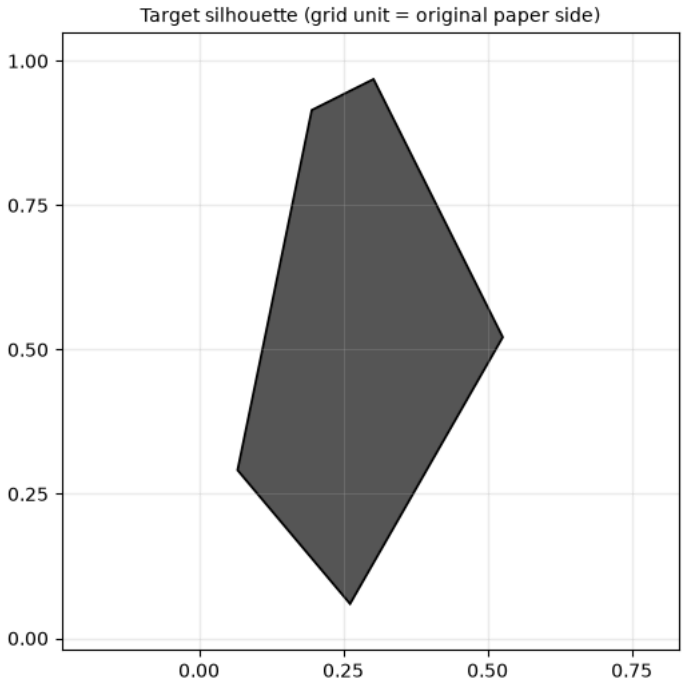
Input `target_silhouette`:

```
[[0.195332,0.91481],[0.195332,0.91481],[0.195332,0.91481],[0.302474,0.968489],[0.302474,0.968489],[0.302474,0.968489],[0.302474,0.968489],[0.302474,0.968489],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.526443,0.521457],[0.462716,0.410218],[0.41458,0.326193],[0.28063,0.092375],[0.261774,0.059462],[0.261774,0.059462],[0.261774,0.059462],[0.261774,0.059462],[0.143924,0.199465],[0.143924,0.199465],[0.143924,0.199465],[0.143924,0.199465],[0.143924,0.199465],[0.143924,0.199465],[0.143924,0.199465],[0.066751,0.291145],[0.066751,0.291145],[0.066751,0.291145],[0.066751,0.291145],[0.066751,0.291145],[0.195332,0.91481]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q171

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q171/

Task (design_image, hard):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

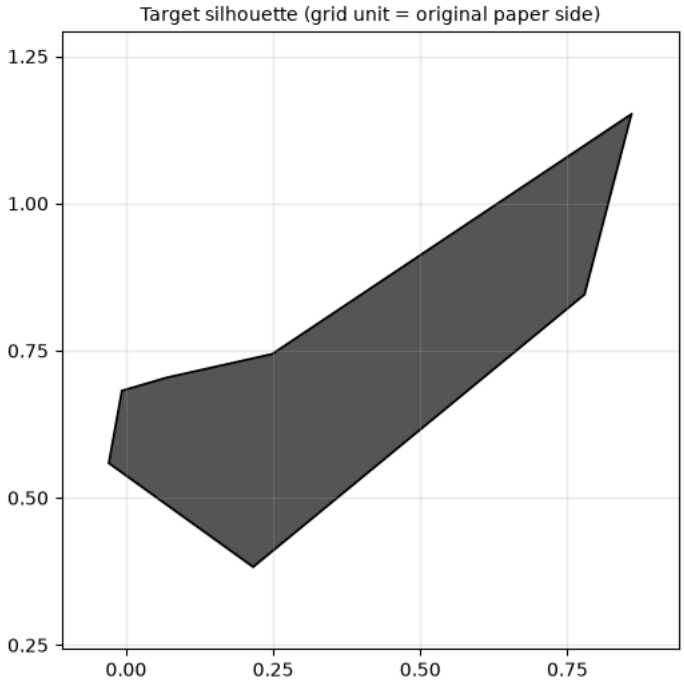
Input `target_silhouette`:

```
[[0.858919,1.151887],[0.858919,1.151887],[0.858919,1.151887],[0.858919,1.151887],[0.858919,1.151884],[0.779095,0.845289],[0.779095,0.845289],[0.779095,0.845289],[0.779095,0.845289],[0.779095,0.845289],[0.779095,0.845289],[0.779095,0.845289],[0.779094,0.845288],[0.705206,0.784568],[0.705206,0.784568],[0.705206,0.784568],[0.613716,0.709382],[0.613716,0.709382],[0.613716,0.709382],[0.613716,0.709382],[0.613716,0.709382],[0.613716,0.709382],[0.237968,0.400593],[0.215974,0.382518],[0.215974,0.382518],[0.215974,0.382518],[0.215974,0.382518],[0.215974,0.382518],[0.215974,0.382518],[0.215974,0.382518],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.02934,0.558911],[-0.007095,0.681916],[-0.007095,0.681916],[-0.007095,0.681916],[-0.007095,0.681916],[-0.007095,0.681916],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.704208],[0.069069,0.70 ...
[truncated: the full input is in inputs/Q171/]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q172

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q172/

Task (design_image, medium):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

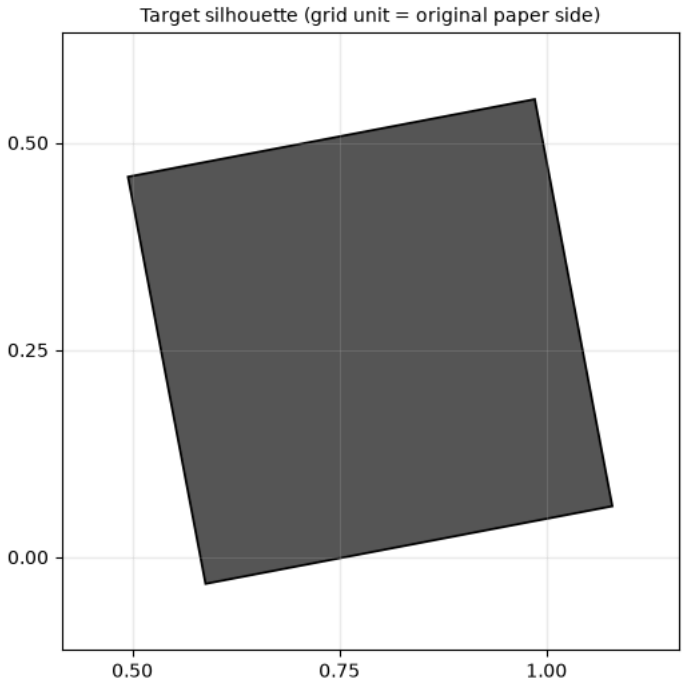
Input `target_silhouette`:

```
[[0.833928,0.014532],[0.833928,0.014532],[0.588351,-0.032287],[0.588351,-0.032287],[0.588351,-0.032287],[0.541532,0.21329],[0.541532,0.21329],[0.494714,0.458867],[0.494714,0.458867],[0.494714,0.458867],[0.74029,0.505685],[0.74029,0.505685],[0.985867,0.552504],[0.985867,0.552504],[0.985867,0.552504],[1.032686,0.306927],[1.032686,0.306927],[1.079505,0.06135],[1.079505,0.06135],[1.079505,0.06135]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q173

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q173/

Task (design_image, medium):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

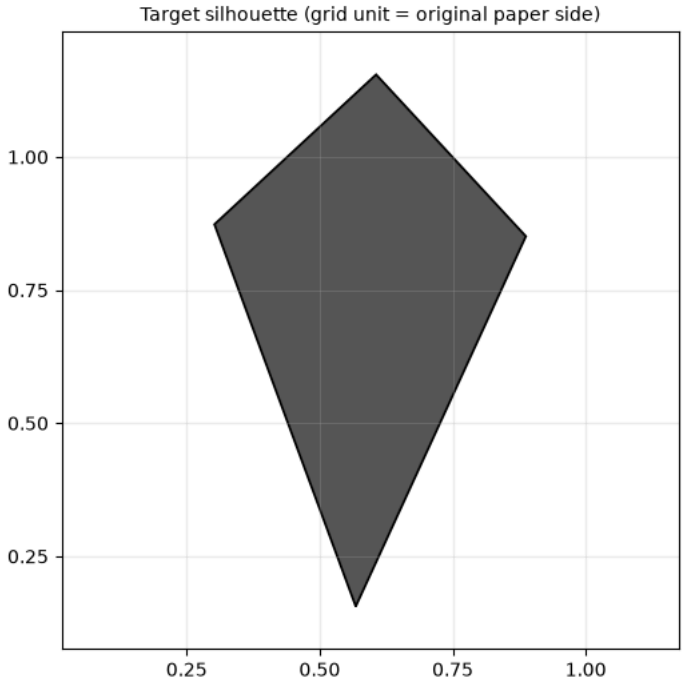
Input `target_silhouette`:

```
[[0.567769,0.155574],[0.302168,0.873378],[0.302168,0.873378],[0.302168,0.873378],[0.606062,1.154841],[0.606062,1.154841],[0.606062,1.154841],[0.887525,0.850946],[0.887525,0.850946],[0.887525,0.850946],[0.567769,0.155574],[0.567769,0.155574]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q174

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q174/

Task (design_image, easy):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

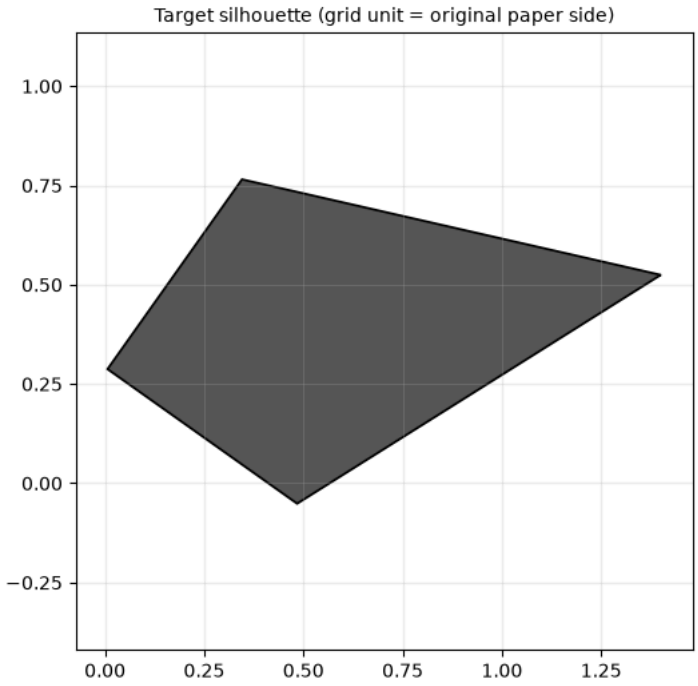
Input `target_silhouette`:


```
[[1.40045,0.52466],[1.40045,0.52466],[1.40045,0.52466],[0.484075,-0.051387],[0.484075,-0.051387],[0.484075,-0.051387],[0.484075,-0.051387],[0.006265,0.287498],[0.006265,0.287498],[0.006265,0.287498],[0.345149,0.765309],[0.345149,0.765309],[0.345149,0.765309],[1.40045,0.52466],[1.40045,0.52466],[1.40045,0.52466]]
```


Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q175

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q175/

Task (design_image, medium):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

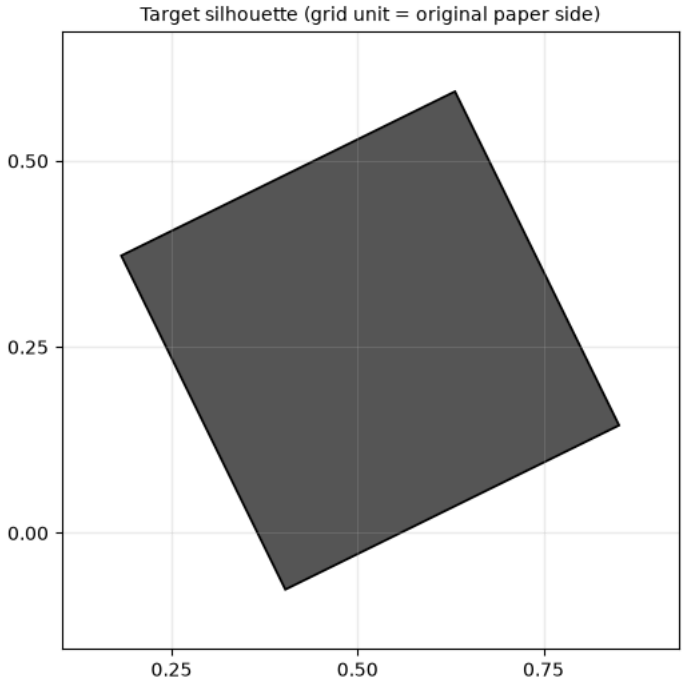
Input `target_silhouette`:

```
[[0.18235,0.372706],[0.18235,0.372706],[0.631043,0.593328],[0.631043,0.593328],[0.631043,0.593328],[0.851665,0.144635],[0.851665,0.144635],[0.851665,0.144635],[0.851665,0.144635],[0.402972,-0.075987],[0.402972,-0.075987],[0.402972,-0.075987],[0.18235,0.372706]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q176

split: warmup · family: design_image · conventions: basic (front matter) · files: inputs/Q176/

Task (design_image, easy):
The image shows the outline (silhouette) of a flat-folded origami model, folded from a unit-square sheet; the grid unit equals the side of the original paper. The outline's polygon coordinates are also given (its position and rotation are arbitrary). Design a crease pattern that folds into this shape. You are graded on the overlap (IoU) between your folded silhouette and the target, after the best rigid alignment (mirror images allowed); scale is NOT normalised.

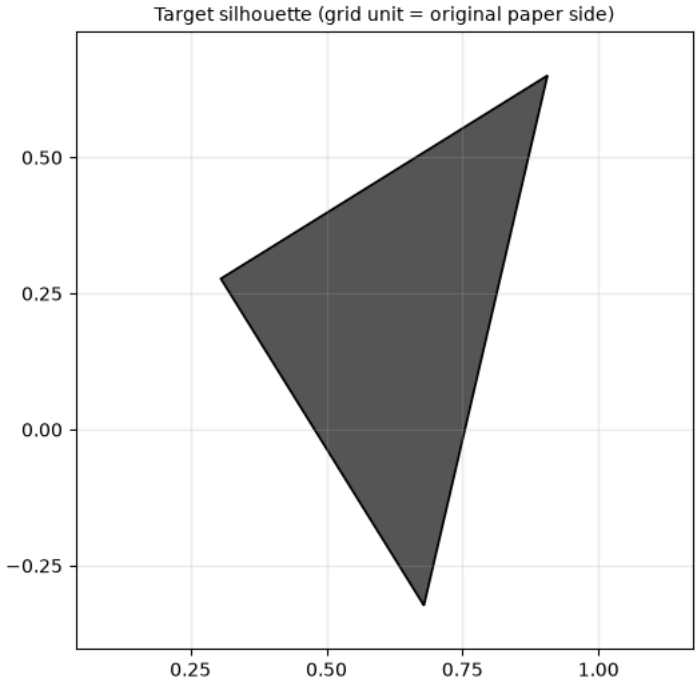
Input `target_silhouette`:

```
[[0.679113,-0.323375],[0.679113,-0.323375],[0.305777,0.277141],[0.305777,0.277141],[0.305777,0.277141],[0.906293,0.650478],[0.906293,0.650478],[0.906293,
0.650478],[0.792703,0.163551],[0.679113,-0.323375]]
```

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.



Working / answer

Q177

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q177/

Task (design_text, hard):
Design an arrow: a long, mirror-symmetric shape with a pointed arrowhead at one end that is wider than the shaft behind it.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q178

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q178/

Task (design_text, easy):
Fold the paper so that the corner at (0, 0) lands exactly on the centre of the paper, (0.5, 0.5). The folded model must have an area of exactly 7/8 of the paper.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q179

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q179/

Task (design_text, medium):
Fold the paper into a square with one quarter of the paper's area, in such a way that all four corners of the paper meet at a single point.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q180

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q180/

Task (design_text, medium):
Fold the paper into an equilateral triangle with an area of at least 0.3.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q181

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q181/

Task (design_text, hard):
Design a simple flat fish: an elongated, mirror-symmetric body with a pointed head or tail and at least one notch or fin making the outline non-convex.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q182

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q182/

Task (design_text, medium):
Fold the paper into the outline of a simple house: a convex, mirror-symmetric pentagon made of a rectangular wall with a triangular roof on top. The area must be at least 0.4.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q183

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q183/

Task (design_text, medium):
Fold the paper into a long thin rectangular strip at least 0.95 long and at most 0.2 wide.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q184

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q184/

Task (design_text, easy):
Fold the paper into a right isosceles triangle whose area is exactly one quarter of the paper.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer

Q185

split: warmup · family: design_text · conventions: basic (front matter) · files: inputs/Q185/

Task (design_text, hard):
Fold the paper into a regular hexagon with an area of at least 0.4.

Answer format: a crease pattern on the unit square, as a list of straight creases.

```
{
  "creases": [[x1, y1, x2, y2, "M"], [x1, y1, x2, y2, "V"], ...]
}
```

- Only list creases that are folded in the final model; the paper boundary is added automatically.
- Creases may cross; they are split at intersections. Creases must end on the paper edge or on other creases (dangling crease ends are removed). Endpoints closer than 1e-4 are merged.
- Assignments must be "M" or "V" ("U" is accepted but costs 10% of the score).
- The pattern is graded by actually folding it: it must be flat-foldable (Maekawa, Kawasaki and a valid layer order), otherwise the score is 0.

Working / answer